Abstract

Efficient Hash- and Lattice-based Proof Systems for Mixed Algebras

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Succinct, Non-interactive Arguments of Knowledge (SNARKs) produce small proofs that can be verified quickly while attesting to the correctness of arbitrary computations. First deployed in 2016, SNARKs have enabled private transactions, decentralized storage, anonymous credentials (e.g., private age verification), proofs about network sessions (e.g., email and TLS session contents), and more. This dissertation advances SNARKs in two ways: (1) it develops arithmetization techniques and interactive oracle proofs (IOPs) for mixed- and small-domain computations, enabling constraints to be written over their natural domains and reducing non-native overhead; and (2) it gives efficient post-quantum (hash- and lattice-based) polynomial commitments (PCs) for these specialized domains.

Zinc+ is a framework for proving computations specified across multiple rings and ideals simultaneously. We define the Universal Constraint System (UCS), capturing constraints over $\mathbb{Z}, \mathbb{Q}$, finite fields, and polynomial rings together with ideal-membership predicates. The Zinc+ compiler lifts finite-field polynomial IOPs (PIOPs) into round-by-round knowledge-sound PIOPs for UCS. We instantiate it with a new IOP of proximity, Zip+, that supports mixed-domain coefficients. *Microlotus* accelerates SNARKs in small-modulus fields and cyclotomic rings. We introduces a packed variant of sumcheck for small fields and a finite-field IOP for cyclotomic ring multiplication that, unlike previous results, has native prover complexity. We also study extensions of linear codes and proximity testing from fields to rings. Finally, *Orbweaver* gives a lattice-based functional commitment for linear relations over cyclotomic rings and the integers, achieving $\tilde{O}_\lambda(\log n)$ verification time and supporting public aggregation. We show it yields univariate and multilinear

PCs, and a SNARK directly.

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Per scientiam ad justitiam

Chapter 1

Introduction

1.1 Motivation

Note on Authorship, Scope of Included Material, and Use of AI Tools

Each chapter of this dissertation is adapted from joint work with co-authors, in which both the writing and the technical development were collaborative. In preparing this dissertation, I have therefore distinguished between expository material and core technical development. The core technical sections included here are limited to material of which I was the sole author or principal author, so that the dissertation's principal technical content reflects my primary research and writing contributions. By contrast, although I contributed to all of the introduction and technical overview sections included here, I was not the principal author of all of them. I have included this expository material to preserve the motivation, structure, and overall technical narrative of the underlying joint works, so that each chapter remains intelligible in context even where the detailed technical presentation is limited to portions that principally reflect my own authorship.

In preparing this manuscript, I used an AI-based tool (Claude) for limited purposes such as grammar review, editing assistance, and latex code-generation assistance. I bear full responsibility for the accuracy, originality, and proper attribution of all content in this dissertation.

Chapter 2

Zinc+: SNARKs for Polynomial Rings

2.1 Introduction

Succinct Non-interactive Arguments of Knowledge (SNARKs) are proof systems that allow a prover to succinctly demonstrate the validity of arbitrary computations. More precisely, given a relation R consisting of input-witness tuples $(\mathbb{x}; \mathbb{w})$ and a public input $\mathbb{x}$, a SNARK is a non-interactive protocol that enables a prover, P, to convince a verifier, V, that they know a witness, $\mathbb{w}$, such that $(\mathbb{x}; \mathbb{w}) \in R$, where communication and verifier runtime are sublinear (i.e. the scheme is *succinct*). The most common approach to building SNARKs is to first choose a finite field $\mathbb{F}$ and then express R as a new relation $R'_{\mathbb{F}}$ consisting of a system of polynomial equations over $\mathbb{F}$ in a certain fixed form (e.g. R1CS, CCS, AIR, Plonkish, or lookups)[1]. This process is called *arithmetization* of the computation Comp, and it is by far the most common paradigm in both the SNARK literature and industry [**Groth16**; **GWC19**; **ethstark**; **Cairo**]. SNARKs based on this approach have experienced remarkable performance improvements over the past years.

A limitation of this approach is that most computations of interest (see table 2.2) are not naturally expressed as systems of equations over a single finite field. Having to express a computation naturally defined over multiple algebraic domains, or with non-algebraic operations, over a single finite field results in large blowups to the circuit size, i.e. given $(\mathbb{x}; \mathbb{w}) \in R$, the size of the arithmetized $(\mathbb{x}'; \mathbb{w}') \in R'_{\mathbb{F}}$ is typically much larger than $(\mathbb{x}; \mathbb{w})$ (as will the corresponding circuit for verifying

[1] Lookup and permutation constraints can be reduced to (high degree) polynomial equations; see section 2.4.2 for details.

membership) [**Zinc**; **jea_codes**; **EFpaper**; **binius**]. Accordingly, state-of-the-art SNARKs end up proving computations that are, in a sense, significantly larger than the original computation Comp. We call this relative size increase the *arithmetization overhead*, and note it detrimentally affects all performance metrics of the SNARK: prover time, proof size, and verifier time. Beyond performance, there are two other problems brought by non-native arithmetizations:

1) Need for expert arithmetization or generic arithmetizers (zkVMs). Designing efficient and correct arithmetizations over a finite field is challenging both conceptually and in terms of implementation [**PailoorUnderconstrained**; **WenZKPSecurity**]. This makes the arithmetization step inaccessible to non-experts, who are then bound to generic tools such as zkVMs [**Cairo**; **Jolt**]. In fact, current zkVM designs even incur a *double* arithmetization overhead: first from compiling the computation into VM cycles (typically composed of RISC-V instructions), and then from arithmetizing those cycles into a finite-field relation. We note that this double overhead is not inherent to the zkVM paradigm—one could design zkVMs whose underlying proof system avoids arithmetization overhead. We envision the present work as a step in this direction.

2) Correctness and auditability. The gap between the original computation and its arithmetized form makes the final proof system harder to audit and more prone to bugs, e.g., due to underconstrained systems.

2.1.1 Contributions

We next list the contributions of this work, which we elaborate on in the technical overview (section 2.2.2).

Universal Constraint Systems (UCS). We propose a new arithmetization framework which is tailored to express most computations of interest (as captured in table 2.2), with significantly reduced arithmetization overhead compared to single-field constraint systems (see table 2.2 for concrete examples). In particular, we design a relation $\mathsf{REL}_{\mathsf{UCS}}$ called *Universal Constraint System* (UCS) that efficiently encapsulates said computations (see section 2.2.1 and the formal definition in definition 2.4.1).

UCS is a generalization of constraints such as R1CS, CCS, AIR, lookups, etc., along two main axes: the algebraic domains over which constraints are defined, and the types of predicates supported.

First, UCS expresses constraints simultaneously over different rings such as $\mathbb{Q}$ (rationals), $\mathbb{Z}$ (integers), $\mathbb{F}_q$ (prime or prime power fields), and, importantly, polynomial rings such as $\mathbb{Q}[X], \mathbb{Z}[X], \mathbb{F}_q[X], R[X]$. In short, UCS can express R1CS, AIR, etc. relations over multiple such rings in the same instance. Additionally, it supports ideal-membership predicates, i.e. predicates of the form $Q(\dots) \in \mathfrak{I}$, where Q is a polynomial expression in the input and the witness, and $\mathfrak{I}$ is an ideal of $\mathbb{Q}[X]$ or $\mathbb{F}_q[X]$. This expressivity allows one to capture operations such as rotation of bitstrings (through the ideal $X^w - 1$), relating bitstrings and integers (through the ideal $(X - 2)$), constraints over multiple rings and fields (through quotienting $\mathbb{Z}[X]$ via an ideal), etc.

A SNARK for the UCS relation. We design a general framework for building SNARKs for $\mathsf{REL}_{\mathsf{UCS}}$, which we call Zinc+ (section 2.2.2). This framework consists of two components:

Zinc+ PIOP compiler. A compiler that takes existing PIOPs over finite fields, and applies a so-called *lifting* to adapt them to our constraint setting, which, as we mentioned, involves domains such as $\mathbb{Z}, \mathbb{Q}$, $\mathbb{F}_q$, and polynomial rings $R[X]$ (for $R \in \{\mathbb{Z}, \mathbb{Q}, \mathbb{F}_q\}$). The compiled PIOP involves polynomial oracles defined over all the domains in the relation. Outside the PIOP model (i.e. when building a final SNARK), the correct typing (i.e., that each oracle's coefficients lie in the declared ring or field and satisfy the prescribed degree and bit-size bounds) and evaluations of the oracles are enforced via Zip+ (explained below). We prove that the Zinc+ PIOP achieves *round-by-round knowledge soundness* (theorem 2.5.7), with knowledge error $2^{-\lambda}$ per round (the PIOR has two rounds of verifier randomness, contributing at most $2 \cdot 2^{-\lambda}$ total knowledge error), where λ is the security parameter. Since round-by-round knowledge soundness composes with the Fiat–Shamir transform in the random oracle model [**BCS16**], this yields a knowledge-sound non-interactive argument (SNARK) when the PIOP is compiled with Zip+ and Fiat–Shamir (**??**).

Zinc+ in a nutshell. In simple terms, the Zinc+ PIOP compiler works as follows: first, the prover sends an oracle (or oracles) $[[\tilde{\mathbb{w}}]]$ to the multilinear extensions (MLE) of the witness vector(s) $\mathbb{w}$. Second, all ideal membership constraints $Q(\dots) \in \mathfrak{I}$ are reduced to strict equalities (over $\mathbb{Q}, \mathbb{Q}[X], \mathbb{F}_q, \mathbb{F}_q[X]$, etc.) by checking that the evaluation on a random point of the MLE of $Q(\dots)$ belongs to $\mathfrak{I}$. This way, $Q(\dots) \in \mathfrak{I}$ is replaced by an MLE evaluation claim, which is a strict equality. Third, the strict equality is projected onto a random finite field $\mathbb{F}_q$ via one of the following maps,

depending on whether our initial constraint is over $\mathbb{Q}[X]$ or $\mathbb{F}_q[X]$, respectively:

$$\begin{aligned} \psi_{q',a} : \mathbb{Q}[X] &\to \mathbb{F}_{q'} & \qquad \psi_b : \mathbb{F}_q[X] &\to \mathbb{F}_q \\ f(X) &\mapsto f(a) \mod q' & \qquad f(X) &\mapsto f(b) \end{aligned} \tag{2.1}$$

where q' is a random prime when working over $\mathbb{Q}[X]$, a is a random integer, and b is a random element of $\mathbb{F}_q$ (or of a field extension of $\mathbb{F}_q$ if q is small).

Fourth, the resulting finite field equality is proved using any suitable PIOP $\Pi_{\mathbb{F}}$ for constraints over a finite field $\mathbb{F}$. During this step, every time the verifier of $\Pi_{\mathbb{F}}$ queries the MLE of the projected witness at a point $\mathbf{r}$, the Zinc+ verifier lifts $\mathbf{r}$ and queries $[[\tilde{\mathfrak{w}}]]$ at the lifted point, and then projects the resulting evaluation onto $\mathbb{F}$.

Zip+. An Interactive Oracle Proof of Proximity (IOPP) for handling commitments to multilinear polynomials whose coefficients belong to the rings mentioned above, i.e. $\mathbb{Q}, \mathbb{Q}[X], \mathbb{F}_q, \mathbb{F}_q[X]$. We call this IOPP Zip+. Zip+ is round-by-round knowledge sound, supports batched evaluations, and it can handle distance parameters up to the Mutual Correlated Agreement (MCA) bound of the underlying code—unlike its predecessor Zip [**Zinc**], a PCS for multilinear polynomials with coefficients in the set $\mathbb{Q}^{<B}$ of rationals of bit-size less than B. Zip+ is a modern version of the Brakedown/Ligero PCS [**Brakedown**; **Ligero**] adapted to our rings and fields. In short, Zip+ allows one to commit to multilinear polynomials with coefficients in $\mathbb{Q}^{<d,B}[X]$, where $\mathbb{Q}^{<d,B}[X]$ denotes the set of rational polynomials of degree less than d and coefficients in $\mathbb{Q}^{<B}$. The commitment is done by using an improved version of Zip to commit to d multilinear polynomials with coefficients in $\mathbb{Q}^{<B}$.

In section 2.2.3 we discuss the efficiency of Zip+ in isolation and in the broader context of our compiled SNARK and use-cases.

Integer Pseudo Reed–Solomon (IPRS) codes. We introduce a new linear code over the field of rational numbers $\mathbb{Q}$ and over the ring of integers $\mathbb{Z}$, which we call *Integer Pseudo Reed–Solomon (IPRS) code*, and which we believe is of independent interest. This code is not a Reed–Solomon (RS) code over the integers, but it presents similar characteristics, namely it is minimum distance separable and admits a FFT encoding process. Additionally, IPRS codes do not blowup the bit-size of encoded messages (see next).

Our IOPP and multilinear PCS Zip+ relies at its core on linear error correcting codes over

$\mathbb{Q}, \mathbb{Q}[X], \mathbb{F}_q$, and $\mathbb{F}_q[X]$. As with other code-based IOPPs, the performance of Zip+ is heavily influenced by both the minimum distance of the underlying code and the efficiency of its encoding. Additionally, when working over $\mathbb{Q}$ (or fields of size much larger than the size of the witness entries), one also needs to ensure that the bit-size of the codewords does not blowup compared to the bit-size of the messages.

IPRS codes in a nutshell. When working over finite fields, Reed-Solomon (RS) codes are a natural choice as underlying code. These codes have optimal minimum distance and support efficient FFT-based encoding. One could try to use RS codes over $\mathbb{Q}$ or $\mathbb{Z}$ as well — for example, by naively defining RS codes over $\mathbb{Q}$ as the set of vectors obtained by evaluating low-degree polynomials on integer points. However, for practical code dimensions, the bit-size of the resulting codewords would be in the thousands or even millions, which would result in an inefficient IOPP. Alternatively, one could "lift" a standard RS code $\mathsf{RS}[\mathbb{F}_q, n, k]$ from a small field $\mathbb{F}_q$ onto $\mathbb{Q}$ by looking at the Vandermonde generating matrix of $\mathsf{RS}[\mathbb{F}_q, n, k]$ as an integer matrix with entries in $\{0, \dots, q-1\}$, and defining the lifted code as the set of rational linear combinations of the rows of this matrix. However, this code does not have, a priori, an efficient FFT encoding.

IPRS codes constitute a middle ground between the two aforementioned RS-based constructions. In short, they are obtained by "lifting" to $\mathbb{Q}$ the FFT encoding algorithm for a standard RS code over a small field $\mathbb{F}_q$. Concretely, we fix such an FFT algorithm (by specifying its radix, twiddle factors, depth, etc.), and then we look at its twiddle factors and other constants as integers from $\{-(q-1)/2, \dots, (q-1)/2\}$, and perform the FFT algorithm over $\mathbb{Q}$, without performing modular reduction. The IPRS code is defined as the set of vectors obtained by applying such lifted FFT algorithm.

RAA and JEA codes. Non RS-based codes such as RAA and JEA codes [**jea_codes**; **blaze**] can be defined over $\mathbb{Q}$ and $\mathbb{Z}$ without blowup issues and with efficient encoding. However, their *proved* minimum distance is significantly smaller than that of RS codes and our IPRS codes.

Implementation and Evaluation. We partially implement our proof system (section 2.2.3). Concretely, we implement the Zip+ IOPP and a component of our PIOP. We arithmetize the SHA-256 compression function and the ECDSA verification algorithm as an instance of our UCS relation. Then we evaluate the implemented components on inputs corresponding to said arithmetizations.

We expect the IOPP to constitute the most expensive part of our scheme, in prover time, proof size, and verifier time. Hence our benchmarks for the IOPP may be taken as a proxy for the performance of the full proof system on the arithmetized computations.

Our partial benchmarks provide preliminary evidence that, when run on computations of practical interest (such as SHA-256 hashing followed by ECDSA signature verification—see table 2.2), a fully optimized end-to-end implementation of our schemes may approach the performance of state-of-the-art SNARKs in prover time, with comparable or smaller proof size, and a fast verifier. We emphasize that our current benchmarks are partial: they cover the Zip+ IOPP (which we expect to dominate the overall cost), together with most steps of the Zinc+ PIOP, but not the full end-to-end pipeline. Therefore, the figures reported below should be understood as *indicative estimates*, not direct apples-to-apples comparisons with fully deployed systems. The next table describes the costs of our Zip+ IOPP compiled with Merkle tree commitments, which yields a hash-based polynomial commitment scheme. Throughout this paper, all reported Zip+ IOPP times include the cost of Merkle tree hashing. In table 2.7 we provide almost full prover performance for the Zinc+ PIOP. See **??** for a detailed description of our experiments.

Zip+ IOPP on trace of 8x SHA-256 + ECDSA		
Prover (ms)	Verifier (ms)	Proof (KB)
10.71	2.30	211

Table 2.1: Benchmarks for our Zip+ IOPP compiled with Merkle tree commitments, on a batched set of multilinear polynomials whose sizes and types correspond to our arithmetization of 8 SHA-256 compressions, followed by one ECDSA signature verification. All reported times include the cost of Merkle tree hashing. The compiled IOPP is expected to dominate all costs of our proof system (see section 2.2.3), and so we view these benchmarks as a rough estimate of times Zinc+ could achieve on this task. We note that ours is a PoC implementation.

The following table lists some computations of practical interest in the context of SNARKs, and the types of operations they involve.

Computation	$\mathbb{Z}/2^w\mathbb{Z}$	**Bit**	$\mathbb{F}_q$	$\mathbb{F}_{2^w}$	$\mathbb{Z}/n\mathbb{Z}$	$\mathbb{Z}$ or $\mathbb{Q}$	R_{latt}
Cryptographic primitives							
Classic hashes (SHA, etc.)	✓	✓					
AES / symmetric encr.	✓	✓		✓			
Elliptic curve crypt.			✓				
Lattice crypt.	✓						✓
RSA crypt.					✓		
Applications							
zkVMs / CPU emulation	✓	✓	✓				
zkID	✓	✓	✓		✓		
Blockchain transactions	✓	✓	✓				
TLS (AEAD)	✓	✓	✓	✓			
Proof recursion (classic hash)	✓	✓	✓		✓		
Hash-to-curve	✓	✓	✓				
FHE	✓					✓	✓
Machine learning						✓	
Finance						✓	

Table 2.2: Common computations and the types of arithmetic they require. Columns named with a ring R refer to "addition and multiplication R". The Bit column refers to operations such as XOR, AND, rotations, shifts, etc. For an integer $m \geq 1$, $\mathbb{Z}/m\mathbb{Z}$ denotes integers modulo m (where composite m are relevant in RSA cryptography and lattices, FHE, etc.). Further, $\mathbb{F}_q$ is a finite field (often prime); $\mathbb{F}_{2^w}$ denotes fields of size 2^w; and R_{latt} denotes rings relevant to lattice-based cryptography such as cyclotomic rings $\mathbb{Z}[X]/(X^n + 1)$.
zkID refers to performing a classic hash followed by a signature verification with, e.g. ECDSA or RSA. Proof recursion refers to proving acceptance of a SNARK verifier in a context where the original proof system uses classic hashes (due to Fiat-Shamir or Merkle tree commitments); it has $\mathbb{Z}/n\mathbb{Z}$ checkmarked as it sometimes requires using different moduli, e.g. in IVC recursion [**Nova**]. Hash-to-curve refers to algorithms that iteratively perform hashes and check membership of points on an elliptic curve. zkVMs may use so-called precompiles/builtins to perform computations such as ECDSA verification. In this case they also involve prime field arithmetic.

2.1.2 Related work

SNARKs for integer constraints. Zinc [**Zinc**] presents a framework for building proof systems for constraints over $\mathbb{Z}$ or $\mathbb{Q}$ and introduces a hash-based PCS for polynomials with coefficients in $\mathbb{Q}$, built with a tensor IOPP similar to the one introduced in Brakedown [**Brakedown**].

Our work improves upon [**Zinc**] significantly in the following ways. (1) our UCS constraints strictly generalize Zinc's, which is restricted to constraints over $\mathbb{Z}$ and $\mathbb{Q}$, by additionally supporting

multiple finite fields, polynomial rings $R[X]$, and ideal-membership predicates. UCS supports simultaneously multiple rings and fields of the form $R^{<d}[X]$ and ideal membership constraints, which, as we demonstrate, enriches the expressivity of the constraint system substantially. (2) We design a PIOP framework for these more general constraints. Additionally, we prove round-by-round knowledge soundness, allowing for Fiat-Shamir compilation. (3) Our IOPP Zip+ is an improvement to the PCS from [**Zinc**] in that it supports polynomials over a wide range of rings and fields, can handle proximity parameters up to the MCA bound of the underlying code (**??**) (this greatly helps reduce proof size and verifier time), merges the so-called "test" and "evaluate" phases into one, is proved to be round-by-round knowledge sound, and has batching capabilities. (4) We improve upon the underlying linear code with our IPRS codes. (5) We demonstrate the practicality of our approach with our implementation and evaluation.

Zartan [**Zartan**] builds SNARKs over $\mathbb{Z}$ by compiling so-called mod-AHPs with a PCS for integral polynomials based on hidden-order groups [**BlockDARK**; **BunzFisch22**]. We adopt the idea from [**Zartan**] of working modulo a random prime, but replace the hidden-order-group PCS with a hash-based PCS (Zip+), thereby avoiding the stronger cryptographic assumptions (e.g. the adaptive root assumption) that hidden-order-group commitments require. Moreover, as in the comparison with Zinc [**Zinc**], our scheme works with constraint systems that are substantially more expressive than Zartan's.

Binary-field SNARKs. Binius [**binius**; **fribinius**] is a series of SNARKs for relations over binary fields $\mathbb{F}_{2^k}$. Binary fields are more friendly to bitstring operations than prime fields. For example, bitwise XOR can be modeled as addition in $\mathbb{F}_{2^k}$. However, AND, rotation, and modular-2^w arithmetic remain non-native in binary fields. Despite this, Binius achieves significant improvements on computations involving bitstring operations and arithmetic modulo 2^w, when compared to prime-field SNARKs [**ethproofs**]. We believe this is in part due to $\mathbb{F}_{2^k}$ being more amenable to some of these operations, and due to the *pay-per-bit* capacity of Binius' IOPP.

Our benchmarks show that, for computations dominated by bitstring and modular-integer operations (e.g. SHA-256 hashing), Zinc+ achieves performance competitive with Binius. For computations that additionally involve prime-field arithmetic (e.g. SHA-256 followed by ECDSA signature verification), our results indicate a clear advantage, since Zinc+ handles prime-field

constraints natively whereas Binius must emulate them over binary fields. While not evaluated in the benchmarks of this paper, Zinc+ can natively handle further constraints such as lattice-based encryption combined with classic hashing (see section 2.2.1).

Proof systems over rings. There are a number of works describing SNARKs over different types of specific rings, such as cyclotomic rings [**beullens2023labrador**], Galois rings [**new_rings_work_one**; **new_rings_work_two**], and abstract rings with large exceptional sets [**bootle2021sumcheck**; **attema2022vector**; **soria2022doubly**]. Approaches for proving statements modulo 2^n are based on VOLE techniques (see, e.g. [**C:LinXinYao24**; **baum2021appenzeller**; **baum2022moz**]) but yield non-succinct schemes. Unlike our work, none of these approaches support the wide range of rings and fields that we can handle with our framework, including constraints over $\mathbb{Z}$ and $\mathbb{Z}[X]$. The recent works FREPack [**FREPack**] and Swirl [**swirl**] treat the problem of allowing different constraint domains within the same proof system. FREPack focuses on highly repetitive computations and packs multiple field operations into a single CCS/R1CS constraint via the Chinese Remainder Theorem, achieving significant speedups for structured circuits but limiting its flexibility: the rings used must factor the chosen CRS modulus, and the approach does not natively support polynomial-ring constraints or ideal-membership predicates as UCS does. Swirl stacks WHIR with LogUp to handle lookups across different subfields within a single proof, but remains in a finite-field setting and does not address constraints over $\mathbb{Z}$, $\mathbb{Q}$, or polynomial rings $R[X]$. In contrast, Zinc+ supports constraints over $\mathbb{Z}, \mathbb{Q}, \mathbb{F}_q$ and their polynomial ring counterparts simultaneously, with ideal-membership predicates that natively capture operations such as bitstring rotation and modular reduction without additional encoding.

Applications. In this work we discuss the example of SHA-256 hashing followed by ECDSA signature verification. Two recent approaches to this, which additionally include document parsing, are [**longfellow**; **vega**]. As future work, we plan to include document parsing in our arithmetizations and compare our scheme with these approaches.

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2.2 Technical Overview

We now provide an overview of the main contributions of this work.

2.2.1 Universal Constraint Systems (UCS)

To describe and motivate our framework we focus on two running examples. The first involves arithmetic modulo 2^w, bitstring operations such as XOR and rotation, and arithmetic modulo an arbitrary integer (possibly non-prime, e.g. a power of two). The second revolves around Hamming weight computations, lattice based operations, and a toy electronic voting application.

We introduce some notation. Further background can be found in section 4.2. Given a ring or field R, we write $R^{<d}[X]$ to denote polynomials on the variable X and of degree less than d. Note that $R^{<1}[X] = R$. Given a bit size $w \geq 1$ we let $[0\,..\,2^w - 1]$ denote the set of integers $\{0, \dots, 2^w - 1\}$. We define the set of so-called *bit-polynomials* of width w as

$$\{0,1\}^{<w}[X] \;:=\; \Big\{ \sum_{i=0}^{w-1} b_i\, X^i \;\Big|\; b_i \in \{0,1\} \Big\} \;\subseteq\; \mathbb{Z}[X]. \tag{2.2}$$

The set $\{0,1\}^{<w}[X]$ is in bijective correspondence with $[0\,..\,2^w - 1]$ via the evaluation map

$$\psi_2 \colon \mathbb{Z}[X] \to \mathbb{Z} \qquad \text{defined by} \qquad \psi_2(u(X)) := u(2),$$

which restricts to a bijection $\{0,1\}^{<w}[X] \xrightarrow{\sim} [0\,..\,2^w - 1]$. Note that ψ_2 is the canonical surjective ring homomorphism $\mathbb{Z}[X] \twoheadrightarrow \mathbb{Z}[X]/(X-2) \cong \mathbb{Z}$. Here, $(X-2)$ denotes the ideal generated by $X-2$ in the ring $\mathbb{Z}[X]$. See section 4.2 for background. We also let

$$\phi_2 \colon \mathbb{Z}[X] \to \mathbb{F}_2[X]$$

denote reduction modulo 2 coefficient-wise.

We emphasize that $\{0,1\}^{<w}[X]$ is formally a set of *integer* polynomials, and for the purposes of this paper this is not the same as $\mathbb{F}_2^{<w}[X]$. It is important that commitments to polynomials in $\{0,1\}^{<w}[X]$ are over $\mathbb{Z}[X]$, and cannot be taken modulo 2.

A motivating example: Simultaneous bitstring operations and arithmetic mod 2^w and mod primes

Our first example has the following function f at its core. Let $r \in [w-1]$ be a rotation parameter. Let $n \in [0\,..\,2^w]$ be an integer, possibly non-prime. Now define

$$\begin{aligned} f : [0\,..\,2^w - 1] &\to [0\,..\,2^w - 1], \\ x &\mapsto (x^2 \bmod n) \ \ \mathsf{XOR} \ \ \mathsf{ROTR}^r(x), \end{aligned} \tag{2.3}$$

where XOR is the "bitwise exclusive or" (XOR) operation, and ROTR^r is right rotation by r bits. Precisely, for two integers $u, v \in [0\,..\,2^w - 1]$ with binary representations $u = \sum_{i=0}^{w-1} u_i 2^i$ and $v = \sum_{i=0}^{w-1} v_i 2^i$, we have $u \ \mathsf{XOR} \ v = \sum_{i=0}^{w-1} (u_i + v_i \bmod 2) 2^i$, and $\mathsf{ROTR}^r(u) = \sum_{i=0}^{w-1} u_{(i+r) \bmod w} 2^i$. We extend these operations to the set $\{0,1\}^{<w}[X]$ by applying them coefficient wise.In the definition of f (eq. (2.3)), the expression $x^2 \bmod n$ means that x is understood as an integer in $[0\,..\,2^w - 1]$, squared, and then the canonical representative of x^2 modulo 2^w in $[0, n-1]$ is returned.

We view f as a toy function roughly representative of computations involving some or all of the following: arithmetic modulo 2^w, bitstring operations, and prime field operations. Concretely, we think of f as a proxy for some of the subroutines in computations such as classic hashing, CPU and zkVM instructions, zkID authentication systems, TLS, etc. See table 2.2.

Given a number of iterations $n \geq 1$, our goal is to design a proof system for the relation REL_f defined as

$$\mathsf{REL}_f = \left\{ (\mathbb{x}; \mathbb{w}) \;\middle|\; \begin{array}{l} \mathbb{x} = (u_1, u_n) \in [0\,..\,2^w - 1]^2, \\ \mathbb{w} = (u_2, \dots, u_{n-1}) \in [0\,..\,2^w - 1]^{n-2}, \\ u_{i+1} = f(u_i) \text{ for all } i \in [n-1] \end{array} \right\}. \tag{2.4}$$

Before diving into arithmetizing REL_f, we make a series of observations about the operations used by the function f, which will motivate our arithmetization framework.

XOR and AND For any $u, v \in \{0,1\}^{<w}[X]$, the following identity holds in $\mathbb{Z}[X]$:

$$u + v \;=\; (u \;\mathsf{XOR}\; v) + 2 \cdot (u \;\mathsf{AND}\; v)\,. \qquad \text{in } \mathbb{Z}[X], \tag{2.5}$$

The converse also holds: if $u, v, z, a \in \{0,1\}^{<w}[X]$ satisfy $u + v = z + 2 \cdot a$ in $\mathbb{Z}[X]$, then $z = u \;\mathsf{XOR}\; v$ and $a = u \;\mathsf{AND}\; v$. Both directions follow from the fact that $b + b' = (b \;\mathsf{XOR}\; b') + 2\,(b \;\mathsf{AND}\; b')$ for any two $b, b' \in \{0,1\}$, applied coefficient-wise.

Hence, informally speaking, provided the witness elements belong to $\{0,1\}^{<w}[X]$, the identity (2.5) arithmetizes both XOR and AND over $\mathbb{Z}[X]$ simultaneously.

A further consequence of (2.5) is that for any $u, v, z \in \{0,1\}^{<w}[X]$,

$$z = u \;\mathsf{XOR}\; v \quad \Longleftrightarrow z = u + v \mod 2, \tag{2.6}$$

In summary, one can expose the XOR of two elements of $\{0,1\}^{<w}[X]$ either via a single equality over $\mathbb{Z}[X]$ (which additionally yields the AND) or through addition modulo 2. We leverage both approaches in our arithmetization of SHA-256.

Rotation through ideal membership For any $u \in \{0,1\}^{<w}[X]$, the following ideal-membership relation holds in $\mathbb{Z}[X]$:

$$\mathsf{ROTR}^r(u) - X^{w-r} \cdot u \in (X^w - 1). \tag{2.7}$$

where $(X^w - 1)$ denotes the ideal in $\mathbb{Z}[X]$ generated by the polynomial $X^w - 1$. Conversely, for any two $u, v \in \{0,1\}^{<w}[X]$, if $v - X^{w-r} \cdot u \in (X^w - 1)$ then $v = \mathsf{ROTR}^r(u)$ (**??**). Hence rotation *for elements in* $\{0,1\}^{<w}[X]$ can be enforced by the ideal-membership predicate $v - X^{w-r}u \in (X^w - 1)$.

At this point one may now wonder whether such a predicate can be efficiently handled in proof systems. The answer is that this can be achieved by checking that certain appropriate multilinear extension, when evaluated at a random point, returns a value that belongs to the ideal. This is analogous to the technique in classical proof systems where the MLE of the polynomial constraints composed with the MLE of the witnesses evaluates to zero at a randomly sampled point. Here, instead, evaluation to zero (i.e. membership to the zero ideal) is replaced by ideal membership. See

lemma 2.2.5.

Mixing XOR and rotation Given elements $u, v, z \in \{0,1\}^{<w}[X]$, we want to arithmetize the constraint $z = u \ \mathsf{XOR}\ \mathsf{ROTR}^r(v)$. To do so, observe that as long as u, v, z belong to $\{0,1\}^{<w}[X]$, the following equivalence holds:

$$z = u \ \mathsf{XOR}\ \mathsf{ROTR}^r(v) \quad \iff \quad \phi_2(u) + \phi_2(v) \cdot X^{w-r} - \phi_2(z) \in (X^w - 1) \qquad \text{in } \mathbb{F}_2[X], \tag{2.8}$$

The equivalence in (2.8) can be generalized to more complex expressions involving several XORs and rotations. It is particularly useful for arithmetizing some inner functions of SHA-256 and related hashes and ciphers.

Linking bit-polynomials to integers, and squaring modulo an arbitrary integer n We would now like to, given $u, v \in \{0,1\}^{<w}[X]$, express the constraint that the integer represented by u, i.e. $\psi_2(u)$, is the square of the integer represented by v, i.e. $\psi_2(v)$, modulo an arbitrary integer n. We can do so by observing the following:

$$\psi_2(v) = \psi_2(u)^2 \bmod n \quad \iff \quad v - u^2 + n \cdot \mu \in (X - 2) \qquad \text{in } \mathbb{Z}[X], \tag{2.9}$$

for some $\mu \in \mathbb{Z}$. To see why, assume there exists $\mu \in \mathbb{Z}$ such that $v(X) - u(X)^2 + n \cdot \mu \in (X-2)$ in $\mathbb{Z}[X]$. Then $v(X) - u(X)^2 + n \cdot \mu$ vanishes at $X = 2$, so $v(2) - u(2)^2 + n \cdot \mu = 0$ in $\mathbb{Z}$, which gives $\psi_2(v) = \psi_2(u)^2 \bmod n$. The forward direction is analogous.

A similar approach arithmetizes other algebraic modular operations. An important underlying principle is that membership to the ideal $(X-2)$ links bit-representations (modeled as elements from $\{0,1\}^{<w}[X]$) to the integer they represent, as long as elements are constrained to belong to $\{0,1\}^{<w}[X]$ and $[0 \,..\, 2^w - 1]$.

Remark 2.2.1 (Lookup or typing constraints)**.** In all examples above, the typing of elements is crucial. I.e. most of the ideal membership constraints described above lose their intended effect if one does not pair them with *lookup* or *typing* constraints, e.g. the constraint (2.9) must be accompanied of the lookup constraints $v, u \in \{0,1\}^{<w}[X]$ and $\mu \in \mathbb{Z}$. Because of this, our arithmetizations consist mostly of algebraic constraints involving witness elements (these may be zero equalities or ideal

membership constraints), plus lookup or typing constraints.

Arithmetization of REL_f We now put together a full arithmetization for the relation REL_f. Recall that each application of the function f computes

$$u_{t+1} \;=\; (u_t^2 \bmod n) \;\; \mathsf{XOR} \;\; \mathsf{ROTR}^r(u_t).$$

To arithmetize the full $(n-1)$-fold iteration, we introduce witness vectors for each step and impose constraints that enforce the transition given by f. The entire arithmetization uses three witness elements per step, two typed in $\{0,1\}^{<w}[X]$ and one in $\mathbb{Z}$, and two constraints per transition.

<u>*Witness.*</u> For each step $t \in [n]$ we introduce three witness elements. We think of the witness as a $n \times 3$ matrix, or *trace* (even though our definition of UCS does not restrict to working with traces and AIR-like systems). We denote the inverse of ψ_2 with $\hat{\cdot}$, so that for all $u \in [0\,..\,2^w - 1]$ we have $\hat{u} = \hat{u}(X) \in \{0,1\}^{<w}[X]$ and $\hat{u}(2) = u$. The following table describes each witness element, its intended meaning, and its domain .

Entry	Intended meaning	Lookup constraint
$\hat{u}_t$	bit-polynomial representation of u_t at step t	$\in \{0,1\}^{<w}[X] \subset \mathbb{Z}[X]$
$\hat{s}_t$	bit-polynomial representation of $(u_t)^2 \bmod n$	$\in \{0,1\}^{<w}[X] \subset \mathbb{Z}[X]$
μ_t	the quotient such that $(u_t)^2 = s_t + \mu_t\, n$	$\in \mathbb{Z}$

<u>*Transition constraints.*</u> For each step $t \in [n-1]$ we impose two constraints.

1. **Squaring modulo n.** $\hat{s}_t - (\hat{u}_t)^2 + n \cdot \mu_t \;\in\; (X-2)$ in $\mathbb{Z}[X]$. By eq. (2.9), this forces $s_t \equiv (u_t)^2 \pmod{n}$.

2. **XOR and rotation.** $\phi_2(\hat{u}_{t+1}) - \phi_2(\hat{s}_t) - X^{w-r} \cdot \phi_2(\hat{u}_t) \;\in\; (X^w - 1)$ in $\mathbb{F}_2[X]$. By (2.8), this enforces that $\hat{u}_{t+1} = \hat{s}_t \;\mathsf{XOR}\; \mathsf{ROTR}^r(\hat{u}_t)$.

We omit boundary constraints for brevity.

The reader may appreciate the proposed arithmetization as being both conceptually simple and small (in the sense of it requiring a small number of constraints given the relation sought to arithmetize). The question now is whether such arithmetization *is useful in practice*, i.e. whether it can be efficiently handled by proof systems. One of the main contributions of this work is to

show that the answer is yes. We outline why and how in the next sections of the present technical overview.

Further examples

The same ideal-membership approach extends to other settings. To give a couple more examples:

- **Lattice-based operations.** Constraints in, for example, cyclotomic rings $R_q = \mathbb{F}_q[X]/(\Phi(X))$ can be enforced through ideal-membership predicates over $\mathbb{F}_q[X]$ where the ideal is $(\Phi(X))$. The typing of the witnesses can either be in some subset of $\mathbb{Z}[X]$ or in $\mathbb{F}_q[X]$, depending on the application (see the discussion after definition 2.4.1 for more details).

- **Hamming weight.** For any $u \in \{0,1\}^{<w}[X]$, let $\mathsf{wt}(u)$ denote the number of nonzero coefficients of u. Then, given $c \in \mathbb{Z}$ we have that $\mathsf{wt}(u) = c \iff u - c \in (X-1)$ in $\mathbb{Z}[X]$. This is because evaluation at $X = 1$ sums coefficients.

Universal Constraint Systems (UCS)

We now introduce the main relation we will analyse. We call it the *Universal Constraint System* (UCS) relation $\mathsf{REL}_{\mathsf{UCS}}$.

A UCS instance is specified by an *index* $\mathbb{i}$ that fixes a witness-vector length m, a number of rows n, a tuple of prime powers $\mathbf{q} = (q_1, \dots, q_{|\mathbf{q}|})$, a bit-size bound B, degree bounds $\mathbf{d} = (d_0, \dots, d_{|\mathbf{q}|})$, and a set of constraints $\mathcal{C}$ described below.

The witness consists of $|\mathbf{q}| + 1$ vectors $\mathbf{f}_0, \dots, \mathbf{f}_{|\mathbf{q}|}$, each of length m. The vectors are typed as

$$\mathbf{f}_0 \in \left(\mathbb{Q}^{<d_0,B}[X]\right)^m, \qquad \mathbf{f}_i \in \left(\mathbb{F}_{q_i}^{<d_i}[X]\right)^m \text{ for all } i \in [|\mathbf{q}|],$$

where, recall, $R^{<d,B}[X]$ denotes the set of polynomials of degree $< d$ with coefficients in R, of bit-length less than B (when R is a field, we drop the bit bound and allow coefficients in all R). The entries of the witness vectors are thus *polynomials themselves*. In particular, the multilinear extension (MLE) (cf. section 2.3.1) of $\mathbf{f}_i$ is a multilinear polynomial whose coefficients lie in $\mathbb{Q}[X]$ or $\mathbb{F}_{q_i}[X]$—that is, a polynomial whose coefficients are polynomials. Regarding why $\mathbf{f}_0$ is typed in $\mathbb{Q}^{<d_0,B}[X]$ rather than $\mathbb{Z}^{<d_0,B}[X]$, see the technical observations after definition 2.4.1.

The public input is a vector $\mathbf{y} \in (\mathbb{Q}^{<d_0,B}[X])^{\ell}$.

Each constraint in $\mathcal{C}$ is a tuple $(Q, \mathfrak{I})$, where Q is a polynomial with coefficients either in $\mathbb{Q}$ or $\mathbb{F}_{q_i}$ for some $i \in [|\mathbf{q}|]$, and $\mathfrak{I}$ is an ideal of $\mathbb{Q}[X]$ or $\mathbb{F}_{q_i}[X]$, respectively. In the first case, $Q = Q(K, \mathbf{y}, \mathbf{f}_0)$ is a polynomial on $\mathbf{f}_0$ and $\mathbf{y}$ and a row variable K intended to range in $[n]$. We call $(Q, \mathfrak{I})$ a $\mathbb{Q}[X]$-*constraint*. In the later case, $Q = Q(K, \phi_{q_i}(\mathbf{y}), \phi_{q_i}(\mathbf{f}_0), \mathbf{f}_i)$ is a polynomial on $\mathbf{f}_i$, $\phi_{q_i}(\mathbf{f}_0)$, $\phi_{q_i}(\mathbf{y})$ and K, where, recall, ϕ_{q_i} denotes (coefficient-wise) reduction modulo q_i. We call this later $(Q, \mathfrak{I})$ a $\mathbb{F}_{q_i}$-*constraint*. See remark 2.2.3 for a short comment on having $\phi_{q_i}(\mathbf{f}_0)$, $\phi_{q_i}(\mathbf{y})$ be well-defined.

The relation $\mathsf{REL}_{\mathsf{UCS}}$ consists of all triples $(\mathbb{i}, \mathbb{x}; \mathbb{w})$ such that, for every constraint $(Q, \mathfrak{I}) \in \mathcal{C}$ and every row $k \in [n]$, the following holds:

$$
\begin{aligned}
&Q(k, \mathbf{y}, \mathbf{f}_0) \in \mathfrak{I} && \text{if } (Q, \mathfrak{I}) \text{ is a } \mathbb{Q}[X]\text{-constraint,} \\
&Q(k, \phi_{q_i}(\mathbf{y}), \phi_{q_i}(\mathbf{f}_0), \mathbf{f}_i) \in \mathfrak{I} && \text{if } (Q, \mathfrak{I}) \text{ is a } \mathbb{F}_{q_i}[X]\text{-constraint}, i \in [|\mathbf{q}|].
\end{aligned}
$$

Remark 2.2.2 (Shared witness principle)**.** The witness $\mathbf{f}_0$ acts as a shared witness across different modular arithmetic. This allowed in eq. (2.3) to have a single bit-polynomial column $\hat{u} \in \{0,1\}^{<w}[X] \subset \mathbb{Z}[X]$ simultaneously participate in a squaring of integers modulo some integer, and XOR operations expressed as additions modulo 2.

The formal definition of UCS is given in definition 2.4.1. When restricted to a single finite field and the zero ideal, UCS subsumes standard constraint systems—R1CS, AIR, CCS, Plonkish, and lookup constraints—as special cases (cf. sections 2.4.2 and 2.4.3).

After definition 2.4.1 we give a detailed discussion of the design choices behind the definition of UCS, and further technical observations about the definition.

Remark 2.2.3 (Well-defined projections of $\mathbf{f}_0$ and $\mathbf{y}$)**.** In the UCS definition, we also require that the reductions $\phi_{q_i}(\mathbf{f}_0)$, $\phi_{q_i}(\mathbf{y})$ are well-defined. Namely, we ask that $\mathbf{f}_0$ and $\mathbf{y}$ have entries in $\mathbb{Z}_{(q_i)}^{<d_0,B}[X]$, where, in short, $\mathbb{Z}_{(q_i)}$ is the subring of $\mathbb{Q}$ formed by rationals whose denominator is not divisible by q_i (see section 2.3.1).

In practice, this requirement is usually subsumed (i.e. implied) by the lookup or typing constraints on $\mathbf{f}_0$ and $\mathbf{y}$. For example, in our arithmetization of REL_f, the witness $\mathbf{f}_0$ contains bit-polynomial entries, which are polynomials with integer coefficients, which belong to $\mathbb{Z}_{(q_i)}$ for all prime or prime power q_i.

2.2.2 Zinc+: a PIOP for UCS from PIOPs over finite fields

We now describe the Zinc+ compiler, which constructs a PIOP for our UCS relation $\mathsf{REL}_{\mathsf{UCS}}$ from PIOPs for standard algebraic relations over finite fields. The compiler generalizes the Zinc compiler from [**Zinc**], which handles algebraic relations over $\mathbb{Q}$.

For clarity, we present a simplified version that focuses on UCS instances that only have $\mathbb{Q}[X]$-constraints (i.e. instances where $\mathbf{q} = \emptyset$, so there are no $\mathbb{F}_{q_i}[X]$-constraints). The general case is discussed in remark 2.2.6 below and treated fully in section 2.5.

Recall that for a prime q, $\mathbb{Z}_{(q)}$ denotes the subring of $\mathbb{Q}$ formed by rationals whose denominator (in lowest form) is not divisible by q, cf. section 2.3.1. We let

$$\phi_q : \mathbb{Z}_{(q)}[X] \to \mathbb{F}_q[X]$$

be the natural projection that reduces coefficients modulo q. We also let

$$\begin{aligned} \psi_a : \mathbb{F}_q[X] &\to \mathbb{F}_q \\ f &\mapsto f(a) \end{aligned}$$

be the evaluation map that replaces X by a. Both ϕ_q and ψ_a are ring homomorphisms. We let

$$\psi_{q,a} = \psi_a \circ \phi_q : \mathbb{Z}_{(q)}[X] \to \mathbb{F}_q$$

be the composition of the two maps.

From UCS constraints over $\mathbb{Q}[X]$ to algebraic constraints over a finite field. Recall that a $\mathbb{Q}[X]$-constraint in $\mathsf{REL}_{\mathsf{UCS}}$ takes the form

$$Q(k, \mathbf{y}, \mathbf{f}_0) \in \mathfrak{I}, \qquad \text{for all rows } k \in [n], \tag{2.10}$$

where Q is a polynomial with coefficients in $\mathbb{Q}$ and involving k, the public input $\mathbf{y} \in \mathbb{Q}^{<d_0,B}[X]^\ell$ and the witness $\mathbf{f}_0 \in \mathbb{Q}^{<d_0,B}[X]^m$, and $\mathfrak{I} \subseteq \mathbb{Q}[X]$ is an ideal. Assume $n = 2^\mu$ and identify $[n]$ with $\{0,1\}^\mu$. We assume without loss of generality that $\mathfrak{I} \neq \mathbb{Q}[X]$, since otherwise the constraint is always satisfied.

Our Zinc+ compiler first reduces the constraint (2.10) to standard algebraic constraints over a finite field $\mathbb{F}_q$ via a three-step process. After that, it invokes any suitable PIOP working over $\mathbb{F}_q$ to certify said constraints. The three-step process is formed by (1) a randomized projection from $\mathbb{Z}_{(q)}[X] \subseteq \mathbb{Q}[X]$ onto $\mathbb{F}_q[X]$, where q is a randomly sampled prime, which reduces (2.10) to an ideal membership constraint over $\mathbb{F}_q[X]$ of the form

$$\phi_q(Q)(k, \phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0)) \in \mathfrak{I}', \qquad \text{for all rows } k \in [n], \tag{2.11}$$

where $\mathfrak{I}' = \phi_q(\mathfrak{I} \cap \mathbb{Z}_{(q)})$; (2) a randomized *ideal check* that replaces the 2^μ ideal membership constraints in (2.11) with a strict equality over $\mathbb{F}_q[X]$; and (3) a randomized *projection* from $\mathbb{F}_q[X]$ onto the finite field $\mathbb{F}_q$, which turns the strict equality over $\mathbb{F}_q[X]$ from Step (2) into an equality over $\mathbb{F}_q$.

We note that ϕ_q is only well-defined on the subring $\mathbb{Z}_{(q)}[X]$ of $\mathbb{Q}[X]$. In our completeness and security proofs, we show that for random q the probability that the relevant elements do not belong to this subring is negligible. For brevity, we do not consider this issue further in the present overview. We refer to **??** for the technical details, or to the technical overview of the Zinc paper [**Zinc**], where this matter is discussed thoroughly.

We remark that both Zartan [**Zartan**] and Zinc [**Zinc**] already used a randomized projections onto finite fields. In particular, [**Zartan**] projects $\mathbb{Z}$ onto $\mathbb{F}_q$ for q a random prime, and [**Zinc**] projects $\mathbb{Z}_{(q)}$ onto $\mathbb{F}_q$. Overall, Zinc+ employs the projection given by $\psi_{q,a} = \psi_a \circ \phi_q$, which is applied in a two-stage manner: first ϕ_q is applied in Step (1), and then ψ_a is applied in Step (3).

We next describe the Zinc+ compiler in more detail.

Step 1: Prime projection from $\mathbb{Q}[X]$ onto $\mathbb{F}_q[X]$. The verifier samples a random $\Omega(\lambda)$-bit prime q. The constraint (2.10) is replaced by

$$\phi_q(Q)(\mathbf{b}, \phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0)) \in \mathfrak{I}', \qquad \text{in } \mathbb{F}_q[X], \quad \text{for all } \mathbf{b} \in \{0,1\}^\mu,$$

where $\mathfrak{I}' = \phi_q(\mathfrak{I} \cap \mathbb{Z}_{(q)}[X])$.

Proof sketch of soundness for Step 1. When $\mathfrak{I} = (0)$, if $Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0) = 0$ fails over $\mathbb{Q}[X]$ for some $\mathbf{b}$,

the element $Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0) \in \mathbb{Q}[X]$ is a nonzero polynomial in $\mathbb{Q}[X]$, and it vanishes modulo q only when q divides the numerators of its coefficients in lowest form. Say that such a prime q is "bad". By a counting argument (cf. lemma 2.5.8 and [**Zinc**]), only polynomially many primes cause this, so the probability of sampling a bad prime is negligible provided the set of sampleable primes is large enough (see remark 2.2.4 for some comments on this).

Now assume $\mathfrak{I} \neq (0)$. We want to show that if $Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0) \notin \mathfrak{I}$ for some $\mathbf{b}$, then $\phi_q(Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0)) \notin \mathfrak{I}'$ except with negligible probability over the choice of q. Every ideal of $\mathbb{Q}[X]$ (and of $\mathbb{F}_q[X]$) is principal. Let g denote the monic generator of $\mathfrak{I}$. Since $\mathfrak{I} \neq \mathbb{Q}[X]$, we have $\deg\ g \geq 1$. Because g is monic, $\phi_q(g)$ has the same degree as g whenever $g \in \mathbb{Z}_{(q)}[X]$, which fails only for the finitely many primes dividing a denominator of some coefficient of g; in particular $\mathfrak{I}' = (\phi_q(g))$ with $\deg\ \phi_q(g) = \deg\ g$ except with negligible probability.

If $\deg\ Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0) < \deg\ g$, consider two cases. If $\phi_q(Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0)) = 0$ in $\mathbb{F}_q[X]$, then $Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0)$ is a nonzero element of $\mathbb{Q}[X]$ that vanishes modulo q, and the counting argument from the $\mathfrak{I} = (0)$ case shows this happens with negligible probability. Otherwise $\phi_q(Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0)) \neq 0$, and since its degree is less than $\deg\ \phi_q(g)$, it cannot belong to $(\phi_q(g)) = \mathfrak{I}'$, so soundness holds.

If $\deg\ Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0) \geq \deg\ g$, we write $Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0) = g \cdot h + r$ via Euclidean division in $\mathbb{Q}[X]$, where $\deg\ r < \deg\ g$. Euclidean division is exact over $\mathbb{Q}[X]$ (since $\mathbb{Q}$ is a field), and the quotient h and remainder r are uniquely determined. Because $Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0) \notin (g)$, the remainder r is nonzero. Moreover, since g is monic, the Euclidean division algorithm introduces no new denominator primes: each step subtracts a monomial multiple of g from the current polynomial, and since g has leading coefficient 1, no denominator prime factors, other than the ones already present in Q and g, are used. Consequently, the coefficients of both h and r lie in $\mathbb{Z}_{(q)}$ whenever those of $Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0)$ and g do (which holds except for negligibly many primes q). Once we have $r \neq 0$ with $\deg\ r < \deg\ g$, the low-degree argument above applies to r after projection: $\phi_q(Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0)) = \phi_q(g) \cdot \phi_q(h) + \phi_q(r)$, and since $\deg\ \phi_q(r) < \deg\ \phi_q(g)$, the element $\phi_q(Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0))$ can belong to $(\phi_q(g)) = \mathfrak{I}'$ only if $\phi_q(r) = 0$, which happens with negligible probability by the counting argument for the zero-ideal case. See lemma 2.5.8 for the precise bound on the number of bad primes. □

Remark 2.2.4 (Correct typing of $\mathbf{y}$ and $\mathbf{f}_0$)**.** In the above argument, it is essential that the entries of $\mathbf{y}$ and $\mathbf{f}_0$ belong to $\mathbb{Q}^{<d_0,B}[X]$, i.e. that the entries are polynomials of degree bounded by d_0,

and with coefficients of bit-size less than B. This is necessary to conclude that only polynomially many primes can divide the coefficients of $Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0)$. In our PIOP soundness model, we assume malicious PIOP provers send oracles to polynomials whose coefficients belong to $\mathbb{Q}^{<d_0,B}[X]$ (and, perhaps, other appropriately typed sets). We delegate the task of enforcing this correct typing to the Polynomial Commitment Scheme (PCS) or to the IOPP. See remark 2.4.2 for more details, and the technical overview of Zinc [**Zinc**] for a thorough discussion of this matter.

Step 2: Ideal check via random MLE evaluation. The projected constraint from Step 1 requires $\phi_q(Q)(\mathbf{b}, \phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0)) \in \mathfrak{I}'$ for all $\mathbf{b} \in \{0,1\}^\mu$ over $\mathbb{F}_q[X]$. Let $\bar{v}(K)$ denote the multilinear polynomial on μ variables such that $\bar{v}(\mathbf{b}) = \phi_q(Q)(\mathbf{b}, \phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0))$ for all $\mathbf{b} \in \{0,1\}^\mu$. The verifier samples a random point $\mathbf{r} \leftarrow S^\mu$ for a large subset $S \subseteq \mathbb{F}_q$, and the prover sends

$$e = \bar{v}(\mathbf{r}) = \sum_{\mathbf{b}\in\{0,1\}^\mu} \phi_q(Q)(\mathbf{b}, \phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}; \mathbf{r}) \quad \in \mathbb{F}_q[X].$$

The verifier checks that $e \in \mathfrak{I}'$. Then the original n-row constraint (2.11) is replaced by the constraint

$$\phi_q(\Sigma^{\mathbf{r},e})(\phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0)) = 0, \qquad \text{in } \mathbb{F}_q[X], \tag{2.12}$$

where $\Sigma^{\mathbf{r},e}(\mathbf{Y}, \mathbf{Z})$ is defined as

$$\phi_q(\Sigma^{\mathbf{r},e})(\mathbf{Y}, \mathbf{Z}) := \sum_{\mathbf{b}\in\{0,1\}^\mu} \phi_q(Q)(\mathbf{b}, \mathbf{Y}, \mathbf{Z}) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}; \mathbf{r}) - e \quad \in (\mathbb{F}_q[X])[\mathbf{Y}, \mathbf{Z}],$$

The prover and verifier will now focus on proving the strict equality (2.12) over $\mathbb{F}_q[X]$.

Proof sketch of soundness for Step 2. The following lemma justifies that if $e \in \mathfrak{I}'$, then with high probability over the choice of $\mathbf{r}$, the original ideal membership constraints hold as well.

Lemma 2.2.5 (Testing ideal membership via MLE evaluation; informal version of lemma 2.5.1)**.** *Let $\mathbb{K}$ be a field, let $\mathfrak{I} \subseteq \mathbb{K}[X]$ be an ideal such that $\mathfrak{I} \cap \mathbb{K} = (0)$, and let $n = 2^\mu$. Let $\mathbf{v} \in \mathbb{K}[X]^n$ be a vector and $\tilde{\mathbf{v}}$ its multilinear extension. If not all entries of $\mathbf{v}$ belong to $\mathfrak{I}$, then for any finite $S \subseteq \mathbb{K}$ and uniformly random $\mathbf{r} \leftarrow S^\mu$, $\Pr[\tilde{\mathbf{v}}(\mathbf{r}) \in \mathfrak{I}] \leq \frac{\mu}{|S|}$.*

The result follows by projecting onto the quotient ring $\mathbb{K}[X]/\mathfrak{I}$ and applying the Schwartz–Zippel

lemma for exceptional sets (lemma 2.3.1); a full proof is given in lemma 2.5.1. Note that here we have $\mathfrak{I}' \cap \mathbb{F}_q = (0)$ because $\mathfrak{I} \neq \mathbb{Q}[X]$ and is generated by a monic polynomial. $\square$

Thus ideal-membership predicates can be handled analogously to strict equalities in standard proof systems: just as one checks $\tilde{\mathbf{u}}(\mathbf{r}) = 0$ (i.e. membership in the zero ideal) to test whether all entries of a vector $\mathbf{u}$ are zero, one checks $\tilde{\mathbf{u}}(\mathbf{r}) \in \mathfrak{I}'$ to test whether all entries of $\mathbf{u}$ belong to $\mathfrak{I}$.

Step 3: Evaluation projection from $\mathbb{F}_q[X]$ to $\mathbb{F}_q$. The verifier samples a random element $a \leftarrow \mathbb{F}_q$ and both parties apply the evaluation map ψ_a. Keeping in mind that ψ_a is a ring homomorphism, the claim $\phi_q(\Sigma^{\mathbf{r},e})(\phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0)) = 0$ in $\mathbb{F}_q[X]$ is replaced by

$$\psi_{q,a}(\Sigma^{\mathbf{r},e})(\psi_{q,a}(\mathbf{y}),\, \psi_{q,a}(\mathbf{f}_0)) = 0 \qquad \text{in } \mathbb{F}_q, \tag{2.13}$$

where we used $\psi_a \circ \phi_q = \psi_{q,a}$.

Proof sketch of soundness for Step 3. The field element $\psi_{q,a}(\Sigma^{\mathbf{r},e})(\psi_{q,a}(\mathbf{y}),\, \psi_{q,a}(\mathbf{f}_0))$ is the result of evaluating the polynomial $u(X) = \phi_q(\Sigma^{\mathbf{r},e})(\mathbf{y}, \mathbf{f}_0) \in \mathbb{F}_q[X]$ at $X = a$. If u is nonzero, then the probability that (2.13) holds is at most $\deg\, u/|\mathbb{F}_q|$. This is negligible as long as d_0 and $\deg\, Q$ are much smaller than $|\mathbb{F}_q|$. $\square$

Combining the three steps. The following diagram summarizes the three steps of the reduction from $\mathbb{Q}[X]$-constraints to algebraic constraints over $\mathbb{F}_q$:

$$\mathbb{Q}[X] \xrightarrow[\text{(Step 1)}]{\phi_q} \mathbb{F}_q[X] \xrightarrow[\text{(Step 2: ideal check)}]{\text{MLE eval at } \mathbf{r}} \mathbb{F}_q[X] \xrightarrow[\text{(Step 3)}]{\psi_a} \mathbb{F}_q.$$

Since (2.13) is a standard algebraic claim over a finite field, the prover and verifier can now invoke any PIOP for algebraic relations over $\mathbb{F}_q$ to prove it. In particular, the verifier can query the oracle $[[\psi_{q,a}(\tilde{\mathbf{f}}_0)]]$ at any point $\mathbf{r} \in \mathbb{F}_q^\mu$ by querying $[[\tilde{\mathbf{f}}_0]]$ at $\hat{\mathbf{r}}$ and then applying $\psi_{q,a}$ to the result, where $\hat{\mathbf{r}}$ is a preimage of $\mathbf{r}$ under the map induced by $\psi_{q,a}$.

The overall protocol is summarized informally in Figure 2.1; the formal version is given in Protocol 2.2 in section 2.5.

Figure 2.1: Zinc^+-PIOP (simplified, $\mathbb{Q}[X]$-part only): a PIOP for $\mathsf{REL}_{\mathsf{UCS}}$ from PIOPs over finite fields (informal).

Input: P and V receive $(\mathbb{i}, \mathbb{x}; \mathbb{w}) \in \mathsf{REL}_{\mathsf{UCS}}$ and $(\mathbb{i}, \mathbb{x})$, respectively, with $\mathbb{w} = (\mathbf{f}_0)$, $\mathbf{f}_0 \in (\mathbb{Q}^{<d,B}[X])^m$.

1: P sends the oracle $[[\tilde{\mathbf{f}}_0]]$ to the multilinear extension of $\mathbf{f}_0$.

2: **(Step 1: Prime projection.)** V samples a random $\Omega(\lambda)$-bit prime q. Both parties replace each constraint $Q(\mathbf{b}, \mathbf{y}, \mathbf{f}_0) \in \mathfrak{I}$ $(\forall \mathbf{b} \in \{0,1\}^\mu)$ (cf. (2.10)) in $\mathbb{Q}[X]$, by the constraint

$$\phi_q(Q)(\mathbf{b}, \phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0)) \in \mathfrak{I}' \qquad \forall \mathbf{b} \in \{0,1\}^\mu$$

in $\mathbb{F}_q[X]$ (cf. (2.11)). If some application of ϕ_q is not well defined, the protocol aborts.

3: **(Step 2: Randomized ideal check.)** V samples $\mathbf{r} \leftarrow (\mathbb{F}_q)^\mu$. For each constraint $(Q, \mathfrak{I})$, P computes and sends $e \in \mathbb{F}_q[X]$, which is supposedly the evaluation at $\mathbf{r}$ of the multilinear extension (MLE) of the vector $(\phi_q(Q)(\mathbf{b}, \phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0)))_{\mathbf{b} \in \{0,1\}^\mu}$ in $\mathbb{F}_q[X]$, i.e. supposedly

$$0 = \phi_q(\Sigma^{\mathbf{r},e})(\phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0)) = \sum_{\mathbf{b} \in \{0,1\}^\mu} \phi_q(Q)(\mathbf{b}, \phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}; \mathbf{r}) - e \qquad \text{in } \mathbb{F}_q[X].$$

V checks $e \in \mathfrak{I}'$ (over $\mathbb{F}_q[X]$). The new claim is (2.12), i.e. $\phi_q(\Sigma^{\mathbf{r},e})(\phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0)) = 0$ in $\mathbb{F}_q[X]$.

4: **(Step 3: Evaluation projection.)** V samples $a \leftarrow \mathbb{F}_q$ and both parties evaluate $\phi_q(\Sigma^{\mathbf{r},e})(\phi_q(\mathbf{y}), \phi_q(\mathbf{f}_0)) \in \mathbb{F}_q[X]$ at $X = a$, obtaining the finite-field claim (2.13), i.e.

$$\psi_{q,a}(\Sigma^{\mathbf{r},e})(\psi_{q,a}(\mathbf{y}),\, \psi_{q,a}(\mathbf{f}_0)) = 0 \qquad \text{in } \mathbb{F}_q. \tag{2.14}$$

5: **(Finite-field PIOP.)** P and V execute a PIOP over $\mathbb{F}_q$ to prove the above claim. Oracle queries to the witness are answered by querying $[[\tilde{\mathbf{f}}_0]]$ at an appropriate lifted point, and applying $\psi_{q,a} = \psi_a \circ \phi_q$ to the result. V accepts if and only if the finite-field PIOP accepts.

Remark 2.2.6 (Adaptation to $\mathbb{F}_{q_i}[X]$-constraints)**.** The full Zinc+ compiler (Protocol 2.2 in section 2.5) also handles $\mathbb{F}_{q_i}[X]$-constraints. The treatment is analogous: Step (1) is skipped; the ideal check (Step 2) works identically (since lemma 2.2.5 applies over any field), and the evaluation projection (Step 3) uses a map $\psi_{q_i,a_i} : \mathbb{F}_{q_i}[X] \to \mathbb{F}_{\tilde{q}_i}$ that evaluates at a random element a_i of a sufficiently large extension field $\mathbb{F}_{\tilde{q}_i}$ of $\mathbb{F}_{q_i}$. Choosing the extension degree ℓ_i large enough (so that $|\mathbb{F}_{\tilde{q}_i}| \geq \Omega(2^\lambda)$) ensures negligible soundness error for each $\mathbb{F}_{q_i}[X]$-part.

The $\mathbb{F}_{q_i}[X]$-constraints pose some technical hurdles of their own. For instance, when q_i is small (say $q_i = 2$), the extension $\mathbb{F}_{\tilde{q}_i}$ must be large, and one must analyse the event that $a_i \in \mathbb{F}_{\tilde{q}_i}$ has small extension degree over $\mathbb{F}_{q_i}$ (e.g. a_i happens to belong to the base field $\mathbb{F}_{q_i}$), which prevents the image of the projection ψ_{a_i} from covering all of $\mathbb{F}_{\tilde{q}_i}$ and can cause a soundness loss. Fortunately, this bad event can be shown to occur with negligible probability.

The $\mathbb{Q}[X]$-part and the $\mathbb{F}_{q_i}[X]$-parts are compiled independently: each part is projected to its own finite field ($\mathbb{F}_q$ for the $\mathbb{Q}[X]$-part, $\mathbb{F}_{\tilde{q}_i}$ for each $\mathbb{F}_{q_i}[X]$-part), and independent finite-field PIOPs are invoked for each. The witness vector $\mathbf{f}_0$, shared across all parts, is handled by answering the verifier's finite-field oracle queries via queries to the original $\mathbb{Q}[X]$-oracle $[[\widetilde{\mathbf{f}}_0]]$ composed with the appropriate projection. The knowledge error from each part is bounded separately, and the overall error is obtained by a union bound.

2.2.3 Efficiency and experimental results

Experimental results

In this section we report on a PoC partial implementation of our proof system. Concretely, we provide micro-benchmarks for: 1) Commit, open, and verify times for Zip+ compiled with Merkle tree commitments (i.e. our Polynomial Commitment Scheme, roughly speaking), and 2) Essentially all steps of the Zinc+ PIOP compiler (Figure 2.1). The last Step 4 of Figure 2.1 is only partially implemented and thus we report partial results for it.

Remark 2.2.7. When benchmarking Zinc+ against other schemes, it is important to compare on the basis of a concrete target computation (e.g. SHA-256 compression plus ECDSA verification) rather than raw trace size alone. This is because **Zinc+ deliberately trades some performance (per fixed trace size) for the ability to express computations with significantly smaller traces than other schemes**. Consequently, a naïve comparison at equal trace sizes would understate Zinc+'s advantage. For this reason, we arithmetize SHA-256 and ECDSA and use the resulting trace sizes as the basis for all our benchmarks (see below for more details on what traces are).

We emphasize that while our experiments revolve around SHA-256 and ECDSA, we expect our framework to be useful and competitive for a wide variety of applications, particularly those listed in table 2.2.

UAIR+ constraints As done in section 2.2.1 with the relation REL_f, we arithmetize our computations as a specific type of UCS constraint, which we call *Universal AIR constraint with lookups (UAIR+)*. Specifically, witness vectors are organized into traces (i.e. matrices), and constraints apply consecutively on contiguous rows of the trace (sometimes the separation between constrained rows

can be larger).Additionally, trace columns admit types, which are enforced via lookup constraints, e.g., a column may be constrained to have all its entries in $\{0,1\}^{<32}[X]$, $[0\,..\,2^{32}-1]$, etc.. UAIR+ constraints generalize standard AIR constraints [**ethstark**]. We refer to **??** for a full description and discussion of the UAIR+ relation.

Remark 2.2.8 (Indicative benchmarks)**.** The benchmark in table 2.6 is intended as a rough estimate of the end-to-end performance achievable with our techniques, not as a direct comparison with fully deployed systems. Three caveats apply: first, our experiments do not cover the full proving and verification pipeline, although they include Zip+, which is expected to dominate the overall cost. Second, our implementation is an initial proof of concept and has not been optimized; third, due to an incomplete implementation, we make benchmark choices that are detrimental to the overall performance, as discussed in remark 2.2.9.

We expect the benchmarks to improve in completeness and performance as we further develop our scheme in the coming months.

Technical specifications All reported benchmarks are run on a MacBook Air M4 (16 GB RAM, 2025), multi-threaded. All experiments target 100 bits of security.

Our IPRS codes use rate 1/4, the base IPRS field is of prime size $p = 2^{16} + 1$ (a Fermat prime), the FFT radix is 8, and the FFT depth is 1 when #Vars. is 9, and 2 when #Vars. is 10 or 11. The coefficients of each polynomial are organized in a matrix of size $2 \times 2^{\#\text{Vars}-1}$. This leads to optimal proof size in this setting. We use the Unique Decoding Radius proximity parameter . Run at this rate with a target security of 100 bits, our instantiation of Zip+ requires 147 queries. If IPRS codes were shown to have Mutual Correlated Agreement up to the Johnson bound,then the number of queries would be 100 and proof sizes would decrease substantially.

Table 2.3 consolidates the main parameters used throughout our experiments.

Parameter	Value
Target security (λ)	100 bits
Base IPRS field (p)	$2^{16}+1$ (Fermat prime)
IPRS rate	1/4
FFT radix	8
FFT depth	1 (#Vars. = 9), 2 (#Vars. = 10, 11)
Proximity parameter	Unique Decoding Radius (UDR)
Minimum relative distance (δ)	3/4
Number of queries	147 (100 if MCA up to Johnson bound)
Matrix layout	$2 \times 2^{\#\text{Vars}-1}$
Hash function (Merkle tree)	BLAKE3
Fiat–Shamir hash	BLAKE3
Prime modulus (q_0) bit-size	$\Omega(\lambda)$ ($\geq$ 128-bit prime)

Table 2.3: Consolidated parameter table for all experiments in this section. The proximity parameter and query count affect both proof size and verifier time. The FFT depth and base field determine the IPRS norm growth.

Benchmarks for the Zip+ IOPP

Tables 2.4 and 2.5 provides Zip+ IOPP benchmarks for batches of MLE polynomials of sizes corresponding to our SHA-256 arithmetization, namely 7 polynomials with coefficients in $\{0,1\}^{<32}[X]$ (in fact, two of them are in $\mathbb{F}_2[X]$, see remark 2.2.11) and 3 polynomials with coefficients in $[0\,..\,2^{32}-1]$. Each row corresponds to a different vector length, which in turn corresponds to a different number of SHA-256 compressions.

Table 2.5 provides further benchmarks within parenthesis for our unoptimized arithmetization of ECDSA verification. Table 2.6 aggregates the benchmarks from tables 2.4 and 2.5 to provide an estimate of the Zip+ costs for proving 8 SHA-256 compressions followed by an ECDSA verification.

Our SHA-256 compression arithmetization has 10 columns typed in $\{0,1\}^{<32}[X]$, 3 in $[0\,..\,2^{32}-1]$, and 2 in $\mathbb{F}_2[X]$, the columns length for 2^c compressions is 2^{6+c+1}, cf. table 2.9. Our (non-optimized) ECDSA verification arithmetization has 24 columns typed in $[0\,..\,2^{32}-1]$, each of length 2^{10}, cf. table 2.10.

Remark 2.2.9 (Detrimental benchmark choices)**.** We briefly mention three detrimental benchmark choices that we make due to our incomplete implementation: 1) the three columns typed in $\mathbb{F}_2^w[X]$ of the SHA-256 trace (with w much smaller than 32) are typed in $\{0,1\}^{<32}[X] \subseteq \mathbb{Z}[X]$ (we expect this to be substantially more expensive for Zip+); 2) when combining SHA-256 and ECDSA, we simply add together the costs of the two separate benchmarks, missing on the amortization benefits

of our batching techniques —note that one of the key features of our framework is that it allows mixing constraints expressed in different algebraic structures, so we can combine the SHA-256 and ECDSA arithmetization together with almost no cost; 3) we treat each ECDSA $\mathbb{F}_p$-typed element as 8 copies of a 32-bit integer, with no specific gain. This removes the advantage of using modular arithmetic as opposed to integers.

Note that 1) and 3) can be done without altering the correctness of the arithmetization: one can always "lift" elements from $\mathbb{F}_2[X]$ or $\mathbb{F}_p$ to $\mathbb{Z}[X]$ and $[0\,..\,p-1]$, respectively, and obtain an equivalent arithmetization.

Batched Zip$^+$ on 7 polynomials over $\{0,1\}^{<32}[X]$				
#Vars.	Prover (ms)	Verifier (ms)	Proof (KB)	#Compr. (Proof, KB)
9	3.34	1.28	211	4 (158)
10	5.23	1.50	221	8 (168)
11	9.51	2.70	240	16 (186)

Table 2.4: Zip+ IOPP benchmarks (compiled with Merkle tree commitments) for batches of MLE polynomials of sizes corresponding to the dominant part of our SHA-256 arithmetization (cf. **??**). All rows provide costs of batching 7 MLEs of vectors with entries in $\{0,1\}^{<w}[X]$. The coefficients of each polynomial are organized in a $1 \times 2^{\#\text{Vars}}$ table, which leads to optimal proof size in this setting. The "#Compr." column indicates how many SHA-256 compressions (as per our arithmetization in **??**) the committed polynomials correspond to. We make the unfavorable (to us) choices in remark 2.2.9, namely, we treat two columns typed in $\mathbb{F}_2[X]$ as $\{0,1\}^{<32}[X]$-typed columns. In parentheses next to the number of compressions we indicate the proof size if we kept said columns in their correct $\mathbb{F}_2[X]$ type.

Batched Zip$^+$ on 3 (27) polynomials over $[0\,..\,2^{32}-1]$			
#Vars.	Prover (ms)	Verifier (ms)	Proof (KB)
10	1.10 (5.48)	0.63 (0.80)	15 (43)
11	1.46 (7.03)	0.68 (0.86)	20 (62)
12	1.68 (8.93)	0.87 (0.93)	30 (87)
13	2.73 (14.59)	1.35 (1.58)	41 (125)

Table 2.5: Zip+ IOPP benchmarks (compiled with Merkle tree commitments) for batches of 3 (respectively, 27) MLE polynomials with entries in $[0\,..\,2^{32}-1]$. Our SHA-256 compression arithmetization has 3 such polynomials, with 10 variables for 8 compressions, 11 for 9 compressions, and so on. Our initial, non-optimized, ECDSA arithmetization **??** has 24 such polynomials, each of 10 variables. We use a batch size of 27 in parentheses which corresponds to the aggregate number of polynomials typed in $[0\,..\,2^{32}-1]$ between both our SHA-256 and ECDSA arithmetizations.

Batched Zip$^+$ on trace of SHA-256 ($\times 8$) + ECDSA		
Prover (ms)	Verifier (ms)	Proof (KB)
10.71	2.30	211

Table 2.6: Zip+ IOPP benchmarks (compiled with Merkle tree commitments) for a trace layout matching 8 SHA-256 compressions plus one ECDSA verification, as per our arithmetizations. All times include the cost of Merkle tree hashing. Prover and verifier times are obtained by summing the results from table 2.4 and table 2.5 (cf. remark 2.2.9). Proof sizes correspond to batched Zip+ sizes (i.e. they are not just the sum of proof sizes in tables 2.4 and 2.5). See remark 2.2.7 for further comments on these benchmarks. For context, Binius64's *end-to-end* benchmark for the same task is given in table 2.8.

Benchmarks for the Zinc+ PIOP compiler

In table 2.7 we provide benchmarks for almost all the steps of the Zinc+ PIOP prover (Figure 2.1). For convenience, we include the corresponding Zip+ prover times as well. Our benchmarks use a UAIR+ whose trace matches that of our SHA-256 + ECDSA verification trace. Namely, it has 7 columns typed in $\{0,1\}^{<32}[X]$ and 27 columns typed $[0\,..\,2^{32}-1]$, and the trace has 2^{10} rows. The constraints are described after the table. They do not correspond to the actual constraints of our SHA-256 + ECDSA arithmetization, but we believe are of similar complexity. In remark 2.2.10 we discuss the main unimplemented parts of our Zinc+ PIOP and how they may affect the benchmarks in table 2.7.

The finite field PIOPs used by our Zinc+ PIOP compiler are simple: they (1) run batched sumcheck on the projected MLE evaluation claims (2.14) plus other sumcheck claims arising in the next step; (2) run a lookup argument to enforce the projected typing constraints in the UAIR+ (remark 2.2.10), (3) use a batched sumcheck approach for proving evaluation claims for the MLE of shifted columns. Our final step in table 2.7 only includes numbers for (1)—see remark 2.2.10 for further discussion.

Zip+ IOPP prover	
Zip+ prover	10.71 ms
(Partial) Zinc+ PIOP prover	
Project trace from $\mathbb{Z}[X]$ to $\mathbb{F}_q[X]$	1.75 ms
Ideal elements computation	3.39 ms
Project trace from $\mathbb{F}_q[X]$ to $\mathbb{F}_q$	0.25 ms
Batched sumcheck over $\mathbb{F}_q$	1.72 ms
Subtotal	7.11 ms
Total	**17.82 ms**

Table 2.7: Prover times for the Zip+ IOPP and the main steps of the Zinc+ PIOP compiler Figure 2.1. We use a UAIR+ matching the column profile from table 2.6, namely it has 7 columns in $\{0,1\}^{<32}[X]$ and 27 columns in $[0\,..\,2^{32}-1]$. The trace has 2^{10} rows. This trace layout corresponds to that of our unoptimized arithmetization of SHA-256 + ECDSA (with 8 compressions). The constraints are described after the table.
The steps measured in the table are the following: (1) *Zip+ prover*: the commit and open times for the Zip+ IOPP prover, as per table 2.6; (2) *Project trace from $\mathbb{Z}[X]$ to $\mathbb{F}_q[X]$ – Step 1 of Figure 2.1*: prover reduces all elements in the trace modulo q, via the map $\phi_q : \mathbb{Z}[X] \to \mathbb{F}_q[X]$. This is not a negligible step because the reduction entails computing Montgomery forms of integers (for later efficient operation on $\mathbb{F}_q$); (3) *Ideal elements computation – Step 2 of Figure 2.1*: run the randomized ideal check via MLE evaluation at $\mathbf{r}$, reducing ideal-membership constraints to a strict equality over $\mathbb{F}_q[X]$; (4) *Project to $\mathbb{F}_q$ – Step 3 of Figure 2.1*: apply evaluation projection ψ_a, evaluating the $\mathbb{F}_q[X]$ polynomials at the challenge a to obtain constraints over $\mathbb{F}_q$; (5) *Sumcheck over $\mathbb{F}_q$ – Partial Step 4 of Figure 2.1*: execute the finite-field PIOP, combining all evaluated polynomials into a single opening claim for the PCS, see remark 2.2.10.

The UAIR used in the benchmarks from table 2.7 has 7 binary-polynomial columns $b_0, \dots, b_6$ each in $\{0,1\}^{<32}[X]^{2^{10}}$ and 27 integer columns $z_0, \dots, z_{26}$, each in $[0\,..\,2^{32}-1]^{2^{10}}$, with 14 constraints applied for every row $t \in [2^{10}]$:

$$
\begin{aligned}
b_i[t] - z_i[t] &\in (X-2), & i &\in \{0,1,2\}, && \text{(3 deg-1)}\\
b_{i+3}[t] - z_i[t+1] &\in (X-2), & i &\in \{0,1,2\}, && \text{(3 deg-1)}\\
z_{3+3i}[t] + z_{4+3i}[t] - z_{5+3i}[t] &= 0, & i &\in \{0,\dots,5\}, && \text{(6 deg-1)}\\
z_{21+3i}[t] \cdot z_{22+3i}[t] - z_{23+3i}[t] &= 0, & i &\in \{0,1\},. && \text{(2 deg-2)}
\end{aligned}
$$

Here $v_i[t]$ denotes the t-th entry of a vector v_i. The element $z_i[t+1]$ is defined as zero when $t = 2^{10}$.

Remark 2.2.10 (Unimplemented parts of Zinc+ PIOP)**.** The main unimplemented parts of our Zinc+ PIOP have to do with managing MLE evaluations of shifted columns and running lookup arguments over the finite field $\mathbb{F}_q$. The former is the main reason why the UAIR+ constraints we use for benchmarking are not exactly those of our SHA-256 + ECDSA arithmetization. These

missing parts are purely based on sumchecks over $\mathbb{F}_q$, most of which can be batched with other sumchecks. Other unimplemented parts of the Zinc+ PIOP have to do with boundary constraints and selector polynomials for some constraints, cf. **??**.

Regarding the lookup arguments, we remark that these are performed purely over $\mathbb{F}_q$. In particular, any decomposition vectors required during the argument (in order to handle lookups into large tables) are vectors of $\mathbb{F}_q$ elements. To save on implementation hurdles we have the prover send the decomposition vectors in the clear: for lookup tables $\{0,1\}^{<32}[X]$ and $[0\,..\,2^{32}-1]$, and batching lookup claims, this requires sending a vectors of length 2^{10} (in our example UAIR+) with entries in $\mathbb{F}_q$, which amounts to 12 KB.

We can give a rough estimate of the cost of the missing components. The lookup and shifted-column MLE sumchecks each operate on vectors of length 2^{10} over $\mathbb{F}_q$, i.e. they are of the same size as the batched sumcheck over $\mathbb{F}_q$ already measured in table 2.7 at 1.72 ms. The lookup argument (say LogUp) requires an additional smaller batched sumcheck invocation, which can be batched with the aforementioned batched sumcheck (see remark 2.2.10 for a discussion of how to handle chunk decomposition in lookups). The shifted-column MLE evaluation adds one more batched sumcheck (this cannot be batched with the previous one). Overall, becasue all these extra batched sumchecks are on smaller polynomials (in number of monomials) than the original one, we expect them to, at most, double our current sumcheck costs.

Partial comparisons with other schemes

For context, we next provide end-to-end benchmarks for the Binius64 proof system on SHA-256 hashing and SHA-256+ECDSA [**binius**]. We emphasize that the following comparison is between our partial (IOPP-only) benchmarks and Binius64's end-to-end numbers, and is therefore not apples-to-apples; nonetheless, the gap between the two provides preliminary evidence that a fully optimized end-to-end Zinc+ implementation may outperform state-of-the-art SNARKs for these computations. Binius64 is the most competitive scheme we are aware of for the specific computation our benchmarks target, namely SHA-256 followed by ECDSA verification. For other applications (e.g. the Hamming weight + RLWE encryption example from **??**) our framework may show a larger advantage, though again this remains to be validated with end-to-end benchmarks.

On the other hand, using our work for, say, proving Poseidon2 hashing on a small field would

result in worse performance for Zinc+ than state-of-the-art competitors[2]. Indeed, Zinc+'s value proposition is to allow for significantly smaller arithmetization sizes than other schemes in non-field-based computations.

Remark 2.2.11 (Comparison caveats)**.** The reported benchmarks are both favorable to us because we do not report end-to-end results (cf. remark 2.2.10), and unfavorable to us because of unoptimized implementation and detrimental benchmark choices. Regardless, we hope the reported experiments resolve what is, in our opinion, the most pressing question regarding the performance of our framework: namely whether the smaller arithmetization sizes compensate for the cost of committing to polynomials with coefficients in $\mathbb{Z}[X]$ (see section 2.2.3 for further discussion). The experiments provide preliminary support for a positive answer: for 8×SHA-256 + ECDSA, our Zip+-only prover (the IOPP compiled with Merkle tree commitments, not end-to-end) runs in 10.71 ms with a 211 KB proof, whereas the Binius64 end-to-end prover takes 136 ms with a 291 KB proof (table 2.8). While this is not an apples-to-apples comparison since our numbers cover only the IOPP/PCS component (including Merkle tree hashing)—the over 12× gap in prover time and 27% smaller proof size leave substantial room for the remaining PIOP-related costs. For reference, the (partial) PIOP costs reported in table 2.7 add up to 7.11 ms, which is about 66% of the Zip+ prover time.

table 2.8 provides end-to-end benchmark results for Binius64 on SHA-256 and SHA-256+ECDSA. To aid comparison: the Binius64 row for 8×SHA-256 + ECDSA (prover: 136 ms, proof: 291 KB) targets the same computation as our aggregated Zip+ benchmark in table 2.6 (prover: 10.71 ms, proof: 211 KB). Similarly, the SHA-256-only rows in table 2.8 correspond to the same number of compressions as the rows in table 2.4. Note that our numbers cover only the Zip+ layer (IOPP compiled with Merkle tree commitments), whereas Binius64's are end-to-end.

[2]Technically, Zinc+ can be instantiated so that it degenerates into any existing multilinear-based scheme for finite fields. Hence, technically speaking, a perfectly instantiated and implemented Zinc+ is always roughly at least as performant as the best existing multilinear-based scheme over finite fields

Binius64: SHA-256			
#Compr.	Prover (ms)	Verifier (ms)	Proof (KB)
4	23.1	3.5	156.0
8	25.9	3.8	166.3
16	31.5	4.4	179.7
Binius64: 8×SHA-256 + ECDSA			
8	136.0	10.79	291

Table 2.8: Binius64 *end-to-end* benchmarks. Top: SHA-256 for 4 to 16 compression rounds. Bottom: 8 SHA-256 compressions plus one ECDSA signature verification. Times do not include the witness generation step.

Summary of SHA-256 and ECDSA arithmetizations.

table 2.9 and table 2.10 summarize the trace layout and constraint structure of our SHA-256 and ECDSA arithmetizations in UAIR$^+$ form, respectively. UAIR$^+$ is an AIR-like subclass of our general UCS relation $\mathsf{REL}_{\mathsf{UCS}}$. Its description is in **??**. The main idea of UAIR$^+$ is that witness vectors are organized into tables, called *traces*, and polynomial constraints apply to pairs of consecutive rows (sometimes one allows constraints accessing non-adjacent rows).

A full arithmetization for SHA-256 can be found in **??**, and **??** sketches the arithmetization of ECDSA.

Trace layout			*Constraints*		
#Cols	Lookup set	Domain	#Mon.	Ideal	Ring
10	$\{0,1\}^{<w}[X]$	$\subseteq \mathbb{Q}^{<w}[X]$	10	(X^w-1)	$\subseteq \mathbb{F}_2[X]$
5	$[0\,..\,2^w-1]$	$\subseteq \mathbb{Q}$	6	(0)	$\subseteq \mathbb{F}_2[X]$
4	$\mathbb{F}_2^{<w}[X]$	$\subseteq \mathbb{F}_2^{<w}[X]$	27	$(X-2)$	$\subseteq \mathbb{Q}[X]$

Table 2.9: SHA-256 (1 compression) UAIR$^+$ summary (64 rows, 19 witness columns, 53 monomials total), where $w = 32$. The "#Mon." column counts the total number of monomials across constraint polynomials sharing the same ideal and ring. All constraint polynomials are linear in the witness variables (the degree may increase when multiplied by selector polynomials; see **??**). Of the 5 rational-integer columns, 3 have entries in $[0\,..\,6]$; the remaining 2 have arbitrary entries in $[0\,..\,2^w-1]$. In our proxy benchmarks, we tighten the trace to have size $2^7 \times 8$ cells in $\{0,1\}^{<32}[X]$ and $2^7 \times 3$ in $[0\,..\,2^{32}-1]$, i.e. we make it have 2x more rows and 2x less columns. This favors proof size at the expense of slightly extra PIOP work over the random projected finite field.
Due to an incomplete implementation, the $\mathbb{F}_2^{<w}[X]$ columns are treated as being typed in $\{0,1\}^{<w}[X]$ (which is a detrimental choice for us).

Trace layout			*Constraints*		
#Cols	Lookup set	Domain	#Cstr.	Ideal	Ring
3	$\{0,1\}$	$\subseteq \mathbb{Q}$	9	(0)	$\subseteq \mathbb{F}_p$
9	$\mathbb{F}_p$	$\subseteq \mathbb{F}_p$	2	(0)	$\subseteq \mathbb{F}_n$
2	$\mathbb{F}_n$	$\subseteq \mathbb{F}_n$			

Table 2.10: ECDSA verification UAIR$^+$ summary (258 rows, 14 witness columns). The "#Cstr." column counts the number of non-boundary constraint polynomials per (ideal, ring) group. The maximum constraint degree is 6. Of the 3 rational-integer columns, all belong to $\{0,1\}$.
For our benchmarks, we make the non-optimal choice of representing all columns from $\mathbb{F}_p$ and $\mathbb{F}_n$ as 8 integer columns over $[0\,..\,2^w{-}1]$, where $w = 32$. In our proxy benchmarks, we ignore the 3 columns with entries in $\{0,1\}$.
As with the SHA-256 trace, and due to the same reasons (table 2.9) we tighten the ECDSA trace to have size $2^{10} \times 24$ rather than $2^8 \times 88$.
We believe this arithmetization can be greatly optimized, and we leave this to future work.

Efficiency and opportunities of our framework

The techniques used in Zinc+ in the context of SNARKs are relatively different from standard techniques that revolve around finite field arithmetic and polynomials with coefficients in said fields. We consider the problem of understanding how our framework compares to standard techniques an open problem. We provide an informal discussion of our observations around this question.

In terms of efficiency, we view Zinc+ as a framework that bootstraps the vast majority of SNARK costs to the single task of committing to multilinear polynomials with coefficients in $\mathbb{Q}[X]$, $\mathbb{Z}[X]$, $\mathbb{F}_q[X]$, etc.—in particular, for the applications we discussed, to polynomials with coefficients in $\{0,1\}^{<w}[X]$. Indeed, in many ways, working in $\{0,1\}^{<w}[X]$ creates w copies of a problem to be solved. For example, our commitments to vectors with entries in $\{0,1\}^{<w}[X]$ are effectively w parallel commitments to binary vectors (optimizations apply, see below). But on the other hand, the random projection technique onto finite fields compresses these w copies into a single one, by introducing a small soundness error. Putting together this compression saving and the smaller initial arithmetizations that our UCS constraints produce, we observe in our implementation that the costs once in the finite field are tiny compared to the commitment costs and the projection costs.

The fundamental question is then: *does the tradeoff of smaller arithmetizations, plus the compression savings of projection, outweigh the cost of committing to these polynomials (and also the projection costs)?* We believe that our preliminary benchmarks answer the question positively.

We remark that committing to polynomials with coefficients in $\{0,1\}^{<w}[X]$, while seemingly

a "w-wise" task as already mentioned, can be implemented so that it does not scale linearly in w. Indeed, we observe engineering opportunities here by, for example, representing elements of $\{0,1\}^{<w}[X]$ as `u32` types and using SIMD optimizations to operate on those. The fact that the Zinc+ framework concentrates all the computational burden on these commitments allows one to focus on optimizing a single type of operation.

Zip+, proof sizes, and future work Zip+ is a Ligero/Brakedown-like IOPP (in modern instantiation) for $\mathbb{Q}[X]$ and $\mathbb{F}_q[X]$-valued multilinear polynomials. Ligero/Brakedown on finite fields are known for having large proof sizes. And indeed our benchmarks show significant proof sizes relative to prover and verifier time (which on the other hand are very small). Besides being based on large-proof IOPPs, Zip+ experiments show a further proof size increase due to the fact that encodings of vectors in $\{0,1\}^{<w}[X]$ scale linearly on w in bit-size, being, from a purely size perspective, the same as w commitments to binary vectors. Note that these commitments are over $\mathbb{Z}[X]$ and not over $\mathbb{F}_2[X]$, hence there is no modular reduction that would help keep encoding sizes small.

Ongoing work of ours is focused on designing a WHIR, Ligerito, Blaze, FICS/FACS, etc. [**whir**; **Ligerito**; **Blaze**; **FICS-FACS**]—IOPPs known for their small proofs among others—variant for our framework.

2.3 Preliminaries

2.3.1 Basic notions

For $k \geq 1$, let $[k] := \{1, \dots, k\}$, and let $[0\,..\,k]$ be $\{0\} \cup [k]$. We write $x \leftarrow S$ for uniform sampling from a finite set S, and $x \leftarrow \mathcal{D}$ for sampling from a distribution $\mathcal{D}$.

We use $\mathbb{K}$ to denote (possibly infinite) fields and R to denote rings. We write $\mathbb{F}_q$ for the finite field with q elements (where q is a prime or a prime power), $\mathbb{Q}$ for the field of rational numbers, and $\mathbb{Z}$ for the ring of integers. We use standard terminology and notation: an ideal I generated by a set $S \subseteq R$ is denoted (S), the quotient ring is denoted R/I. Sometimes we write $(S)_R$ to emphasize that the ideal is an ideal of the ring R. A ring is an integral domain if the product of any two nonzero elements is nonzero.

Polynomial rings and commutative algebra. For variables $\mathbf{Y} = (Y_1, \dots, Y_\mu)$, $R[\mathbf{Y}]$ denotes multivariate polynomials over R, and $R^{\mathsf{ml}}[\mathbf{Y}]$ the multilinear polynomials (individual degree ≤ 1 in each variable).

We often work with multivariate polynomials whose coefficients are themselves univariate polynomials. For a ring $\mathcal{R}$ (typically $\mathbb{Z}[X]$, $\mathbb{Q}[X]$, or $\mathbb{Z}_{(q)}[X]$) and variables $\mathbf{Y} = (Y_1, \dots, Y_\mu)$, we write $\mathcal{R}[\mathbf{Y}]$ for polynomials in $\mathbf{Y}$ with coefficients in $\mathcal{R}$. We view elements of $(\mathbb{Z}[X])[\mathbf{Y}]$ as polynomials in $\mathbf{Y}$ (not as elements of $\mathbb{Z}[X, \mathbf{Y}]$), denoting them by $f(\mathbf{Y})$. For polynomial rings, $\mathcal{R}^{<d,<B}[X]$ denotes polynomials of degree $< d$ with coefficients in $\mathcal{R}^{<B}$; we may write $\mathcal{R}^{<d}[X]$ when the bit bound is clear. Ring homomorphisms $f : \mathcal{R}_1 \to \mathcal{R}_2$ extend coefficient-wise to $\mathcal{R}_1[\mathbf{Y}] \to \mathcal{R}_2[\mathbf{Y}]$. More generally, all maps are extended entrywise to vectors and matrices.

For any field $\mathbb{F}_{p^e}$ containing subfield L we can map elements of L into $\mathbb{F}_{p^e}$ via the canonical embedding $\iota_{p^e} : L \hookrightarrow \mathbb{F}_{p^e}$. We extend this embedding to the ring $\mathbb{F}_{p^e}[X]$ its subrings $L[X]$ by applying ι_{p^e} coefficient-wise. If $(\mathrm{j}) = \mathcal{J} \subseteq \mathbb{F}_q[X]$ is an ideal, we note $(\iota_{q^\ell}(\mathrm{j})) = \mathcal{J}' \subseteq \mathbb{F}_{q^\ell}[X]$ is also.

The subset $\mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : p \nmid b\}$ of $\mathbb{Q}$ is a ring called the *localization of* $\mathbb{Z}$ *at the prime* p. This ring admits a surjective homomorphism $\phi_p : \mathbb{Z}_{(p)} \to \mathbb{F}_p$, $a/b \mapsto a \cdot b^{-1}$, with kernel $\ker(\phi_p) = p\mathbb{Z}_{(p)}$.

For prime powers $q = p_i^{e_i}$, we set $\mathbb{Z}_{(q_i)} := \mathbb{Z}_{(p_i)}$, and extend ϕ by defining $\phi_{q_i} = \iota_{q_i} \circ \phi_{p_i}$. The *intersection localization ring* is

$$\mathbb{Z}_{(q_1 \cdots q_n)} \;:=\; \bigcap_{i \in [n]} \mathbb{Z}_{(p_i)} \;=\; \Big\{ a/b \in \mathbb{Q} \;\Big|\; \gcd\big(b, \textstyle\prod_i p_i\big) = 1 \Big\}. \tag{2.15}$$

Bitstring representation. We use standard representations: integers are represented in two's complement, so an integer $n \in \mathbb{Z}$ has bit-size $\lceil \log_2(|n|+1) \rceil + 1$; rationals $a/b \in \mathbb{Q}$ are represented as pairs (a, b) in *lowest terms* (i.e., $\gcd(a, b) = 1$ and $b > 0$), with bit-size the sum of the bit-sizes of a and b; polynomials are represented as tuples of coefficients (in increasing degree order), with bit-size the sum of the coefficient bit-sizes. We write $\mathbb{Z}^{<B}$, $\mathbb{Q}^{<B}$, $\mathbb{Z}^{<d,<B}[X]$, $\mathbb{Q}^{<d,<B}[X]$, and $\mathbb{Z}_{(q)}^{<d,<B}[X]$ for elements with bit-size less than B (and degree $< d$ for polynomials). For a ring $\mathcal{R}$, we write $\mathcal{R}^{<B}$ for elements representable with fewer than B bits.

Multilinear extensions (MLE). A multilinear polynomial $f(\mathbf{Y}) \in R^{\mathsf{ml}}[\mathbf{Y}]$ is uniquely determined by its values on $\{0,1\}^\mu$. For any map $f : \{0,1\}^\mu \to R$, we write $\tilde{f}(\mathbf{Y})$ for its *multilinear extension*

(MLE), the unique multilinear polynomial that agrees with f on the Boolean hypercube. For a vector $\mathbf{v} \in R^{2^\mu}$, we write $\tilde{\mathbf{v}}(\mathbf{Y})$ for the MLE of the map $\mathbf{y} \mapsto v_{\mathbf{y}}$ (interpreting $\mathbf{y} \in \{0,1\}^\mu$ as an integer in binary). We also use the standard *equality polynomial*

$$\widetilde{\mathsf{eq}}(\mathbf{y};\mathbf{Y}) := \prod_{i\in[\mu]} \left(y_i Y_i + (1-y_i)(1-Y_i)\right), \tag{2.16}$$

which satisfies $\widetilde{\mathsf{eq}}(\mathbf{y};\mathbf{z}) = 1$ if $\mathbf{y} = \mathbf{z}$ and $\widetilde{\mathsf{eq}}(\mathbf{y};\mathbf{z}) = 0$ otherwise for $\mathbf{y}, \mathbf{z} \in \{0,1\}^\mu$.

Schwartz–Zippel lemma for exceptional sets. We will need the following general form of Schwartz-Zippel lemma in order to address certain ideal-membership constraints in our proof system. A subset $S \subseteq \mathcal{R}$ is *exceptional* if $a - b$ has a multiplicative inverse for all distinct $a, b \in S$.

Lemma 2.3.1 (Schwartz–Zippel lemma for exceptional sets)**.** *Let $S \subseteq \mathcal{R}$ be a finite exceptional set and $f(\mathbf{Y}) \in \mathcal{R}[\mathbf{Y}]$ a nonzero multilinear polynomial in μ variables. Then* $\Pr_{\mathbf{r}\leftarrow S^\mu}[f(\mathbf{r}) = 0] \le \mu/|S|$.

2.3.2 Indexed relations and oracles.

An *indexed relation* REL is a set of index-input-witness triples $(\mathbb{i}, \mathbb{x}; \mathbb{w})$. The *language* associated with REL is $\mathsf{LANG}(\mathsf{REL}) := \{(\mathbb{i}, \mathbb{x}) \mid \exists\, \mathbb{w} \text{ such that } (\mathbb{i}, \mathbb{x}; \mathbb{w}) \in \mathsf{REL}\}$.

An *oracle* to a string or vector π of elements from a ring R is a black-box that allows querying any position j and obtaining π_j. We denote it by $[[\pi]]$. An *oracle to a polynomial* $f(\mathbf{X})$ over R is a black-box that allows querying any point $\mathbf{r}$ in the domain of f and obtaining $f(\mathbf{r})$. We denote it by $[[f]]$.

Remark 2.3.2 (Oracle typing assumption)**.** Soundness and knowledge soundness for PIOPs are required to hold only for malicious provers P^* that send *correctly typed* oracles—i.e., oracles that actually correspond to polynomials with the prescribed number of variables, degrees, and coefficient domains. Informally speaking, enforcing that a (computationally bounded) prover sends such oracles is the responsibility of the polynomial commitment scheme (PCS) or other cryptographic mechanisms (such as IOPP) used to compile the PIOP into a concrete argument system.

2.3.3 Polynomial Interactive Oracle Reductions (PIORs)

Polynomial Interactive Oracle Reductions (PIORs) generalize PIOPs. A *public-coin polynomial IOR* from a relation $\mathcal{R}$ to a relation $\mathcal{R}'$ is a pair $\Pi = (\mathsf{P}, \mathsf{V})$ specified as follows.

The prover P is an interactive algorithm that receives as input $(\mathbb{i}, \mathbb{x}, \mathbb{w}) \in \mathcal{R}$. The verifier V is an interactive polynomial oracle algorithm begins with inputs $(\mathbb{i}, \mathbb{x})$, which may include both polynomial oracles and explicit values. They interact for k rounds. In round $i \in [k]$, the prover sends a polynomial oracle π_i and the verifier sends a uniformly random message ρ_i. After k rounds of interaction, the verifier makes evaluation queries to the polynomial oracles in its inputs and sent in prover messages $\pi_1, \dots, \pi_k$ as oracles. Then:

- V either rejects or outputs a new index-instance $(\mathbb{i}', \mathbb{x}')$, and
- P outputs a new witness $\mathbb{w}'$ such that $(\mathbb{i}', \mathbb{x}', \mathbb{w}') \in \mathcal{R}'$.

Without loss of generality, we may view V's output as a deterministic function of its input and the interaction transcript. Note that oracles in $\mathbb{x}'$ are specified implicitly in terms of oracles given in $\mathbb{x}$ and $\pi_1, \dots, \pi_k$.

Definition 2.3.3 (Completeness)**.** We say that Π has *completeness* with error ϵ if for every $(\mathbb{i}, \mathbb{x}, \mathbb{w}) \in \mathcal{R}$, letting the interaction proceed honestly and writing $(\mathbb{i}', \mathbb{x}') \leftarrow \mathsf{V}(\mathbb{i}, \mathbb{x}, \pi_1, \rho_1, \dots, \pi_k, \rho_k)$ and $(\pi_k, \mathbb{w}') \leftarrow \mathsf{P}(\mathbb{i}, \mathbb{x}, \mathbb{w}, \rho_1, \dots, \rho_{k-1})$, we have

$$\Pr_{\rho_1, \dots, \rho_k} \left[(\mathbb{i}', \mathbb{x}', \mathbb{w}') \notin \mathcal{R}' \right] \leq \epsilon(\mathbb{i}, \mathbb{x}).$$

We omit a standard soundness definition as we will prove the stronger notion of round-by-round knowledge soundness.

We will use the definition of Round-by-Round knowledge soundness from [**BCFW25**].

Definition 2.3.4 (Knowledge state function)**.** Let Π be an IOR from a relation $\mathcal{R}$ to a relation $\mathcal{R}'$. A *knowledge state function* for Π is a (possibly inefficient) function KState, that takes as input a statement $(\mathbb{i}, \mathbb{x}, \mathbb{y})$, an interaction transcript tr, a knowledge witness w, and outputs a bit. It must satisfy:

- **Empty transcript.** If $\mathsf{tr} = \emptyset$, then $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathbb{y}, \mathsf{tr}, \mathsf{w}) = 1$ if and only if $(\mathbb{i}, \mathbb{x}, \mathbb{y}, \mathsf{w}) \in \mathcal{R}$.

- **Prover moves.** If tr is a transcript where the prover is about to move and $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathbb{y}, \mathsf{tr}, \mathsf{w}) = 0$, then for every prover message π,

$$\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathbb{y}, \mathsf{tr}\|\pi, \mathsf{w}) = 0.$$

- **Full transcript.** If $\mathsf{tr} = (\pi_1, \rho_1, \dots, \pi_k, \rho_k)$ is a full transcript, then $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathbb{y}, \mathsf{tr}, \mathsf{w}) = 1$ if and only if $\mathsf{V}^{\mathbb{y}, \pi_1, \dots, \pi_k}(\mathbb{i}, \mathbb{x}, \rho_1, \dots, \rho_k)$ outputs $(\mathbb{x}', \mathbb{y}')$ such that $(\mathbb{i}, \mathbb{x}', \mathbb{y}', \mathsf{w}) \in \mathcal{R}'$.

Definition 2.3.5 (Round-by-round knowledge soundness)**.** An IOR Π from $\mathcal{R}$ to $\mathcal{R}'$ has *round-by-round knowledge soundness* with errors $(\kappa_1, \dots, \kappa_k)$ if there exists a knowledge state function KState for Π and a deterministic extractor $\mathcal{E}$ such that for every statement $(\mathbb{i}, \mathbb{x}, \mathbb{y})$, every round index $i \in [k]$, and every interaction transcript where the verifier is about to move it holds:

$$\Pr_{\rho_i}\left[\begin{array}{l} \exists \mathsf{w} = 0 \text{ s.t.} \\ \wedge\ \mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathbb{y}, \mathsf{tr}, \mathcal{E}(\mathbb{i}, \mathbb{x}, \mathbb{y}, \mathsf{tr}\|\rho_i, \mathsf{w})) = 1 \\ \wedge\ \mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathbb{y}, \mathsf{tr}\|\rho_i, \mathsf{w}) = 1 \end{array}\right] \leq \kappa_i(\mathbb{i}, \mathbb{x}, \mathbb{y}).$$

At times we will also use a prior, stronger notion of round-by-round knowledge soundness, which is known to imply the round-by-round knowledge soundness according to definition 2.3.5 with the same errors [**BCFW25**].

Definition 2.3.6 (Knowledge state function)**.** Let Π be an IOR from a relation $\mathcal{R}$ to a relation $\mathcal{R}'$. A *knowledge state function* for Π is a (possibly inefficient) function KState, that takes as input a statement $(\mathbb{i}, \mathbb{x})$ and an interaction transcript tr, and outputs a bit. It must satisfy:

- **Empty transcript.** If $\mathsf{tr} = \emptyset$, then $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr}) = 0$.
- **Prover moves.** If tr is a transcript where the prover is about to move and $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr}) = 0$, then for every prover message π,

$$\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr}\|\pi) = 0.$$

- **Full transcript.** If $\mathsf{tr} = (\pi_1, \rho_1, \dots, \pi_k, \rho_k)$ is a full transcript, then $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr}) = 1$ if and only if $\mathsf{V}(\mathbb{i}, \mathbb{x}, \pi_1, \rho_1, \dots, \pi_k, \rho_k)$ outputs $(\mathbb{i}', \mathbb{x}')$ such that there exists $\mathbb{w}$' with $(\mathbb{i}', \mathbb{x}', \mathbb{w}') \in \mathcal{R}'$.

Definition 2.3.7 (Round-by-round knowledge soundness)**.** An IOR Π from $\mathcal{R}$ to $\mathcal{R}'$ has *round-by-round knowledge soundness* with errors $(\kappa_1, \dots, \kappa_k)$ if there exists a knowledge state function KState

for Π and a deterministic extractor $\mathcal{E}$ such that for every statement $(\mathbb{i}, \mathbb{x})$, every round index $i \in [k]$, and every interaction transcript where the verifier is about to move it holds:

$$\Pr_{\rho_i}\left[\begin{array}{l} \mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr}) = 0 \\ \wedge\ \mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr}\|\rho_i) = 1 \\ \wedge\ (\mathbb{i}, \mathbb{x}, \mathcal{E}(\mathbb{i}, \mathbb{x}, \mathsf{tr})) \notin \mathcal{R} \end{array}\right] \leq \kappa_i(\mathbb{i}, \mathbb{x}).$$

2.4 Universal Constraint Systems (UCS)

We start this section by presenting the main relation we are interested in proving, which we call the *universal constraint system* (UCS) relation. After that, we discuss some structural semantics of UCS, namely how one can write lookup and permutation constraints within a UCS.

2.4.1 General UCS definition

In this section we define the *universal constraint system* (UCS) relation $\mathsf{REL}_{\mathsf{UCS}}$. This relation combines algebraic constraints over $\mathbb{Q}[X]$ with algebraic constraints over finite-field polynomial rings $\mathbb{F}_{q_i}[X]$ (or, simply, finite fields $\mathbb{F}_{q_i}$).

Definition 2.4.1 (Universal constraint system (UCS))**.** The relation $\mathsf{REL}_{\mathsf{UCS}}$ is the set of all triples $(\mathbb{i}, \mathbb{x}; \mathbb{w})$ of the following form.

Index. An index is a tuple

$$\mathbb{i} = (m, \ell, n, \mathbf{q}, B, \mathbf{d}, \mathcal{C}),$$

where $m \geq 0$ is the *witness length*, $\ell \geq 0$ is the *input length*, $n \geq 0$ is the *number of rows*, $\mathbf{q} = (q_1, \dots, q_{|\mathbf{q}|})$ is a tuple of prime powers, $B \geq 0$ is a *bit-size bound*, $\mathbf{d} = (d_0, \dots, d_{|\mathbf{q}|})$ is a tuple of *degree bounds*, and $\mathcal{C}$ is a family of *constraints* (defined below). Let K be a formal variable (interpreted as ranging over rows in $[n]$), let $\mathbf{Y}$ be a tuple of ℓ formal variables (interpreted as placeholders for public input entries), and let $\mathbf{Z}_1, \mathbf{Z}_2$ be tuples of m formal variables (interpreted as placeholders for witness vectors). $\mathbf{Z}_1$ serves as placeholder for a witness with entries in $\mathbb{Q}^{<d,B}[X]$ and $\mathbf{Z}_2$ for witnesses with entries in $\mathbb{F}_{q_i}^{<d_i}[X]$. Additionally, in the finite-field constraints, $\mathbf{Z}_1$ serves as well as a placeholder for the projected integer witness $\phi_{q_i}(\mathbf{f}_0)$.

The constraint family $\mathcal{C}$ is a tuple

$$\mathcal{C} = \Big((Q_{0t}, \mathfrak{I}_{0t})_{t \in [|\mathcal{C}|]},\ (Q_{it}, \mathfrak{I}_{it})_{i \in [|\mathbf{q}|], t \in [|\mathcal{C}|]} \Big),$$

where, for every $t \in [|\mathcal{C}|]$ and every $i \in [|\mathbf{q}|]$,

$$Q_{0t} \in (\mathbb{Q}^{<d_0,B}[X])[K, \mathbf{Y}, \mathbf{Z}_1], \qquad Q_{it} \in (\mathbb{F}_{q_i}[X])[K, \mathbf{Y}, \mathbf{Z}_1, \mathbf{Z}_2],$$

and $\mathfrak{I}_{0t} = \langle \mathsf{j}_{0t} \rangle$ and $\mathfrak{I}_{it} = \langle \mathsf{j}_{it} \rangle$ are ideals given by generators

$$\mathsf{j}_{0t} \in \mathbb{Q}^{<d_0,B}[X], \qquad \mathsf{j}_{it} \in \mathbb{F}_{q_i}[X].$$

We call $(Q_{it}, \mathfrak{I}_{it})$ a $\mathbb{Q}[X]$ (or $\mathbb{F}_{q_i}[X]$) *constraint*.

Input and witness. The public input is $\mathbb{x} = \mathbf{y}$ with

$$\mathbf{y} \in (\mathbb{Z}^{<d_0,B}_{(q_1 \cdots q_{|\mathbf{q}|})}[X])^{\ell},$$

(recall that $\mathbb{Z}_{(q_1 \cdots q_{|\mathbf{q}|})}$ denotes the intersection localization ring from eq. (2.15)). The witness is $\mathbb{w} = (\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{|\mathbf{q}|})$ with

$$\mathbf{f}_0 \in (\mathbb{Q}^{<d_0,B}[X])^m, \qquad \mathbf{f}_i \in (\mathbb{F}_{q_i}^{<d_i}[X])^m \ \text{ for all } \ i \in [|\mathbf{q}|]. \tag{2.17}$$

Constraints. We say that $(\mathbb{i}, \mathbb{x}; \mathbb{w}) \in \mathsf{REL}_{\mathsf{UCS}}$ if all of the following conditions hold.

(i) $\mathbb{Q}[X]$**-constraints.** For every $k \in [n]$ and every $t \in [|\mathcal{C}|]$,

$$Q_{0t}(k, \mathbf{y}, \mathbf{f}_0) \in \mathfrak{I}_{0t} \subseteq \mathbb{Q}[X].$$

(ii) $\mathbb{F}_{q_i}[X]$**-constraints.** For every $i \in [|\mathbf{q}|]$, every $k \in [n]$, and every $t \in [|\mathcal{C}|]$,

$$Q_{it}(k, \phi_{q_i}(\mathbf{y}), \phi_{q_i}(\mathbf{f}_0), \mathbf{f}_i) \in \mathfrak{I}_{it} \subseteq \mathbb{F}_{q_i}[X].$$

(iii) **Well-definedness of ϕ_{q_i}.** The homomorphisms ϕ_{q_i} from section 2.3.1 are defined only on the intersection localization ring $\mathbb{Z}_{(q_1 \cdots q_{|\mathbf{q}|})}$. Since the previous item applies ϕ_{q_i} to $\mathbf{f}_0$, we require the following well-definedness condition:

$$\mathbf{f}_0 \in \left(\mathbb{Z}^{<d_0,B}_{(q_1 \cdots q_{|\mathbf{q}|})}[X]\right)^m. \tag{2.18}$$

When $|\mathbf{q}| = 0$, this condition is vacuous. We show in section 2.4.2 that the well-definedness condition can be expressed as a $\mathbb{Q}[X]$-constraint.

When an ideal $\mathfrak{I}_{it}$ is zero, the corresponding constraint $Q_{it}(\cdots) \in \mathfrak{I}_{it}$ reduces to the equality $Q_{it}(\cdots) = 0$. When $|\mathbf{q}| = 0$, the tuple $\mathbf{q}$ is empty and UCS reduces to a purely $\mathbb{Q}[X]$-constraint system with no $\mathbb{F}[X]$-constraints.

Remark 2.4.2 (Separation of PIOP and IOPP enforcement)**.** The satisfaction conditions in definition 2.4.1 (Items (i)–(iii)) will be enforced at the *PIOP level*, while the typing constraints for the witness (2.17) will be enforced at the *IOPP* (or *PCS*) *level*. More precisely, we design PIOPs that prove the $\mathbb{Q}[X]$-constraints, the $\mathbb{F}_{q_i}[X]$-constraints, and the well-definedness constraint, under the assumption that malicious provers send oracles of the correct type—i.e., oracles to multilinear polynomials with coefficients in $\mathbb{Q}^{<d_0,B}[X]$ or $\mathbb{F}^{<d_i}_{q_i}[X]$, respectively. Enforcing that the oracles are actually of this declared type will be the job of the IOPP (or PCS) compiler.

We remark that, in particular, the well-definedness condition (2.18) is not enforced at the IOPP level, but rather at the PIOP level (cf. section 2.4.2).

Remark 2.4.3 (The well-definedness condition)**.** We show in section 2.4.2 that the well-definedness condition (2.18) can be written as a $\mathbb{Q}[X]$-constraint. Hence, (2.18) does not require its own separate syntax, as long as one of the $\mathbb{Q}[X]$-constraints in the UCS instance enforces it.

In practice, the computations we are interested in proving already require constraining $\mathbf{f}_0$ in a way that guarantees the well-definedness condition automatically. For example, in all examples from **??**, the witness $\mathbf{f}_0$ is constrained to belong to some subset of $\mathbb{Z}^{<d}[X]^m$ (via a $\mathbb{Q}[X]$-constraint). This constraint implies the well-definedness condition for any choice of primes $q_1, \dots, q_{|\mathbf{q}|}$.

For ease of reference, we restate the definition of $\mathsf{REL}_{\mathsf{UCS}}$ below.

$$\mathsf{REL}_{\mathsf{UCS}} = \left\{ \begin{pmatrix} \mathbb{i}, \\ \mathbb{x}; \\ \mathbb{w} \end{pmatrix} \;\middle|\; \begin{array}{l}
\textbf{Index-input-witness} \\
\quad \textbf{Index} \\
\qquad \mathbb{i} = (m, \ell, n, \mathbf{q}, B, \mathbf{d}, \mathcal{C}), \\
\qquad \mathbf{q} = (q_1, \dots, q_{|\mathbf{q}|}), \quad \mathbf{d} = (d_0, \dots, d_{|\mathbf{q}|}), \\
\qquad \mathcal{C} = ((Q_{0t}, \mathfrak{I}_{0t} = \langle \mathsf{j}_{0t} \rangle), (Q_{it}, \mathfrak{I}_{it} = \langle \mathsf{j}_{it} \rangle))_{i \in [|\mathbf{q}|], t \in [|\mathcal{C}|]}, \\
\qquad \mathsf{j}_{0t} \in \mathbb{Q}^{<d_0, B}[X],\ \mathsf{j}_{it} \in \mathbb{F}_{q_i}[X], \\
\qquad Q_{0t} \in (\mathbb{Q}^{<d_0, B}[X])[K, \mathbf{Y}, \mathbf{Z}_1],\ Q_{it} \in (\mathbb{F}_{q_i}[X])[K, \mathbf{Y}, \mathbf{Z}_1, \mathbf{Z}_2], \\
\qquad |\mathbf{Z}_1| = |\mathbf{Z}_2| = m \\
\quad \textbf{Input} \\
\qquad \mathbb{x} = (\mathbf{y}) \in (\mathbb{Z}^{<d_0, B}_{(q_1 \cdots q_{|\mathbf{q}|})}[X])^{\ell} \\
\quad \textbf{Witness} \\
\qquad \mathbb{w} = (\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{|\mathbf{q}|}),\ \mathbf{f}_i \in (\mathbb{F}^{<d_i}_{q_i}[X])^m \quad \text{for all } i \in [|\mathbf{q}|] \\
\qquad \mathbf{f}_0 \in (\mathbb{Q}^{<d_0, B}[X])^m, \\
\\
\textbf{Constraints} \\
\quad \textbf{Algebraic constraints over } \mathbb{Q}[X] \textbf{ mod ideals} \\
\qquad Q_{0t}\,(k, \mathbf{y}, \mathbf{f}_0) \in \mathfrak{I}_{0t} \quad \text{for all } k \in [n],\ t \in [|\mathcal{C}|] \\
\\
\quad \textbf{Algebraic constraints over } \mathbb{F}_{q_i}[X] \\
\qquad Q_{it}\left(k, \phi_{q_i}(\mathbf{y}), \phi_{q_i}(\mathbf{f}_0), \mathbf{f}_i\right) \in \mathfrak{I}_{it}, \\
\qquad\qquad \text{for all } k \in [n],\ i \in [|\mathbf{q}|],\ t \in [|\mathcal{C}|]. \\
\quad \textbf{Well-definedness} \\
\qquad \mathbf{f}_0 \in (\mathbb{Z}^{<d_0, B}_{(q_1 \cdots q_{|\mathbf{q}|})}[X])^m.
\end{array} \right\}$$

Design choices. We discuss some of the design choices behind the definition of UCS.

Why are witness and input entries typed in polynomial rings $R^{<w}[X]$ (e.g. $R = \mathbb{Z}, \mathbb{F}_q$) rather than in R (or some other rings)?

One may instead try to reinterpret $R^{<w}[X]$ as w copies of R and treat each witness entry as an element of R^w. We see three advantages in preserving the ring structure of $R[X]$:

First, $R[X]$ makes ideal membership constraints natural and efficient to reason about. It is not clear how to translate ideal membership predicates into R^w without resorting to lookups, and even when lookups are available, the translation loses the algebraic structure of the ideals. Our PIOP framework leverages this algebraic structure by resolving ideal membership predicates with a single MLE randomized check, analogously to how standard proof systems check strict equalities.

Second, $R[X]$ enables multiple ideal membership predicates in the same relation. If there were only one ideal $\mathfrak{I}$, one could work over $R[X]/\mathfrak{I}$ directly. In the presence of membership predicates involving more than one ideal, staying in $R[X]/\mathfrak{I}$ would complicate the arithmetization of all but one of the predicates. Working in $R[X]$ glues together constraints that would otherwise be separated into different quotient rings. For example, in the previous example from section 2.2.1 we have ideal membership constraints simultaneously in $(X-2)$ and (X^w-1) (Example 1), and in $(X-1)$ and $(\Phi(X))$ (Example 2).

Third, our PIOP projects constraints over $\mathbb{Q}^{<d,B}[X]$ and $\mathbb{F}_q^{<w}[X]$ onto a random finite field by, respectively: 1) ($\mathbb{Q}$ case) replacing X by a random integer $a \in \mathbb{Z}$ and then reducing modulo a random prime; 2) ($\mathbb{F}_q$ case) replacing X by a random element a in $\mathbb{F}_q$ or a field extension of $\mathbb{F}_q$. When $R^{<w,B}[X]$ is large (for $R \in \{\mathbb{Q}, \mathbb{F}_q\}$) this projection provides equally large compression benefits, in that elements from $R^{<w,B}[X]$, which may be thousands of bits long (for example, in lattice-based settings), become 128–256 bits long.

Overall, it is possible to reformulate our definition of UCS as one operating over $\mathbb{Q}$ and $\mathbb{F}_q$ rather than $\mathbb{Q}^{<d_0,B}[X]$ and $\mathbb{F}_q^{<d_i}[X]$, but we think that the polynomial-ring formulation is more natural and better captures the algebraic structure of the computations we are interested in proving.

Why allow integers? Why not work entirely over $\mathbb{F}_q$ or $\mathbb{F}_q[X]$ for a sufficiently large prime q?

A plausible alternative to having constraints over $\mathbb{Z}$ or $\mathbb{Z}[X]$ is to work over $\mathbb{F}_q$ or $\mathbb{F}_q[X]$ for q larger than any value that the integer computation ever produces (let's call such a value the *computation norm*), and avoid modular overflow via lookups or range checks. We see several

advantages of staying over $\mathbb{Z}/\mathbb{Z}[X]$, though some use-cases may benefit from working on $\mathbb{F}_q$ or $\mathbb{F}_q[X]$.

First, by the same reasoning as the second point above, staying in $\mathbb{Z}$ or $\mathbb{Z}[X]$ allows projecting onto fields and rings $\mathbb{F}_q$ or $\mathbb{F}_q[X]$ for different values of q within the same relation. For example, our SHA-256 (+ECDSA verification) arithmetization showcases simultaneous constraints modulo 2^{32}, 2, and p for a 256-bit prime p.

Second, the third point above also applies: in applications such as verifiable FHE (e.g. the CKKS scheme) or RSA cryptography, one would require working with a very large prime q (i.e. the computation norm in such applications is large). The random-projection technique may provide significant compression benefits in these cases.

Third, when q is large, one faces the familiar overhead of field arithmetic in $\mathbb{F}_q$ or $\mathbb{F}_q[X]$ throughout the proof. For instance, Reed–Solomon encoding becomes expensive over large fields. Our Integer Pseudo Reed–Solomon (IPRS) codes operate over $\mathbb{Z}$ or $\mathbb{Q}$ and show promising practical performance (**??**). We think that specifically designed IPRS codes can be set up with good distance over $\mathbb{F}_q$ for a fixed large q, but we lack theoretical guarantees for this; we pose this as an open question in **??**.

When the computation norm q is small, say of 32 bits, then working over $\mathbb{F}_q[X]$ would be a better option. But in most applications it is not the case that 32 bits suffices to capture all integer values without overflowing the modulus. (Unless one pays arithmetization overheads by decomposing into limbs, etc. which is what we want to avoid in the first place.)

Technical observations. We make some technical observations about the definition of UCS.

Lookup constraints. In our examples section 2.2.1 and **??** we extensively rely on constraining witness elements in certain sets, e.g. $\{0,1\}^{<w}[X]$ and $[0\,..\,2^w-1]$. Our definition of UCS is expressive enough to subsume such constraints. See section 2.4.2 for an explanation of how to express lookup constraints in a UCS instance.

Rationals and Integers. In our examples, our witness vectors were either typed in $\mathbb{Z}^{<d}[X]$ or in $\mathbb{F}_q^{<d}[X]$. On the other hand, in our definition of UCS we used $\mathbb{Q}^{<d}[X]$ rather than $\mathbb{Z}^{<d}[X]$ as typing choice. Besides the possible independent interest of typing witnesses in $\mathbb{Q}^{<d}[X]$, this choice is motivated by technical reasons related to the proof system.

The main idea is that our IOPP (or PCS) is only capable of extracting witnesses in $\mathbb{Q}^{<d}[X]$, and

not in $\mathbb{Z}^{<d}[X]$. It is not clear to us how to build an IOPP or PCS that 1) guarantees typing $\mathbb{Z}^{<d}[X]$, and 2) does not incur large performance losses in doing so. Indeed, the only integer-based PCSs we are aware of [**Zartan**; **BlockDARK**] leverage hidden-order-group and expensive repeated checks (besides the one we mention below). Here we make the design choice of allowing rational coefficients in the witness, and delegate the task of enforcing integrality to the PIOP through a lookup PIOP. This subtlety is explained at length in the technical overview of the Zinc paper [**Zinc**]. Because it is not novel, we do not dwell on it further.

We note that combining an IOPP (or PCS) for $\mathbb{Q}^{<d_0,B}[X]$ with a lookup PIOP for $\mathbb{Z}^{<d}[X]$ provides an IOPP (or PCS) for $\mathbb{Z}^{<d_0,B}[X]$, and hence our work provides that.

Modular reductions of rationals Observe that the $\mathbb{F}_{q_i}[X]$-constraints involve the reduction of the vector $\mathbf{f}_0$ modulo q_i (via the map ϕ_q). However, the entries of $\mathbf{f}_0$ are polynomials with rational coefficients. The reduction of a rational a/b modulo a prime q is only well-defined when b is not zero modulo q, and in that case it is defined as $a \cdot b^{-1} \mod q$ where b^{-1} is the modular inverse of b section 4.2. We understand the triple $(\mathbb{i}, \mathbb{x}; \mathbb{w})$ to be invalid if some modular reduction is not well-defined. This is not a problem for our proof system, as the reduction can be performed by the prover and the verifier can check that the reduction was done correctly by checking that the difference between the reduced and non-reduced polynomials belongs to the ideal generated by q_i in $\mathbb{Q}[X]$.

2.4.2 Lookup and permutation constraints as UCS constraints

In this section and in section 2.4.3, we show how several common relations and constraints—namely lookups, permutations, R1CS, CCS, and AIR—can be written as UCS constraints. For clarity, we present these reductions in isolation; however, in practice one can combine multiple constraint types within a single UCS instance. See our UAIR$^+$ relation in **??** for an example.

We begin with lookup relations. The construction follows Section 4.4 of [**Zinc**], which proves the result for any integral domain; we include it here for completeness but omit a detailed proof. We work over rings R of the form $\mathbb{Q}^{<d,B}[X]$ or $\mathbb{F}_q^{<d}[X]$ for some prime power q, bit-size bound B, and degree bound d.

We define the *lookup relation* over R as:

$$\mathsf{REL}_{\mathsf{Look}} = \left\{ (\mathbb{i}, \mathbb{x}; \mathbb{w}) \;\middle|\; \begin{array}{l} \mathbb{i} = (m, \tau, \mathbf{t}, R), \\ m, \tau \geq 0,\ \mathbf{t} \in R^{\tau}, \\ \mathbb{x} = \emptyset,\ \mathbb{w} = \mathbf{w} \in R^{m}, \\ \mathbf{w}_i \in \{\mathbf{t}_j \mid j \in [\tau]\} \text{ for all } i \in [m]. \end{array} \right\}.$$

Let $\mathbb{i}_{\mathsf{Look}} := (m, \tau, \mathbf{t}, R)$ denote an index for $\mathsf{REL}_{\mathsf{Look}}$. We now construct a UCS index $\mathbb{i}_{\mathsf{UCS}}$ such that $(\mathbb{i}_{\mathsf{Look}}, \emptyset; \mathbf{w}) \in \mathsf{REL}_{\mathsf{Look}}$ if and only if $(\mathbb{i}_{\mathsf{UCS}}, \emptyset; \mathbb{w}') \in \mathsf{REL}_{\mathsf{UCS}}$ for an explicit transformation $\mathbb{w}'$ of $\mathbf{w}$.

We set $n := m$ and $|\mathcal{C}| := 1$. Recall that the *Lagrange basis polynomial* for $i \in [N]$ is

$$\mathsf{Lag}_{[N],i}(T) := \prod_{j \neq i} \frac{T - j}{i - j}. \tag{2.19}$$

It is well-defined over $\mathbb{Q}$ or any $\mathbb{F}_q$ with $q > N$. Define

$$L(K, \mathbf{Z}) \;:=\; \sum_{k \in [m]} \mathbf{Z}_k \cdot \mathsf{Lag}_{[m],k}(K) \tag{2.20}$$

so that $L(k, \mathbf{z}) = \mathbf{z}_k$ for all $k \in [m]$. Define

$$Q_{\mathsf{Look}}(K, \mathbf{Z}) \;:=\; \prod_{j \in [\tau]} (L(K, \mathbf{Z}) - \mathbf{t}_j). \tag{2.21}$$

If $R = \mathbb{Q}^{<d,B}[X]$, define $\mathbb{i}_{\mathsf{UCS}} = (m, \ell{=}0, n, \mathbf{q}{=}\emptyset, B, \mathbf{d}{=}d, \mathcal{C})$ with $\mathcal{C} = (Q_{0,1}, \mathfrak{I}_{0,1}) := (Q_{\mathsf{Look}}, (0))$. Here the subscript $(0, 1)$ denotes prime-index $i = 0$ (the $\mathbb{Q}[X]$-part) and constraint-index $t = 1$ (the single constraint).

If $R = \mathbb{F}_q^{<d}[X]$, define $\mathbb{i}_{\mathsf{UCS}} = (m, \ell{=}0, n, \mathbf{q}{=}q, B, \mathbf{d}{=}(0, d), \mathcal{C})$ with $\mathcal{C} = ((Q_{0,1}, \mathfrak{I}_{0,1}), (Q_{1,1}, \mathfrak{I}_{1,1}))$ where $Q_{0,1} \equiv 0$, $\mathfrak{I}_{0,1} = (0) \subseteq \mathbb{Q}[X]$, $\mathfrak{I}_{1,1} = (0) \subseteq \mathbb{F}_q[X]$, and $Q_{1,1}(K, \mathbf{Y}, \mathbf{Z}_1, \mathbf{Z}_2) := Q_{\mathsf{Look}}(K, \mathbf{Z}_2)$.

Lemma 2.4.4 (Lookup as UCS)**.** *Let $\mathbb{i}_{\mathsf{Look}}$ and $\mathbb{i}_{\mathsf{UCS}}$ be defined as above. For every $\mathbf{w} \in R^m$, letting $\mathbb{x} = \emptyset$, we have:*

- *If $R = \mathbb{Q}^{<d,B}[X]$ then $(\mathbb{i}_{\mathsf{Look}}, \emptyset; \mathbf{w}) \in \mathsf{REL}_{\mathsf{Look}}$ if and only if $(\mathbb{i}_{\mathsf{UCS}}, \emptyset; \mathbf{w}) \in \mathsf{REL}_{\mathsf{UCS}}$.*
- *If $R = \mathbb{F}_q^{<d}[X]$ then, writing $\mathbf{0}$ for the all-zero $\mathbb{Q}$-witness component of length m, $(\mathbb{i}_{\mathsf{Look}}, \emptyset; \mathbf{w}) \in$*

$\mathsf{REL}_{\mathsf{Look}}$ *if and only if* $(\mathbb{i}_{\mathsf{UCS}}, \emptyset; (\mathbf{0}, \mathbf{w})) \in \mathsf{REL}_{\mathsf{UCS}}$.

The proof is straightforward and we omit it. The key observation is that R is an integral domain (both $\mathbb{Q}^{<d,B}[X]$ and $\mathbb{F}_q^{<d}[X]$ are polynomial rings over fields, hence have no zero divisors). Thus $Q_{\mathsf{Look}}(i, \mathbf{w}) = \prod_j (\mathbf{w}_i - \mathbf{t}_j) = 0$ if and only if $\mathbf{w}_i \in \{\mathbf{t}_j\}$.

Remark 2.4.5 (Degree blowup and practical lookup arguments)**.** The polynomial Q_{Look} in (2.21) has degree τ in the selector value $L(K, \mathbf{Z})$. The purpose of the above construction is to exhibit lookups as UCS instances, so that generic UCS compilers and soundness statements apply. In protocols, we will not prove (2.21) via a high-degree polynomial identity; instead, we will use lookup arguments that exploit the lookup semantics directly, following the approach of [**Zinc**].

Remark 2.4.6 (Lookup constraints on polynomial expressions of witness entries)**.** The lookup relation $\mathsf{REL}_{\mathsf{Look}}$ defined above requires each entry of a witness vector $\mathbf{w}$ to belong to a table $\mathbf{t}$. In applications, one sometimes needs to constrain *polynomial expressions* of witness entries to belong to a lookup table. Concretely, given a witness vector $\mathbf{w} \in R^m$ and a polynomial $P \in R[\mathbf{Z}]$, one may require that

$$P(\mathbf{w}_i) \ \in \ \{\mathbf{t}_j \mid j \in [\tau]\} \quad \text{for all } i \in [m].$$

The extension to this setting is straightforward: one replaces the selector polynomial $L(K, \mathbf{Z})$ in (2.20) with

$$L'(K, \mathbf{Z}) \ := \ P\Big(\sum_{i \in [m]} \mathbf{Z}_i \cdot \mathsf{Lag}_{[m],i}(K)\Big),$$

and defines Q'_{Look} analogously. This generalization is needed in our definition of the relation UAIR$^+$ (**??**) and in some of our applications in **??**, notably in the SHA-256 arithmetization (**??**); see **??** for further discussion.

Remark 2.4.7 (Well-definedness as a lookup constraint)**.** Recall that definition 2.4.1 includes a well-definedness condition (2.18) requiring $\mathbf{f}_0 \in (\mathbb{Z}^{<d_0,B}_{(q_1 \cdots q_{|\mathbf{q}|})}[X])^m$. This is simply a lookup constraint: the table is $\mathbf{t} := \mathbb{Z}^{<d_0,B}_{(q_1 \cdots q_{|\mathbf{q}|})}[X]$, and each entry of $\mathbf{f}_0$ must belong to this finite set. Thus, the well-definedness condition can be enforced via a $\mathbb{Q}[X]$-constraint in UCS and does not require separate syntax in the definition of UCS.

Definition 2.4.8 (Well-formed UCS index)**.** A UCS index $\mathbb{i}$ is *well-formed* if it includes a $\mathbb{Q}[X]$-constraint that enforces the well-definedness condition (2.18).

In many applications, other lookup constraints placed on $\mathbf{f}_0$ already subsume the well-definedness condition (see **??**), so it does not need to be explicitly included. Note that while $\mathbf{t}$ may be very large (or even infinite in the limit), in practice the table size is bounded by the constraint parameters.

Remark 2.4.9 (Permutation constraints as UCS instances)**.** So-called *copy constraints* of the form $\mathbf{u}_i = \mathbf{v}_{\sigma(i)}$, where the permutation $\sigma : [n] \to [n]$ is fixed by the index, can also be encoded as UCS instances. Such constraints capture copy relations that arise from wiring in arithmetizations such as Plonk [**GWC19**]. We define the *permutation relation* over a ring R as

$$\mathsf{REL}_{\mathsf{perm}} \;=\; \left\{ (\mathbb{i}, \mathbb{x}; \mathbb{w}) \;\middle|\; \begin{array}{l} \mathbb{i} = (n, \sigma, R), \\ \mathbb{x} = \emptyset, \quad \mathbb{w} = (\mathbf{u}, \mathbf{v}) \in R^n \times R^n, \\ \mathbf{u}_i = \mathbf{v}_{\sigma(i)} \text{ for all } i \in [n]. \end{array} \right\}.$$

To write this relation as a UCS instance, we view the witness as a length-$2n$ vector $\mathbf{w} = (\mathbf{u}||\mathbf{v})$ and define the selector polynomial

$$Q_{\mathsf{Perm}}(K, \mathbf{Z}) \;:=\; \sum_{i \in [n]} (\mathbf{Z}_i - \mathbf{Z}_{n+\sigma(i)}) \cdot \mathsf{Lag}_{[n],i}(K).$$

Then evaluating at $K = i \in [n]$ yields $Q_{\mathsf{Perm}}(i, \mathbf{w}) = \mathbf{u}_i - \mathbf{v}_{\sigma(i)}$. Thus, $Q_{\mathsf{Perm}}(i, \mathbf{w}) = 0$ for all $i \in [n]$ if and only if $(\mathbf{u}, \mathbf{v}) \in \mathsf{REL}_{\mathsf{perm}}$. The UCS construction then proceeds exactly as in lemma 2.4.4: for $R = \mathbb{Q}^{<d,B}[X]$, one sets $\mathcal{C} = (Q_{\mathsf{Perm}}, (0))$; for $R = \mathbb{F}_q^{<d}[X]$, one places the constraint in the finite-field component with $Q_{1,1}(K, \mathbf{Y}, \mathbf{Z}_1, \mathbf{Z}_2) := Q_{\mathsf{Perm}}(K, \mathbf{Z}_2)$.

2.4.3 R1CS, CCS, and AIR as UCS instances

In this section we explain how several standard relations—R1CS, CCS, AIR (over fields, rationals, or polynomial rings)—can be written as $\mathsf{REL}_{\mathsf{UCS}}$ relations. Since the rings $\mathbb{Q}^{<d,B}[X]$ and $\mathbb{F}_q^{<d}[X]$ are integral domains, the constructions and proofs follow directly from [**Zinc**], which establishes the result for arbitrary integral domains; we summarize the key ideas below.

All transformations below work in a *strictly algebraic regime*: for every constraint we set $\mathfrak{I}_{it} = (0)$, so that a constraint $Q_{it}(\cdots) \in \mathfrak{I}_{it}$ reduces to the polynomial identity $Q_{it}(\cdots) = 0$.

R1CS. Recall that the R1CS relation over a ring R requires $(M_A \cdot \mathbf{z}) \circ (M_B \cdot \mathbf{z}) - (M_C \cdot \mathbf{z}) = \mathbf{0}$, where $M_A, M_B, M_C \in R^{n \times (\ell+m)}$ are constraint matrices and $\mathbf{z} = (\mathbf{y}||\mathbf{w})$ concatenates the public input and witness. The transformation to UCS proceeds as follows: first, use bivariate Lagrange interpolation to encode each matrix $M \in \{M_A, M_B, M_C\}$ as a polynomial $\hat{M}(K, Z)$; then define a linear selector $L_M(K; \mathbf{Y}, \mathbf{Z}_1)$ that evaluates to the k-th entry of $M \cdot \mathbf{z}$ when $K = k$; finally, set the UCS constraint polynomial to

$$Q_{\mathsf{R1CS}}(K, \mathbf{Y}, \mathbf{Z}_1) \;:=\; L_{M_A} \cdot L_{M_B} - L_{M_C}.$$

For R1CS over $\mathbb{Q}^{<d,B}[X]$, the UCS index has $\mathbf{q} = \emptyset$ and a single $\mathbb{Q}[X]$-constraint $(Q_{\mathsf{R1CS}}, (0))$. For R1CS over $\mathbb{F}_q^{<d}[X]$, one instead uses an $\mathbb{F}_q[X]$-constraint with a trivial $\mathbb{Q}[X]$-part. The equivalence $(\mathbb{i}_{\mathsf{R1CS}}, \mathbb{x}; \mathbb{w}) \in \mathsf{REL}_{\mathsf{R1CS}} \Leftrightarrow (\mathbb{i}_{\mathsf{UCS}}, \mathbb{x}; \mathbb{w}') \in \mathsf{REL}_{\mathsf{UCS}}$ follows from the defining property of Lagrange interpolation; see [**Zinc**] for the full proof.

CCS and AIR. CCS relations [**ccs**] admit an analogous transformation as R1CS relations. Further, since AIR relations [**ethstark**] can be written as CCS relations [**ccs**], they too can be instantiated as UCS instances. For an AIR-like instantiation tailored to the UCS framework, see **??**.

2.5 Zinc+ Compiler: from finite field PIOPs to PIOPs for UCS

In this section we present the Zinc+ compiler, which can be seen as a Polynomial Interactive Oracle Reduction (PIOR) from the UCS relation (definition 2.4.1) to a family of polynomial satisfiability relations over finite fields. In particular, given PIOPs for specific subclasses of finite-field relations (e.g., AIR-like constraints, R1CS, or CCS), one obtains a PIOP for the corresponding subclass of $\mathsf{REL}_{\mathsf{UCS}}$ via the PIOR. Before getting to the Zinc+ PIOR in section 2.5.4 we introduce a slight generalization of polynomial oracles called (possibly) incomplete polynomial oracles in section 2.5.1, introduce our techniques for batching ideal membership constraints and projecting the constraints onto finite fields in section 2.5.2, and introduce the strictly algebraic UCS relation we reduce to in section 2.5.3.

2.5.1 Incomplete polynomial oracles

Recall at the end of a PIOR that the verifier outputs a new instance $\mathbb{x}' \leftarrow \mathsf{V}(\mathbb{i}, \mathbb{x}, \pi_1, \rho_1, \dots, \pi_k, \rho_k)$, and any oracles therein are specified implicitly in terms of oracles given in $\mathbb{x}$ and $\pi_1, \dots, \pi_k$. For example, the new instance could contain an oracle $[[f']] = [[f]] + \rho_1 \cdot [[\pi_1]]$, for some $[[f]]$ included in the old instance. Assuming ρ_1 and the coefficients of f and π_1 all reside over the same ring, this new polynomial oracle is complete, in that it can be evaluated at any set of points in that ring.

In Zinc+ the verifier outputs a new instance with polynomial oracles of the form $\psi_\rho([[f(\psi_\rho^{-1}(\cdot))]])$, where we can always compute $v' = [[f(\psi_\rho^{-1}(\cdot))]]$, but for malicious provers v' may lie outside the domain ψ_ρ is defined on. We model this via the (possibly) *incomplete polynomial oracle* $\{\{f'\}\} = \psi_\rho([[f(\psi_\rho^{-1}(\cdot))]])$, which either outputs a correct evaluation or $\perp$ if an undefined operation is encountered during the evaluation, and which we syntactically distinguish from complete polynomial oracles using braces rather than brackets.

We can modify any PIOP to work with possibly incomplete oracles as follows. If the oracle outputs $\perp$, the verifier automatically rejects, and otherwise the protocol proceeds normally. If we can show that during the execution of the modified PIOP incomplete oracles that are not complete output $\perp$ whp, then whenever the verifier accepts we can assume whp that the possibly incomplete oracle is in fact complete. For the possibly incomplete oracle $\{\{f'\}\}$ used in Zinc+, we show that when queried on a random point it outputs $\perp$ whp. This means that the very weak requirement that the PIOP queries $\{\{f'\}\}$ on a random point suffices for the security of our reduction, and in practice this does not rule out any public-coin PIOPs of interest.

2.5.2 Batching ideal membership checks and projecting to finite fields

In this section, we provide lemmas for reducing ideal membership constraints in $\mathsf{REL}_{\mathsf{UCS}}$ to strict equalities in $\mathbb{Q}[X]$ and $\mathbb{F}_{q_i}[X]$, and subsequently to zerocheck-type polynomial relations over fields $\mathbb{F}_{q_0}$ and $\mathbb{F}_{\tilde{q}_i}$.

Ideal batching. We first give our ideal batching lemma, which captures the fact that w.h.p.a vector $\mathbf{v}$ belongs to an ideal $\mathfrak{I}$ if the evaluation of the multilinear extension of $\mathbf{v}$ at a random point belongs to $\mathfrak{I}$. This is a consequence of the Schwartz-Zippel lemma over rings (lemma 2.3.1), which follows by looking at membership in $\mathfrak{I}$ as equality to zero over an appropriate quotient by $\mathfrak{I}$.

Lemma 2.5.1 (Testing ideal membership in $\mathbb{K}[X]$ via one random MLE evaluation)**.** *Let $\mathbb{K}$ be a field, and let $\mathfrak{I} \subseteq \mathbb{K}[X]$ be an ideal such that $\mathfrak{I} \cap \mathbb{K} = (0)$. Let $\mu \geq 1$, let $\mathbf{v} : \{0,1\}^\mu \to \mathbb{K}[X]$ be a vector indexed by $\{0,1\}^\mu$, and let $\tilde{\mathbf{v}}(\mathbf{Y}) \in (\mathbb{K}[X])[\mathbf{Y}]$ be its multilinear extension (as in (2.16)). The following hold:*

1. *If $\mathbf{v}(\mathbf{x}) \in \mathfrak{I}$ for all $\mathbf{x} \in \{0,1\}^\mu$, then $\tilde{\mathbf{v}}(\mathbf{r}) \in \mathfrak{I}$ for every $\mathbf{r} \in \mathbb{K}^\mu$.*

2. *If there exists $\mathbf{x}_0 \in \{0,1\}^\mu$ such that $\mathbf{v}(\mathbf{x}_0) \notin \mathfrak{I}$, then, for every nonempty finite set $S \subseteq \mathbb{K}$,*

$$\Pr_{\mathbf{r} \leftarrow S^\mu}\left[\tilde{\mathbf{v}}(\mathbf{r}) \in \mathfrak{I}\right] \;\leq\; \frac{\mu}{|S|}.$$

Proof. Let $\mathcal{R} := \mathbb{K}[X]/\mathfrak{I}$ and let $\pi : \mathbb{K}[X] \to \mathcal{R}$ be the natural projection map. Since $\mathfrak{I} \cap \mathbb{K} = (0)$, the induced map $\mathbb{K} \hookrightarrow \mathcal{R}$ is injective.

For Item 1, assume $\mathbf{v}(\mathbf{x}) \in \mathfrak{I}$ for all $\mathbf{x} \in \{0,1\}^\mu$ and fix $\mathbf{r} \in \mathbb{K}^\mu$. By definition of the multilinear extension and closure of $\mathfrak{I}$ under $\mathbb{K}[X]$-linear combinations,

$$\tilde{\mathbf{v}}(\mathbf{r}) \;=\; \sum_{\mathbf{x} \in \{0,1\}^\mu} \mathbf{v}(\mathbf{x}) \cdot \widetilde{\mathsf{eq}}(\mathbf{x};\mathbf{r}) \;\in\; \mathfrak{I},$$

because each coefficient $\widetilde{\mathsf{eq}}(\mathbf{x};\mathbf{r})$ lies in $\mathbb{K} \subseteq \mathbb{K}[X]$.

For Item 2, assume $\mathbf{v}(\mathbf{x}_0) \notin \mathfrak{I}$ for some $\mathbf{x}_0 \in \{0,1\}^\mu$. Define $\bar{\mathbf{v}} : \{0,1\}^\mu \to \mathcal{R}$ by $\bar{\mathbf{v}}(\mathbf{x}) := \pi(\mathbf{v}(\mathbf{x}))$. Then $\bar{\mathbf{v}}(\mathbf{x}_0) \neq 0$, so the multilinear polynomial $\tilde{\bar{\mathbf{v}}}(\mathbf{Y}) \in \mathcal{R}[\mathbf{Y}]$ is nonzero.

Let $S := S \subseteq \mathbb{K} \subseteq \mathcal{R}$. For distinct $a, b \in S$, the difference $a - b$ is a nonzero element of the field $\mathbb{K}$, and hence a unit in $\mathcal{R}$. Therefore S is an exceptional set in $\mathcal{R}$, and Lemma 2.3.1 yields

$$\Pr_{\mathbf{r} \leftarrow S^\mu}\left[\tilde{\bar{\mathbf{v}}}(\mathbf{r}) = 0\right] \;\leq\; \frac{\mu}{|S|}.$$

Finally, for every $\mathbf{r} \in \mathbb{K}^\mu$, coefficient-wise application of π gives $\tilde{\bar{\mathbf{v}}}(\mathbf{r}) = \pi(\tilde{\mathbf{v}}(\mathbf{r}))$. Thus $\tilde{\bar{\mathbf{v}}}(\mathbf{r}) = 0$ holds if and only if $\tilde{\mathbf{v}}(\mathbf{r}) \in \mathfrak{I}$, which proves the claimed bound. $\square$

Extend map. Let $\mathbb{P}$ be the set of all prime powers. Throughout the rest of the section, we fix the extend map

$$\mathsf{Extend} : \mathbb{P} \to \mathbb{P}$$

that will parameterize our protocols and definitions. This map is defined so that, given a prime power q, $\mathsf{Extend}(q) = q^\ell$ is a sufficiently large power of q (where precise requirements in terms of the security parameter are given in the round-by-round knowledge proof of Zinc+). We will generally write $\tilde{q}$ to mean the output of $\mathsf{Extend}(q)$.

Projecting from rings to fields. We now show how to sample a random projection from rings $\mathbb{Q}[X]$ and $\mathbb{F}_q[X]$ to sufficiently large finite fields.

Definition 2.5.2 (Evaluation projections)**.** Let $q = p^e$ be a prime power, $\tilde{q} = q^\ell$. Let $a \in \mathsf{Prim}(\mathbb{F}_{\tilde{q}})$ such that $\langle a \rangle = \mathbb{F}_{\tilde{q}}$. We define the surjective ring homomorphism

$$\psi_{q,a} : \mathbb{F}_q[X] \to \mathbb{F}_{\tilde{q}}, \qquad f(X) \mapsto \iota_{\tilde{q}}(f)(a),$$

where the canonical embedding is first applied coefficient-wise to f to lift it to the extension ring $\mathbb{F}_{\tilde{q}}$ and then the resulting polynomial is evaluated at a. We also define the right-inverse

$$\psi_{q,a}^{-1} : \mathbb{F}_{\tilde{q}} \to \mathbb{F}_q^{<\ell}[X]$$

which sends $v \in \mathbb{F}_{\tilde{q}}$ to the unique $g \in \mathbb{F}_q^{<\ell}[X]$ such that $\psi_{q,a}(g) = v$. To compute this explicitly, one can fix an irreducible degree ℓ polynomial b and represent elements of $\mathbb{F}_{\tilde{q}}$ as polynomials in $\mathbb{F}_q[X]/(b(X)) \cong \mathbb{F}_{\tilde{q}}$, where the coefficients in this representation are exactly the coefficients of g. We stress that g is unique and choice of b only affects arithmetic efficiency. While $\psi_{q,a} \circ \psi_{q,a}^{-1}$ is the identity function $\mathsf{id}_{\mathbb{F}_{\tilde{q}}}$ on $\mathbb{F}_{\tilde{q}}$, we have that $\psi_{q,a}^{-1} \circ \psi_{q,a}$ is only the identity function on the restricted domain $\mathbb{F}_q^{<\ell}[X] \subset \mathbb{F}_q[X]$.

Next, let q_0 be a prime, $a \in \mathbb{F}_{q_0}$. We extend $\psi_{q_0,a}$ to a surjective ring homomorphism

$$\psi_{q_0,a} : \mathbb{Z}_{(q_0)}[X] \to \mathbb{F}_{q_0}, \qquad f(X) \mapsto \phi_{q_0}(f)(a),$$

where first we apply the ring homomorphism ϕ_{q_0} coefficient-wise to f to project it to $\mathbb{F}_q$ and then evaluate at a. We also define the right-inverse

$$\psi_{q_0,a}^{-1} : \mathbb{F}_{q_0} \to \mathbb{Z}_{(q_0)}[X]$$

which simply returns $v \in \mathbb{F}_{q_0}$ as a constant integral polynomial in $\mathbb{Z}_{(q_0)}[X]$.

Lemma 2.5.3. *Let $\psi : \mathcal{R} \to \mathcal{R}'$ be a surjective ring homomorphism. Let $n = 2^\mu$ and let $\mathbf{v} \in \mathcal{R}^n$ be a vector and $\tilde{\mathbf{v}}$ its multilinear extension. Let $\widetilde{\psi(\mathbf{v})}$ be the multilinear extension of $\psi(\mathbf{v}) \in \mathcal{R}'^n$. Then, for any* inverse map *$\psi^{-1} : \mathcal{R}' \to \mathcal{R}$, i.e. any map such that $\psi \circ \psi^{-1}$ is the identity map on $\mathcal{R}'$, we have*

$$\psi(\tilde{\mathbf{v}}(\psi^{-1}(\mathbf{u}))) = \widetilde{\psi(\mathbf{v})}(\mathbf{u}) \qquad \textit{for all } \mathbf{u} \in \mathcal{R}'^\mu.$$

Note that an inverse map ψ^{-1} exists since ψ is surjective.

The proof is deferred to **??**.

2.5.3 Strictly algebraic UCS

We introduce the strictly algebraic UCS relation, defined over a single prime-power field, which we will reduce UCS to. More precisely, Zinc+ reduces UCS to the joint relation $\wedge_{i \in [0,|\mathbf{q}|]} \mathsf{REL}_{\mathsf{UCS-ALG}}(\tilde{q}_i)$, where the new index-instance is of the form $\mathbb{i}' = (\mathbb{i}'_0, \dots, \mathbb{i}'_{|\mathbf{q}|})$, $\mathbb{x}' = (\mathbb{x}'_0, \dots, \mathbb{x}'_{|\mathbf{q}|})$ and

$$(\mathbb{i}', \mathbb{x}') \in \mathcal{L}\left(\wedge_{i \in [0,|\mathbf{q}|]} \mathsf{REL}_{\mathsf{UCS-ALG}}(\tilde{q}_i)\right) \iff (\mathbb{i}'_i, \mathbb{x}'_i) \in \mathcal{L}(\mathsf{REL}_{\mathsf{UCS-ALG}}(\tilde{q}_i)) \ (\forall i \in [0, |\mathbf{q}|]).$$

By combining the Zinc+ PIOR with PIOPs for each $\mathsf{REL}_{\mathsf{UCS-ALG}}(\tilde{q}_i)$ (or a single field-agnostic PIOP that can work for any prime-power), we obtain a PIOP for UCS.

Definition 2.5.4 (The relation $\mathsf{REL}_{\mathsf{UCS-ALG}}(q)$)**.** Let q be a prime power. The *strictly algebraic UCS relation* $\mathsf{REL}_{\mathsf{UCS-ALG}}(q) \subseteq \mathsf{REL}_{\mathsf{UCS}}$ is the subset of $\mathsf{REL}_{\mathsf{UCS}}$ formed by all triples $(\mathbb{i}, \mathbb{x}; \mathbb{w}) \in \mathsf{REL}_{\mathsf{UCS}}$ where all constraints are set over the same finite field and every ideal appearing in the index is the

zero ideal.

$$\mathsf{REL}_{\mathsf{UCS-ALG}}(q) = \left\{ \begin{pmatrix} \mathbb{i}, \\ \mathbb{x}, \\ \mathbb{w} \end{pmatrix} \middle| \begin{array}{l} \textbf{Index} \\ \quad \mathbb{i} = (m = 2^\nu, \ell, \mathcal{C}),\ \ \mathcal{C} = [Q_t]_{t \in [|\mathcal{C}|]},\ \ Q_t \in \mathbb{F}_q[\mathbf{Y}, \mathbf{Z}_1, \mathbf{Z}_2], \\ \quad |\mathbf{Y}| = \ell,\ \ |\mathbf{Z}_1| = |\mathbf{Z}_2| = m \\ \textbf{Input} \\ \quad \mathbb{x} = (\mathbf{y}, [[\tilde{f}_1]], [[\tilde{f}_2]]),\ \mathbf{y} \in \mathbb{F}_q^\ell,\ \tilde{f}_1, \tilde{f}_2 \in \mathbb{F}_q[\mathbf{W}],\ |\mathbf{W}| = \nu \\ \textbf{Witness} \\ \quad \mathbb{w} = (\mathbf{f}_1, \mathbf{f}_2) \in \mathbb{F}_q^m \times \mathbb{F}_q^m, \\ \textbf{Algebraic constraints over } \mathbb{F}_q \\ \quad Q_t\left(\mathbf{y}, [\tilde{f}_1(\mathbf{w})]_{\mathbf{w} \in \{0,1\}^\nu}, [\tilde{f}_2(\mathbf{w})]_{\mathbf{w} \in \{0,1\}^\nu}\right) = 0 \text{ for } t \in [|\mathcal{C}|] \end{array} \right\}.$$

Remark 2.5.5 (Mapping from UCS witnesses to $\mathsf{REL}_{\mathsf{UCS-ALG}}$ witnesses)**.** The two witness vectors $\mathbf{f}_1, \mathbf{f}_2 \in \mathbb{F}_q^m$ in $\mathsf{REL}_{\mathsf{UCS-ALG}}(q)$ arise from the Zinc+ PIOR (Protocol 2.2) as follows. For the i-th target field $\mathbb{F}_{\tilde{q}_i}$, the PIOR sets $\mathbf{f}_1 = \psi_{q_i,a_i}(\mathbf{f}_0)$, which is the shared integer witness projected to $\mathbb{F}_{\tilde{q}_i}$, and $\mathbf{f}_2 = \psi_{q_i,a_i}(\mathbf{f}_i)$, which is the dedicated $\mathbb{F}_{q_i}[X]$-witness projected to $\mathbb{F}_{\tilde{q}_i}$. For the $\mathbb{Q}[X]$-part ($i = 0$), the constraints involve only $\mathbf{f}_1$ (i.e., only the projected shared witness), and $\mathbf{f}_2$ is unused.

2.5.4 Zinc+ Compiler

In this section, we present the Zinc+ PIOR, which together with a field-agnostic PIOP for $\mathsf{REL}_{\mathsf{UCS-ALG}}$ gives us a PIOP for $\mathsf{REL}_{\mathsf{UCS}}$. We then prove its completeness and round-by-round knowledge soundness. Below we sometimes write $\tilde{q}_0$ to mean q_0 for the sake of compactness.

The well-defined condition in $\mathsf{REL}_{\mathsf{UCS}}$ as a $\mathbb{Q}[X]$-constraint Recall that $\mathsf{REL}_{\mathsf{UCS}}$ specifies a well-defined constraint, which states that one of the witness vectors, namely $\mathbf{f}_0$, lies in the intersection of certain localization rings (definition 2.4.1, specifically eq. (2.18)). As demonstrated in remark 2.4.7, this condition can be written in the form of an algebraic constraint over $\mathbb{Q}[X]$. When such a constraint is included in the index, we call it a *well-formed* index, and only specify Zinc+ to work with well-formed $\mathsf{REL}_{\mathsf{UCS}}$ instances.

Completeness and round-by-round knowledge soundness. Let $B_{\mathcal{P}}$ be a lower bound on the

Figure 2.2: Protocol $\Pi^{\mathsf{UCS}} = (\mathsf{P}_{\mathsf{UCS}}, \mathsf{V}_{\mathsf{UCS}})$: Zinc+ PIOR from $\mathsf{REL}_{\mathsf{UCS}}$ to $\wedge_{i\in[0,|\mathbf{q}|]}\mathsf{REL}_{\mathsf{UCS-ALG}}(\tilde{q}_i)$

Input: A well-formed index (definition 2.4.8) and input $(\mathbb{i}, \mathbb{x})$ for $\mathsf{REL}_{\mathsf{UCS}}$ (definition 2.4.1). $\mathsf{P}_{\mathsf{UCS}}$ additionally receives a witness $\mathbb{w}$ such that $(\mathbb{i}, \mathbb{x}, \mathbb{w}) \in \mathsf{REL}_{\mathsf{UCS}}$. Write

$$\mathbb{i} = (m = 2^\nu, \ell, n = 2^\mu, \mathbf{q}, B, \mathbf{d}, \mathcal{C} = [(Q_{it}, \langle j_{it}\rangle)]_{i\in[0,|\mathbf{q}|], t\in[|\mathcal{C}|]}),\ \ \mathbb{x} = (\mathbf{y}) \in (\mathbb{Z}_{(\mathbf{q})}^{<d_0,B}[X])^\ell,$$
$$\mathbb{w} = (\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{|\mathbf{q}|}),\ \mathbf{f}_0 \in (\mathbb{Z}_{(\mathbf{q})}^{<d_0,B}[X])^m,\ \mathbf{f}_i \in (\mathbb{F}_{q_i}^{<d_i}[X])^m \text{ for } i \in [|\mathbf{q}|].$$

1: $\mathsf{P}_{\mathsf{UCS}}$ sends polynomial oracles $[[\tilde{f}_0]], \dots, [[\tilde{f}_{|\mathbf{q}|}]]$, where $\tilde{f}_0 \in (\mathbb{Q}^{<d_0,B}[X])[\mathbf{W}]$, $\tilde{f}_1, \dots, \tilde{f}_{|\mathbf{q}|} \in (\mathbb{F}_{q_i}^{<d_i}[X])[\mathbf{W}]$, and $|\mathbf{W}| = \nu$.
2: $\mathsf{V}_{\mathsf{UCS}}$ sends a random prime $q_0 \leftarrow \mathcal{P}$ and field elements $\mathbf{r}_0 \leftarrow \mathbb{F}_{q_0}^\mu$, $\mathbf{r}_i \leftarrow \mathbb{F}_{\tilde{q}_i}^\mu$ for $i \in [|\mathbf{q}|]$.
3: If $Q_{0t} \notin (\mathbb{Z}_{(q_0)}[X])[K, \mathbf{Y}, \mathbf{Z}_1]$ or $\mathsf{j}_{0t}, \mathbf{y}, \mathbf{f}_0 \notin \mathbb{Z}_{(q_0)}[X]$ for any $t \in [|\mathcal{C}|]$, the protocol aborts.
4: $\mathsf{P}_{\mathsf{UCS}}$ sends for $i \in [|\mathbf{q}|], t \in [|\mathcal{C}|]$:

$$\begin{aligned} e_{0t} &= \textstyle\sum_{\mathbf{b}\in\{0,1\}^\mu} \phi_{q_0}(Q_{0t}(\mathbf{b}, \mathbf{y}, \mathbf{f}_0)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_0) && \in \mathbb{F}_{q_0}[X] \\ e_{it} &= \textstyle\sum_{\mathbf{b}\in\{0,1\}^\mu} \iota_{\tilde{q}_i}(Q_{it})(\mathbf{b}, \phi_{\tilde{q}_i}(\mathbf{y}), \phi_{\tilde{q}_i}(\mathbf{f}_0), \iota_{\tilde{q}_i}(\mathbf{f}_i)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_i) && \in \mathbb{F}_{\tilde{q}_i}[X] \end{aligned}$$

5: $\mathsf{V}_{\mathsf{UCS}}$ rejects if $e_{0t} \notin \langle \phi_{q_0}(\mathsf{j}_{0t})\rangle$ or $e_{it} \notin \langle \iota_{\tilde{q}_i}(\mathsf{j}_{it})\rangle$ for any $i \in [|\mathbf{q}|], t \in [|\mathcal{C}|]$.
6: $\mathsf{V}_{\mathsf{UCS}}$ sends $\mathbf{a} = (a_0, a_1, \dots, a_{|\mathbf{q}|})$ by sampling $a_i \leftarrow \mathbb{F}_{\tilde{q}_i}$ for $i \in [0, |\mathbf{q}|]$, aborting if any $a_i \notin \mathsf{Prim}(\mathbb{F}_{\tilde{q}_i})$.
7: **for** $i \in [0, |\mathbf{q}|]$ **do**
8: Let

$$\begin{aligned} &Q'_{0t} = \textstyle\sum_{\mathbf{b}\in\{0,1\}^\mu} \psi_{q_0,a_0}(Q_{0t})(\mathbf{b}, \mathbf{Y}, \mathbf{Z}_1) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_0) - e_{0t}(a_0) \in \mathbb{F}_{q_0}[\mathbf{Y}, \mathbf{Z}_1] \\ &Q'_{it} = \textstyle\sum_{\mathbf{b}\in\{0,1\}^\mu} \psi_{q_i,a_i}(Q_{it})(\mathbf{b}, \mathbf{Y}, \mathbf{Z}_1, \mathbf{Z}_2) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_i) - e_{it}(a_i) \in \mathbb{F}_{\tilde{q}_i}[\mathbf{Y}, \mathbf{Z}_1, \mathbf{Z}_2] \\ &\mathbf{y}'_0 = \psi_{q_0,a_0}(\mathbf{y}) \in \mathbb{F}_{q_0}^\ell,\ \mathbf{y}'_i = \psi_{q_i,a_i}(\mathbf{y}) \in \mathbb{F}_{\tilde{q}_i}^\ell \\ &\{\{\tilde{f}'_{i0}\}\} = \psi_{q_i,a_i}([[\tilde{f}_0(\psi_{q_i,a_i}^{-1}(\mathbf{W}))]]),\ [[\tilde{f}'_{i1}]] = \psi_{q_i,a_i}([[\tilde{f}_i(\psi_{q_i,a_i}^{-1}(\mathbf{W}))]]) \in \mathbb{F}_{\tilde{q}_i}[\mathbf{W}] \\ &\mathbf{f}'_{i0} = \psi_{q_i,a_i}(\mathbf{f}_0),\ \mathbf{f}'_{i1} = \psi_{q_i,a_i}(\mathbf{f}_i) \in (\mathbb{F}_{\tilde{q}_i})^m \end{aligned}$$

9: $\mathsf{V}_{\mathsf{UCS}}$ outputs for $i \in [|\mathbf{q}|]$:

$$\begin{aligned} \mathbb{i}'_0 &= \left(m, \ell, q_0, \mathcal{C}'_0 = [Q'_{0t}]_{t\in[|\mathcal{C}|]}\right) && \mathbb{x}'_0 = \left(\mathbf{y}_0,\ \{\{\tilde{f}'_{00}\}\}\right) \\ \mathbb{i}'_i &= \left(m, \ell, \tilde{q}_i, \mathcal{C}'_i = [Q'_{it}]_{t\in[|\mathcal{C}|]}\right) && \mathbb{x}'_i = \left(\mathbf{y}'_i,\ \{\{\tilde{f}'_{i0}\}\},\ [[\tilde{f}'_{i1}]]\right) \end{aligned}$$

10: $\mathsf{P}_{\mathsf{UCS}}$ outputs $\mathbb{w}'_i = (\mathbf{f}'_{i0}, \mathbf{f}'_{i1})$.
11: **end for**

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bit length of primes in $\mathcal{P}$. Recall that the ℓ_0-norm of a polynomial indicates the number of nonzero rational coefficients.

As constraint polynomials are often very sparse, our completeness bound is written in terms of a completeness parameter ξ. After setting an upper bound on the density of the constraint

polynomials, one may pick $\mathcal{P}$ that meets their desired completeness guarantees for all circuits that may be defined by such constraint polynomials. That said, it may still turn out $\mathcal{P}$ needs to be even larger to guarantee λ bits of security.

Theorem 2.5.6 (Completeness of the Zinc+ PIOR)**.** *Let*

$$|\mathcal{P}|(B_{\mathcal{P}}-1) \geq 2^{\xi}(B-1)\left(d_0 \cdot (\ell+m) + \sum_{t\in[|\mathcal{C}|]} \|\mathsf{j}_{0t}\|_0 + \|Q_{0t}\|_0\right),$$

and let $\mathsf{Extend}(q) = q^k$ *pick* k *such that* $q^k \geq 2^{2\xi}\cdot(8(\xi+1))^2$*. Then* Π_{UCS} *(Figure 2.2) is complete (definition 2.3.3) with error* $\epsilon < (|\mathbf{q}|+2)/2^{\xi}$*.*

Proof. First we show that unless the protocol halts in step 3 or 6 that $(\mathbb{i}', \mathbb{x}', \mathbb{w}') \in \wedge_{i\in[0,|\mathbf{q}|]}\mathsf{REL}_{\mathsf{UCS-ALG}}(\tilde{q}_i)$. Then we bound the probability of this happening. We begin with $(\mathbb{i}, \mathbb{x}, \mathbb{w}) \in \mathsf{REL}_{\mathsf{UCS}}$, where

$$\begin{aligned}
&\mathbb{i} = (m, \ell, n = 2^{\mu}, \mathbf{q}, B, \mathbf{d}, \mathcal{C} = [(Q_{it}, \langle j_{it}\rangle)]_{i\in[0,|\mathbf{q}|], t\in[|\mathcal{C}|]}),\ \ \mathbb{x} = (\mathbf{y}) \in (\mathbb{Z}_{(\mathbf{q})}^{<d_0,B}[X])^{\ell}\\
&\mathbb{w} = (\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{|\mathbf{q}|}),\ \ \mathbf{f}_0 \in \mathbb{Q}^{<d_0,B}[X]^m,\ \ \mathbf{f}_i \in \mathbb{F}_{q_i}^{<d_i}[X]^m \text{ for } i \in [|\mathbf{q}|].
\end{aligned}$$

In step 1, the prover sends the oracles for $\tilde{f}_0 \in (\mathbb{Q}^{<d_0,B}[X])[\mathbf{W}]$ and $\tilde{f}_1, \dots, \tilde{f}_{|\mathbf{q}|} \in (\mathbb{F}_{q_i}^{<d_i}[X])[\mathbf{W}]$, which are the MLEs of the corresponding witness vectors. In step 4, the prover sends

$$\begin{aligned}
e_{0t} &= \textstyle\sum_{\mathbf{b}\in\{0,1\}^{\mu}} \phi_{q_0}(Q_{0t})(\mathbf{b}, \phi_{q_0}(\mathbf{y}), \phi_{q_0}(\mathbf{f}_0)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_0) && \in \mathbb{F}_{q_0}[X]\\
e_{it} &= \textstyle\sum_{\mathbf{b}\in\{0,1\}^{\mu}} \iota_{\tilde{q}_i}(Q_{it})(\mathbf{b}, \phi_{\tilde{q}_i}(\mathbf{y}), \phi_{\tilde{q}_i}(\mathbf{f}_0), \iota_{\tilde{q}_i}(\mathbf{f}_i)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_i) && \in \mathbb{F}_{\tilde{q}_i}[X]
\end{aligned}$$

for $i \in [|\mathbf{q}|], t \in [|\mathcal{C}|]$. The verifier defines

$$\begin{aligned}
Q'_{0t} &= \textstyle\sum_{\mathbf{b}\in\{0,1\}^{\mu}} \psi_{q_0,a_0}(Q_{0t})(\mathbf{b}, \mathbf{Y}, \mathbf{Z}_1) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_0) - e_{0t}(a_0) \in \mathbb{F}_{q_0}[\mathbf{Y}, \mathbf{Z}_1]\\
Q'_{it} &= \textstyle\sum_{\mathbf{b}\in\{0,1\}^{\mu}} \psi_{q_i,a_i}(Q_{it})(\mathbf{b}, \mathbf{Y}, \mathbf{Z}_1, \mathbf{Z}_2) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_i) - e_{it}(a_i) \in \mathbb{F}_{\tilde{q}_i}[\mathbf{Y}, \mathbf{Z}_1, \mathbf{Z}_2]
\end{aligned}$$

Then plugging in $\mathbf{y}'_i = \psi_{q_i,a_i}(\mathbf{y})$, $\mathbf{f}'_{i0} = \psi_{q_i,a_i}(\mathbf{f}_0)$, $\mathbf{f}'_{i1} =$ we obtain

$$\begin{aligned}
Q'_{0t} = \ & \textstyle\sum_{\mathbf{b}\in\{0,1\}^{\mu}} \left(\psi_{q_0,a_0}(Q_{0t})(\mathbf{b}, \psi_{q_0,a_0}(\mathbf{y}), \psi_{q_0,a_0}(\mathbf{f}_0)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_0)\right)\\
& - \left(\textstyle\sum_{\mathbf{b}\in\{0,1\}^{\mu}} \phi_{q_0}(Q_{0t})(\mathbf{b}, \phi_{q_0}(\mathbf{y}), \phi_{q_0}(\mathbf{f}_0)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_0)\right)(a_0)\\
= \ & \textstyle\sum_{\mathbf{b}\in\{0,1\}^{\mu}} \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_0) \cdot [\psi_{q_0,a_0}(Q_{0t})(\mathbf{b}, \psi_{q_0,a_0}(\mathbf{y}), \psi_{q_0,a_0}(\mathbf{f}_0))\\
& \qquad - \phi_{q_0}(Q_{0t})(\mathbf{b}, \phi_{q_0}(\mathbf{y}), \phi_{q_0}(\mathbf{f}_0))(a_0)] = 0,
\end{aligned}$$

and similarly

$$\begin{aligned} Q'_{it} = & \sum_{\mathbf{b}\in\{0,1\}^\mu} \left(\psi_{q_i,a_i}(Q_{it})(\mathbf{b}, \psi_{q_i,a_i}(\mathbf{y}), \psi_{q_i,a_i}(\mathbf{f}_0), \psi_{q_i,a_i}(\mathbf{f}_i)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_i)\right) \\ & - \left(\sum_{\mathbf{b}\in\{0,1\}^\mu} \iota_{\tilde{q}_i}(Q_{it})(\mathbf{b}, \phi_{\tilde{q}_i}(\mathbf{y}), \phi_{\tilde{q}_i}(\mathbf{f}_0), \iota_{\tilde{q}_i}(\mathbf{f}_i)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_i)\right)(a_i) \\ = & \sum_{\mathbf{b}\in\{0,1\}^\mu} \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_i) \cdot [\psi_{q_i,a_i}(Q_{it})(\mathbf{b}, \psi_{q_i,a_i}(\mathbf{y}), \psi_{q_i,a_i}(\mathbf{f}_i), \psi_{q_i,a_i}(\mathbf{f}_i)) \\ & \qquad - \iota_{\tilde{q}_i}(Q_{it})(\mathbf{b}, \phi_{\tilde{q}_i}(\mathbf{y}), \phi_{\tilde{q}_i}(\mathbf{f}_0), \iota_{\tilde{q}_i}(\mathbf{f}_i))(a_i)] = 0. \end{aligned}$$

Finally, we conclude our claim of conditional correctness section with the observation that since each $a_i \in \mathsf{Prim}(\mathbb{F}_{\tilde{q}_i})$ by hypothesis, it holds by lemma 2.5.3 that $\{\{\tilde{f}'_{i0}\}\}$ and $[[\tilde{f}'_{i1}]]$ are (complete) polynomial oracles for the MLEs of the corresponding witness vectors.

Now it's left to bound the probability the protocol terminates early. The total number of rational coefficients we apply ϕ_{q_0} to is at most $N_{\text{rat}} = d_0 \cdot (\ell + m + |\mathcal{C}| + \sum_{t\in[|\mathcal{C}|]} \|Q_{0t}\|_0)$, where $\|Q\|_0$ denotes the size of the support of Q as a polynomial in $K, \mathbf{Y}$, and $\mathbf{Z}_1$. Then using lemma 2.5.8 and the union bound we can conclude the probability q_0 divides the denominator of any of these rational coefficients is less than

$$\frac{N_{\text{rat}}(B-1)}{|\mathcal{P}|(B_{\mathcal{P}}-1)}.$$

So given our hypothesis on $|\mathcal{P}|(B_{\mathcal{P}}-1)$ this happens with probability at most $2^{-\xi}$. Similarly, given our hypothesis on Extend we have that the event E_i that $a_i \notin \mathsf{Prim}(F_{\tilde{q}})$ occurs with probability at most $2^{-\xi}$ (by lemma 2.5.10). Taking the union bound over all E_i and also the event q_0 divides some denominator gives our completeness bound. □

Theorem 2.5.7 (Round-by-round knowledge soundness of the Zinc+ PIOR)**.** *Let the following hold for all $i \in [|\mathbf{q}|], t \in [|\mathcal{C}|]$:*

$$\begin{aligned} |\mathcal{P}|(B_{\mathcal{P}}-1) & \geq 2^{\lambda+4} d_0^2 B, \\ B_{\mathcal{P}} & \geq \lambda + \log(m \cdot \ell \cdot d_0 \cdot \mathsf{deg}(Q_{0t})), \\ |\mathsf{Prim}(\mathbb{F}_{\tilde{q}_i})| & \geq 2^{\lambda+1} m \cdot \ell \cdot d_i \cdot \mathsf{deg}(Q_{it}). \end{aligned}$$

Then Π_{UCS} (Figure 2.2) has round-by-round knowledge soundness (definition 2.3.7) with errors $\kappa_1 = \kappa_2 = 2^{-\lambda}$.

Proof. We make the simplifying assumption that $(\mathbb{i}', \mathbb{x}') \notin \mathcal{R}'$ unless $\{\{\tilde{f}'_{00}\}\}$ is complete, which effectively forces $\mathbf{f}_0 \in \mathbb{Z}_{(q)}[X]$. We believe this is justified based on the fact this can be enforced via asking for a single random evaluation of $\{\{\tilde{f}'_{00}\}\}$, which will abort with probability greater than

$1 - \nu/2^{B_{\mathcal{P}}}$ unless complete.

First we define the extractor and knowledge state function, then we upper bound the round-by-round knowledge soundness errors. Throughout we use

$$\mathbb{i} = (m = 2^\nu, \ell, n = 2^\mu, \mathbf{q}, B, \mathbf{d}, \mathcal{C} = [(Q_{it}, \langle j_{it} \rangle)]_{i \in [0,|\mathbf{q}|], t \in [|\mathcal{C}|]}), \;\; \mathbb{x} = (\mathbf{y}) \in (\mathbb{Z}_{(\mathbf{q})}^{<d_0,B}[X])^\ell,$$

and assume $\mathbb{i}$ is well-formed.

Extractor. For any transcript tr where the verifier is about to move, the extractor $\mathcal{E}$ queries each polynomial oracle $[[\tilde{f}_i]]$ at each point in the $\log m$-dimensional hypercube, and outputs the resulting vectors $(\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{|\mathbf{q}|})$, where $\mathbf{f}_0 \in (\mathbb{Q}^{<d_0,B}[X])^m$ and $\mathbf{f}_i \in (\mathbb{F}_{q_i}^{<d_i}[X])^m$ for $i \in [|\mathbf{q}|]$.

Knowledge state function. We define the knowledge state function's behavior in each round:

1. **Ring projection and ideal batching randomness:** At this stage the transcript tr has the form $\mathrm{tr} = ([[\tilde{f}_0]] \| \cdots \| [[\tilde{f}_{|\mathbf{q}|}]])$. The verifier samples a random prime $q_0 \leftarrow \mathcal{P}$ and field elements $\mathbf{r}_i \leftarrow \mathbb{F}_{\tilde{q}_i}^\mu$ for all $i \in [0, |\mathbf{q}|]$. We set $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr} \| q_0 \| \mathbf{r}) = 1$ if and only if running the extractor returns $(\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{|\mathbf{q}|})$ such that $\mathbf{f}_0 \in (\mathbb{Z}_{(q_0)}[X])^m$ and for $i \in [|\mathbf{q}|], t \in [|\mathcal{C}|]$:

$$\begin{array}{lll} \sum_{\mathbf{b} \in \{0,1\}^\mu} \phi_{q_0}(Q_{0t}(\mathbf{b}, \mathbf{y}, \mathbf{f}_0)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_0) & \in \langle \phi_{q_0}(\mathsf{j}_{0t}) \rangle & \subseteq \mathbb{F}_{q_0}[X], \\ \sum_{\mathbf{b} \in \{0,1\}^\mu} \iota_{\tilde{q}_i}(Q_{it})(\mathbf{b}, \phi_{\tilde{q}_i}(\mathbf{y}), \phi_{\tilde{q}_i}(\mathbf{f}_0), \phi_{\tilde{q}_i}(\mathbf{f}_i)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_i) & \in \langle \iota_{\tilde{q}_i}(\mathsf{j}_{it}) \rangle & \subseteq \mathbb{F}_{\tilde{q}_i}[X]. \end{array} \tag{2.22}$$

   Recall that since the index is well-formed this also implies $\mathbf{f}_0 \in (\mathbb{Z}_{(q_1,\dots,q_{|\mathbf{q}|})}[X])^m$.

2. **Field projection randomness:** At this stage the transcript tr has the form $\mathrm{tr} = ([[\tilde{f}_0]] \| \cdots \| [[\tilde{f}_{|\mathbf{q}|}]], q_0 \| \mathbf{r}, [e_{it}]_{i \in [0,|\mathbf{q}|], t \in [|\mathcal{C}|]})$. The verifier samples random field elements $a_i \leftarrow \mathbb{F}_{\tilde{q}_i}$, for $i \in [0, |\mathbf{q}|]$. We set $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr} \| q_0 \| \mathbf{r}) = 1$ if and only if running the extractor returns $(\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{|\mathbf{q}|})$ such that $\mathbf{f}_0 \in (\mathbb{Z}_{(q_0)}[X])^m$ and for $i \in [|\mathbf{q}|], t \in [|\mathcal{C}|]$:

$$\begin{array}{ll} \left(\sum_{\mathbf{b} \in \{0,1\}^\mu} \phi_{q_0}(Q_{0t})(\mathbf{b}, \phi_{q_0}(\mathbf{y}), \phi_{q_0}(\mathbf{f}_0)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_0) - e_{0t} \right)(a_0) = 0 & \text{over } \mathbb{F}_{q_0} \\ \left(\sum_{\mathbf{b} \in \{0,1\}^\mu} \iota_{\tilde{q}_i}(Q_{it})(\mathbf{b}, \phi_{\tilde{q}_i}(\mathbf{y}), \phi_{\tilde{q}_i}(\mathbf{f}_0), \phi_{\tilde{q}_i}(\mathbf{f}_i)) \cdot \widetilde{\mathsf{eq}}(\mathbf{b}, \mathbf{r}_i) - e_{it} \right)(a_i) = 0 & \text{over } \mathbb{F}_{\tilde{q}_i} \end{array} \tag{2.23}$$

3. **Full transcript:** It remains to show for a full transcript $\mathsf{tr} = ([[\tilde{f}_0]] \| \cdots \| [[\tilde{f}_{|\mathbf{q}|}]], q_0 \| \mathbf{r}, \mathbf{e}, \mathbf{a})$ that $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr}) = 1$ if and only if $\mathsf{V}(\mathbb{i}, \mathbb{x}, \mathsf{tr})$ outputs $(\mathbb{i}', \mathbb{x}')$ such that there exists $\mathbb{w}$' with $(\mathbb{i}', \mathbb{x}', \mathbb{w}') \in \mathcal{R}'$.

   First assume $\mathsf{V}(\mathbb{i}, \mathbb{x}, \mathsf{tr})$ outputs $(\mathbb{i}', \mathbb{x}')$ such that there exists $\mathbb{w}$' with $(\mathbb{i}', \mathbb{x}', \mathbb{w}') \in \mathcal{R}'$. Since

$(\mathbb{i}', \mathbb{x}') \notin \mathcal{L}(\mathcal{R}')$ unless $\{\{\tilde{f}'_{00}\}\}$ is complete by assumption, we have that $\mathbf{f}_0 \in \mathbb{Z}_{(q_0)}$. The $\mathbb{w}$' is by definition of UCS and by lemma 2.5.3 comprised of the vectors obtained by applying ψ to the $\mathbf{f}_i$ as in an honest run of the protocol. Then eq. (2.23) is exactly checking if that unique $\mathbb{w}$' satisfies $(\mathbb{i}', \mathbb{x}', \mathbb{w}') \in \mathcal{R}'$, outputting $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr}) = 1$ if so.

Next assume $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr}) = 1$. In this case it is straightforward to see we can explicitly produce a $\mathbb{w}$' such that $(\mathbb{i}', \mathbb{x}', \mathbb{w}') \in \mathcal{R}'$ using the extracted vectors.

RBR knowledge soundness. We upper bound the round-by-round knowledge soundness errors based on the knowledge state function and extractor described above.

1. **Ring projection and ideal batching randomness:** We show that

$$\Pr_{q_0, \mathbf{r}} \left[\begin{array}{l} \mathsf{KState}(\mathbb{i}, \mathbb{x}, ([[f_i]])_{i=0}^{|\mathbf{q}|} \| (q_0, \mathbf{r})) = 1 \\ \wedge\ (\mathbb{i}, \mathbb{x}, \mathcal{E}(\mathbb{i}, \mathbb{x}, ([[f_i]])_{i=0}^{|\mathbf{q}|})) \notin \mathcal{R} \end{array} \right] \ \leq\ 2^{-\lambda}.$$

By lemma 2.5.9 we have that

$$\Pr_{q_0 \leftarrow \mathcal{P}} \left[\phi_{q_0}\left(Q_{0t}(k, \mathbf{y}, \mathbf{f}_0)\right) \in \langle \phi_{q_0}(\mathsf{j}_{0t}) \rangle \ \wedge\ Q_{0t}(k, \mathbf{y}, \mathbf{f}_0) \notin \langle \mathsf{j}_{0t} \rangle \right] < \frac{8 d_0^2 B}{|\mathcal{P}|(B_{\mathcal{P}} - 1)} < 2^{-\lambda - 1}.$$

We note applying the canonical embedding cannot result in an unsatisfied constraint over $\mathbb{F}_{q_i}$ being satisfied in $\mathbb{F}_{\tilde{q}_i}$.

Else suppose for some (k, i, t) a constraint doesn't hold over $\mathbb{F}_{q_0}$ or $\mathbb{F}_{\tilde{q}_i}$. In this case lemma 2.5.1 tells us that eq. (2.22) will hold with probability at most $\mu / |\mathbb{F}_{\tilde{q}_i}| \leq 2^{-\lambda - 1}$. A union bound over these two cases gives $\kappa_1 = 2^{-\lambda}$.

We now turn to the second source of randomness, namely the field-projection points $\mathbf{a}$, and bound the probability that the knowledge state transitions from 0 to 1 in this round.

2. **Field projection randomness:** We show that for all transcripts of the form $\mathsf{tr} = ([[f_i]])_{i=0}^{|\mathbf{q}|} \| (q_0, \mathbf{r}) \| [e_{it}]_{it}$ that

$$\Pr_{\mathbf{a}} \left[\begin{array}{l} \mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr}) = 0 \\ \wedge\ \mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr} \| \mathbf{a}) = 1 \\ \wedge\ (\mathbb{i}, \mathbb{x}, \mathcal{E}(\mathbb{i}, \mathbb{x}, \mathsf{tr})) \notin \mathcal{R} \end{array} \right] \ \leq\ 2^{-\lambda}.$$

To do this we must consider the case that for some $i \in [0, |\mathbf{q}|], t \in [|\mathcal{C}|]$ lemma 2.5.1 doesn't hold, but when projected to $\mathbb{F}_{\tilde{q}_i}$ via ψ_{a_i} it does. This happens with probability at most $2 m \ell d_i \mathsf{deg}(Q_{it}) / |\mathsf{Prim}(\mathbb{F}_{\tilde{q}_i})| \leq 2^{-\lambda} = \kappa_2$.

□

Supplementary lemmas. Here we include some lemmas we draw on in our proofs of completeness and RBR knowledge.

Lemma 2.5.8. *Let $B > B_{\mathcal{P}}$. For any $a \in \mathbb{Z}^{<B}\ \{0\}$ of bit length less than B, the number of primes of bit length at least $B_{\mathcal{P}}$ dividing it is less than $(B-1)/(B_{\mathcal{P}}-1)$.*

Proof. Say t primes $p_1, \dots, p_t$ divide a. By assumption $|a| < 2^{B-1}$ and that $p_i \geq 2^{B_{\mathcal{P}}-1}$ for all $i \in [t]$. Then we have $2^{t(B_{\mathcal{P}}-1)} \leq |a| < 2^{B-1}$, which implies $t < (B-1)/(B_{\mathcal{P}}-1)$. □

Lemma 2.5.9. *Let $f, \mathsf{j} \in \mathbb{Q}^{<d,<B}[X]$, $d \geq 1$, $B > B_{\mathcal{P}} \geq 2$. Then*

$$\Pr_{p \leftarrow \mathcal{P}}\left[\phi_p(f) \in \langle \phi_p(\mathsf{j}) \rangle \ \wedge\ f \notin \langle \mathsf{j} \rangle \ \middle|\ f, \mathsf{j} \in \mathbb{Z}_{(p)}\right] < \frac{8d^2B}{|\mathcal{P}|(B_{\mathcal{P}}-1)}.$$

Proof. Let $(q, r) \in \mathbb{Q}[X] \times \mathbb{Q}[X]$ be the unique quotient–remainder pair given by Euclidean division in $\mathbb{Q}[X]$:

$$f = \mathsf{j}q + r, \qquad \deg\ (r) < \deg\ (\mathsf{j}).$$

Then $f \in \langle \mathsf{j} \rangle$ if and only if $r = 0$.

Write r in lowest terms as $r = n_r/d_r$, where $n_r \in \mathbb{Z}[X]$ and $d_r \in \mathbb{Z}^+$ satisfy $\gcd(d_r, c) = 1$, where $c = \mathsf{cont}(n_r)$ is the content of n_r. Similarly write a, the leading coefficient of j, in lowest terms $a = n_a/d_a$. Define $N := c \cdot n_a$. We show that for every prime p,

$$\left(\phi_p(f) \in \langle \phi_p(\mathsf{j}) \rangle \ \wedge\ f \notin \langle \mathsf{j} \rangle \ \wedge\ f, \mathsf{j} \in \mathbb{Z}_{(p)}\right) \Longrightarrow p \mid N.$$

Fix p and assume $\phi_p(f) \in \langle \phi_p(\mathsf{j}) \rangle$, $f \notin \langle \mathsf{j} \rangle$, and $f, \mathsf{j} \in \mathbb{Z}_{(p)}$. Define $N := c \cdot n_a$. We show $p \mid N$. First assume $p \nmid n_a$. Then the leading coefficient of j is a unit in $\mathbb{Z}_{(p)}$, so Euclidean division of f by j can be performed in $\mathbb{Z}_{(p)}[X]$. Let $(q_p, r_p) \in \mathbb{Z}_{(p)}[X] \times \mathbb{Z}_{(p)}[X]$ be the resulting quotient–remainder pair, so

$$f = \mathsf{j}q_p + r_p, \qquad \deg\ (r_p) < \deg\ (\mathsf{j}).$$

By uniqueness of Euclidean division in $\mathbb{Q}[X]$, we have $r_p = r$, and in particular $r \in \mathbb{Z}_{(p)}[X]$. Applying

ϕ_p gives

$$\phi_p(f) = \phi_p(\mathsf{j}) \cdot \phi_p(q_p) + \phi_p(r).$$

Since $p \nmid n_a$, we have deg $(\phi_p(\mathsf{j}))$ = deg (j), hence deg $(\phi_p(r))$ < deg $(\phi_p(\mathsf{j}))$. Together with $\phi_p(f) \in \langle \phi_p(\mathsf{j}) \rangle$, this implies $\phi_p(r) = 0$. Because $r \in \mathbb{Z}_{(p)}[X]$, we have $\phi_p(r) = 0$ if and only if $p \mid \mathsf{cont}(n_r) = c$. Hence $p \mid c$, and therefore $p \mid N$.

Next assume $p \nmid c$. If also $p \nmid n_a$, then the same argument shows $r \in \mathbb{Z}_{(p)}[X]$ and $\phi_p(r) = 0 \iff p \mid c$, so $p \nmid c$ implies $\phi_p(r) \neq 0$ while deg $(\phi_p(r))$ < deg $(\phi_p(\mathsf{j}))$. A nonzero polynomial of degree strictly less than deg $(\phi_p(\mathsf{j}))$ cannot lie in $\langle \phi_p(\mathsf{j}) \rangle$, contradicting $\phi_p(f) \in \langle \phi_p(\mathsf{j}) \rangle$. Therefore $p \mid n_a$, and hence $p \mid N$.

Let $\mathcal{P}_{\text{bad}} = \{p \in \mathcal{P} : p \mid N\}$. The above implication yields

$$\Pr_{p \leftarrow \mathcal{P}} \left[\phi_p(f) \in \langle \phi_p(\mathsf{j}) \rangle \ \wedge \ f \notin \langle \mathsf{j} \rangle \ \middle| \ f, \mathsf{j} \in \mathbb{Z}_{(p)} \right] \leq \frac{|\mathcal{P}_{\text{bad}}|}{|\mathcal{P}|}.$$

Let $B_N = \lceil \log(|N| + 1) \rceil$. If $B_N \leq B_{\mathcal{P}}$ then $|\mathcal{P}_{\text{bad}}| = 0$. Otherwise, by lemma 2.5.8,

$$|\mathcal{P}_{\text{bad}}| < \frac{B_N - 1}{B_{\mathcal{P}} - 1},$$

so

$$\Pr_{p \leftarrow \mathcal{P}}[\cdots] < \frac{B_N - 1}{|\mathcal{P}|(B_{\mathcal{P}} - 1)}. \tag{2.24}$$

We now bound B_N. Since the coefficients of f and j have bitlength $< B$, we may write each coefficient in lowest terms with numerator and denominator of magnitude $< 2^B$. Let $\Delta \in \mathbb{Z}^+$ be the product of all coefficient denominators appearing in f and j. Then $\Delta < 2^{2dB}$, and the polynomials

$$F := \Delta f \in \mathbb{Z}[X], \qquad J := \Delta \mathsf{j} \in \mathbb{Z}[X]$$

satisfy $\|F\|_\infty, \|J\|_\infty < 2^B \Delta < 2^{(2d+1)B}$.

Let $m :=$ deg (F) and $n :=$ deg (J), so $m, n < d$. Let b be the leading coefficient of J. By pseudo-division, there exist $Q', R' \in \mathbb{Z}[X]$ with deg $(R') < n$ such that

$$b^{m-n+1} F = JQ' + R'. \tag{2.25}$$

Dividing (2.25) by Δ and comparing with $f = \mathsf{j}q + r$ shows that

$$r = \frac{R'}{b^{m-n+1}\Delta}.$$

In particular, there exists $\alpha \in \mathbb{Z}^+$ such that $R' = \alpha n_r \in \mathbb{Z}[X]$ and $b^{m-n+1}\Delta = \alpha d_r \in \mathbb{Z}\ \{0\}$, hence

$$c = \mathsf{cont}(n_r) \leq \|n_r\|_\infty \leq \|R'\|_\infty.$$

Each coefficient of R' can be expressed as a determinant of a matrix of dimension at most $m+n < 2d$ whose entries are coefficients of F and J. By Hadamard's inequality,

$$\|R'\|_\infty \leq (2d)^d \cdot \left(\max(\|F\|_\infty, \|J\|_\infty)\right)^{2d} < (2d)^d \cdot \left(2^{(2d+1)B}\right)^{2d} = (2d)^d \cdot 2^{(4d^2+2d)B}.$$

Therefore $c < (2d)^d \cdot 2^{(4d^2+2d)B}$. Since $|n_a| < 2^B$, we have

$$|N| < (2d)^d \cdot 2^{(4d^2+2d+1)B}.$$

It follows that

$$B_N = \lceil \log(|N| + 1) \rceil \leq (4d^2 + 2d + 1)B + d\log(2d) + 1 < 8d^2 B.$$

Substituting this bound into eq. (2.24) completes the proof. □

Lemma 2.5.10. *Let q be a prime power, $k \geq 2$ an integer. Let $\mathsf{Extend}(q) = q^k$ pick k such that $q^k \geq 2^{2\xi} \cdot (8(\xi+1))^2$. Define $\mathsf{Prim}(\mathbb{F}_{q^k}) = \{\alpha \in \mathbb{F}_{q^k} \mid \mathbb{F}_q(\alpha) = \mathbb{F}_{q^k}\}$. Then*

$$\frac{|\mathsf{Prim}(\mathbb{F}_{q^k})|}{q^k} \geq 1 - 2^{-\xi}.$$

Proof. An element $\alpha \in \mathbb{F}_{q^k}$ fails to generate $\mathbb{F}_{q^k}$ over $\mathbb{F}_q$ if and only if α lies in a proper intermediate subfield $\mathbb{F}_{q^d}$ for some divisor $d \mid k$ with $d < k$. Thus

$$\mathbb{F}_{q^k}\ \mathsf{Prim}(\mathbb{F}_{q^k}) \subseteq \bigcup_{\substack{d \mid k \\ d<k}} \mathbb{F}_{q^d}.$$

Taking cardinalities and applying the union bound yields

$$|\mathbb{F}_{q^k}\ \mathsf{Prim}(\mathbb{F}_{q^k})| \leq \sum_{\substack{d|k \\ d<k}} |\mathbb{F}_{q^d}| = \sum_{\substack{d|k \\ d<k}} q^d.$$

Dividing by q^k gives

$$1 - \frac{|\mathsf{Prim}(\mathbb{F}_{q^k})|}{q^k} \leq \sum_{\substack{d|k \\ d<k}} q^{d-k}.$$

Let ℓ denote the smallest prime divisor of k. Then every proper divisor d of k satisfies $d \leq k/\ell$, so $q^{d-k} \leq q^{k/\ell-k} = q^{-(k-k/\ell)}$. Since k has at most $k-1$ proper divisors,

$$1 - \frac{|\mathsf{Prim}(\mathbb{F}_{q^k})|}{q^k} \leq (k-1)\, q^{-(k-k/\ell)}.$$

Using $\ell \geq 2$, we have $k - k/\ell \geq k/2$, hence

$$(k-1)\, q^{-(k-k/\ell)} \leq (k-1)\, q^{-k/2}.$$

Write $N := q^k$. Then $q^{-k/2} = N^{-1/2}$, so the previous equation becomes

$$1 - \frac{|\mathsf{Prim}(\mathbb{F}_{q^k})|}{q^k} \leq (k-1)\, N^{-1/2}.$$

Moreover, since $q \geq 2$ and $k \geq 2$, we have $N = q^k \geq 2^k$, and thus $k - 1 \leq \log_2 N$. Therefore

$$1 - \frac{|\mathsf{Prim}(\mathbb{F}_{q^k})|}{q^k} \leq \frac{\log_2 N}{\sqrt{N}}.$$

By the hypothesis $N = q^k \geq 2^{2\xi} \cdot (8(\xi+1))^2$, we have $\sqrt{N} \geq 2^{\xi} \cdot 8(\xi+1)$. Also,

$$\log_2 N \leq \log_2\Big(2^{2\xi} \cdot (8(\xi+1))^2\Big) = 2\xi + 2\log_2(8(\xi+1)) \leq 8(\xi+1),$$

where the last inequality uses $\xi \geq 1$ and $\log_2(8(\xi+1)) \leq 3 + (\xi+1)$. Combining the above bounds gives

$$\frac{\log_2 N}{\sqrt{N}} \leq \frac{8(\xi+1)}{2^{\xi} \cdot 8(\xi+1)} = 2^{-\xi}.$$

Hence $1 - \frac{|\mathsf{Prim}(\mathbb{F}_{q^k})|}{q^k} \leq 2^{-\xi}$, which is equivalent to the claimed bound. $\square$

Chapter 3

Microlotus: Efficient Hash-based Proofs for Cyclotomic Rings

Nearly 30 years after the introduction of the first lattice-based cryptosystem [**Ajtai96**], lattice-based cryptography is now in widespread deployment. Following NIST standardization of the ML-KEM key exchange mechanism and the ML-DSA and Falcon signatures, lattice-based primitives have been integrated into applications used by billions daily—including TLS, SSH, Signal and iMessage, Zoom, WireGuard and IKEv2, and MLS—while lattice-based fully homomorphic encryption sees increasing real-world deployment as well.

As lattice-based cryptography and succinct arguments have both matured, interest in *proving statements about lattice-based cryptography* has grown in tandem. One important use case is building efficient versions of more complex primitives—aggregate, blind, group, multi, threshold, and ring signatures—by verifying simpler lattice-based signatures inside a SNARK [**PQCRYPTO:BCOS20**; **AC:LyuNgu22**; **C:AABKT24**]. Proving FHE operations, Ajtai hash commitments, and lattice-based VRFs share the same algebraic core: proving statements of the form $As = t$ over a cyclotomic ring $R_q = \mathbb{Z}_q[X]/(X^n + 1)$.

The ring dimension barrier. Prior approaches to proving ring arithmetic with hash-based SNARKs face a fundamental tradeoff between the ring dimension n and either prover cost or proof size. Arguments that work directly over cyclotomic rings R_q pay a linear proof size cost in n, where concretely $n \geq 512$ at the 128-bit security level. On the other hand, the most direct approach

to simulating ring arithmetic over fields—decomposing ring elements into n coefficients over $\mathbb{F}_p$ and encoding ring multiplication as an NTT butterfly circuit [**PQCRYPTO:BCOS20**]—incurs $O(Mn \log n)$ constraints per inner product of M ring elements, inflating the prover by a $\log n$ factor beyond the witness size. Sumcheck-based approaches that compose FFT, Hadamard-product, and inverse-FFT sub-protocols via GKR [**CCS:LiuXieZha21**] match our prover cost $O(NMn)$ but achieve a lower $O(\log(NMn))$ verifier by paying only $O(\log n)$ per batched NTT instance. The innovative ring-switching approach of Huang, Mao, and Zhang [**EPRINT:HuaMaoZha25**] avoids both costs by recasting the statement over a Galois ring, but requires a quotient polynomial oracle $\tilde{Q}$ whose degree scales with n, inflating commitment costs from $O(nM)$ to $O(nNM)$ in the holographic (public-A) setting—a $4\times$ blowup for typical R1CS instances where $M \approx N$. Their oracles are typed over Galois rings rather than fields, and the resulting polynomial commitments still lag behind field-based counterparts.

Our approach: a CRT-domain PIOR for ring arithmetic. We introduce an efficient polynomial interactive oracle reduction (PIOR) for proving $As = t$ over the fully splitting cyclotomic ring $R_p = \mathbb{Z}_p[X]/(X^n + 1)$. Our starting observation is that since the CRT isomorphism $R_p \cong \mathbb{F}_p^n$ converts ring multiplication to pointwise multiplication, we can fold the CRT matrix $\tilde{C}$ directly into the virtual polynomial of a *single* sumcheck expressing the full matrix-vector product, rather than composing separate transform sub-protocols. The CRT-domain virtual polynomials A_C, s_C, t_C are never materialized as committed oracles: they are defined implicitly from the coefficient-domain oracles $\tilde{A}$, $\tilde{s}$, $\tilde{t}$ and the CRT-matrix multilinear extension $\tilde{C}$, and a second sumcheck resolves the CRT-domain claims back to evaluation claims on the committed coefficient-domain oracles. We show that the CRT evaluation tables needed for the first sumcheck can be built via the NTT in $O(Mn \log n)$ time, yielding:

- *Prover:* $O(NMn + Mn \log n)$, which is $O(NMn)$ whenever $N \geq \log n$.
- *Verifier:* $O(n + \log NM)$ field operations plus 3 oracle queries.
- *Proof size:* $2\mu + 5\nu + 3$ elements of $\mathbb{K}$, i.e., $O(\log Mn)$ with no dependence on n.

A holographic specialization for public A eliminates the second sumcheck entirely, reducing the proof size to $2(\mu + \nu) + 1$ elements (all degree-2 rounds) and leaving a single evaluation claim on the committed witness oracle.

Compared to the prior state of the art, this protocol removes n from the proof size (versus ring-native approaches), removes a factor of $\log n$ from the prover (versus NTT-circuit unrolling), and reduces the ring dimension's contribution to the verifier from multiplicative to additive. The resulting $O(n)$ additive verifier term arises from two evaluations of $\tilde{C}$ (lemma 3.4.5), which we show can each be computed in $O(n)$ field operations by exploiting the dyadic periodicity of the CRT roots. Before arriving at this CRT-domain protocol, we developed a coefficient-domain sumcheck that proves ring multiplication directly as a negacyclic circulant matrix–vector product, bypassing the CRT decomposition entirely and recovering an $O(\log n)$ verifier. However, the coefficient-domain convolution entangles the column-index and coefficient-index variables, producing folding challenges that fix the oracle evaluation points at distinct last-ν coordinates. This precludes the shared-coordinate PCS consolidation that our CRT-domain protocol enables. In our ring Plonkish SpeedySpartan example (section 3.4.4), these PCS optimizations—including selector-polynomial merging, free multi-point reduction, ring-switch packing, and interleaved BaseFold opening—save the verifier significantly more time than the single $O(n)$ CRT evaluation costs, given that n typically ranges from 64 to 512 at the 128-bit security level.

Composability via the PIOR framework. We prove round-by-round knowledge soundness (RBR-KS) for our protocol by decomposing it into four composable PIOR building blocks—$\Pi_{\alpha,\alpha'}$, Π_{sc1}, Π_{ξ}, Π_{sc2}—whose knowledge errors are bounded independently and concatenated via the composition theorem of [**BCFW25**]. This decomposition, visualized as a claim-flow DAG (Figure 3.3), makes the security analysis modular: each PIOR box has a clearly defined source and target relation, and new reductions can be attached to the DAG's output without revisiting the inner protocol's soundness proof. In particular, the evaluation claims produced by our ring-multiplication PIOR feed directly into further sumcheck-based reductions (packing, multi-point-to-single-point, BaseFold folding), and in principle into any other RBR-KS protocol that consumes evaluation claims. This composability means our ring-arithmetic PIOR could be used as a subroutine within a larger proof system that combines ring-arithmetic checks with other types of constraints, with the sumchecks potentially sharing challenges or folding into a unified protocol.

Field universality: from binary to Mersenne primes. Our PIOR operates over any extension field $\mathbb{K}/\mathbb{F}_p$ of sufficient size, making it compatible with a wide variety of polynomial commitment

schemes.

- When $p^k - 1$ has high 2-adicity (the "NTT-friendly" case, which includes Goldilocks and BabyBear primes), the PCS can be instantiated with Reed–Solomon codes via FRI [**BBHR18**] or WHIR [**EPRINT:ACFY24**], yielding the best concrete proof sizes and verification times.
- When instead $p^k + 1$ has high 2-adicity (e.g., the Mersenne prime $p = 2^{31} - 1$, where $p + 1 = 2^{31}$), Circle STARK polynomial commitments [**EPRINT:HabLevPap24**] can be used: they exploit the circle group of order $p^k + 1$ over $\mathbb{F}_{p^k}$ in place of the multiplicative group.
- For $p = 2$, the Binius PCS [**EPRINT:DiaPos24**] commits multilinear polynomials over binary tower fields with zero embedding overhead, and is particularly efficient for Boolean-heavy computations.
- For arbitrary fields where none of the above apply, random foldable linear codes (as in Basefold [**basefold**] or Brakedown [**C:GLSTW23**]) provide a universal baseline that works over any field.

Our construction thus covers the widest set of field characteristics of any ring-arithmetic IOP we are aware of. That said, the choice of base field should be driven primarily by the computation being arithmetized: the arithmetization overhead from a mismatched base field will almost always dominate the efficiency difference between polynomial commitment schemes, so choosing the right arithmetization domain trumps choosing the best PCS.

An odd-prime tower for packing. To benefit from the ring-switch packing technique of Diamond and Posen [**EC:DiaPos25**]—which reduces the polynomial dimension by $s = \log_2[\mathbb{K} : \mathbb{F}_p]$ at the cost of an s-round degree-1 sumcheck—we require a tower of quadratic extensions $\mathbb{F}_p \subset \mathbb{F}_{p^2} \subset \cdots \subset \mathbb{K}$ with an explicit multilinear basis at each level. Diamond and Posen's original construction targets the binary tower over $\mathbb{F}_2$. We give a new tower construction for all odd primes p (section 3.3), built from iterated quadratic extensions $\tau_\iota = \tau_{\iota-1}[X_{\iota-1}]/(X_{\iota-1}^2 - X_{\iota-2})$ with a carefully chosen non-square $\alpha \in \mathbb{F}_p$ at the base. Choosing α as small as possible (e.g., $\alpha = 3$ for $p = 257$) makes the "soft" multiplications (by compile-time constants) essentially free, and the Karatsuba-recursive tower multiplication achieves $k^{\log_2 3}$ hard $\mathbb{F}_p$-multiplications for a degree-k extension—matching the binary tower's hard-multiplication count. This tower is needed specifically for the packing step; the core ring-arithmetic PIOR itself only requires a sufficiently large extension field.

Application beyond lattice cryptography: SWIFFT as an in-circuit hash. Our protocol is useful even when the circuit being proved was not designed to verify lattice cryptography. As a motivating example, one can replace the in-circuit hash function in an existing SNARK with SWIFFT [**FSE:LMPR08**], the lattice-based hash whose collision resistance reduces to worst-case ideal-lattice problems.[1] To do so, one reparameterizes SWIFFT to use the modulus p of the field one already works over for the rest of the circuit, and sets the ring dimension n and norm bound β to achieve the desired RSIS security level. For large moduli, inputs to SWIFFT need not be represented in binary: one can use a higher-radix representation, with each digit a value in $\{0, \dots, B-1\}$ for an appropriate base B, reducing the number of ring elements in the RSIS instance accordingly. This approach is particularly attractive in light of ongoing cryptanalytic concerns regarding algebraic hash functions optimized for proof systems: recent Gröbner-basis analyses [**EPRINT:GKR25**; **EPRINT:BBLP22**] have identified inaccuracies in Poseidon's original security model that lead to misestimated round requirements, while SWIFFT's security rests on the well-studied worst-case hardness of ideal lattice problems and admits no known structural attacks. Moreover, SWIFFT is one of the fastest hash functions both in-circuit (when the field matches the SWIFFT modulus) and out-of-circuit, where NTT-based evaluation is competitive with SHA-256 throughput.

Systems like SWIRL [**swirl**], which stack WHIR with LogUp to handle lookups across AIR traces of different sizes within a single proof, may provide further opportunities in this direction. In particular, if one picks the SWIFFT modulus to be p^t for t large enough that RSIS over $\mathbb{Z}_{p^t}[X]/(X^n+1)$ is secure for norm bound $\beta = p$, then every $\mathbb{F}_p$-element fed to SWIFFT already satisfies the norm constraint without any digit decomposition. Cross-subfield lookup mechanisms could then handle the interaction between $\mathbb{F}_p$-arithmetic in the main circuit and $\mathbb{F}_{p^t}$-arithmetic in the SWIFFT sub-circuit within a unified proof.

Comparison with prior work. We briefly compare with the main prior approaches for proving ring arithmetic, with a detailed survey in section 4.1.2. Unrolling ring multiplication into an NTT butterfly circuit [**PQCRYPTO:BCOS20**; **EC:BCRSVW19**] contributes $O(Mn \log n)$ constraints regardless of the backend, a $\log n$ factor beyond the witness size; our prover is linear in

[1] SWIFFT provides collision resistance, not random-oracle security. Following the BHP-BLAKE2s construction of [**plumo**], one can compose SWIFFT with a single iteration of the BLAKE3 compression function—applied to the (short) SWIFFT output—to obtain a hash suitable for instantiating a random oracle.

the witness. The GKR-based sumcheck composition of [**CCS:LiuXieZha21**] matches our $O(NMn)$ prover but achieves $O(\log(NMn))$ verification, whereas our verifier pays $O(n+\log NM)$; as discussed above, the additive $O(n)$ is a worthwhile trade for the PCS consolidation it enables. The ring-switching approach of [**EPRINT:HuaMaoZha25**] requires committing to quotient polynomial oracles typed over Galois rings, inflating commitment costs and precluding the DP25 packing reduction. Galois-ring proof systems [**C:JLXYY25**; **PKC:WeiZhaDen25**] target a complementary setting—general-purpose $\mathbb{Z}_{2^k}$-arithmetic—and cannot directly exploit the field-tower packing that our fully-splitting-field approach enables. The FHE-oriented GKR protocol of [**EPRINT:PLFT25**] amortizes across CRT slots in parallel, committing in the CRT basis, whereas our commitments are in the coefficient basis and our CRT-domain polynomials remain virtual.

Implementation: Microlotus PCS and ring Plonkish SpeedySpartan. We implement Microlotus, a multilinear polynomial commitment scheme combining Basefold, random linear codes, our odd-prime tower construction, ring-switch packing, and oracle skipping. Instantiated over the degree-16 tower extension of $\mathbb{F}_{257}$—the natural setting for SWIFFT—Microlotus achieves a prover over 10× faster than WHIR and Basefold, a verifier over 10× faster than Brakedown and 2× faster than Basefold for larger polynomials, with proof sizes roughly 2× those of WHIR and Basefold but far smaller than Brakedown's $O(\sqrt{n})$. These benchmarks reinforce the principle that the right choice of arithmetization domain (here, an odd-prime tower matching the computation's native modulus) dominates the PCS-level efficiency differences.

We illustrate the full SNARK pipeline with a ring Plonkish SpeedySpartan [**speedyspartan**] that verifies arithmetic circuit satisfiability over R_p. The holographic variant of our PIOR, combined with the CRT-inlining optimization that eliminates the second sumcheck, produces evaluation claims whose shared coordinates enable a chain of zero-cost or low-cost reductions: selector-polynomial merging consolidates four indexed oracles into one, multi-point reduction exploits the full Boolean-hypercube covering for free, ring-switch packing compresses dimensions by $s = \log_2[\mathbb{K} : \mathbb{F}_p]$, and interleaved BaseFold opening shares Merkle authentication paths across the two surviving claims. The final PIOR chain reduces five evaluation claims on five oracles to two packed Basefold openings with shared folding. The efficiency and cleanliness of this pipeline is a direct consequence of our CRT-domain protocol design: by ensuring that all evaluation claims share a common set of

coordinates, each downstream optimization applies at full strength.

3.0.1 Contributions

This work makes the following contributions:

1. **A PIOR for cyclotomic ring multiplication** (section 3.4.1). We present a sumcheck-based PIOR for proving $As = t$ over fully splitting cyclotomic rings $\mathbb{Z}_p[X]/(X^n + 1)$, with $O(NMn)$ prover complexity, $O(n + \log NM)$ verifier complexity, and $O(\log Mn)$ proof size. An NTT-based evaluation-table construction eliminates the $O(Mn^2)$ bottleneck of naïve CRT-domain expansion. A holographic variant for public A eliminates the second sumcheck, reducing proof size to $2(\mu + \nu) + 1$ with all degree-2 rounds and a single committed-oracle evaluation claim. We prove round-by-round knowledge soundness via a compositional PIOR decomposition. The f-circulant formulation also covers odd cyclotomics via the Chinese remainder transform.
2. **Codes and proximity gaps for cyclotomic rings** (**??**). We show that any linear $[n, k, d]$-code over a field $\mathbb{F}_q$ gives a linear $[n, k, d]$-code for all cyclotomic rings $\mathbb{Z}_q[X]/(\Phi_m(X))$. We prove that linear codes over all finite rings R satisfy a $(d/3, E)$-proximity gap in the unique decoding radius regime, where E is determined by the residue fields of R. This yields an IOP of proximity to the ring, which in turn gives a hash-based multilinear PCS over cyclotomic rings.
3. **An odd-prime tower construction** (section 3.3). We construct a tower of quadratic extensions over any odd prime p, generalizing the Diamond–Posen packing technique [**EC:DiaPos25**] from binary to odd-characteristic fields. The tower achieves $k^{\log_2 3}$ hard $\mathbb{F}_p$-multiplications for degree-k extensions.
4. **Microlotus: a multilinear PCS implementation** (**??**). We implement a PCS combining Basefold, random linear codes, our odd-prime tower, ring-switch packing, and oracle skipping. Over a degree-16 tower of $\mathbb{F}_{257}$, the prover is over 10× faster than WHIR and Basefold for polynomials of practical size.
5. **A ring Plonkish SpeedySpartan** (section 3.4.4). We instantiate a SpeedySpartan-style SNARK for ring arithmetic and show how the CRT-domain protocol design enables a chain of zero-cost reductions—selector merging, free multi-point reduction, ring-switch packing, and interleaved BaseFold opening—that consolidates five evaluation claims on five oracles into two

packed Basefold openings with shared folding.

3.0.2 Related work

We survey prior work on proving statements involving ring arithmetic, structured linear algebra, and lattice-based relations, organized by the approach taken to handle the ring structure.

Unrolling ring arithmetic into field circuits. The most direct approach is to decompose each ring element $a \in R_q$ into its n coefficients over $\mathbb{F}_p$ and encode ring multiplication as an explicit NTT butterfly circuit. Boschini, Camenisch, Ovsiankin, and Spooner [**PQCRYPTO:BCOS20**] instantiate this with the Aurora IOP [**EC:BCRSVW19**], proving RSIS and RLWE relations by compiling them into R1CS over $\mathbb{F}_p$; the same strategy underlies vCNN [**EPRINT:LKKO20**] and ZEN [**EPRINT:FQZDC21**]. The fundamental limitation is that each ring multiplication contributes $O(n \log n)$ constraints, so an inner product $\langle a, s \rangle = t$ with M ring elements yields $O(Mn \log n)$ constraints regardless of the backend proof system used.

Sumcheck-based FFT and convolution proofs. Liu, Xie, and Zhang [**CCS:LiuXieZha21**] observe that the DFT entry ω^{jm} has a multilinear extension that factors as a product of $\log n$ terms over the binary digits of j and m. This lets a single sumcheck verify a length-n FFT with $O(n)$ prover work, faster than computing the transform, and the verifier evaluates the factored MLE in $O(\log n)$. A ring convolution is proved by composing FFT, Hadamard-product, and inverse-FFT sumchecks inside a GKR framework [**JACM:GolKalRot15**]. These can be amortized: the M forward transforms are batched into one GKR sumcheck, and likewise for the N inverse transforms. The dominant prover cost $O(NMn)$ matches ours; the verifier cost $O(\log(NMn))$ is lower, since each batched instance costs $O(\log n)$.

Our protocol folds the CRT matrix $\tilde{C}$ directly into the virtual polynomial of a single sumcheck expressing $As = t$, rather than composing separate transform sub-protocols via GKR. Because $\tilde{C}$ enters as a factor, the verifier encounters it exactly once, evaluating $\tilde{C}(\alpha', \gamma)$ in $O(n)$; this covers all NM ring products and does not scale with the matrix dimensions, giving verifier cost $O(n + \log(NM))$.

Before arriving at this solution, we developed a coefficient-domain sumcheck that proves ring

multiplication directly as a negacyclic circulant matrix-vector product, bypassing the CRT decomposition entirely and recovering $O(\log n)$. However, in the coefficient domain the convolution entangles the column-index and coefficient-index variables: the first sumcheck must range over all $\mu + \nu$ of them simultaneously, producing folding challenges that fix the oracle evaluation points at distinct last-ν coordinates. In section 3.4.4, we sketch a SpeedySpartan for ring arithmetic and show how shared evaluation coordinates yield a consolidation of these claims essentially for free. Concretely, these PCS optimizations save the verifier significantly more time than a single $O(n)$ evaluation takes, considering n generally ranges from 64 to 512 for protocols at the 128-bit security level.

SNARKs for ring arithmetic. Jia, Li, Xing, Yao, and Yuan [**C:JLXYY25**] establish proximity gaps for Reed–Solomon codes over Galois rings $\mathrm{GR}(2^k, d)$ and build a FRI-based polynomial commitment scheme, yielding the first publicly verifiable SNARKs for $\mathbb{Z}_{2^k}$-arithmetic with polylogarithmic proof size. Their machinery targets power-of-two moduli, not the NTT-friendly setting we consider. Similarly, Wei, Zhang, and Deng [**PKC:WeiZhaDen25**] construct transparent SNARKs over arbitrary Galois rings by extending Brakedown's [**C:GLSTW23**] linear-time encodable codes to the ring setting, achieving $O(n)$ prover time, $O(\sqrt{n})$ proof size, and $O(\sqrt{n})$ verifier time for log-space uniform circuits.

These Galois-ring works address a complementary problem to ours. They build general-purpose proof systems whose native computation domain is a ring $\mathbb{Z}_{2^k}$ (or its Galois extension), motivated by matching machine-word arithmetic or MPC-over-$\mathbb{Z}_{2^k}$ protocols. Our work targets a different setting: proving structured linear-algebraic relations (matrix-vector products) over *cyclotomic* rings $\mathbb{Z}_p[X]/(X^n + 1)$ with splitting primes, as arise in lattice-based cryptography. We exploit the CRT factorization specific to this setting to obtain a specialized PIOP whose prover complexity $O(NMn)$ is linear in the witness size and a $\log n$ factor better than any approach that treats ring multiplication as a black-box circuit gate. A further practical advantage of working over splitting fields (rather than Galois rings) is compatibility with the DP25 packing reduction [**EC:DiaPos25**]: since our sumcheck and oracles live over the extension field $\mathbb{K}$, which admits an explicit tower $\mathbb{F}_p \subset \mathbb{F}_{p^2} \subset \cdots \subset \mathbb{K}$, we can apply the s-round dimension reduction to compress PCS costs. The Galois-ring systems cannot directly exploit this tower structure, as their native domain is a ring rather than a field.

Ring-switching proofs over polynomial rings. Huang, Mao, and Zhang [**EPRINT:HuaMaoZha25**] propose a proof system for Ring-R1CS over prime-power modulus cyclotomic rings. Their key technique is *ring switching*: for each row i of $As = t$, the prover computes the quotient $Q_i(X) = (\sum_j A_{ij}(X) \cdot s_j(X) - t_i(X))/(X^n + 1)$, lifting the mod-$(X^n + 1)$ congruence to an exact identity in $\mathbb{Z}_q[X]$. Assuming preprocessed A, they would still send oracles for $\tilde{s} \in (\mathbb{F}_q^{<n}[X])[Y_1, \dots, Y_{\log M}]$ and $\tilde{Q} \in (\mathbb{F}_q^{<3n}[X])[Y_1, \dots, Y_{\log N}]$. They consider the extension field $\mathbb{K} = \mathbb{Z}_{q^k}$ such that $q^k \geq 2^\lambda$. The identity is evaluated at a random $X = \alpha \in K$, collapsing each ring equation to a scalar identity in $\mathbb{K}$, and then a standard multilinear sumcheck is used to reduce this to verifying PCS openings.

While in some ways our protocols are similar, we believe our scheme presents a few benefits. First, since our scheme works with the cyclotomic structure in the sumcheck argument machinery rather than collapsing it, we don't need to compute or send $\tilde{Q}$, reducing commitment costs from $O(nNM)$ to $O(nM)$ in the holographic setting. In their full quadratic R1CS protocol Q's degree in X becomes $3N$; practically speaking, since $M \approxeq N$ for R1CS circuits their protocol requires committing to $4\times$ the base field elements[2]. Second, typing oracles over Galois rings $\mathsf{GR}_q^n(\mathbb{F}_q^{<n}[X])$ is non-standard and has required extending many theorems about codes [**ESORICS:WZDWZY25**; **C:JLXYY25**]; these results, and thus the efficiency of the resulting polynomial commitments, still lag behind those for finite fields. Hachi [**hachi**] introduces a generic reduction from polynomial evaluation over extension fields to cyclotomic rings (so this is relevant to our PIOP as well), but state-of-the-art lattice PCs still lag behind their hash-based counterparts. At the information-theoretic level, it is unknown whether many PIORs can be applied. One important example of this is the ring-switch based *packing* technique of Diamond and Posen [**EC:DiaPos25**] that we show can be applied in our setting to reduce $\hat{s}$ to just $\log M - \log[\mathbb{K} : \mathbb{F}_p]$ variables.

Works proving FHE. Laminate [**EPRINT:PLFT25**] presents a verifiable FHE construction. It proves a layered payload circuit via a GKR-style blind hIOP while preserving BFV/BGV's native SIMD layout; the protocol can be understood as amortizing over running a proof for each CRT slot in parallel, using standard extension field techniques to amplify soundness, whereas our

[2]Empirically, $M/N \approx 1$ across diverse practical R1CS instantiations, including both algebraically-oriented circuits (Aptos keyless OpenID, https://github.com/aptos-labs/keyless-zk-proofs) and bitwise-dominated circuits (circomlib SHA-256, https://github.com/iden3/circomlib/blob/master/circuits/sha256/sha256.circom).

commitments are in the coefficient basis.

Wei et al. [**EPRINT:WWXZDZ25**] give a new sumcheck argument over fields of small characteristic that asymptotically reduces prover multiplications, while requiring sending $2kd$ oracles in each of dk rounds. As such, it remains impractical.

DP25 ring-switch packing. Diamond and Posen [**EC:DiaPos25**] introduce a packing technique that exploits the tower of quadratic extensions $\mathbb{F}_p \subset \mathbb{F}_{p^2} \subset \cdots \subset \mathbb{K}$ to compress multilinear polynomial dimensions by $s = \log[\mathbb{K} : \mathbb{F}_p]$ via an s-round degree-1 sumcheck. We use this as a post-processing step (section 3.4.4) to reduce the polynomial commitment cost, and we provide an odd-prime tower construction (section 3.3) that extends DP25's original even-characteristic instantiation.

As mentioned before, DP25 packing requires a tower of *field* extensions with an explicit multilinear basis at each level. The Galois-ring proof systems [**C:JLXYY25**; **PKC:WeiZhaDen25**; **ESORICS:WZDWZY25**] work over the ring $\mathrm{GR}(p^k, d) = \mathbb{Z}_{p^k}[X]/(h(X))$, so DP25's tower construction does not directly apply. These systems instead use the canonical embedding $\mathbb{Z}_{p^k} \hookrightarrow \mathrm{GR}(p^k, d)$.

3.1 Technical Overview

3.1.1 PIOP for cyclotomic ring arithmetic

We give an overview of our PIOP for proving matrix-vector products $As = t$ over the cyclotomic ring $R_p = \mathbb{Z}_p[X]/(X^n + 1)$, where $n = 2^\nu$ and $p \equiv 1 \pmod{2n}$. This is the R-algebraic core of lattice-based cryptography including MLWE/RLWE encryption, Dilithium/Falcon/Hawk signatures, FHE operation proofs, and Ajtai hash commitments. The full protocol is given in section 3.4.1.

A CRT-domain sumcheck for ring arithmetic. Our starting observation is that since we target fully splitting rings, we can use the CRT isomorphism $R_p \cong \mathbb{F}_p^n$ to reduce ring multiplication to pointwise multiplication. If $C \in \mathbb{F}_p^{n \times n}$ denotes the CRT matrix with entries $C_{k,\ell} = \omega^{(2k+1)\ell}$ (where ω is a primitive $2n$-th root of unity), then $As = t$ in R_p^N is equivalent to the pointwise identity

$$t_C(i,k) = \sum_{j=0}^{M-1} A_C(i,j,k) \cdot s_C(j,k) \qquad \text{for all } (i,k) \in [N] \times [n], \tag{3.1}$$

where $A_C = CA$, $s_C = Cs$, and $t_C = Ct$ denote the CRT transforms applied to coefficient vectors.

As in the main construction, we never materialize these CRT transforms as committed polynomials. Instead, from the coefficient-domain multilinear extensions $\tilde{A}$, $\tilde{s}$, $\tilde{t}$ and the CRT-matrix multilinear extension $\tilde{C}$, we define the virtual polynomials

$$A_C(i,j,k) := \sum_{\ell \in \{0,1\}^\nu} \tilde{C}(k,\ell) \cdot \tilde{A}(i,j,\ell), \qquad s_C(j,k) := \sum_{\ell \in \{0,1\}^\nu} \tilde{C}(k,\ell) \cdot \tilde{s}(j,\ell), \qquad t_C(i,k) := \sum_{\ell \in \{0,1\}^\nu} \tilde{C}(k,\ell) \cdot \tilde{t}(i,\ell).$$

On Boolean inputs, these agree with the actual CRT-domain values by linearity.

The protocol instance includes oracles for $\tilde{A}$, $\tilde{s}$, $\tilde{t}$ (you may think of these as either prover or indexer-provided, depending on the setting). The verifier moves first, sampling $\alpha \in \mathbb{K}^\tau$ and $\alpha' \in \mathbb{K}^\nu$, and the prover sends e_t claimed to equal $t_C(\alpha, \alpha')$. Next sumcheck is run on

$$e_t = \sum_{(j,k) \in \{0,1\}^{\mu+\nu}} A_C(\alpha, j, k) \cdot s_C(j,k) \cdot \widetilde{\mathsf{eq}}(k, \alpha').$$

This is a $(\mu + \nu)$-round sumcheck: the **J**-rounds are degree 2, while the **K**-rounds are degree 3. At the end, the verifier has random challenges $(\beta, \gamma) \in \mathbb{K}^{\mu+\nu}$ and a reduced claim $\sigma' = A_C(\alpha, \beta, \gamma) \cdot s_C(\beta, \gamma) \cdot \widetilde{\mathsf{eq}}(\gamma, \alpha')$. The prover then sends explicit claims $e_A = A_C(\alpha, \beta, \gamma)$ and $e_s = s_C(\beta, \gamma)$, and the verifier checks $\sigma' = e_A \cdot e_s \cdot \widetilde{\mathsf{eq}}(\gamma, \alpha')$.

To connect the virtual CRT-domain claims back to the committed coefficient-domain oracles, the verifier samples a batching challenge $\xi \in \mathbb{K}$ and the parties run another ν round sumcheck on the claim

$$e_A + \xi \cdot e_s + \xi^2 \cdot e_t = \sum_{\ell \in \{0,1\}^\nu} \Big[\tilde{C}(\gamma, \ell) \cdot (\tilde{A}(\alpha, \beta, \ell) + \xi \cdot \tilde{s}(\beta, \ell)) + \xi^2 \cdot \tilde{C}(\alpha', \ell) \cdot \tilde{t}(\alpha, \ell) \Big].$$

Notice that A_C and s_C are opened at the CRT point γ produced by the first sumcheck, while t_C remains tied to the original point α'. After the verifier samples a final challenge $\delta \in \mathbb{K}^\nu$, the protocol reduces to three ordinary evaluation claims on $\tilde{A}$, $\tilde{s}$, and $\tilde{t}$, together with two evaluation claims $\tilde{C}(\gamma, \delta)$ and $\tilde{C}(\alpha', \delta)$ which we show can be computed in $O(n)$ time (lemma 3.4.5).

Holographic variant and reduced sumcheck. When A is public, the indexer can supply the CRT-domain oracle A_C directly. If in addition the claimed value $e_t = t_C(\alpha, \alpha')$ is public or the

claimed evaluation of a virtual polynomial, then the only remaining committed CRT-consistency obligation is for $\tilde{s}$. In that case, we can inline the CRT expansion of s_C directly into the first sumcheck and eliminate the second sumcheck entirely:

$$e_t = \sum_{(j,\ell)\in\{0,1\}^{\mu+\nu}} A_C(\alpha, j, \alpha') \cdot \tilde{C}(\alpha', \ell) \cdot \tilde{s}(j, \ell).$$

This merged sumcheck has degree 2 in every round, saves $3\nu + 2$ field elements and $\nu + 1$ rounds relative to the base protocol, and leaves only a single evaluation claim on the committed oracle $\tilde{s}$.

Ring Plonkish SNARK example. We informally sketch how our PIOR can yield a SNARK for ring Plonkish, building on SpeedySpartan. The highly optimized final variant we arrive at reflects the strength and composability of our protocol design as well as the PIOR framework.

Security. We prove round-by-round knowledge soundness by decomposing the protocol into PIOR building blocks and bounding the knowledge error of each. This modular viewpoint makes the ring-arithmetic subprotocol compose cleanly with other RBRKS arguments. Applying Fiat–Shamir to the resulting public-coin protocol then yields a non-interactive argument with the usual QROM knowledge guarantees.

Odd-prime tower fields. The core ring-arithmetic PIOP only requires a sufficiently large extension field $\mathbb{K}/\mathbb{F}_p$. The odd-prime tower from section 3.3 is needed specifically for the DP25 ring-switch packing step in the oracle-reduction pipeline: it realizes the quadratic tower $\mathbb{F}_p \subset \mathbb{F}_{p^2} \subset \cdots \subset \mathbb{K}$ over an odd prime base field, so the packing reduction can run natively without changing the underlying arithmetization field.

3.1.2 Polynomial commitment scheme implementation

3.1.3 sec:microlotus

We implement *microlotus*, a multilinear polynomial commitment scheme that combines the Basefold IOPP, random linear codes, our tower field construction for odd primes, ring-switch packing, and oracle skipping. Microlotus supports evaluating v-variate polynomials $f \in \mathbb{F}^{\mathsf{ml}}[\mathbf{X}]$, for $\mathrm{char}(\mathbb{F}) > 2$, at points $\mathbf{x} \in \mathbb{K}^v$ for any extension field $\mathbb{K}/\mathbb{F}$ such that $[\mathbb{K} : \mathbb{F}]$ is a power of two.

In the context of arithmetization, the choice of base field is driven primarily by the computation being arithmetized. The arithmetization overhead from a mismatched base field will almost always dominate the efficiency gains between being able to use one PCS versus another, so in this sense Microlotus is not directly comparable with schemes that don't support odd prime towers. With this caveat, we compare the performance of Microlotus against other schemes (Basefold, WHIR, Brakedown). Microlotus is instantiated over the degree 16 odd prime tower extension $\mathbb{K}$ of $\mathbb{F} = \mathbb{F}_{257}$, which corresponds to SWIFFT[3]. Basefold below, refers to Basefold with Reed-Solomon using code rate $\frac{1}{2}$ and 100 queries. Our benchmarks for WHIR are over a degree 3 Goldilocks extension. For Brakedown, they are over a 255 bit prime field. We also compare our Basefold implementation over degree 3 Goldilocks for a fair comparison with WHIR. The following observations can be observed from our benchmarks of concrete performance:

- The Microlotus prover is over 10x faster than WHIR and Basefold in prover time, and comparable to Brakedown for polynomial sizes of interest.
- The Microlotus verifier is over 10x faster than Brakedown and 2x faster than Basefold for larger polynomials, while still being about 6x slower than WHIR.
- Our proof size is a little less than twice as big as Basefold and WHIR, while being significantly smaller than Brakedown (the only scheme with square root proof size).

The benchmarks presented in this section were run *single-threaded* on an AWS r7iz.8xlarge EC2 instance with 256 GB RAM.

3.2 Preliminaries

Notation. We write $[n]$ for the set $\{1, \dots, n\}$ and $\{0, 1\}$ for $\{0, 1\}$. We denote by λ the security parameter, by $\mathrm{negl}(\lambda)$ a function negligible in λ, and by $\mathrm{poly}(\lambda)$ an unspecified polynomial in λ. For a finite set S, we write $x \leftarrow\$ S$ to denote sampling x uniformly at random from S. We let $\mathbb{F}$ denote a finite field and $\mathbb{K}$ an extension field of $\mathbb{F}$. For a positive integer q, we write $\mathbb{Z}_q$ for the ring $\mathbb{Z}/q\mathbb{Z}$.

[3]In practice one should instead reparametrize SWIFFT around the modulus that works best for the rest of the circuit.

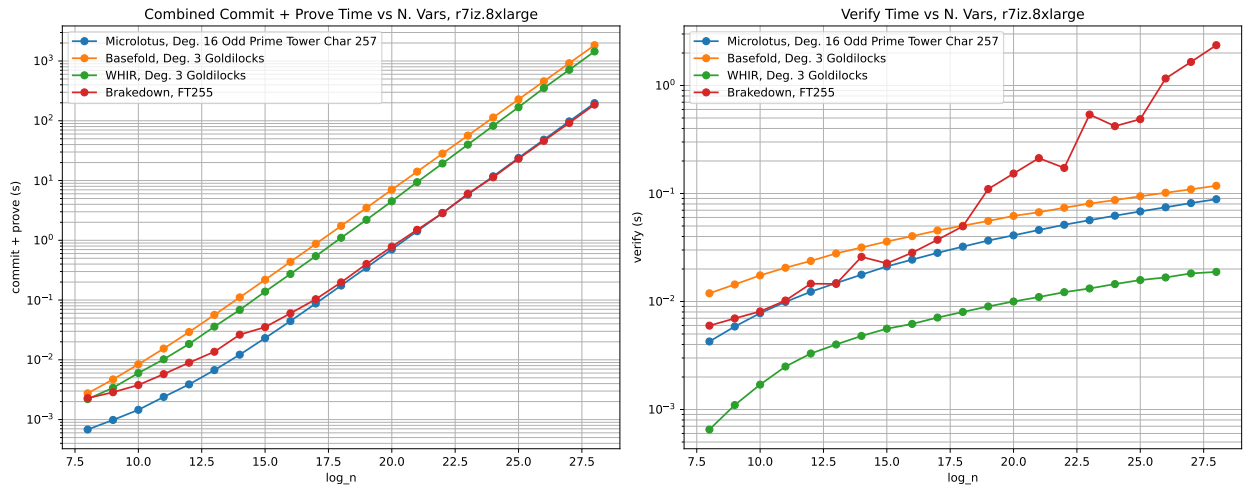


Figure 3.1: Total Commit + Prove, Verify Times vs. $\log n$.

An *oracle* to a string or vector π of elements from a ring R is a black-box that allows querying any position j and obtaining π_j. We denote it by $[[\pi]]$. An *oracle to a polynomial* $f(\mathbf{X})$ over R is a black-box that allows querying any point $\mathbf{r}$ in the domain of f and obtaining $f(\mathbf{r})$. We denote it by $[[f]]$.

3.2.1 Multilinear polynomials

Lemma 3.2.1 (Lagrange interpolation of multilinear polynomials)**.** *Let $f : \{0,1\}^v \to \mathbb{F}$ be any function. Then the following multilinear polynomial $\hat{f}$ extends f:*

$$\hat{f}(X_1, \dots, X_v) = \sum_{w \in \{0,1\}^v} f(w) \cdot \chi_w(X_1, \dots, X_v)$$

where, for any $w = (w_1, \dots, w_v)$,

$$\chi_w(X_1, \dots, X_v) := \prod_{i=1}^{v} (X_i w_i + (1 - X_i)(1 - w_i))$$

The set $\{\chi_w : w \in \{0,1\}^v\}$ is referred to as the set of multilinear Lagrange basis polynomials with interpolating set $\{0,1\}^v$.

We will also sometimes write $\widetilde{\mathsf{eq}}(w, X)$ instead of $\chi_w(X)$ to denote the same polynomial.

Lemma 3.2.2. *Let $p : \mathbb{F}^\ell \to \mathbb{F}$ be a multilinear polynomial. Then for any $0 \le i \le \ell$, any*

N. Vars	Commit + Prove	Verify	Proof Size
16	44.5ms	24.4ms	259.0 KiB
20	703.9ms	41.0ms	542.0 KiB
24	11.8s	62.3ms	931.2 KiB
28	197.4s	88.6ms	1430.8 KiB

Table 3.1: Microlotus PCS, Odd Prime Tower

N. Vars	Commit + Prove	Verify	Proof Size
16	434.9ms	40.3ms	568.2 KiB
20	7.0s	62.0ms	973.1 KiB
24	113.5s	86.9ms	1482.4 KiB
28	1840.5s	117.9ms	2114.8 KiB

Table 3.2: Basefold, Deg. 3 Goldilocks

N. Vars	Commit + Prove	Verify	Proof Size
16	273.1 ms	6.2 ms	287.5 KiB
20	4.5 s	10.0 ms	413.3 KiB
24	82.3 s	14.5 ms	558.8 KiB
28	1444.8 s	18.8 ms	712.8 KiB

Table 3.3: WHIR, Deg. 3 Goldilocks

N. Vars	Commit + Prove	Verify	Proof Size
16	59.9 ms	28.3 ms	4.04 MiB
20	780.1 ms	153.0 ms	9.11 MiB
24	11.3 s	420.3 ms	19.3 MiB
28	186.1 s	2.4 s	41.7 MiB

Table 3.4: Brakedown, Ft255

$(\alpha_1, \dots, \alpha_i) \in \mathbb{F}^i$, *and any* $x' \in \{0,1\}^{\ell-i}$,

$$p(\alpha_1, \dots, \alpha_i, x') = \sum_{y \in \{0,1\}^i} \widetilde{\mathsf{eq}}((\alpha_1, \dots, \alpha_i), y) \cdot p(y, x')$$

Setting $i = 1$, the identity in lemma 3.2.2 simplifies to

$$p(\alpha_1, x') = (1 - \alpha_1) \cdot p(0, x') + \alpha_1 \cdot p(1, x')$$

Setting $i = \ell$, we obtain the multilinear extension formula

$$p(\alpha_1, \dots, \alpha_\ell) = \sum_{y \in \{0,1\}^\ell} \widetilde{\mathsf{eq}}((\alpha_1, \dots, \alpha_\ell), y) \cdot p(y)$$

3.2.2 Cyclotomic Rings

Let $\zeta = \zeta_m$ be a primitive mth root of unity (i.e., $\zeta^m = 1$, but $\zeta^i \neq 1$ for $i \in [m-1]$). For any positive integer m, the mth cyclotomic polynomial Φ_m is the minimal polynomial of ζ over the

integers:

$$\Phi_m(X) = \prod_{\substack{k \in [m], \\ \gcd(k,m)=1}} (X - \zeta^k)$$

where we observe its conjugates are the other primitive mth roots of unity. There is a natural isomorphism between $\mathbb{Z}[\zeta]$ and $R = \mathbb{Z}[X]/\Phi_m(X)$ given by $\zeta \mapsto X$. Let $n = \phi(m)$. Then $\mathbf{p} = (\zeta^i)_{i=0}^{n-1}$ defines a $\mathbb{Z}$-basis for R, called the *power basis*. We note that any $\mathbb{Z}$-basis for R is a $\mathbb{Z}_q$-basis for $R_q = \mathbb{Z}_q[X]/\Phi_m(X)$.

Another important $\mathbb{Z}_q$-basis for R_q is the Chinese remainder transform basis, or *CRT basis*, where ring multiplication simplifies to computing the Hadamard product of the $\mathbb{Z}_q^n$ vectors. The Chinese remainder transform can be seen as a specialization of the discrete Fourier transform to cyclotomic rings, and is also called the number theoretic transform (NTT). By virtue of being the rings of integers of algebraic number fields, cyclotomic rings are Dedekind, which means ideals can be factored into a product of prime ideals. A particular case of interest is $q \equiv 1 \mod m$, which corresponds to the "fully splitting" case. Notice that $\mathbb{Z}_q$ contains a primitive mth root of unity ω_m. Then the principal ideal $\langle q \rangle$ in R can be written as the product of the prime ideals $\mathfrak{p}_i = \langle q \rangle + \langle \zeta_m - \omega_m^i \rangle$ for $i \in \mathbb{Z}_m^*$, i.e., it splits completely into n ideals of norm q each. Each quotient ring $R/\mathfrak{p}_i$ is isomorphic to the field $\mathbb{Z}_q$ via the map $\zeta_m \mapsto \omega_m^i$. Then by the Chinese remainder theorem

$$R/\mathfrak{n} \cong \prod_i R/\mathfrak{p}_i \cong (\mathbb{Z}_q)^n$$

The CRT basis $\mathbf{c}$ is then defined such that $c_i \equiv 1 \bmod \mathfrak{p}_i$ and $c_i \equiv 0 \bmod \mathfrak{p}_j$ for all $j \neq i$.

The Chinese remainder transform and its inverse convert between the power and CRT bases. The linear transform, given as a matrix $\mathsf{CRT}_{m,q}$ below, has the identity $\mathbf{p}^\top = \mathbf{c}^\top \cdot \mathsf{CRT}_{m,q}$.

Definition 3.2.3 ([**EC:LyuPeiReg13**])**.** Let m be a prime power and let R denote any commutative ring containing some element ω_m of multiplicative order m, i.e., a primitive mth root of unity.

- The *discrete Fourier transform* DFT_m over R is the $\mathbb{Z}_m \times \mathbb{Z}_m$ matrix whose (i,j)th entry is ω_m^{ij}.
- The *Chinese remainder transform* CRT_m over R is the (square) submatrix of DFT_m obtained by restricting to the rows indexed by $\mathbb{Z}_m^*$ and the columns indexed by $[\varphi(m)]$.

Applying DFT_m corresponds to evaluating a polynomial of degree less than m over $R[Y]$

(represented by the vector of coefficients in the natural order) at all the mth roots of unity. Similarly, CRT_m corresponds to evaluating a polynomial of degree less than $\phi(m)$ at the primitive mth roots of unity ω_m^i for $i \in \mathbb{Z}_m^*$.

3.2.3 Indexed relations

An *indexed relation* is a set $\mathcal{R} \subseteq \{0,1\}^* \times \{0,1\}^* \times \{0,1\}^*$ of triples $(\mathbb{i}, \mathbb{x}, \mathbb{w})$, where $\mathbb{i}$ is the *index* (encoding structural parameters such as field characteristic, dimensions, and degree bounds), $\mathbb{x}$ is the *instance* (the public input, which in the polynomial oracle model may include oracle access to polynomials), and $\mathbb{w}$ is the *witness* (the private input). The *language* of $\mathcal{R}$ is $\mathcal{L}(\mathcal{R}) = \{(\mathbb{i}, \mathbb{x}) : \exists\, \mathbb{w} \text{ s.t. } (\mathbb{i}, \mathbb{x}, \mathbb{w}) \in \mathcal{R}\}$.

3.2.4 Polynomial interactive oracle reductions

We define holographic polynomial interactive oracle reductions, specializing the definition of interactive oracle reductions in [**BCFW25**] to the polynomial oracle model.

Definition 3.2.4 (Holographic PIOR)**.** A *public-coin holographic polynomial interactive oracle reduction* (PIOR) from an indexed relation $\mathcal{R}$ to an indexed relation $\mathcal{R}'$ is a triple $\Pi = (\mathsf{I}, \mathsf{P}, \mathsf{V})$ that works as follows.

- *Offline phase.* The *indexer* I takes the index $\mathbb{i}$ and outputs a new index $\mathbb{i}'$, which may contain both explicit values and oracles.
- *Online phase.* The prover P is an interactive algorithm that receives as input $(\mathbb{i}', \mathbb{x}, \mathbb{w}) \in \mathcal{R}$, and the verifier V is an interactive oracle algorithm that receives $(\mathbb{i}', \mathbb{x})$. They interact over k rounds as follows. In each round $i \in [k]$, P sends a message π_i that may contain both explicit inputs and oracles, and V sends a uniformly random message $\rho_i \leftarrow\$ \mathcal{C}_i$ for some challenge space $\mathcal{C}_i$. After k rounds of interaction, V queries any oracles in $\mathbb{i}', \mathbb{x}$, and the messages sent by P. After that:
  - V either rejects or outputs a new instance $\mathbb{x}$', and
  - P outputs a new witness $\mathbb{w}$' such that $(\mathbb{i}', \mathbb{x}', \mathbb{w}') \in \mathcal{R}'$.

  Without loss of generality, V's output is a deterministic function of its input and the transcript. Note that V may specify oracles in its output implicitly, e.g., V could output an oracle

$[[f]] := [[g_0]] + \rho_1[[g_1]]$, where $[[g_0]], [[g_1]]$ are oracles it has access to.

A *polynomial IOP* (PIOP) is the special case where $\mathcal{R}'$ is trivial: the verifier either accepts or rejects.

An *i-round partial transcript* is $\mathsf{tr}_i = (\pi_1, \rho_1, \dots, \pi_i, \rho_i)$; a *0-round partial transcript* is $\mathsf{tr}_0 = \emptyset$. We write $\mathsf{tr}_{i-1} \| \pi_i$ for the partial transcript after the prover's round-i message but before the verifier's round-i challenge.

Definition 3.2.5 (PIOR completeness)**.** We say that a PIOR $(\mathsf{I}, \mathsf{P}, \mathsf{V})$ has (perfect) completeness if for every $(\mathbb{i}, \mathbb{x}, \mathbb{w}) \in \mathcal{R}$,

$$\Pr\left[(\mathbb{i}', \mathbb{x}', \mathbb{w}') \in \mathcal{R}' \;\middle|\; \begin{array}{l} \mathbb{i}' \leftarrow \mathsf{I}(\mathbb{i}) \\ \pi_1 \leftarrow \mathsf{P}(\mathbb{i}', \mathbb{x}, \mathbb{w}) \\ \rho_1 \leftarrow \mathcal{C}_1 \\ \vdots \\ (\pi_k, \mathbb{w}') \leftarrow \mathsf{P}(\mathbb{i}', \mathbb{x}, \mathbb{w}, \rho_1, \dots, \rho_{k-1}) \\ \rho_k \leftarrow \mathcal{C}_k \\ \mathbb{x}' \leftarrow \mathsf{V}(\mathbb{i}', \mathbb{x}, \rho_1, \dots, \rho_k) \end{array} \right] = 1.$$

We now specialize the round-by-round knowledge soundness framework of [**BCFW25**] for IORs to PIORs, which enables modular composition of PIOR chains.

Definition 3.2.6 (Knowledge state function [**BCFW25**])**.** Let $\Pi = (\mathsf{I}, \mathsf{P}, \mathsf{V})$ be a (holographic) PIOR from $\mathcal{R}$ to $\mathcal{R}'$. A *knowledge state function* for Π is a (possibly inefficient) function KState that takes as input a statement $(\mathbb{i}, \mathbb{x})$, a partial transcript tr, and a *knowledge state witness* w, and outputs a bit. It must satisfy:

1. **Empty transcript.** $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \emptyset, \mathsf{w}) = 1$ if and only if $(\mathbb{i}, \mathbb{x}, \mathsf{w}) \in \mathcal{R}$.
2. **Prover moves.** If $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr}, \mathsf{w}) = 0$ at a point where the prover is about to move, then $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr} \| \pi, \mathsf{w}) = 0$ for every prover message π.
3. **Full transcript.** For a full transcript tr_k: $\mathsf{KState}(\mathbb{i}, \mathbb{x}, \mathsf{tr}_k, \mathsf{w}) = 1$ if and only if V outputs $\mathbb{x}'$ such that $(\mathbb{i}', \mathbb{x}', \mathsf{w}) \in \mathcal{R}'$.

Definition 3.2.7 (RBR knowledge soundness [**BCFW25**])**.** A PIOR Π from $\mathcal{R}$ to $\mathcal{R}'$ has *round-by-round knowledge soundness* with errors $(\kappa_1, \dots, \kappa_k)$ if there exist a knowledge state function KState and a deterministic extractor $\mathcal{E}$ such that for every statement $(\mathbb{i}, \mathbb{x})$, every $i \in [k]$, and every partial

transcript tr where the verifier is about to move,

$$\Pr_{\rho_i}\left[\exists \mathsf{w}: \begin{array}{l} \mathsf{KState}\,(\mathbb{i}, \mathbb{x}, \mathsf{tr}, \mathcal{E}(\mathbb{i}, \mathbb{x}, \mathsf{tr}\|\rho_i, \mathsf{w})) = 0 \\ \wedge\ \mathsf{KState}\,(\mathbb{i}, \mathbb{x}, \mathsf{tr}\|\rho_i, \mathsf{w}) = 1 \end{array}\right] \leq \kappa_i(\mathbb{i}, \mathbb{x}).$$

Informally, if a knowledge state witness w certifies $\mathsf{KState} = 1$ after the challenge ρ_i, the extractor must produce a witness certifying $\mathsf{KState} = 1$ *before* the challenge; the error κ_i bounds the probability this fails.

Remark 3.2.8 (Composition of PIORs)**.** The composition theorem of [**BCFW25**] shows that if Π_1 is a PIOR from $\mathcal{R}$ to $\mathcal{R}'$ with RBR-KS errors $(\kappa_1, \dots, \kappa_k)$ and Π_2 is a PIOR from $\mathcal{R}'$ to $\mathcal{R}''$ with errors $(\kappa'_1, \dots, \kappa'_{k'})$, then the composed protocol has RBR-KS with errors $(\kappa_1, \dots, \kappa_k, \kappa'_1, \dots, \kappa'_{k'})$.

Remark 3.2.9 (Implied RBR and generalized special soundness)**.** RBR knowledge soundness implies *round-by-round soundness* (the weaker notion where only false statements start doomed, and no extractor is required) with the same errors [**BGTZ24**]. It also implies $(k_1, \dots, k_\mu)$-special soundness [**BGTZ24**].

Remark 3.2.10 (Compilation to non-interactive arguments)**.** Any PIOR with RBR (knowledge) soundness can be compiled into a non-interactive argument via the Fiat-Shamir transformation [**BCS16**] that is adaptively (knowledge) sound in both the random oracle model [**BCS16**] and the quantum random oracle model [**CMS19**].

3.2.5 Sumcheck

Variations of the sumcheck protocol [**JACM:LFKN92**] have become central to the construction of many of today's state-of-the-art PIOPs. Here we consider sumcheck instances over a field $\mathbb{F}$ to be claims of the form (σ, g, v, d), where g is a v-variate polynomial of degree at most d in each variable, and σ is its claimed sum over the v-dimensional hypercube:

$$\sigma = \sum_{x_1, \dots, x_v \in \{0,1\}^v} g(x_1, \dots, x_v)$$

The sumcheck protocol gives a reduction of this claim to checking the evaluation of g at a random point $g(\alpha_1, \dots, \alpha_v) = \sigma_v$. For self-containedness, we have included in appendix .1 an optimized

sumcheck protocol (Figure 3) and a linear time and space prover algorithm for it (Figure 4) from [**BagadDaoDombThaler25**].

3.3 A tower construction for odd primes

For odd primes we can construct the following tower of fields

$$\begin{aligned} & \tau_0 = \mathbb{F}_p & \\ & \tau_1 = \tau_0[X_0]/(X_0^2 - \alpha) & \cong \mathbb{F}_{p^2} \\ (\forall \iota \geq 2) \quad & \tau_\iota = \tau_{\iota-1}[X_{\iota-1}]/(X_{\iota-1}^2 - X_{\iota-2}) & \cong \mathbb{F}_{p^{2^\iota}} \end{aligned}$$

where α is any nonsquare in $\mathbb{F}_p$, guaranteed to exist by the condition $p \equiv 1 \mod 2$.

Choice of α. Since multiplication by α appears in every reduction step of the tower, α should be chosen as small as possible (e.g. $\alpha = 3$ for $p = 257$) so that it can be implemented via a short addition chain. We call such multiplications by compile-time constants *soft*, in contrast to *hard* multiplications between two runtime-variable $\mathbb{F}_p$ elements.[4]

Tower multiplication. Write elements of τ_ι as $a + bX_{\iota-1}$ with $a, b \in \tau_{\iota-1}$. Karatsuba multiplication computes $z_1 = aa'$, $z_2 = bb'$, $z_3 = (a+b)(a'+b') - z_1 - z_2$ in $\tau_{\iota-1}$, then reduces via $X_{\iota-1}^2 = X_{\iota-2}$:

$$(a + bX_{\iota-1})(a' + b'X_{\iota-1}) = (z_1 + z_2 X_{\iota-2}) + z_3 X_{\iota-1}.$$

The product $z_2 X_{\iota-2}$ in $\tau_{\iota-1}$ is a permutation of tower coefficients plus one soft α-multiply (since at each level $X_j^2 = X_{j-1}$ reduces to a swap, bottoming out at $X_0^2 = \alpha$). The recurrence $M(\iota) = 3M(\iota - 1) + 1$ with $M(0) = 1$ gives

$$M(\iota) = \frac{3^{\iota+1} - 1}{2} = \lfloor 1.5 \cdot k^{\log_2 3} \rfloor \quad \mathbb{F}_p\text{-multiplications}$$

for a degree-k extension ($k = 2^\iota$), of which $M_h(\iota) = 3^\iota = k^{\log_2 3}$ are hard and $M_s(\iota) = (3^\iota - 1)/2$ are soft.

Squaring. We adapt the Gauss complex-multiplication trick. Compute $u = ab$, $w = X_{\iota-2} \cdot b$ (soft),

[4] Furthermore, for primes of the form $p = 2^n + 1$ (e.g., $p = 257 = 2^8 + 1$), elements fit in $(n+1)$-bit words and reduction modulo p is a single conditional subtraction. With w-bit accumulators for $w \gg n$, all intermediate additions in the Karatsuba recursion can be accumulated without modular reduction.

and $v = (a+b)(a+w)$. Expanding:

$$v = a^2 + X_{\iota-2} \cdot b^2 + (1 + X_{\iota-2}) \cdot u$$

so $(a + bX_{\iota-1})^2 = (v - u - X_{\iota-2} \cdot u) + 2u \cdot X_{\iota-1}$, where $X_{\iota-2} \cdot u$ is soft. This uses two recursive multiplications instead of two squarings plus a multiplication, giving $S(\iota) = 2M(\iota-1) + 2 + 2^{\iota-1}$ with hard/soft costs:

$$S_h(\iota) = \tfrac{2}{3}\, k^{\log_2 3}, \qquad S_s(\iota) = 3^{\iota-1} + 2^{\iota-1} + 1$$

where the $2^{\iota-1}$ term accounts for the coefficient-wise $\times 2$ scaling of u.

Inversion. Using the norm $\delta = a^2 - b^2 X_{\iota-2} \in \tau_{\iota-1}$, we have

$$(a + bX_{\iota-1})^{-1} = a\delta^{-1} - b\delta^{-1} X_{\iota-1}.$$

This requires two squarings and two multiplications in $\tau_{\iota-1}$ plus one recursive inversion, giving the recurrence $I_h(\iota) = I_h(\iota-1) + 2S_h(\iota-1) + 2M_h(\iota-1)$ with $I_h(0) = 0$ (plus one $\mathbb{F}_p$-inversion), so that $I_h(\iota) \to \frac{5}{3}\, M_h(\iota)$ for large ι.

Concrete costs. Table 3.5 summarizes costs for small tower heights.

ι	$k = 2^\iota$	Multiplication		Squaring		Inversion*	
		hard	soft	hard	soft	hard	soft
1	2	3	1	2	3	4	1
2	4	9	4	6	6	14	10
3	8	27	13	18	14	44	31
4	16	81	40	54	36	134	86
5	32	243	121	162	98	404	239
6	64	729	364	486	276	1214	678

Table 3.5: Odd-prime tower costs in $\mathbb{F}_p$-operations. Hard multiplications are between runtime-variable elements; soft multiplications (by α or small constants) cost $\sim$1 addition each. *Inversion costs are in addition to one $\mathbb{F}_p$-inversion.

Comparison to binary tower fields. In the context of arithmetization, the choice of base field is driven primarily by the computation being arithmetized: binary towers over $\mathbb{F}_2$ (e.g. the Wiedemann tower [**wiedemann1988iterated**]) are natural when the computation is expressed as a Boolean circuit over XOR, NOT, and other bitwise gates, whereas odd-prime towers are natural for circuits over $\mathbb{F}_p$ involving mostly algebraic constraints. The arithmetization overhead from a mismatched

base field will almost always dominate any difference in tower arithmetic efficiency, so the two settings are not directly comparable. With that caveat, the tower-level costs are similar: both towers use Karatsuba and achieve $3^\iota = k^{\log_2 3}$ hard base-field multiplications for degree-k extension multiplication, with the remaining overhead in additions or soft operations. Our tower's reduction ($X_{\iota-1}^2 = X_{\iota-2}$, a permutation plus one soft α-multiply) is particularly cheap; the Wiedemann reduction (multiply by a tower constant $c_{\iota-1}$) is $\mathbb{F}_2$-linear and thus also addition-only, but operates on increasingly large vectors. The binary tower's main structural advantage is free squaring via the Frobenius ($x \mapsto x^2$ is $\mathbb{F}_2$-linear), versus $\frac{2}{3}\, k^{\log_2 3}$ hard multiplications in our tower.

3.4 PIOPs over fields for cyclotomic ring arithmetic

We first give the core matrix-vector PIOR in section 3.4.1, which reduces proving $As \equiv t$ over R_p to a small number of evaluation claims on the coefficient-domain oracles. Its prover and verifier complexity are $O(NMn)$ and $O(n + \log NM)$, and its proof size is $O(\log Mn)$. We show completeness, round-by-round knowledge soundness in section 3.4.2. Next, section 3.4.3 presents a holographic specialization for the public A setting, where its commitment is output by an indexer. We show this allows a significantly simplified and more efficient protocol. Finally, section 3.4.4, using a ring-Plonkish SpeedySpartan running example, describes how a chain of PIORs can reduce verification to two query claims: one the MLE of the prover's witness and the other on an indexed oracle, wrapping up with a couple final optimizations in the PCS domain.

3.4.1 A PIOR for cyclotomic matrix-vector multiplication

Fix a prime p, $n = 2^\nu$ with $p \equiv 1 \pmod{2n}$, $N = 2^\tau$, $M = 2^\mu$, and let $\mathcal{R}_p = \mathbb{Z}_p[X]/(X^n + 1)$.

Definition 3.4.1. The *ring multiplication relation* $\mathcal{R}_{\mathsf{RM}}$ consists of triples $(\mathbb{i}, \mathbb{x}, \mathbb{w})$ where:

- Index: $\mathbb{i} = (p, \nu, \tau, \mu)$.
- Instance: $\mathbb{x} = ([[\tilde{A}]], [[\tilde{s}]], [[\tilde{t}]])$, three multilinear polynomial oracles over $\mathbb{F}_p$ of arities $\tau{+}\mu{+}\nu$, $\mu{+}\nu$, and $\tau{+}\nu$ respectively.
- Witness: $\mathbb{w} = (A, s, t) \in \mathcal{R}_p^{N\times M} \times \mathcal{R}_p^M \times \mathcal{R}_p^N$.
- Membership: $As = t$ in $\mathcal{R}_p^N$, and the oracles are the multilinear extensions of the coefficient representations of A, s, t.

The target relation is the polynomial evaluation relation $\mathcal{R}_{\mathsf{eval}}$, whose triples $(\mathbb{i}', \mathbb{x}', \mathbb{w}')$ consist of a list of evaluation claims $\mathbb{x}' = \{([[f_j]], z_j, v_j)\}_{j=1}^{m}$ on polynomial oracles, with membership requiring $f_j(z_j) = v_j$ for all j. The witness $\mathbb{w}'$ is unconstrained: membership depends only on the oracles and the claimed evaluations.

Let $\omega \in \mathbb{F}_p$ be a primitive $2n$-th root of unity and let $\tilde{C} \in \mathbb{F}_p^{\mathsf{ml}}[\mathbf{K}, \mathbf{L}]$ denote the multilinear extension of the CRT matrix $C = \mathsf{CRT}_{n,p}$,

$$C_{k,\ell} = \omega^{(2k+1)\ell}, \qquad k, \ell \in \{0,1\}^{\nu},$$

which maps coefficient vectors to evaluations at the roots of $X^n + 1$. Define virtual polynomials

$$\begin{aligned} A_C(i,j,k) &:= \textstyle\sum_{\ell \in \{0,1\}^{\nu}} \tilde{C}(k,\ell) \cdot \tilde{A}(i,j,\ell), \\ s_C(j,k) &:= \textstyle\sum_{\ell \in \{0,1\}^{\nu}} \tilde{C}(k,\ell) \cdot \tilde{s}(j,\ell), \\ t_C(i,k) &:= \textstyle\sum_{\ell \in \{0,1\}^{\nu}} \tilde{C}(k,\ell) \cdot \tilde{t}(i,\ell). \end{aligned}$$

When $t \equiv As \mod p$, the CRT transform gives the pointwise relation $t_C(i,k) = \sum_{j \in \{0,1\}^{\mu}} A_C(i,j,k) \cdot s_C(j,k)$ for all $(i,k) \in \{0,1\}^{\tau+\nu}$. Since A_C and s_C are individually multilinear in $\mathbf{K}$, the product $\sum_j A_C \cdot s_C$ has individual degree 2 in $\mathbf{K}$; the sumcheck therefore includes an $\widetilde{\mathsf{eq}}(\mathbf{K}, \alpha')$ kernel to project onto the multilinear extension in $\mathbf{K}$ and ensure completeness.

Figure 3.2: Protocol $\Pi_{\mathsf{RingMul}} = (\mathsf{P}, \mathsf{V})$: a PIOR for checking matrix-vector multiplication in a cyclotomic ring $R_p = \mathbb{Z}_p[X]/(X^{2^\nu} + 1)$ over $\mathbb{F}_p$

Input: The verifier receives oracles $[[\tilde{A}]] \in \mathbb{F}_p^{\mathsf{ml}}[\mathbf{I}, \mathbf{J}, \mathbf{K}]$, $[[\tilde{s}]] \in \mathbb{F}_p^{\mathsf{ml}}[\mathbf{J}, \mathbf{K}]$, and $[[\tilde{t}]] \in \mathbb{F}_p^{\mathsf{ml}}[\mathbf{I}, \mathbf{K}]$. The prover receives matrix $A \in R_p^{N \times M}$ and vectors $s \in R_p^M$, $t \in R_p^N$ such that

$$\tilde{A}(i, j, k) = \mathsf{cf}_k(A_{i,j}), \qquad \tilde{s}(j, k) = \mathsf{cf}_k(s_j), \qquad \tilde{t}(i, k) = \mathsf{cf}_k(t_i)$$

holds for $i \in \{0,1\}^\tau, j \in \{0,1\}^\mu, k \in \{0,1\}^\nu$, where $\tau = \log N$, $\mu = \log M$, $\nu = \log n$, and $n = 2^\nu$ is the degree of the cyclotomic polynomial.

1: The verifier samples $\alpha \leftarrow\$ \mathbb{K}^\tau$, $\alpha' \leftarrow\$ \mathbb{K}^\nu$.

2: The prover sends e_t claimed to equal $t_C(\alpha, \alpha')$.

3: P and V engage in a sumcheck for the claim

$$e_t = \sum_{(j,k) \in \{0,1\}^{\mu+\nu}} A_C(\alpha, j, k) \cdot s_C(j, k) \cdot \widetilde{\mathsf{eq}}(k, \alpha')$$

with challenges $(\beta, \gamma) \leftarrow\$ \mathbb{K}^{\mu+\nu}$, reducing to the claim $\sigma' = A_C(\alpha, \beta, \gamma) \cdot s_C(\beta, \gamma) \cdot \widetilde{\mathsf{eq}}(\gamma, \alpha')$.

4: The prover sends e_A and e_s claimed to equal $A_C(\alpha, \beta, \gamma)$ and $s_C(\beta, \gamma)$ respectively. The verifier evaluates $\widetilde{\mathsf{eq}}(\gamma, \alpha')$ in $O(\nu)$ operations and checks

$$\sigma' \;=\; e_A \cdot e_s \cdot \widetilde{\mathsf{eq}}(\gamma, \alpha').$$

5: The verifier samples batching challenge $\xi \leftarrow\$ \mathbb{K}$.

6: P and V engage in a sumcheck for the claim

$$e_A + \xi \cdot e_s + \xi^2 \cdot e_t = \sum_{\ell \in \{0,1\}^\nu} \Big[\tilde{C}(\gamma, \ell) \cdot (\tilde{A}(\alpha, \beta, \ell) + \xi \cdot \tilde{s}(\beta, \ell)) + \xi^2 \cdot \tilde{C}(\alpha', \ell) \cdot \tilde{t}(\alpha, \ell)\Big]$$

with folding challenges $\delta \leftarrow\$ \mathbb{K}^\nu$, reducing to the claim

$$\sigma'' = \tilde{C}(\gamma, \delta) \cdot (\tilde{A}(\alpha, \beta, \delta) + \xi \cdot \tilde{s}(\beta, \delta)) + \xi^2 \cdot \tilde{C}(\alpha', \delta) \cdot \tilde{t}(\alpha, \delta).$$

7: The verifier evaluates $\tilde{C}(\gamma, \delta)$ and $\tilde{C}(\alpha', \delta)$ in $O(n)$ field operations (lemma 3.4.5), queries $e'_A \leftarrow [[\tilde{A}(\alpha, \beta, \delta)]]$, $e'_s \leftarrow [[\tilde{s}(\beta, \delta)]]$, $e'_t \leftarrow [[\tilde{t}(\alpha, \delta)]]$, and checks

$$\sigma'' = \tilde{C}(\gamma, \delta) \cdot (e'_A + \xi \cdot e'_s) + \xi^2 \cdot \tilde{C}(\alpha', \delta) \cdot e'_t.$$

3.4.2 Completeness, round-by-round knowledge soundness, and efficiency

Completeness.

Theorem 3.4.2 (Completeness of Π_{RingMul})**.** Π_{RingMul} *(Figure 3.2) satisfies perfect completeness (Definition 3.2.5).*

Proof. Since $As = t$, the CRT pointwise identity $t_C(i,k) = \sum_j A_C(i,j,k)\, s_C(j,k)$ holds for all $(i,k) \in \{0,1\}^{\tau+\nu}$. Both $t_C(I,K)$ and $\sum_{k\in\{0,1\}^\nu} \widetilde{\mathsf{eq}}(k,K) \sum_j A_C(I,j,k)\, s_C(j,k)$ are multilinear in (I,K) and agree on the Boolean hypercube, so they agree everywhere; in particular the honest claim $e_t = t_C(\alpha,\alpha')$ equals the Sumcheck 1 sum $\sum_{(j,k)} A_C(\alpha,j,k)\, s_C(j,k)\, \widetilde{\mathsf{eq}}(k,\alpha')$. By sumcheck completeness, the reduced claim $\sigma' = A_C(\alpha,\beta,\gamma) \cdot s_C(\beta,\gamma) \cdot \widetilde{\mathsf{eq}}(\gamma,\alpha')$ holds.

The honest prover sets $e_A = A_C(\alpha,\beta,\gamma)$ and $e_s = s_C(\beta,\gamma)$, so the multiplication check $\sigma' = e_A \cdot e_s \cdot \widetilde{\mathsf{eq}}(\gamma,\alpha')$ passes. The Sumcheck 2 sum identity then holds by the definitions of A_C, s_C, t_C, and sumcheck completeness yields $\sigma'' = f_{\mathsf{sc2}}(\delta)$ where $f_{\mathsf{sc2}}(\ell) := \tilde{C}(\gamma,\ell) \cdot [\tilde{A}(\alpha,\beta,\ell) + \xi\, \tilde{s}(\beta,\ell)] + \xi^2\, \tilde{C}(\alpha',\ell) \cdot \tilde{t}(\alpha,\ell)$.

Since the oracles are the honest multilinear extensions, the queried values e'_A, e'_s, e'_t equal the true evaluations $\tilde{A}(\alpha,\beta,\delta)$, $\tilde{s}(\beta,\delta)$, $\tilde{t}(\alpha,\delta)$, and the final evaluation check $\sigma'' = \tilde{C}(\gamma,\delta) \cdot (e'_A + \xi\, e'_s) + \xi^2\, \tilde{C}(\alpha',\delta) \cdot e'_t$ holds. The verifier accepts and outputs the three evaluation claims $([[\tilde{A}]], (\alpha,\beta,\delta))$, $([[\tilde{s}]], (\beta,\delta))$, $([[\tilde{t}]], (\alpha,\delta))$ with correct values, which lie in $\mathcal{R}_{\mathsf{eval}}$. $\square$

PIOR composition formulation. We prove RBR-KS compositionally using the claim-flow DAG in Figure 3.3. Each box is a PIOR. For any nonterminal box, its local source relation is the conjunction of the incoming arrow labels, and its local target relation is the conjunction of the outgoing arrow labels, with the relevant oracle handles included in the local instance. Terminal verifier-side checks that are consumed inside a box (rather than forwarded) are written in the box itself and contribute zero additional knowledge error.

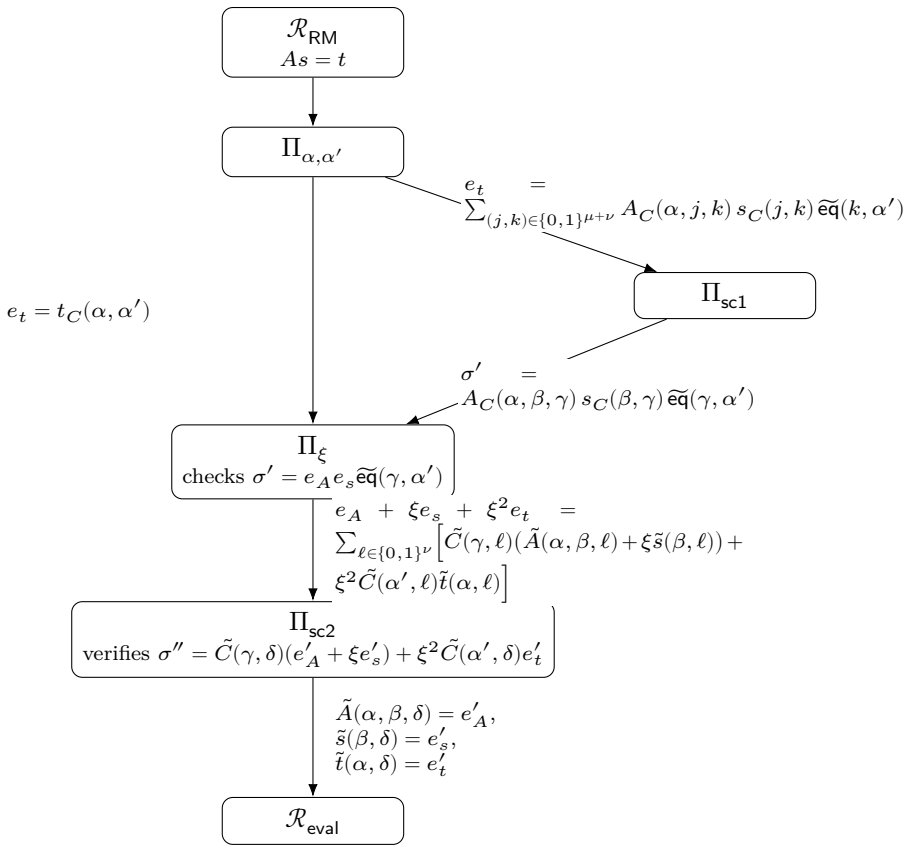


Figure 3.3: Claim flow for the compositional RBR proof.

Round-by-round knowledge soundness.

Theorem 3.4.3 (RBR knowledge soundness of Π_{RingMul})**.** *Viewed as a PIOR* $\Pi_{\mathsf{RingMul}} : \mathcal{R}_{\mathsf{RM}} \to \mathcal{R}_{\mathsf{eval}}$*, the protocol has round-by-round knowledge soundness with error vector*

$$(\kappa_{\alpha,\alpha'},\ \underbrace{2/|\mathbb{K}|,\dots,2/|\mathbb{K}|}_{\mu\ \text{copies}},\ \underbrace{3/|\mathbb{K}|,\dots,3/|\mathbb{K}|}_{\nu\ \text{copies}},\ \kappa_\xi,\ \underbrace{2/|\mathbb{K}|,\dots,2/|\mathbb{K}|}_{\nu\ \text{copies}}),$$

where $\kappa_{\alpha,\alpha'} \le (\tau+\nu)/|\mathbb{K}|$ *and* $\kappa_\xi \le 2/|\mathbb{K}|$.

Proof. We prove the four boxes in Figure 3.3 separately and concatenate the resulting error vectors.

For the two nonsumcheck PIORs, fix one deterministic extractor: query $[[\tilde{A}]], [[\tilde{s}]], [[\tilde{t}]]$ on the Boolean hypercubes and recover the unique triple (A, s, t) encoded by the oracles. For the two sumcheck PIORs, we use the standard RBR-KS bound for sumcheck: fixing a variable of degree d contributes error $d/|\mathbb{K}|$ in that round. Here Π_{sc1} has degree 2 in each J-round and degree 3 in each K-round, while Π_{sc2} has degree 2 in every round.

$\Pi_{\alpha,\alpha'}$ (1 round, error $\kappa_{\alpha,\alpha'}$)**.** The knowledge state moves from 0 to 1 at the verifier challenge (α, α') exactly when

$$t_C(\alpha, \alpha') = \sum_{k \in \{0,1\}^\nu} \widetilde{\mathsf{eq}}(k, \alpha') \sum_{j \in \{0,1\}^\mu} A_C(\alpha, j, k)\, s_C(j, k)$$

even though the multilinear identity

$$t_C(I, K) = \sum_{k \in \{0,1\}^\nu} \widetilde{\mathsf{eq}}(k, K) \sum_{j \in \{0,1\}^\mu} A_C(I, j, k)\, s_C(j, k)$$

fails. Since both sides are multilinear in (I, K) and disagree on some Boolean point whenever $As \neq t$, this happens with probability at most $(\tau + \nu)/|\mathbb{K}|$. When it happens, the prover can choose e_t to satisfy both outgoing claims. Therefore $\kappa_{\alpha,\alpha'} \leq (\tau + \nu)/|\mathbb{K}|$.

Π_ξ (1 round, error κ_ξ)**.** Define

$$Q(Z) := (e_A - A_C(\alpha, \beta, \gamma)) + Z(e_s - s_C(\beta, \gamma)) + Z^2(e_t - t_C(\alpha, \alpha')).$$

A $0 \to 1$ transition at the verifier challenge ξ can occur only if the outgoing batched check holds, that is, only if $Q(\xi) = 0$, while Q is not the zero polynomial.

If $e_t \neq t_C(\alpha, \alpha')$, then the Z^2 coefficient is nonzero, so Q is not the zero polynomial. Otherwise $e_t = t_C(\alpha, \alpha')$, and since by hypothesis the knowledge state is 0, the incoming product claim must fail: $\sigma' \neq A_C(\alpha, \beta, \gamma) s_C(\beta, \gamma) \widetilde{\mathsf{eq}}(\gamma, \alpha')$. But transitioning to 1 requires $\sigma' = e_A e_s \widetilde{\mathsf{eq}}(\gamma, \alpha')$. If $\widetilde{\mathsf{eq}}(\gamma, \alpha') = 0$, these two conditions are incompatible, so any $0 \to 1$ transition must satisfy $\widetilde{\mathsf{eq}}(\gamma, \alpha') \neq 0$. In that case, $e_A e_s \neq A_C(\alpha, \beta, \gamma) s_C(\beta, \gamma)$, so at least one of $e_A \neq A_C(\alpha, \beta, \gamma)$ or $e_s \neq s_C(\beta, \gamma)$ holds. Thus the constant or linear coefficient of Q is nonzero, and again Q is not the zero polynomial.

Therefore every $0 \to 1$ transition forces a nonzero quadratic Q to vanish at $Z = \xi$, which happens with probability at most $2/|\mathbb{K}|$. Hence $\kappa_\xi \leq 2/|\mathbb{K}|$. □

Efficiency.

Theorem 3.4.4 (Efficiency of Π_{RingMul})**.** *Let $N \geq \log n$. The PIOR $\Pi_{\mathsf{RingMul}} : \mathcal{R}_{\mathsf{RM}} \to \mathcal{R}_{\mathsf{eval}}$ (Figure 3.2) with parameters $N = 2^\tau$, $M = 2^\mu$, $n = 2^\nu$ has:*

- *Rounds:* $\mu + 2\nu + 2$.

- *Prover complexity:* $O(NMn)$ *field operations over* $\mathbb{K}$.
- *Verifier complexity:* $O(n + \log NM)$ *field operations over* $\mathbb{K}$*, plus* 3 *oracle queries.*
- *Proof size:* $2\mu + 5\nu + 3$ *elements of* $\mathbb{K}$*, which is* $O(\log Mn)$.

Proof. Rounds. $\Pi_{\alpha,\alpha'}$ (1 round), Π_{sc1} ($\mu + \nu$ rounds), Π_{ξ} (1 round), and Π_{sc2} (ν rounds) total $\mu + 2\nu + 2$.

Proof size. The prover sends e_t (1 element), Sumcheck 1 round polynomials (2 coefficients $\times$ μ **J**-rounds, 3 coefficients $\times$ ν **K**-rounds), e_A and e_s (2 elements), and Sumcheck 2 round polynomials (2 coefficients $\times$ ν rounds): $1 + 2\mu + 3\nu + 2 + 2\nu = 2\mu + 5\nu + 3$.

Prover complexity. Given the evaluation table of a degree-d virtual polynomial over v variables, the sumcheck prover runs in $O(d \cdot 2^v)$ time (Figure 4).

Sumcheck 1 operates on $A_C(\alpha, j, k) \cdot s_C(j, k) \cdot \widetilde{\mathsf{eq}}(k, \alpha')$ over $(j, k) \in \{0,1\}^{\mu+\nu}$, requiring three evaluation tables of size Mn:

- Partial-evaluate $\tilde{A}$ at $\mathbf{I} = \alpha$ (standard halving, $O(NMn)$), yielding $\tilde{A}(\alpha, j, \ell)$ for all $(j, \ell) \in \{0,1\}^{\mu+\nu}$.
- Form $[A_C(\alpha, j, k)]_{k\in\{0,1\}^\nu}$ for each $j \in \{0,1\}^\mu$ via an NTT. At Boolean k, we have $\tilde{C}(k, \ell) = C_{k,\ell} = \omega^{(2k+1)\ell}$, so

$$A_C(\alpha, j, k) = \sum_{\ell\in\{0,1\}^\nu} \omega^{(2k+1)\ell}\, \tilde{A}(\alpha, j, \ell) = \sum_{\ell\in\{0,1\}^\nu} (\omega^2)^{k\ell}\, [\omega^\ell\, \tilde{A}(\alpha, j, \ell)].$$

  Since ω^2 is a primitive n-th root of unity, the map $k \mapsto A_C(\alpha, j, k)$ is the length-n NTT (with respect to ω^2) of the twisted vector $[\omega^\ell\, \tilde{A}(\alpha, j, \ell)]_\ell$. Each NTT costs $O(n \log n)$; over all M slices, the total is $O(Mn \log n)$. The table for $s_C(j, k)$ is formed analogously in $O(Mn \log n)$.

Run Sumcheck 1 in $O(Mn)$ via Figure 4 (degree 3, Mn evaluations).

Between sumchecks, partial-evaluate $\tilde{A}(\alpha, \cdot, \ell)$ and $\tilde{s}(\cdot, \ell)$ at $\mathbf{J} = \beta$ ($O(Mn)$ each), $\tilde{t}$ at $\mathbf{I} = \alpha$ ($O(Nn)$), and evaluate $\tilde{C}(\gamma, \cdot)$ ($O(n)$ by lemma 3.4.5). All five length-n coefficient vectors for Sumcheck 2 are now in hand.

Sumcheck 2 operates on $\tilde{C}(\gamma, \ell) \cdot [\tilde{A}(\alpha, \beta, \ell) + \xi\, \tilde{s}(\beta, \ell)] + \xi^2\, \tilde{C}(\alpha', \ell) \cdot \tilde{t}(\alpha, \ell)$ over $\ell \in \{0,1\}^\nu$ (degree 2, n evaluations): $O(n)$.

The total is $O(NMn + Mn \log n)$. This simplifies to $O(NMn)$ by our assumption $N \geq \log n$.

Verifier complexity. The dominant cost is two evaluations of $\tilde{C}$ (lemma 3.4.5), each $O(n)$.

Verifying $\mu+2\nu$ round polynomials and evaluating $\widetilde{\mathsf{eq}}(\gamma, \alpha')$ costs $O(\mu+\nu)$. Including 3 oracle queries and $O(1)$ field operations for the multiplication and final checks, the total is $O(n + \log(NM))$. $\square$

Lemma 3.4.5. *Let $\omega \in \mathbb{F}_p$ be a primitive $2n$-th root of unity with $n = 2^\nu$, and let $\tilde{C} \in \mathbb{F}_p^{\mathsf{ml}}[\mathbf{K}, \mathbf{L}]$ denote the multilinear extension of $C = \mathsf{CRT}_{n,p}$, given by $C_{k,\ell} = \omega^{(2k+1)\ell}$ for $k, \ell \in \{0,1\}^\nu$. Then for any $\alpha', \gamma \in \mathbb{K}^\nu$, the evaluation $\tilde{C}(\alpha', \gamma)$ can be computed in $O(n)$ field operations.*

Proof. We first derive a factored representation by separating the sum over ℓ. Since $\omega^{(2k+1)\ell} = \prod_{j=0}^{\nu-1} \omega^{(2k+1)\ell_j 2^j}$ and $\widetilde{\mathsf{eq}}(\gamma, \ell) = \prod_j \big((1-\gamma_j)(1-\ell_j) + \gamma_j \ell_j\big)$, the inner sum over ℓ factors across coordinates:

$$\begin{aligned}\sum_{\ell \in \{0,1\}^\nu} \omega^{(2k+1)\ell}\, \widetilde{\mathsf{eq}}(\gamma, \ell) \;&=\; \prod_{j=0}^{\nu-1} \sum_{\ell_j \in \{0,1\}} \omega^{(2k+1)\ell_j 2^j} \;\big((1-\gamma_j)(1-\ell_j) + \gamma_j \ell_j\big) \\ &=\; \prod_{j=0}^{\nu-1} \Big[(1-\gamma_j) + \gamma_j\, \omega^{(2k+1)2^j}\Big]. \qquad (3.2)\end{aligned}$$

Define the univariate polynomial

$$g(X) \;:=\; \prod_{j=0}^{\nu-1} \Big[(1-\gamma_j) + \gamma_j\, X^{2^j}\Big],$$

which has degree $\sum_{j=0}^{\nu-1} 2^j = n-1$. By (3.2), $g(\omega^{2k+1}) = \prod_j [(1-\gamma_j) + \gamma_j\, \omega^{(2k+1)2^j}]$, so

$$\tilde{C}(\alpha', \gamma) \;=\; \sum_{k \in \{0,1\}^\nu} \widetilde{\mathsf{eq}}(\alpha', k) \cdot g(\omega^{2k+1}). \qquad (3.3)$$

To evaluate the right-hand side in $O(n)$ time, we exploit the periodicity of each factor of g. Write $g(\omega^{2k+1}) = \prod_{j=0}^{\nu-1} f_j(k)$ where

$$f_j(k) \;:=\; (1-\gamma_j) + \gamma_j\, \omega^{(2k+1)2^j}.$$

Since ω has order $2n = 2^{\nu+1}$, the element $\omega^{2^{j+1}}$ has order $2^{\nu-j}$, and therefore $\omega^{(2k+1)2^j} = \omega^{2^j} \cdot (\omega^{2^{j+1}})^k$ is periodic in k with period $2^{\nu-j}$. In particular:

- $f_{\nu-1}$ has period 2 in k ($\omega^{2^\nu} = -1$, so $f_{\nu-1}(k)$ depends only on $k \bmod 2$),
- $f_{\nu-2}$ has period 4, ..., f_0 has period n.

We compute the product table $[g(\omega^{2k+1})]_{k=0}^{n-1}$ iteratively. Initialize with the partial product $P_{\nu-1}(k) := f_{\nu-1}(k)$ for $k \in \{0,1\}$, at a cost of $O(1)$. For $s = \nu-2, \nu-3, \dots, 0$: extend P_{s+1} from its period $2^{\nu-s-1}$ to period $2^{\nu-s}$ by replication, and compute $P_s(k) := P_{s+1}(k) \cdot f_s(k)$ for $k = 0, \dots, 2^{\nu-s}-1$. The cost at stage s is $O(2^{\nu-s})$ multiplications, so the total is

$$\sum_{s=0}^{\nu-1} 2^{\nu-s} \;=\; 2+4+\cdots+2^{\nu} \;=\; 2(2^{\nu}-1) \;=\; O(n).$$

After the final stage, $P_0(k) = g(\omega^{2k+1})$ for all $k \in \{0, \dots, n-1\}$. The table $[\widetilde{\mathsf{eq}}(\alpha', k)]_{k\in\{0,1\}^{\nu}}$ is computed in $O(n)$ by the standard recursive formula. The evaluation (3.3) is then an inner product of two length-n vectors, costing $O(n)$. □

3.4.3 A holographic variant

We describe a holographic variant that corresponds to how this protocol would be used as a subroutine in a more general PIOP for NP (e.g., a CCS or R1CS proof system over cyclotomic rings).

In the main protocol (Figure 3.2), Sumcheck 1 includes ν rounds for the $\mathbf{K}$-variables at degree 3, because the product $A_C \cdot s_C$ is degree 2 in $\mathbf{K}$ and the $\widetilde{\mathsf{eq}}(\mathbf{K}, \alpha')$ kernel adds another factor. Moreover, Sumcheck 2 must evaluate $\tilde{C}$ at two CRT points (γ for A_C, s_C and α' for t_C). Both costs stem from the fact that the prover constructs A_C and s_C as virtual polynomials $\sum_{\ell} \tilde{C}(k,\ell)\tilde{A}(\cdot)$ and $\sum_{\ell} \tilde{C}(k,\ell)\tilde{s}(\cdot)$, making k a free variable. In the holographic setting, these extra rounds and the second sumcheck can be eliminated entirely.

We make two assumptions:

1. A is public and the indexer supplies the CRT-domain oracle $[[A_C]] \in \mathbb{K}^{\mathsf{ml}}[\mathbf{I}, \mathbf{J}, \mathbf{K}]$ directly.
2. t_C is a virtual polynomial whose evaluations are determined by the outer protocol: the prover sends e_t claimed to equal $t_C(\alpha, \alpha')$, and the verifier can check this claim without a CRT consistency sumcheck for t.

The second assumption covers the case where t is public (including the common $t = 0$ SIS setting, where $e_t = 0$ is trivially verified), and also the case where t_C arises from the constraint structure of an outer proof system (see remark 3.4.6).

Under these assumptions, the only committed polynomial requiring CRT consistency verification

is $\tilde{s}$. Because A_C is now an oracle rather than a virtual polynomial, the verifier can evaluate it directly at the fixed CRT point α': the quantity $A_C(\alpha, j, \alpha')$ depends on $\mathbf{J}$ alone, and there is no need to sum over $\mathbf{K}$ with an $\widetilde{\mathsf{eq}}$ kernel. This eliminates the degree-3 $\mathbf{K}$-variable rounds from Sumcheck 1. In their place, we inline the CRT expansion $s_C(j, \alpha') = \sum_{\ell \in \{0,1\}^\nu} \tilde{C}(\alpha', \ell) \cdot \tilde{s}(j, \ell)$ directly into the sumcheck, yielding degree-2 rounds for the coefficient variable $\mathbf{L}$. Since A_C is an oracle and t_C is externally verified, Sumcheck 2 is eliminated as well, along with the intermediate evaluation claims e_A, e_s and their multiplication check. The merged claim is

$$e_t \;=\; \sum_{(j,\,\ell) \in \{0,1\}^{\mu+\nu}} A_C(\alpha, j, \alpha') \cdot \tilde{C}(\alpha', \ell) \cdot \tilde{s}(j, \ell). \tag{3.4}$$

The virtual polynomial is a product of three multilinears. In each $\mathbf{J}$-variable round, $A_C(\alpha, \cdot, \alpha')$ and $\tilde{s}$ depend on the current variable while $\tilde{C}(\alpha', \cdot)$ does not; in each $\mathbf{L}$-variable round, $\tilde{C}(\alpha', \cdot)$ and $\tilde{s}$ depend on the current variable while A_C does not. The round polynomial therefore has degree 2 throughout, and the prover sends $2(\mu + \nu)$ field elements. Compared with the main protocol's $2\mu + 5\nu + 3$ elements (which includes the degree-3 rounds in $\mathbf{K}$, the explicit evaluation claims e_A and e_s, and the full Sumcheck 2), this saves $3\nu + 2$ field elements and $\nu + 1$ rounds.

After $\mu + \nu$ rounds with challenges $(\beta, \gamma) \leftarrow\$\ \mathbb{K}^{\mu+\nu}$, the verifier checks the reduced claim

$$\sigma \;=\; A_C(\alpha, \beta, \alpha') \cdot \tilde{C}(\alpha', \gamma) \cdot \tilde{s}(\beta, \gamma)$$

by querying $A_C(\alpha, \beta, \alpha')$ from the indexer oracle, evaluating $\tilde{C}(\alpha', \gamma)$ in $O(n)$ operations (lemma 3.4.5), and querying $e'_s \leftarrow [[\tilde{s}(\beta, \gamma)]]$. The output is a single evaluation claim $\tilde{s}(\beta, \gamma) = e'_s$ on the committed oracle. No $\widetilde{\mathsf{eq}}$ kernel, evaluation claims e_A or e_s, multiplication check, batching challenge, or second sumcheck is needed.

Remark 3.4.6 (Analogy with R1CS second sumcheck)**.** The merged sumcheck (3.4) is the cyclotomic-ring analogue of the "memory-checking" or "oracle-evaluation" sumcheck that appears in holographic proof systems for R1CS (e.g., Spartan [**spartan**]). In a Spartan-style R1CS proof, the witness is z with committed oracle $[[\tilde{z}]]$, and the constraint matrices $M \in \{A, B, C\}$ are public with indexed oracles $[[\tilde{M}]]$. The first sumcheck verifies $0 = \sum_x \widetilde{\mathsf{eq}}(\tau, x) \cdot [z_A(x) \cdot z_B(x) - z_C(x)]$ where $z_M(x) := \sum_y \tilde{M}(x, y) \cdot \tilde{z}(y)$ are virtual polynomials, producing challenges r and claimed evaluations

$e_A = z_A(r)$, $e_B = z_B(r)$, $e_C = z_C(r)$. The second sumcheck then verifies these claims via the batched sum

$$e_A + \rho\, e_B + \rho^2 e_C \;=\; \sum_y [\tilde{A}(r,y) + \rho\, \tilde{B}(r,y) + \rho^2\, \tilde{C}(r,y)] \cdot \tilde{z}(y).$$

The structural correspondence is: $\tilde{s}$ plays the role of $\tilde{z}$ (the committed oracle), $\tilde{C}(\alpha', \cdot)$ plays the role of $\tilde{M}(r, \cdot)$ (a verifier-computable weight function), and the indexed CRT-domain polynomial $A_C(\alpha, \cdot, \alpha')$ plays the role of the constraint-selecting weight in the first sumcheck.

3.4.4 Reducing commitment size and evaluations

After sumchecks, the verifier holds evaluation claims on committed (prover) and indexed (preprocessed) oracles. Each oracle naively requires its own Merkle tree and PCS opening. We describe two standard PIOR optimizations—selector-polynomial merging and multi-point reduction—that consolidate these claims, and illustrate them with a concrete example. We then describe two ways to shorten the proofs of these consolidated claims: the ring-switch packing from [**EC:DiaPos25**], and technique of sumcheck challenge sharing with interleaved Merkle paths from Basefold [**basefold**].

Running example: SpeedySpartan for Ring Arithmetic. Consider a SpeedySpartan-style protocol [**speedyspartan**] proving arithmetic circuit satisfiability over R_p. The indexed (preprocessed) oracles are three wiring matrices $\tilde{W}_a, \tilde{W}_b, \tilde{W}_c \in \mathbb{K}^{\mathsf{ml}}[\mathbf{X}, \mathbf{J}, \mathbf{L}]$ of dimension $m + \mu + \nu$, and a gate-type selector $\tilde{q}_M \in \mathbb{K}^{\mathsf{ml}}[\mathbf{X}]$ of dimension m. The witness oracle $[[\tilde{s}]] \in \mathbb{K}^{\mathsf{ml}}[\mathbf{J}, \mathbf{L}]$ of dimension $\mu + \nu$ is committed by the prover. After an outer (constraint) sumcheck and an inner (memory-checking) sumcheck with challenges $r \in \mathbb{K}^m$ and $(\beta, \gamma) \in \mathbb{K}^{\mu+\nu}$—where the inner sumcheck uses the CRT-inlining of section 3.4.3—the verifier holds five evaluation claims:

$$\begin{aligned} &\tilde{W}_w(r, \beta, \gamma) = e'_w \;\text{ for } w \in \{a, b, c\}, \qquad \tilde{q}_M(r) = e'_q, \\ &\tilde{s}(\beta, \gamma) = e'_s. \end{aligned}$$

The first four oracles are output by the indexer, while $\tilde{s}$ is prover-committed.

SelectorMerge. The indexer merges the four indexed oracles into a single oracle $h \in \mathbb{K}^{\mathsf{ml}}[B_1, B_2, \mathbf{X}, \mathbf{J}, \mathbf{L}]$ of dimension $2 + m + \mu + \nu$ using two selector bits: $h(0,0,\cdot) = \tilde{W}_a$, $h(0,1,\cdot) = \tilde{W}_b$, $h(1,0,\cdot) = \tilde{W}_c$, and $h(1,1,x,j,\ell) = \tilde{q}_{M,\mathrm{pad}}(x)$, padded to be constant in $(\mathbf{J}, \mathbf{L})$. Setting the padded coordinates

of $\tilde{q}_M$'s evaluation point to (β, γ) aligns all four claims to a common data point (r, β, γ), with evaluation points differing only in the two selector bits:

$$h(0,0,r,\beta,\gamma) = e'_a, \quad h(0,1,r,\beta,\gamma) = e'_b,$$
$$h(1,0,r,\beta,\gamma) = e'_c, \quad h(1,1,r,\beta,\gamma) = e'_q.$$

MultiPoint. The multi-point-to-single-point PIOR [**BCHO22**] reduces multiple evaluation claims on one polynomial to a single claim. Given ℓ claims $f(z_1) = v_1, \dots, f(z_\ell) = v_\ell$, the key observation is that whenever the claims cover the full Boolean hypercube $\{0,1\}^c$ of a set of c shared coordinates—with all other coordinates identical—the multilinear extension in those c variables is completely determined by the 2^c known values, and no prover messages are needed.

In our setting, $c = 2$ and the four claims cover all of $\{0,1\}^2$. The verifier computes

$$h^*(b_1, b_2) = (1-b_1)(1-b_2)e'_a + (1-b_1)b_2 e'_b + b_1(1-b_2)e'_c + b_1 b_2 e'_q,$$

samples $(r_1, r_2) \leftarrow\$ \mathbb{K}^2$, and reduces to the single claim $h(r_1, r_2, r, \beta, \gamma) = h^*(r_1, r_2)$. In this special exact hypercube covering case the prover needs send *zero* field elements. Note that because SelectorMerge aligned all data coordinates, every coordinate the claims share is identical and the selector bits span the full hypercube, so this all-at-once reduction applies directly.

At this point, the 5 evaluation claims on 5 oracles have been consolidated into 2 claims: $h(z_h) = v_h$ (indexed, dimension $2 + m + \mu + \nu$) and $\tilde{s}(\beta, \gamma) = e'_s$ (committed, dimension $\mu + \nu$).

Ring-switch packing. Each of the two evaluation claims are fed independently into the ring-switch packing PIOR of [**EC:DiaPos25**], which exploits the tower $\mathbb{F}_p \subset \mathbb{F}_{p^2} \subset \cdots \subset \mathbb{K}$ (with $s = \log_2[\mathbb{K} : \mathbb{F}_p]$) to produce a claim on a packed polynomial of dimension $R - s$, where R is the original dimension. The reduction is an s-round degree-1 sumcheck with round-i challenges in $\mathbb{F}_{p^{2^i}}$; the prover sends s field elements per polynomial. Applied to our two claims, packing produces $\hat{h}$ of dimension $R_h - s$ (where $R_h = 2 + m + \mu + \nu$) and $\hat{s}$ of dimension $R_s - s$ (where $R_s = \mu + \nu$). We use the tower construction for odd primes given in section 3.3.

Interleaved BaseFold opening. Both packed claims are opened via BaseFold [**basefold**], which interleaves an $\widetilde{\mathsf{eq}}$-sumcheck with IOPP folding: at each round, the verifier sends one challenge, the

prover folds the codeword and sends the next sumcheck round polynomial, and commits the folded codeword via a Merkle tree.

Since $\hat{h}$ has $R_h - R_s = 2 + m$ more variables than $\hat{s}$, the prover first folds $\hat{h}$ solo for $2 + m$ rounds, bringing its codeword to the same length as $\hat{s}$'s. For the remaining $R_s - s$ rounds, both polynomials fold with the same challenge: a single uniform $r_i \leftarrow\$ \mathbb{K}$ serves both IOPP instances (soundness holds by a union bound). At each shared round, the prover builds one MMCS-style Merkle tree whose leaf j contains the pair $(\pi_i^{(h)}[j], \pi_i^{(s)}[j])$ of folded codeword values. The prover sends one tree root and two sumcheck round polynomials (one per evaluation target). During the query phase, a single authentication path per shared round opens both values simultaneously, and fold consistency is checked independently for each polynomial. The saving is one full set of Merkle authentication paths over the shared rounds: $(R_s - s) \cdot t \cdot \lambda_{\text{hash}}$ bits, where t is the number of query repetitions.

Full PIOR chain. Starting from 5 evaluation claims on 5 oracles:

$$\text{SelectorMerge} \;\rightarrow\; \text{MultiPoint} \;\rightarrow\; \text{DP24 Packing} \;\rightarrow\; \text{BaseFold (solo + interleaved)}.$$

The prover maintains two original Merkle commitments (T_h from preprocessing, T_s from the prover's first message). After packing, the prover commits $T_{\hat{h}}$ and $T_{\hat{s}}$. The solo-fold phase produces $2 + m$ folding commitments for $\hat{h}$; the interleaved phase produces $R_s - s$ shared commitments. The PIOP-level reductions cost 0 prover field elements for MultiPoint (the four claims cover $\{0,1\}^2$) and $2s$ for packing (s per polynomial).

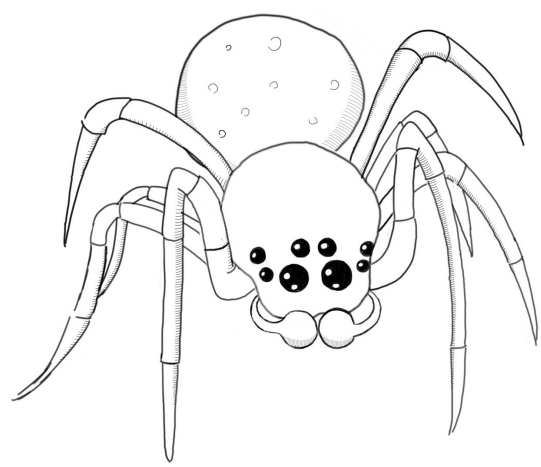
Chapter 4

Orbweaver: Succinct Linear Functional Commitments from Lattices

4.1 Introduction

Over the last decade there has been tremendous progress in the development succinct non-interactive argument of knowledge protocols, called SNARKs [**ITCS:BCCT12**]. For a public arithmetic circuit C and a public input x, these SNARKs enable a prover to convince a verifier that the prover knows a witness w such that $C(\mathsf{x}, \mathsf{w}) = 1$, where:

- the proof is short, its length is at most $\mathrm{poly}(\log|C|,\ |\mathsf{x}|,\ \lambda)$;
- the proof can be verified in time $\mathrm{poly}(\log|C|,\ |\mathsf{x}|,\ \lambda)$; and
- generating the proofs takes quasilinear time in $|C|$.



Lattice Orbweaver (Araneus thaddeus) by Jackie P (CC BY 4.0)

Here $|C|$ is the number of gates in C, and λ is a security parameter. The logarithmic dependence on $|C|$ makes these proofs remarkably short and fast to verify. Since the verifier needs to at least read C, there is also a one-time pre-processing phase applied to the circuit C. Once pre-processed, the prover can provide proofs for many x.

Beyond knowledge of a witness, many of these proof systems can be efficiently extended to also provide zero-knowledge. This progress generated considerable real-world interest, and there are now implemented SNARKs and zkSNARKs capable of handling statements involving many millions of arithmetic gates that are now deployed in real-world applications (e.g., [**SP:BCGGMT14**]).

There are multiple techniques for constructing a pre-processing SNARK. Some require a trusted setup [**AC:Groth10a**; **TCC:Lipmaa12**; **STOC:BCCT13**; **EC:GGPR13**; **SP:PHGR13**; **TCC:BCIOP13**; **EC:Groth16**; **C:GroMal17**; **CCS:MBKM19**; **EPRINT:GabWilCio19**; **EC:CHMMVW20**] where a trusted party must honestly generate public parameters. This setup is called *universal* if it is not specific to the circuit C, i.e., it can be done once to generate parameters that can be used to preprocess any circuit C in a publicly verifiable way. Other systems, called transparent SNARKs, require no trusted setup [**C:BBHR19**; **EPRINT:BunFisSze19**; **EPRINT:ChiOjhSpo19**; **EPRINT:BowGriHop19**].

Many of the existing SNARKs make use of a commitment scheme called a *polynomial commitment* [**AC:KatZavGol10**], or PC. A polynomial commitment enables a prover to commit to a polynomial $f \in \mathbb{F}[x]$ of degree w using a short commitment. Later, given two public values $\alpha, \beta \in \mathbb{F}$, the prover can convince a verifier that the committed polynomial f satisfies $\beta = f(\alpha)$ and that f has degree at most w. Ideally, the prover can verifiably open the polynomial at any point $\alpha \in \mathbb{F}$ using a short proof that can be efficiently checked. In fact, it has been shown that given polynomial commitment scheme with proof size $S(w)$ and verification time $T(w)$ it is possible to construct SNARKs with the same complexity characteristics in w (the length of the witness), and additional complexity dependent only on a security parameter λ [**EC:BunFisSze20**; **EC:CHMMVW20**]. This compilation is in the random oracle model and relies on the Fiat-Shamir transform.

Polynomial commitment schemes are a special case of *linear functional vector commitments*, where the prover has a commitment $C = \mathsf{com}(\mathbf{x})$ to a vector $\mathbf{x} \in \mathbb{Z}_p^w$ and is able to open any linear form $f(\mathbf{x}) = \sum_{i=1}^{w} \mathbf{x}_i f_i \bmod p$. Polynomial commitments, and the more general linear vector

[0]Named for the lattice orbweaver spider (araneus thaddeus).

commitments, have been built from bilinear pairings [**AC:KatZavGol10**], groups of unknown order [**EPRINT:BunFisSze19**], proofs of proximity for Reed-Solomon codes [**ICALP:BBHR18**], and to some degree from lattice-based assumptions [**C:ACLMT22**]. However, thus far all lattice-based constructions have had significant drawbacks, and none have achieved asymptotically (in the polynomial's degree) a quasilinear prover time with logarithmic proof size and logarithmic verification time. There are lattice-based generalizations of Bulletproofs [**EC:BCCGP16**; **SP:BBBPWM18**] based on the Ring-SIS assumption, which achieve quasilinear prover time and a polylogarithmic proof size, but have linear verification time [**C:BLNS20**; **C:AttCraKoh21**; **C:AlbLai21**]. Recently, [**C:ACLMT22**] constructed general degree-d polynomial map vector commitments (which include linear functions $d = 1$ as a special case), which achieve logarithmic proof size and verification time, but have a CRS size and prover runtime that is at least $\Omega(w^{2d})$, and thus quadratic in the vector length for linear functions. This construction requires a trusted setup using lattice trapdoor sampling [**EC:MicPei12**], and is based on k-R-ISIS, a new family of lattice-based knowledge assumptions related to Ring-SIS. Another recent work, LaBRADOR [**BS22**], uses recursion to achieve very compact proof sizes, but has verification time linear in w.

Polynomial commitment schemes based on codes and lattice assumptions are of particular importance due to their plausible post-quantum security. Constructions based on Reed-Solomon codes have quasilinear prover time and both polylogarithmic proof size and verifier time. So far they outperform any lattice-based construction not only asympotically, but also concretely by orders of magnitude in overall size and verification time, even when compared with recent lattice-based constructions that sacrifice prover performance for shorter proofs. Moreover, in the random oracle model, code-based constructions use weaker assumptions than lattice-based constructions. Nonetheless, we are optimistic that the additional structure lattices provide vs. generic (i.e., hash and code-based) constructions can be exploited such that lattice SNARKs eventually surpass these results. As a point of reference, hash-based signatures were originally more efficient, but after over a decade of development lattice-based signatures are an order of magnitude smaller and two faster.

A primary motivation of recent lattice-based constructions [**C:ACLMT22**; **WW22**] of vector commitments supporting higher degree polynomial map openings is that, unlike linear function commitments, they can be used to build SNARKs more directly (e.g., by opening an R1CS form) without invoking the Fiat-Shamir transform. Unfortunately, current approaches to supporting

higher degree polynomial maps seem to fundamentally require a quadratic prover time. Given the additional structure of lattices compared with code-based proof systems that rely only on hash functions, one might expect that it would be possible to obtain smaller proof sizes and faster verification times. The recent work LaBRADOR [**BS22**] was the first lattice-based system to achieve proof sizes smaller (both concretely and asymptotically) than code-based systems, but this has not yet been done for combined proof size and verification time. The results of this paper make progress in this direction.

4.1.1 Our results

Building off of the techniques in [**C:ACLMT22**], we present the first functional commitment for linear relations from lattices that asymptotically has quasilinear prover time, logarithmic proof size, and polylogarithmic verifier time. The scheme supports commitments to vectors $\mathbf{x} \in \mathcal{R}^w$ over a cyclotomic ring $\mathcal{R}$ of degree n and openings of the form $\sum_{i=1}^{w} x_i \cdot f_i$ for any $\mathbf{f} \in \mathcal{R}^w \bmod q$ for a prime $q \gg \alpha$, where both $\mathbf{x}$ and $\mathbf{f}$ have norm bounded by α. In particular, proofs are $O(\log w\alpha)$ and verification time is $O(\log w\alpha \cdot \log\log w\alpha \cdot \log\log\log w\alpha)$.

We present several extensions to Orbweaver and show how it can be used to build other primitives including:

- A public proof aggregation protocol achieving $O(\log t)$ size aggregate proofs for t linear maps evaluations where each linear map being evaluated and committed input may be distinct.
- An inner product argument.
- A linear functional commitment for the non-cyclotomic ring $\mathcal{R} = \mathbb{Z}$ that combines ternary decomposition and proof aggregation to achieve sublogarithmic proofs in the input norm of size $O(\log w + \log\log \alpha)$.
- Univariate and multilinear polynomial commitments. For degree d univariate polynomials we achieve $O(\log d)$ openings and polylogarithmic verifier time.

We also provide concrete proof size comparisons to previous lattice-based proof systems. While succinct hash-based arguments provide better proof sizes for smaller statements, the most efficient of these systems still have $O(\log^2 n)$ proofs. Especially in settings where the input norm is small, Orbweaver provides better proof sizes for larger-but-practical circuits than state-of-the-art hash-based

arguments.

4.1.2 Related work

Efficient lattice-based proof systems have been constructed in the designated verifier setting [**GMNO18**; **ISW21**]. In the publicly verifiable setting, there are many results that achieve succinct proof sizes but require linear verifier time [**BBCPGL18**; **C:BLNS20**; **C:AttCraKoh21**; **BCS21**; **BS22**]. There are also works focusing on practical lattice-based ZK proofs, which have proof size linear in the witness size, but are concretely efficient for small statements [**ESLL19**; **ALS20**; **ENS20**; **ESLR22**; **LNP22**].

Recently, Albrecht et al. [**C:ACLMT22**] constructed extractable lattice-based vector commitments supporting arbitrary degree polynomial maps. For degree d maps their prover time and CRS length are $\tilde{O}(w^{2d})$, while proof size and verifier time are $O(d\log(w))$. Wee and Wu [**WW22**] construct non-extractable lattice-based vector commitments for linear functions including polynomial commitments with $\mathrm{polylog}(w)$ openings and $\tilde{O}(w^2)$ CRS size and prover time. Castro and Peikert [**CP22**] construct non-extractable vector commitments for functions of bounded complexity based on the standard SIS assumption. Their polynomial commitments achieve proof size $O(\log^4 w)$. Balbás, Catalano, Fiore, and Lai construct extractable lattice-based vector commitments supporting arithmetic circuits of width w and depth d that achieve proof size $O(d\log^2 w)$, but have a CRS of size $O(w^5)$ [**EPRINT:BCFL22**]. LaBRADOR [**BS22**] designs a SNARK for quadratic relations with quasilinear prover and verifier time, but via a complex recursive argument achieve $O(\log^2 w)$ proofs that concretely surpass the best results from hash-based proof systems.

There is a much longer history of succinct arguments constructed from Merkle hashes that begins with [**Kilian92**] and more recently includes FRI [**C:BBHR19**], Ligero [**Ligero**], Aurora [**Aurora**], Brakedown [**Brakedown**], and Orion [**XZS22**] among others.

4.1.3 Technical overview

At a high level, our construction uses an assumption from the k-R-ISIS assumption family introduced in [**C:ACLMT22**] to translate the knowledge-of-exponent (KEA) based linear vector commitment scheme sketched in [**ac20**] (which is in turn based on [**AC:Groth10a**], and also independently similar to ideas in [**ICALP:LibRamYun16**]) to the lattice setting. The construction should enable a prover

to commit to a vector $\mathbf{x}$ and later open any linear function $f(\mathbf{x}) = \langle [f_i]_{i=1}^{w},\ [x_i]_{i=1}^{w} \rangle = \sum_{i=1}^{w} f_i x_i$.

As observed in [**ac20**], given a function $x(v) = \sum_{i=1}^{w} x_i v^i$, and a function $f(v) = \sum_{i=0}^{w-1} f_i v^{-i}$, then $h(v) = x(v)f(v) = \sum_{i=-w+1}^{w-1} a_i v^i$, where $a_0 = \sum_{i=1}^{w} f_i x_i$. The idea is that the prover sends $c = x(v)$, $y = f(\mathbf{x})$, and $\pi = \sum_{i=-w+1, i\neq 0}^{w-1} a_i v^i$, and then the verifier computes $f(v) = \mathsf{ck}_f$ and checks $c \cdot \mathsf{ck}_f - y = \pi$.

In order to achieve both binding and succinctness, we situate this abstract protocol in lattice setting using the techniques of [**C:ACLMT22**]. A trusted setup generates a universal (i.e., the same setup works for any linear function up to a given size) structured reference string (SRS) $(\mathbf{a}, v, \mathbf{u}_{-w+1}, \dots, \mathbf{u}_{-1}, \mathbf{u}_1, \dots, \mathbf{u}_{w-1}) \in \mathcal{R}_q \times \mathcal{R}_q^{\times} \times (\mathcal{R}_q)^{2w-2}$, where for each $\mathbf{u}_i$ it holds that $\langle \mathbf{a},\ \mathbf{u}_i \rangle \equiv v^i \bmod q$ and $\mathbf{u}_i$ is "short" relative to q. Note that no preimage $\mathbf{u}_0$ is given out such that $\langle \mathbf{a},\ \mathbf{u}_0 \rangle \equiv 1 \bmod q$. Our k-R-ISIS assumption states that it should be hard to find short $(s, \mathbf{u}) \neq (0, \mathbf{0}) \in \mathcal{R} \times \mathcal{R}$ such that $\langle \mathbf{a},\ \mathbf{u} \rangle \equiv s \bmod q$ even given these preimages for other powers of v. In general, it should be hard to find the preimage of any target not in the linear span of $[v_i]_{i=-w+1, i\neq 0}^{w-1}$.

Let α be a bound on the ℓ-infinity norm of both the witness and function. The computation of commitment c, commitment key ck_f, and y are exactly as in the abstract protocol, but taken mod q for prime $q \gg \alpha$. The prover now computes

$$\pi_0 := \sum_{i=-w+1, i\neq 0}^{w-1} a_i \mathbf{u}_i \bmod q \ ,$$

and the verifier checks

$$\langle \mathbf{a},\ \pi_0 \rangle \equiv c \cdot \mathsf{ck}_f - y \bmod q \ , \tag{4.1}$$

and that π_0 and y have small norm. When correctly executing the protocol it follows from the fact that $\alpha \ll q$ and $\beta \ll q$, where β is a norm bound on the $\mathbf{u}_i$, that the proof and output will have small norm. Fixing $\mathcal{R}_q$ with respect to the security parameter, one can see that the size of π_0 is of size $O(\log w + \log \alpha)$. We also achieve (almost) the same verifier time [1] in the preprocessing setting: while computing the commitment key $\mathsf{ck}_f = \sum_{i=1}^{w} f_i v^{-i}$ takes linear time, it may be computed once in advance then subsequently used to verify openings with respect to any c and y.

Our scheme thus far achieves evaluation binding: an adversary who can open a commitment c to

[1] Our verifier time is $O(\log(w \cdot \alpha) \log\log(w \cdot \alpha))$ in the preprocessing setting.

two different outputs for the same function can be used to break k-R-ISIS. To see this, we subtract one verifying eq. (4.1) from the other to obtain

$$\langle \mathbf{a},\ \pi_0 - \pi_0' \rangle \equiv y' - y \bmod q$$

Since both proofs and outputs are small, this gives k-R-ISIS solution $s = y' - y, \mathbf{u} = \pi_0 - \pi_0'$.

Achieving extractability. While this construction is evaluation binding, this is insufficient to guarantee that the prover knows some witness consistent with the commitment. For example, the prover can randomly sample a short proof π_0, and compute the commitment $c = \langle \mathbf{a},\ \pi_0 \rangle / (\sum_{i=1}^{w} f_i v^{-i})$. In this case, the prover can pass the verifier checks without knowledge of the input.

To achieve extractability we require an additional knowledge assumption from [**C:ACLMT22**], one from the knowledge k-R-ISIS assumption family. Our knowledge k-R-ISIS assumption states that there is an extractor that extracts short $\mathbf{x}^*$ s.t. $c = \sum_{i=1}^{w} x_i^* v^i$ from any prover who, given $[\mathbf{u}_i']_{i \in [w]}$ s.t. $\langle \mathbf{a},\ \mathbf{u}_i' \rangle \equiv v^i \cdot t \bmod q$ (for some t drawn from a special subset of $\mathcal{R}$), outputs commitment c and a short knowledge proof π_1 s.t. $\langle \mathbf{a},\ \pi_1 \rangle \equiv c \cdot t \bmod q$.

Using this assumption, it suffices to include as part of the commitment an additional value $\pi_1 = \sum_{i=1}^{w} x_i \mathbf{u}_i$ in order to achieve extractability. Observe that if for a verifying (c, y, π_0, π_1) it held that $f(\mathbf{x}^*) \neq y$, then $\mathbf{x}^*$ could be used to create verifying $(c, y' = f(\mathbf{x}^*), \pi_0', \pi_1')$, which violates binding and could be used to break plain k-R-ISIS as explained above. Therefore, the extracted witness must be consistent with the evaluation.

Lastly, we note our extracted witness has "stretch", i.e., the best norm bound we can hope to enforce on $\mathbf{x}^*$ is the norm bound on π_1, which in order to achieve correctness is necessarily bigger than the norm bound we impose on $\mathbf{x}$.

Public proof aggregation. An untrusted prover, input t verifying tuples $(c_i, \pi_{0,i}, \pi_{1,i}, y_i)$ tuples, can combine the evaluation proofs into a single proof $\pi_0 := \sum_{i=1}^{t} h_i \pi 0_i$ using some short challenges h_i. The verifier then checks

$$\langle \mathbf{a},\ \pi_0 \rangle \equiv \mathsf{ck} \cdot \sum_{i=1}^{t} (h_i c_i) - \sum_{i=1}^{t} (h_i y_i) \bmod q\ .$$

Since the h_i and $\pi_{0,i}$ are short, π_0 will be as well. If we consider the norm bound on the h_i to be

dependent on the security parameter, then we can see the bit length $|\pi_0|$ is $\log t$ times the bit length of the original proofs $|\pi_{0,i}|$.

Without modifying our knowledge assumption, we cannot take linear combinations of the knowledge proofs. So we only get partial aggregation.

Polynomial commitments for integers. Furthermore, for the special case of polynomial commitments, where $f_i = z^{i-1} \bmod p$ for $i \in [1, w]$ and w is a power of 2 and $q > p^{\log w}$, $\mathsf{ck}_f = \prod_{i=0}^{\log w-1}(z_i + v^{-2^i})v^{-(w+1)} - z_{\log w} \bmod q$ where $z_i = z^{2^i} \bmod p$, which can be computed in time $O(\log w)$ operation over $\mathcal{R}_q$. Given this value the verifier can compute $c \cdot \mathsf{ck}_f = (\sum_{i=1}^{w} x_i v^i)(\sum_{i=1}^{w} f_i v^{-i}) = \sum_{i=1}^{w} f_i x_i + \sum_{i=-w+1}^{w-1} a_i v^i = f(\mathbf{x}) + \sum_{i=-w+1}^{w-1} a_i v^i$, for some ring elements a_i. The opening proof will include information that allows the verifier to cancel out the term dependent on the a_i values. To prevent the prover from cheating, we invoke the k-R-ISIS assumption by involving $\mathbf{a}$ in the opening. More specifically, the opening is computed as $\pi_0 = \sum_{i \in \mathbb{Z}(w)\{0\}} a_i \mathbf{u}_i$, where a_i's can be computed by the prover. The verifier of the opening checks $\langle \mathbf{a},\ \pi_0 \rangle = c \cdot \mathsf{ck}_f - y$, and that f, y, π_0 are short (i.e., their norm is bounded). The k-R-ISIS hardness assumption guarantees that the commitment is evaluation binding when f and x are both short.

4.2 Preliminaries

Let $\mathbb{Z}(b) := (-b, b) \cap \mathbb{Z}$ and $[a] := [1, a] \cap \mathbb{Z}$. For a ring $\mathcal{R}$ of degree n, let $\mathsf{vec}(r) \in \mathbb{Z}^n$ denote the coefficient vector of $r \in \mathcal{R}$ in the integral basis.

For $m \in \mathbb{N}$, let $\zeta_m \in \mathbb{C}$ be any fixed primitive m-th root of unity. Let $\mathcal{R} = \mathbb{Z}[\zeta_m]$ denote its ring of integers, called a cyclotomic ring. We have $\mathcal{R} \cong \mathbb{Z}[x]/\langle \Phi_m(x) \rangle$, where $\Phi_m(x)$ is the m-th cyclotomic polynomial. If m is a power of 2, we call $\mathcal{R}$ a power-of-2 cyclotomic ring. In this paper, we exclusively use power-of-2 cylotomic rings. Let $q \in \mathbb{N}$ be a prime number, we let $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$ and let $\mathcal{R}_q^\times$ denote all invertible elements in $\mathcal{R}_q$. For any $f \in \mathcal{R}$, let $\mathsf{ct}(f)$ denote the constant term of f (i.e., $\mathsf{ct}(f) = \mathsf{vec}(f)_0$).

For $x \in \mathcal{R}$, let $\|x\|$ denote the ℓ-infinity norm of its coefficient vector, i.e., $\|x\| = \max_{i \in [n]} \mathsf{vec}(x)$. We use $\|\cdot\|_p$ for the ℓ_p-norm (e.g., $\|\cdot\|_2$ for the ℓ_2 norm).

Definition 4.2.1 (Ring expansion factor)**.** Let $\mathcal{R}$ be a ring. The *expansion factor* of $\mathcal{R}$ is defined as $\gamma_{\mathcal{R}} := \max_{a,b \in \mathcal{R}} \frac{\|a \cdot b\|}{\|a\| \cdot \|b\|}$.

Theorem 4.2.2 ([**C:AlbLai21**]). *If $\mathcal{R} = \mathbb{Z}[\zeta_m]$ is a power-of-2 cyclotomic ring, then $\gamma_{\mathcal{R}} \leq n$.*

Theorem 4.2.3 ([**C:ACLMT22**]). *Let $q = \omega((w \cdot f)^{f/\phi(m)})$ be a rational prime such that $\mathcal{R}_q$ splits into f fields each of size $q^{\phi(m)/f}$. For $v \leftarrow\$ \mathcal{R}_q^w$, we have $v \in (\mathcal{R}_q^\times)^w$ with probability $(1-1/q^{\phi(m)/f})^{w \cdot f}$.*

Subsequently in this work, we set q large enough so that uniformly random $v \leftarrow\$ \mathcal{R}_q$ satisfies $v \in \mathcal{R}_q^\times$ with non-negligible probability.

The conjugation automorphism. The cyclotomic ring $\mathcal{R}$ has a group of automorphisms $\sigma_{-1}(\mathcal{R})$ that is isomorphic to $\mathbb{Z}_{2n}^\times$,

$$i \to \sigma_i : \mathbb{Z}_{2n}^\times \to \sigma_{-1}(\mathcal{R})$$

We make use of the following property: given two vectors $\mathsf{vec}(a) = (a_0, \dots, a_{kn-1}) \in \mathbb{Z}^{kn}$, $\mathsf{vec}(b) = (b_0, \dots, b_{kn-1}) \in \mathbb{Z}^{kn}$ where $k \geq 1$, let $c \leftarrow \sum_{i=0}^{k-1} \sigma_{-1}(\sum_{j=0}^{n-1} a_{i \cdot n+j} X^j) \cdot (\sum_{j=0}^{n-1} b_{i \cdot n+j} X^j) \in \mathcal{R}$; then the constant coefficient denoted as $\mathsf{ct}(c)$ has the property that $\mathsf{ct}(c) = \langle \mathsf{vec}(a),\ \mathsf{vec}(b) \rangle$ (adapted from [**LNP22**], see also [**ENS20**]).

4.2.1 Exceptional challenge sets

For a cyclotomic ring $\mathcal{R}$ of degree n, let $\mathcal{H} \subset \mathcal{R}$ be the set of ring elements with c plus or minus one coefficients and $n - c$ zero coefficients. Then $|\mathcal{H}| = 2^c \binom{n}{c}$, and if $\mathcal{R}_q$ is a power-of-two cyclotomic, the operator norm of $\mathcal{H}$ with respect to the ℓ-infinity norm is c, i.e., $\mathsf{op}(\mathcal{H}) = c$ (lemma 4.2.5). By [**ls20**] for n a power of two and $q \equiv 2n + 1 \bmod 4n$ prime the $2n$-th cyclotomic polynomial $X^n + 1$ splits completely into linear factors modulo q.[2] Further, infinitely many such primes q exist and for any $h \in \mathcal{R}$ that satisfies $0 < \|h\|_2 < q^{1/n}$ has an inverse in $\mathcal{R}_q$. We wish to ensure that the difference between any two distinct elements in the challenge set is not a zero divisor, i.e., $\mathcal{H}$ is *exceptional*, in order to invoke the following theorem.

Theorem 4.2.4 (Generalized Alon–Füredi Theorem [**BCPS18**]). *Let $\mathcal{R}$ be a ring and let $\mathcal{H}$ be an exceptional, non-empty, finite subset of $\mathcal{R}$. Then for any non-zero polynomial $f \in \mathcal{R}[Y_1, \dots, Y_t]$ such that* deg $(f) < |H|$ *it holds that*

$$\Pr[f(h_1, \dots, h_t) = 0 \mid h_1, \dots, h_t \leftarrow\$ \mathcal{H}] \leq \frac{\deg\ (f)}{|\mathcal{H}|} \ .$$

[2] A close reading of the proof reveals for the fully splitting case that the simpler condition $q \equiv 1 \bmod n$ suffices.

Concretely, for deg $(f) = 1$ and $c = 19$ the right-hand of side of the inequality above is $< 2^{-132}$. We know that differences $c_i - c_j$ for $c_i \neq c_j \in \mathcal{H}$ with 19 ± 2 coefficients can be reduced to the invertibility of 2 and polynomials with 18 ± 1 coefficients. Differences with 18 ± 2 coefficients and 2 ± 1 coefficients have ℓ_2 norm $\sqrt{74}$. We must then have $q \geq 2^{1590} > \sqrt{74}^{512}$. It turns out even larger q are optimal for proof size for concrete instantiations of our scheme capable of handling large circuits.

Lemma 4.2.5 (Operator norm of sparse ternary sets)**.** *For power-of-two n, let $\mathcal{R} = \mathbb{Z}[X]/(X^n + 1)$ and $1 \leq c \leq n$. The set*

$$\mathcal{H} = \left\{ h \in \mathcal{R}_q \mid \|h\|_\infty = 1 \wedge \|h\|_1 = c \right\}.$$

has operator norm

$$\mathsf{op}_\infty(\mathcal{H}) = \max_{h \in \mathcal{H}, r \in \mathcal{R}} \frac{\|hr\|_\infty}{\|h\|_\infty \|r\|_\infty} = c$$

Proof. Let $\mathbf{r}, \mathbf{c}$ be the coefficients of r and c. Let $g = rc$ with coefficient vector $\mathbf{g}$. Then

$$g_k = \sum_{i+j=k} r_i c_j - \sum_{i+j=n+k} r_i c_j$$

Fixing any $j, k \in \{0, \dots, n-1\}$, there is exactly one $i \in \{0, \dots, n-1\}$ such that either $i + j = n$ or $i + j = n + k$. Therefore, each coefficient g_k can be bounded by $c \|r\|_\infty$. □

4.2.2 Functional commitments

Definition 4.2.6 (Functional commitment)**.** A (pre-processing non-interactive) functional commitment (FC) scheme is parameterized by a function family

$$\mathcal{F} \quad := \{\mathcal{F}_w \subseteq \{f : \mathcal{X}^w \to \mathcal{Y}\}\}_{w \in \mathbb{N}}$$

over a ring $\mathcal{R}$ for input alphabet $\mathcal{X} \subseteq \mathcal{R}$ and image space $\mathcal{Y} \subseteq \mathcal{R}$, where w is the dimension of the secret (committed) input. The FC scheme is defined by a 5-tuple of PPT algorithms ($\mathsf{Setup}, \mathsf{Com}, \mathsf{Open}, \mathsf{PreVerify}, \mathsf{Verify}$), working as follows:

- $\mathsf{Setup}(1^\lambda, 1^w) \to (\mathsf{ck}, \mathsf{vk})$: Input (in unary) security parameter λ and secret input dimension w, samples commitment key ck and verification key vk.

- $\mathsf{Com}(\mathsf{ck}, \mathbf{x}) \to (c, \pi_1)$: Input commitment key ck and secret input $\mathbf{x} \in \mathcal{X}^w$, computes commitment c and a proof of knowledge π_1 of vector $\mathbf{x}$ such that $c = \mathsf{Com}(\mathsf{ck}, \mathbf{x})$.
- $\mathsf{Open}(\mathsf{ck}, f, \mathbf{x}) \to \pi_0$: Input commitment key ck, function $f \in \mathcal{F}_w$, and secret input $\mathbf{x} \in \mathcal{R}^w$, computes opening proof π_0 for the evaluation $f(\mathbf{x})$.
- $\mathsf{PreVerify}(\mathsf{ck}, f) \to \mathsf{vk}_f$: Input verification key vk and function $f \in \mathcal{F}_w$, computes preprocessed commitment key vk_f. Preprocessing only needs to be performed once per function and allows Verify to run in time sublinear in f.
- $\mathsf{Verify}(\mathsf{vk}_f, c, y, \pi_1, \pi_0) \to \{0, 1\}$: Input preprocessed verification key vk_f, commitment c, output $y \in \mathcal{Y}$, and proofs π_1 and π_0, the verifier returns 1 if the proofs convince them the verifier knows some $\mathbf{x} \in \mathcal{X}^w$ such that $\mathsf{Com}(\mathsf{ck}, \mathbf{x}) = c$ and $f(\mathbf{x}) = y$ (else 0).

We require that functional commitments satisfy correctness, extractability, and succinctness as defined below.

Definition 4.2.7 (Correctness)**.** An FC scheme for $(\mathcal{F}, \mathcal{X}, \mathcal{Y})$ is *correct* if for any $\lambda, w \in \mathbb{N}$, any $\mathsf{ck} \leftarrow \mathsf{Setup}(1^\lambda, 1^w)$, and for any $(f, \mathbf{x}, y) \in \mathcal{F} \times \mathcal{X}^w \times \mathcal{Y}$ satisfying $f(\mathbf{x}) = y$, any $(c, \pi_1) \leftarrow \mathsf{Com}(\mathsf{ck}, \mathbf{x})$, any $\pi_0 \leftarrow \mathsf{Open}(\mathsf{ck}, f, \mathbf{x})$, and any $\mathsf{vk}_f \leftarrow \mathsf{PreVerify}(\mathsf{ck}, f)$, it holds that $\Pr[\mathsf{Verify}(\mathsf{vk}_f, c, y, \pi_1, \pi_0) = 1] = 1$.

At a high level, extractability of an FC scheme requires that if an adversary can produce a commitment c and a valid opening (π_1, π_0) for some function f and some evaluation y, it must know $\mathbf{x} \in \mathcal{X}^w$ satisfying $\mathsf{Com}(\mathsf{ck}, \mathbf{x}) = c$ and $f(\mathbf{x}) = y$. Note that extractability implies *weak binding* (meaning that it is not possible to open a commitment to two different evaluations for the same function).

Definition 4.2.8 (Extractability)**.** A FC scheme for $(\mathcal{F}, \mathcal{X}, \mathcal{Y})$ is $(\kappa, \mathcal{X}^*)$-*extractable* if for any PPT adversary $\mathcal{A}$, there exists a PPT extractor $\mathcal{E}_{\mathcal{A}}$ that, input $(\mathsf{ck}, \mathsf{vk})$ and given black-box access to $\mathcal{A}$ and any randomness it uses, returns $\mathbf{x}^* \in (\mathcal{X}^*)^w$ such that

$$\Pr\left[\begin{array}{rr} & \mathsf{Verify}(\mathsf{vk}_f, c, y, \pi_1, \pi_0) = 1 \\ \wedge & \left(\begin{array}{rr} & \mathbf{x} \notin (\mathcal{X}^*)^w \\ \vee & c \neq \mathsf{Com}(\mathsf{ck}, \mathbf{x},) \\ \vee & f(\mathbf{x}) \neq y \end{array}\right) \end{array} \middle| \begin{array}{r} (\mathsf{ck}, \mathsf{vk}) \leftarrow \mathsf{Setup}(1^\lambda, 1^w) \\ (f, c, y, \pi_1, \pi_0) \leftarrow \mathcal{A}(\mathsf{ck}, \mathsf{vk}) \\ \mathbf{x} \leftarrow \mathcal{E}_{\mathcal{A}}(\mathsf{ck}, \mathsf{vk}) \\ \mathsf{vk}_f \leftarrow \mathsf{PreVerify}(\mathsf{vk}, f) \end{array}\right] \leq \kappa(\lambda, w) \ .$$

We say the scheme is $\mathcal{X}^*$-extractable if its *knowledge error* $\kappa(\lambda, w)$ is negligible in λ for $w = \mathrm{poly}(\lambda)$.

Definition 4.2.9 (Succinctness)**.** Let Π be a VC scheme for the alphabet $\mathcal{X} = \{r \in \mathcal{R} \mid \|r\| \leq \alpha\}$. We say Π is *succinct* if it satisfies the following properties:

- *Proof succinctness*: $|c + \pi_1 + \pi_0| = \text{poly}(\log w + \log \alpha)$.
- *Verifier succinctness*: Verify runs in time $\text{poly}(\log w + \log \alpha)$.

4.2.3 Sampling Algorithm

The following relies on the Leftover Hash Lemma over rings to generate some vector $\mathbf{a}$ that is indistinguishable from a uniformly randomly sample vector, and some trapdoor that makes it possible to easily generate vectors in the kernel of $\mathbf{a}$. We let $\mathsf{lhl}(\mathcal{R}_q, \beta)$ denote an algorithm that outputs the minimum $\ell \in \mathbb{N}$, which ensures that the resulting distribution of the vector $\mathbf{a}$ is indistinguishable from the uniform distribution. It is formally defined as follows, adapted from [**GPV08**; **EC:MicPei12**; **GM18**; **C:ACLMT22**]:

A sampling algorithm has the following three PPT algorithms (taking 1^λ as input implicitly):

- $(\mathbf{a}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(1, q, \mathcal{R}, \beta)$: takes dimension $\ell \in \mathbb{N}$, a modulus $q \in \mathbb{N}$, a ring $\mathcal{R}$, and a norm bound $\beta \in \mathbb{R}$, and outputs a vector $\mathbf{a} \in \mathcal{R}_q^\ell$ and a trapdoor td. For any $n \in \text{poly}(\lambda), \ell \geq \mathsf{lhl}(\mathcal{R}_q, b)$ where $b = O(\beta)$, the distribution of $\mathbf{a}$ is within $\text{negl}(\lambda)$ statistical distance to $U(\mathcal{R}_q^\ell)$.
- $\mathbf{u} \leftarrow \mathsf{SampD}(1^\ell, \mathcal{R}_q, \beta)$: for $\ell \geq \mathsf{lhl}(\mathcal{R}_q, \beta)$, outputs $\mathbf{u}$ such that $\|\mathbf{u}\| \leq \beta$ and the distribution of $\langle \mathbf{a}, \ \mathbf{u} \rangle \bmod q$ is withing $\text{negl}(\lambda)$ statistical distance to $U(\mathcal{R}_q)$.
- $\mathbf{u} \leftarrow \mathsf{SampPre}(\mathsf{td}, v, \beta)$: for $\ell \geq \mathsf{lhl}(\mathcal{R}_q, \beta)$ and $v \in \mathcal{R}_q$, outputs $\mathbf{u} \in \mathcal{R}_q^\ell$ satisfying $\|\mathbf{u}\| \leq \beta$, $\langle \mathbf{a}, \ \mathbf{u} \rangle \equiv v \bmod q$, and that distribution of $\mathbf{u}$ is within $\text{negl}(\lambda)$ statistical distance the distribution of $v' \leftarrow \mathsf{SampD}(1^\lambda, 1^\ell, \mathcal{R}_q, \beta)$ conditioned on $\langle \mathbf{a}, \ \mathbf{u} \rangle \equiv v' \bmod q$.

4.2.4 Cryptographic assumptions

The Short Integer Solution (SIS) problem was first introduced in [**Ajt96**], which asks to find a short element (of ℓ_2 norm) in the kernel of a random matrix over the ring $\mathbb{Z}_q$. An inhomogeneous version, ISIS, instead asks for a short solution to some linear equation system [**Mic07**]. It has been shown that the SIS and ISIS problems are equivalent.

We define the ring version of SIS (R-SIS) from [**Mic07**] as follows.

Definition 4.2.10 (R-SIS [**Mic07**])**.** Let $\mathcal{R}, q, , \beta$ be parameters depending on λ. The R-SIS problem states the following: for $\mathbf{a} \leftarrow\$ \mathcal{R}_q$ sampled uniformly at random, and $t = 0$, find $\mathbf{u} \neq \mathbf{0} \in \mathcal{R}$ such that $\|\mathbf{u}\| \leq \beta$ and $\langle \mathbf{a},\ \mathbf{u} \rangle \equiv t \bmod q$.

When $t \leftarrow\$ \mathcal{R}_q$ this becomes the ring inhomogeneous SIS (R-ISIS) assumption, which is known to be equivalent. For appropriate parameters, there are no known efficient algorithms for solving R-SIS for cyclotomic rings.

k-R-ISIS assumptions

We define a family of assumptions over rings, k-R-ISIS, introduced in [**C:ACLMT22**]. k-R-ISIS assumptions can be trivially broken if some basic conditions are not satisfied, so we begin by defining those via the notion of k-R-ISIS admissibility:

Definition 4.2.11 (k-R-ISIS admissible)**.** Let $g \in \mathcal{R}(X)$ be a Laurent monomial, i.e., $g(X) := \mathbf{X}^{\mathbf{e}} := \prod_{i \in [y]} X_i^{e_i}$ for some exponent vector $\mathbf{e} \in \mathbb{Z}^y$. Let $\mathcal{G} \subset \mathcal{R}(X)$ be a set of Laurent monomials with $k := |\mathcal{G}|$. Let $g^* \in \mathcal{R}(X)$ be a *target* Laurent monomial. We say a *monomial family* $(\mathcal{G}, g^*)$ is k-R-ISIS admissible if the following conditions are satisfied:

1. All $g \in \mathcal{G}$ and g^* have constant degree
2. All $g \in \mathcal{G}$ are distinct
3. $0 \notin \mathcal{G}$
4. $g^* \notin \mathcal{G}$

Remark 4.2.12. Condition 1 rules out the monomials that depend on $\mathcal{R}$. Condition 2 rules out that trivial linear combinations of preimages give a preimage for the target. Condition 3 rules out a trivially producing multiple preimages of the same target. Condition 4 rules out the trivial solution to get the preimage of the target.

We then define the k-R-ISIS assumption as follows:

Definition 4.2.13 (k-R-ISIS assumption)**.** Let $\in \mathbb{N}$. Let q be a rational prime, $\mathcal{R}$ a cyclotomic ring, and $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$. Let $\mathcal{G} \subset \mathcal{R}(X)$ be a set of y-variate Laurent monomials and let $g^* \in \mathcal{R}(X)$ be a target Laurent monomial such that $(\mathcal{G}, g^*)$ is k-R-ISIS-admissible. Let $\beta, \beta^* \geq 1$ be reals. For

$g \in \mathcal{G}$, $\geq \mathsf{lhl}(\mathcal{R}_q, \beta)$, $\mathbf{a} \in \mathcal{R}_q$, and $\mathbf{v} \in (\mathcal{R}_q^\times)^y$, let $\mathcal{D}_{g,\mathbf{a},\mathbf{v}}$ be a distribution over

$$\{\mathbf{u}_g \in \mathcal{R} \mid \langle \mathbf{a}, \mathbf{u}_g \rangle \equiv g(\mathbf{v}) \bmod q \ \wedge \ \|\mathbf{u}_g\| \leq \beta\} \ .$$

Let $\mathcal{D} := \left\{\mathcal{D}_{g,\mathbf{a},\mathbf{v}} : \in \mathbb{N}, \ g \in \mathcal{G}, \ \mathbf{a} \in \mathcal{R}_q, \ \mathbf{v} \in (\mathcal{R}_q^\times)^y\right\}$ be the family of these distributions. Define $\mathsf{pp} := (\mathcal{R}_q, , y, \mathcal{G}, g^*, \mathcal{D}, \beta, \beta^*)$. The k-R-$\mathsf{ISIS}_{\mathsf{pp}}$ assumption states that for any PPT adversary $\mathcal{A}$, we have $\mathsf{Adv}^{k\text{-}R\text{-}\mathsf{ISIS}}_{\mathsf{pp},\mathcal{A}}(\lambda) \leq \mathrm{negl}(\lambda)$, where $\mathsf{Adv}^{k\text{-}R\text{-}\mathsf{ISIS}}_{\mathsf{pp},\mathcal{A}}(\lambda)$ is the following probability:

$$\Pr\left[\begin{array}{ll} & \langle \mathbf{a}, \mathbf{u}_{g^*} \rangle \equiv s^* \cdot g^*(\mathbf{v}) \bmod q \\ \wedge & \|s^*\| \leq \beta^* \\ \wedge & \|\mathbf{u}_{g^*}\| \leq \beta^* \\ \wedge & (s^* \cdot g^*, \mathbf{u}_{g^*}) \neq (0, \mathbf{0}) \end{array} \middle| \begin{array}{r} \mathbf{a} \leftarrow\$ \ \mathcal{R}_q \\ \mathbf{v} \leftarrow\$ \ (\mathcal{R}_q^\times)^y \\ \mathbf{u}_g \leftarrow\$ \ \mathcal{D}_{g,\mathbf{a},\mathbf{v}}, \ \ \forall g \in \mathcal{G} \\ (s^*, \mathbf{u}_{g^*}) \leftarrow \mathcal{A}(\mathbf{a}, t, [\mathbf{u}_g]_{g \in \mathcal{G}}, \mathbf{v}) \end{array}\right] \ .$$

Remark 4.2.14. For simplicity, we set t in [**C:ACLMT22**] to be fixed to 1 and thus have $\langle\{t\}\rangle = \mathcal{R}_q$. The k-R-ISIS assumption requires v to be in $\mathcal{R}_q^\times$. Otherwise, the scheme can be insecure. For example, for a power-of-two $\mathcal{R}$, if $q = 2^k$, the ideal $q\mathcal{R}$ splits into $\mathcal{I}^{k \cdot \phi(n)}$ for some ideal $\mathcal{I}$ with norm 2. Then, for $v \leftarrow\$ \ \mathcal{R}_q$, $\Pr[v = 0 \mod \mathcal{I}] = 1/2$. Thus, $v^{k \cdot \phi(n)} = 0 \bmod q$. Therefore, we have $p(X) = X^{k \cdot \phi(n)}$ being a solution to any R-V-SIS instance over $\mathcal{R}_q$. Further, we require $1/|\mathcal{R}_q^\times|$ to be negligible such that SIS attacks are not possible in the image space. theorem 4.2.3 shows how to pick parameters to guarantee this.

In [**C:ACLMT22**] they also introduce the following meta assumption:

Assumption 1 (k-R-ISIS meta assumption)**.** *For any k-R-ISIS admissible $(\mathcal{G}, g^*)$, k-R-$\mathsf{ISIS}_{\mathcal{R}_q,,y,\mathcal{G},g^*,\mathcal{D},\beta,\beta^*}$ is hard if R-$\mathsf{ISIS}_{\mathcal{R}_q,,\beta^*}$ is hard and if R-$\mathsf{SIS}_{\mathcal{R}_q,k,\beta^*}$ is hard.*

To achieve extractability we also require an additional knowledge assumption, because by [**GW11**], we need an unfalsifiable assumption to prove the extractability, as follows:

Definition 4.2.15 (Knowledge k-R-ISIS)**.** Let the parameters $(\mathcal{R}_q, , y, \mathcal{G}, \beta, \beta^*)$ be defined as in definition 4.2.13. Let $\alpha^* \geq 1$ be a real. Let $\mathcal{T} \subset \mathcal{R}_q$ be such that for all $t \in \mathcal{T}$ it holds that:

1. $|\langle t \rangle|/|\mathcal{R}_q| = \mathrm{negl}(\lambda)$, and
2. finding $s' \in \mathcal{R}_q$ satisfying $s' \cdot t \equiv 0 \bmod q$ and $0 < \|s'\| \leq \alpha^*$ is hard.

For $g \in \mathcal{G}$, $\geq \mathsf{lhl}(\mathcal{R}_q, \beta)$, $\mathbf{a} \in \mathcal{R}_q$, $t \in \mathcal{T}$, and $\mathbf{v} \in (\mathcal{R}_q^\times)^y$, let $\mathcal{D}_{g,\mathbf{a},t,\mathbf{v}}$ be a distribution over

$$\{\mathbf{u}_g \in \mathcal{R} \mid \langle \mathbf{a}, \mathbf{u}_g \rangle \equiv g(\mathbf{v}) \cdot t \bmod q \ \wedge \ \|\mathbf{u}_g\| \leq \beta\} \ .$$

Let $\mathcal{D} := \left\{\mathcal{D}_{g,\mathbf{a},t,\mathbf{v}} : \in \mathbb{N},\ g \in \mathcal{G},\ \mathbf{a} \in \mathcal{R}_q,\ t \in \mathcal{T},\ \mathbf{v} \in (\mathcal{R}_q^\times)^y\right\}$ be the family of these distributions. Define $\mathsf{pp} := (\mathcal{R}_q, , y, \mathcal{G}, \mathcal{D}, \mathcal{T}, \alpha^*, \beta, \beta^*)$. The knowledge k-R-$\mathsf{ISIS}_{\mathsf{pp}}$ assumption states that for any PPT adversary $\mathcal{A}$ there exists a PPT extractor $\mathcal{E}_{\mathcal{A}}$ s.t. $\mathsf{Adv}^{k\text{-}R\text{-}\mathsf{ISIS}}_{\mathsf{pp},\mathcal{A}} \leq \mathrm{negl}(\lambda)$, where $\mathsf{Adv}^{k\text{-}R\text{-}\mathsf{ISIS}}_{\mathsf{pp},\mathcal{A}}$ is the following probability:

$$\Pr\left[\begin{array}{l} \quad \langle \mathbf{a}, \mathbf{u}\rangle \equiv c \cdot t \bmod q \\ \wedge \quad \|\mathbf{u}\| \leq \beta^* \ \wedge \ (c, \mathbf{u}) \neq (0, \mathbf{0}) \\ \wedge \quad \neg\left(\begin{array}{l} \quad c = \sum_{g \in \mathcal{G}} x_g \cdot g(\mathbf{v}) \bmod q \\ \wedge \quad \|x_g\| \leq \alpha^*,\ \ \forall g \in \mathcal{G} \end{array}\right) \end{array} \middle| \begin{array}{r} \mathbf{a} \leftarrow\$\ \mathcal{R}_q \bmod q;\ \ t \leftarrow\$\ \mathcal{T};\ \ \mathbf{v} \leftarrow\$\ (\mathcal{R}_q^\times)^w \\ \mathbf{u}_g \leftarrow\$\ \mathcal{D}_{g,\mathbf{a},\mathbf{v}},\ \ \forall g \in \mathcal{G} \\ ((c, \mathbf{u}), [x_g]_{g \in \mathcal{G}}) \leftarrow (\mathcal{A} \| \mathcal{E}_{\mathcal{A}})(\mathbf{a}, t, [\mathbf{u}_g]_{g \in \mathcal{G}}, \mathbf{v}) \end{array}\right].$$

To understand the first restriction on $\mathcal{T}$, consider an adversary who samples a random short $\mathbf{u}$ and then checks if $\langle \mathbf{a},\ \mathbf{u}\rangle \in \langle t\rangle$, setting the commitment to be c s.t. $\langle \mathbf{a},\ \mathbf{u}\rangle \equiv c \bmod q$ if so. Since $\langle \mathbf{a},\ \mathbf{u}\rangle$ is close to uniformly distributed over $\mathcal{R}_q$, restriction 1 ensures such an adversary succeeds with negligible probability. To understand the second restriction, consider an adversary who finds s' as above and outputs $(c, \mathbf{u}) = (s', s' \cdot \mathbf{u}_g)$ for an arbitrary $g \in \mathcal{G}$. Observe that $\mathbf{u}$ is short since s' and $\mathbf{u}_g$ are, and since $\langle \mathbf{a},\ \mathbf{u}_g\rangle \equiv g(v) \cdot t \bmod q$ it follows $\langle \mathbf{a},\ \mathbf{u}\rangle \equiv s' \cdot g(v) \cdot t \equiv 0 \equiv c \cdot t \bmod q$. However, we have that $s' \cdot g(\mathbf{v}) \not\equiv c \equiv s' \bmod q$ unless $g(\mathbf{v}) \equiv 1 \bmod q$, which should only happen with negligible probability.

In addition to the two contraints on $\mathcal{T}$ in our knowledge k-R-ISIS definition, in [**C:ACLMT22**] they give a third. Namely, that $1/|\langle t\rangle| = \mathrm{negl}(\lambda)$. We have omitted this constraint given it is implied by constraint 2. Let $\phi : \mathcal{R}_q \mapsto \langle t\rangle$. Then $|\mathsf{ker}(\phi)|/|\mathcal{R}_q| = 1/|\langle t\rangle|$. If $1/|\langle t\rangle|$ is not negligible, then $s' \leftarrow\$\ \mathcal{R}_q$ satisfies $s' \cdot t \equiv 0 \bmod q$ with non-negligible probability.

We believe this assumption is suitable as it can be used to prove the extractability of our scheme as in the proof of theorem 4.4.2, and is not trivially broken due to all the restrictions above. As in [**C:ACLMT22**], we can set $\mathcal{T}$ to contain all the elements t such that exactly half of the elements in the NTT representation of t are zero. Note this is only defined when $\langle q\rangle$ is not a prime ideal in $\mathcal{R}$; in this case $\mathcal{R}_q \simeq \mathbb{Z}_{q^n}$, which only has one NTT coefficient. Our restriction that $q \equiv 2n + 1 \bmod\ 4n$ ensures that $\langle q\rangle$ is not a prime ideal, and in particular that $\mathcal{R}_q$ is isomorphic to the direct product of n copies of $\mathbb{Z}_q$.

The *k-P-R-ISIS* assumption

Our construction relies on a particular assumption in the k-R-ISIS assumption family and its knowledge counterpart, parameterized by the monomial sets $[X^i]_{i\in\mathbb{Z}(w)\{0\}}$ and $[X^i]_{i\in[w]}$, respectively. This assumption could be seen as a lattice analogue of the k-Strong-Diffie-Hellman (k-SDH) problem, and we refer to it as the k-Powers R-ISIS (k-P-R-ISIS) assumption.

Assumption 2 (k-P-R-ISIS assumption)**.** *Define* k-P-R-$\mathsf{ISIS}_{\mathcal{R}_q,,\mathcal{D},\beta,\beta^*} := k$-$R$-$\mathsf{ISIS}_{\mathcal{R}_q,,y,\mathcal{G},g^*,\mathcal{D},\beta,\beta^*}$ *with fixed* $\mathcal{G} = [X^i]_{i\in\mathbb{Z}(k)\{0\}}$, $g^*(X) = 1$, *and consequently* $y = 1$. *The* k-P-R-ISIS *assumption is hard if* R-$\mathsf{ISIS}_{\mathcal{R}_q,,\beta^*}$ *is.*

In the discrete log setting, this is akin to asking an adversary to compute g given $(g^{x^{w-1}}, \dots, g^{x^{-1}}, g^x, \dots, g^{x^w})$. Since $(\mathcal{G}, g^*)$ is k-M-ISIS admissible (definition 4.2.11), our assumption follows from the much broader meta assumption of Lai et al. (Assumption 1). We also require the following knowledge assumption:

Assumption 3 (Knowledge k-P-R-ISIS assumption)**.** *Define knowledge* k-P-R-$\mathsf{ISIS}_{\mathcal{R}_q,,\mathcal{D},\mathcal{T},\alpha^*,\beta,\beta^*}$ *as knowledge* k-R-$\mathsf{ISIS}_{\mathcal{R}_q,,y,\mathcal{G},\mathcal{D},\mathcal{T},\alpha^*,\beta,\beta^*}$ *with* $\alpha^* \geq \beta^* \geq 1$, *fixed* $\mathcal{G} = [X^i]_{i\in[k]}$, *and consequently* $y = 1$. *The knowledge* k-P-R-ISIS *assumption holds if* R-$\mathsf{ISIS}_{\mathcal{R}_q,,\beta^*}$ *does.*

4.3 Cryptanalysis of k-P-R-ISIS

In this section, we consider how the k-P-R-ISIS assumption may be attacked. We begin by formulating a special case of R-SIS assumption when $\mathbf{a}$ is generated by the powers of a single ring element, which we call ring Vandermonde SIS (R-V-SIS). Then we show how an attacker who efficiently solves R-V-SIS can break the security of Orbweaver.

Definition 4.3.1. (R-V-SIS) Let $\mathcal{R}, q, \beta_V$ be parameters depending on λ and w. The R-V-SIS problem asks the following: given $\mathbf{v} := [v^i]_{i\in\mathbb{Z}(w)\{0\}} \in \mathcal{R}_q^{2w-3}$ for $v \leftarrow\$ \mathcal{R}_q^\times$, find $(s, \mathbf{z}) \neq (0, \mathbf{0}) \in \mathcal{R} \times \mathcal{R}^{2w-3}$ such that $0 < \|s\| \leq \beta_V, \|\mathbf{z}\| \leq \beta_V$, and $\langle \mathbf{v},\ \mathbf{z}\rangle \equiv s \bmod q$.

We note that similar assumptions have been proposed by prior works (see [**BSS22**] for a summary). Thus far, there are no known algorithms which solve these variants faster than their non-Vandermonde counterparts.

We now show how an adversary who breaks R-V-SIS can break k-P-R-ISIS. Since we are given $\mathbf{a} \cdot \mathbf{u}_i \equiv v^i \bmod q$ for all $i \in \mathbb{Z}(w) \ \{0\}$, and our goal is to find short $(s^*, \mathbf{u}_{g^*}) \neq (0, \mathbf{0})$ such that $\mathbf{a} \cdot \mathbf{u}_{g^*} \equiv s^* \cdot v^0 \bmod q$. We first solve R-V-SIS, obtaining $s, \mathbf{z}$ satisfying the constraints in definition 4.3.1. Then, we let $\mathbf{u}_{g^*} = \sum_{i \in \mathbb{Z}(w)\{0\}} z_i \cdot \mathbf{u}_i$ and $s^* = s$. We observe that $\langle \mathbf{a},\ \mathbf{u}_{g^*} \rangle \equiv \sum_{i \in \mathbb{Z}(w)\{0\}} z_i \cdot \langle \mathbf{a},\ \mathbf{u}_i \rangle \equiv s^* v^0 \bmod q$.

Notice, however, we do get some blowup in the norm, where now $\|\mathbf{u}_{g^*}\| \leq (2w-3) \cdot \beta_V \cdot \beta \cdot \gamma_{\mathcal{R}}$. So we must solve R-V-SIS for a β_V such that $(2w-3) \cdot \beta_V \cdot \beta \cdot \gamma_{\mathcal{R}} \leq \beta^*$.

In [**C:ACLMT22**], they generalize this attack, requiring the attacker find (not necessarily short) $\mathbf{z}$ and short s^* satisfying $\langle \mathbf{a},\ \mathbf{u}_{g^*} \rangle \equiv s^* v^0 \bmod q$ and $\|\mathbf{u}_{g^*}\| \leq \beta^*$ for $\mathbf{u}_{g^*} = \sum_{i \in \mathbb{Z}(w)\{0\}} z_i \cdot \mathbf{u}_i$. This attack applies to our scheme as well. See Section 4.1 of their paper for details.

Setting parameters according to the direct SIS attack. Given there are no specific algorithms for the above problems, when picking parameters we consider the cost of a direct SIS attack that ignores the preimages and their algebraic dependencies. The following analysis is standard. We first reduce R-ISIS to R-SIS: recall that to break the assumption, the attacker needs to find some $\mathbf{u}_g$ such that $\langle \mathbf{a},\ \mathbf{u}_g \rangle \equiv s^* \bmod q$; instead, we ask the attacker to find $\mathbf{u}'_g$ such that $\langle \mathbf{a}',\ \mathbf{u}'_g \rangle \equiv 0 \bmod q$, where $\mathbf{a}' := \mathbf{a} \| -1$. Then, here the last entry of $\mathbf{u}'_g$ becomes s^*.

We then view this as a SIS instance: for $\mathbf{A} \in \mathbb{Z}_q^{n \times n(+1)}$ find $\mathbf{u} \in \mathbb{Z}_q^{n(+1)}$ such that $\mathbf{A} \cdot \mathbf{u} \equiv 0 \bmod q$. Let $\mathbf{L} \in \mathbb{Z}_q^{n(+1) \times n}$ be a basis for the right kernel of $\mathbf{A}$. Then solving this SIS instance is equivalent to finding a short vector in $\Lambda(\mathbf{L})$. Observe that

$$\mathbf{L}' = \left[\begin{array}{c|c} L & \begin{matrix} 0 \\ qI_n \end{matrix} \end{array}\right]$$

is an equivalent basis for this lattice. If we compute the Gram–Schmidt format of $\mathbf{L}'$, we get the form $\left[\begin{array}{c|c} I & \begin{matrix} 0 \\ qI_n \end{matrix} \\ \tilde{\mathbf{L}} & \end{array}\right]$ for some $\tilde{\mathbf{L}} \in \mathbb{Z}_q^{n(+1) \times n}$.

Thus, it is easy to compute the volume $\mathsf{Vol}(\Lambda) = \prod_{i \in [n(+1)]} \|b_i\|_\infty = q^n$, where $[b_1, \ldots, b_{n(+1)}] = \tilde{\mathbf{L}}$. Thus, the goal is to find a vector in $\Lambda(\mathbf{L})^\top$ in a $d \leq n \cdot (\ell + 1)$ dimensional lattice with volume q^n. We thus consider the adversary wins if it finds a vector with ℓ_2 norm of $\sqrt{d} \cdot \beta^*$, where β^* is the ℓ_∞ norm we allow in our hardness assumption.

Recall that lattice reduction with lattice parameter n (i.e., our ring degree) returns a vector with ℓ_2 norm $\approx \delta^d \cdot \mathsf{Vol}(\Lambda)^{1/d}$, where δ is the root Hermite factor. With the state-of-the-art lattice

reduction algorithm [**SODA:BDGL16**], to achieve $\delta \approx (\frac{n}{2\pi e})^{1/(2n)}$, it takes roughly $2^{0.292n+o(n)}$ time. When $d \approx \sqrt{n \cdot log(q)/\log(\delta)}$, the norm of the vector found (i.e., $\delta^d \cdot \mathsf{Vol}(\Lambda)^{1/d}$) is minimized [**MR09**].

Thus, we can simply set $\delta^d \cdot \mathsf{Vol}(\Lambda)^{1/d} \geq \sqrt{d} \cdot \beta^*$ for the d that minimizes the left hand side, and $2^{0.292n+o(n)} \geq 2^\lambda$. Thus, when we set parameters for R-SIS problem with parameter $\mathcal{R}, q, \ell, \beta^*$ (where $\mathcal{R}$ has ring degree n), essentially, we are requiring that the adversary cannot find a vector with ℓ_2 norm $\leq \sqrt{n \cdot log(q)/\log(\delta)} \cdot \beta^*$ in $\leq 2^\lambda$ time using the attack above, except with neglgible probability. [3]

4.4 Orbweaver: linear functional commitments for rings

We present $\mathsf{Orbweaver}$, a vector commitment (VC) for linear functions that is non-interactive, publicly verifiable, preprocessing, and structure-preserving. Together, these features enable efficient recursive composition of our construction.

$\mathsf{Orbweaver}$ supports opening committed vectors with respect to the function family $\mathcal{F} = \{\mathcal{F}_w\}_{w \in \mathbb{N}}$, where $\mathcal{F}_w = \mathcal{X}^w$ and $f \in \mathcal{F}_w$ is computed as $f(\mathbf{x}) \equiv \langle [f_i]_{i=1}^w,\ \mathbf{x} \rangle \bmod q$. The input alphabet $\mathcal{X} = \{x \in \mathcal{R} \mid \|x\| \leq \alpha\}$ is specified by the ring $\mathcal{R}$ and norm bound α. Let $\delta_M = w \cdot \alpha^2 \cdot \gamma_{\mathcal{R}}$, where $\gamma_{\mathcal{R}}$ is the expansion factor of $\mathcal{R}$ (see definition 4.2.1). The image space $\mathcal{Y}$ is then $\mathcal{Y} = \{y \in \mathcal{R} \mid \|y\| \leq \delta_M\}$.

Let $P := \mathbb{Z}(w)\ \{0\}$. We use the monomial set $\mathcal{G}_0 := [g_i(X) = X^i]_{i \in P}$ to generate our opening proof $[\mathbf{u}_{0,i}]_{i \in P}$, the monomial set $\mathcal{G}_1 := [g_i(X) = X^i]_{i \in [w]}$ to generate our knowledge proof SRS $[\mathbf{u}_{1,i}]_{i \in [w]}$, the target monomial $g^*(X) := 1$, and the distribution $\mathcal{T}$ satisfying the requirements of Assumption 3.

$\underline{\mathsf{Setup}(1^\lambda, 1^w) \to \mathsf{ck}}$
$v \leftarrow\$\ \mathcal{R}_q^\times;\ \ t \leftarrow\$\ \mathcal{T}$
$(\mathbf{a}_0, \mathsf{td}_0) \leftarrow \mathsf{TrapGen}(1, 1, \mathcal{R}_q, \beta_0);\ \ (\mathbf{a}_1, \mathsf{td}_1) \leftarrow \mathsf{TrapGen}(1, 1, \mathcal{R}_q, \beta_1)$
$\mathbf{u}_{0,i} \leftarrow \mathsf{SampPre}(\mathsf{td}_0, v^i, \beta_0),\ \ \forall i \in P;\ \ \mathbf{u}_{1,i} \leftarrow \mathsf{SampPre}(\mathsf{td}_1, v^i \cdot t, \beta_1),\ \ \forall i \in [w]$
Return $\mathsf{ck} := (\mathbf{a}_0, [\mathbf{u}_{0,i}]_{i \in P}, \mathbf{a}_1, t, [\mathbf{u}_{1,i}]_{i \in [w]}, v)$

$\underline{\mathsf{Com}(\mathsf{ck}, \mathbf{x}) \to (c, \pi_1)}$
$c := \sum_{i=1}^w x_i v^i \bmod q$

[3]Note that there are known poly-time attacks against some parameter selection for R-SIS, e.g., [**EC:PXWC21**]. Thus, we also need to avoid those parameter selections. In more detail, we follow what is suggested in [**C:ACLMT22**] and pick q such that $\mathcal{R}_q$ fully splits and pick t as specified in definition 4.2.15, which makes the attack from [**EC:PXWC21**] not work.

Table 4.1: A list of parameters used in Orbweaver.

$w \in \mathbb{N}$		Dimension of secret input $\mathbf{x}$
$n \in \mathbb{N}$		Degree of $\mathcal{R}$
$\alpha \in \mathbb{R}$	$\text{poly}(\lambda)$	Norm bound for $\mathbf{x}$ and f
$\beta_0 \in \mathbb{R}$	$\text{poly}(\lambda)$	Norm bound for public preimages $\mathbf{u}_{0,\cdot}$ for π_0
$\beta_1 \in \mathbb{R}$	$\text{poly}(\lambda)$	Norm bound for public preimages $\mathbf{u}_{1,\cdot}$ for π_1
$\delta_0 \in \mathbb{R}$	$w^2 \cdot \alpha^2 \cdot \beta_0 \cdot \gamma_{\mathcal{R}}^2$	Norm bound for opening proof π_0
$\delta_1 \in \mathbb{R}$	$w \cdot \alpha \cdot \beta_1 \cdot \gamma_{\mathcal{R}}$	Norm bound for commitment knowledge proof π_1
$\delta_M \in \mathbb{R}$	$w \cdot \alpha^2 \cdot \gamma_{\mathcal{R}}$	Norm bound for evaluation of a linear function with coefficients of norm bounded by α at a point of norm bounded by α
$q \in \mathbb{R}$		Modulus for $\mathcal{R}_q$
$_i \in \mathbb{N}$	$\geq \mathsf{lhl}(\mathcal{R}_q, \beta_i)$	Number of ring elements in $\mathbf{a}$
$\gamma_{\mathcal{R}}$		Ring expansion factor of $\mathcal{R}$
$\mathcal{X}$	$\{x \in \mathcal{R} : \|x\| \leq \alpha\}$	$\mathcal{R}$ elements with norm bound α
$\mathcal{F}$		family of w-variate linear functions over $\mathcal{X}$

$\pi_1 := \sum_{i=1}^{w} x_i \cdot \mathbf{u}_{1,i} \bmod q$
Return (c, π_1)

$\underline{\mathsf{Open}(\mathsf{ck}, f, \mathbf{x}) \to \pi_0}$
Let $a_{-w+1}, ..., a_{w-1}$ denote the $2w-1$ coefficients of $\mathbf{x}(v)f(v) := \left(\sum_{i=1}^{w} x_i v^i\right)\left(\sum_{i=1}^{w} f_i v^{-i}\right)$
Return $\pi_0 := \sum_{i \in P} a_i \cdot \mathbf{u}_{0,i} \bmod q$

$\underline{\mathsf{PreVerify}(\mathsf{ck}, f) \to \mathsf{vk}_f}$
If $\|f\| > \alpha$, abort
Return $\mathsf{vk}_f := \sum_{i=1}^{w} f_i \cdot v^{-i} \bmod q$

$\underline{\mathsf{Verify}(\mathsf{vk}_f, c, \pi_1, y, \pi_0) \to \{0,1\}}$
Output 1 if the following conditions all hold (else 0):
$\|y\|_\infty \leq \delta_M$; $\|\pi_1\|_\infty \leq \delta_1$; $\|\pi_0\|_\infty \leq \delta_0$
$\langle \mathbf{a}_1, \pi_1 \rangle \equiv c \cdot t \bmod q$
$\langle \mathbf{a}_0, \pi_0 \rangle \equiv \mathsf{vk}_f \cdot c - y \bmod q$

We next prove that Orbweaver satisfies correctness, extractability, and succinctness.

Theorem 4.4.1. *Orbweaver is correct (definition 4.2.7) for $\delta_1 \geq w \cdot \alpha \cdot \beta_0 \cdot \gamma_{\mathcal{R}}$ and $\delta_0 \geq w^2 \cdot \alpha^2 \cdot \beta_1 \cdot \gamma_{\mathcal{R}}^2$.*

Proof. We begin by proving the norm bound checks hold. $y = f(\mathbf{x}) = \sum_{i=1}^{w} f_i \cdot x_i \bmod q$, where $\|f_i \cdot x_i\| \leq \alpha^2 \cdot \gamma_{\mathcal{R}}$ for all i. Therefore, $\|y\| \leq w \cdot \alpha^2 \cdot \gamma_{\mathcal{R}} = \delta_M$. Next, $\pi_1 = \sum_{i=1}^{w} x_i \cdot \mathbf{u}_{1,i} \bmod q$, where $\left\|x_i \cdot \mathbf{u}_{1,i}\right\| \leq \alpha \cdot \beta_1 \cdot \gamma_{\mathcal{R}}$ for all i. It follows $\|\pi_1\| \leq w \cdot \alpha \cdot \beta_1 \cdot \gamma_{\mathcal{R}} = \delta_1$. Last, $\pi_0 = \sum_{i \in P} a_i \cdot \mathbf{u}_{0,i} \bmod q$, where $a_{-w+1}, ..., a_{w-1}$ denote the $2w-2$ coefficients of $\mathbf{x}(v)f(v) = \left(\sum_{i=1}^{w} x_i v^i\right)\left(\sum_{i=1}^{w} f_i v^{-i}\right)$. Since $\left\|\mathbf{u}_{0,i}\right\| \leq \beta_0$ for all i, $\|\pi_0\| \leq \beta_0 \cdot \gamma_{\mathcal{R}} \cdot \sum_{i \in P} a_i$. Then $\left\|a_{|i|}\right\| \leq (w-i) \cdot \alpha^2 \cdot \gamma_{\mathcal{R}}$ for $i \in [w-1]$. Thus, $\|\pi_0\| \leq w^2 \cdot \alpha^2 \cdot \beta_0 \cdot \gamma_{\mathcal{R}}^2 = \delta_0$.

Next, we show the verification equations hold:

$$\begin{aligned}\langle \mathbf{a}_1,\ \pi_1\rangle &\equiv \langle \mathbf{a}_1,\ \textstyle\sum_{i=1}^{w} x_i \cdot \mathbf{u}_{1,i}\rangle \bmod q \\ &\equiv \textstyle\sum_{i=1}^{w} x_i \cdot \langle \mathbf{a}_0,\ \mathbf{u}_{1,i}\rangle \bmod q \\ &\equiv \textstyle\sum_{i=1}^{w} x_i \cdot v^i \cdot t \bmod q \\ &\equiv c \cdot t \bmod q\end{aligned} \qquad \begin{aligned}\langle \mathbf{a}_0,\ \pi_0\rangle &\equiv \langle \mathbf{a}_0,\ \textstyle\sum_{i\in P} a_i \cdot \mathbf{u}_i\rangle \bmod q \\ &\equiv \textstyle\sum_{i\in P} a_i \cdot v^i \bmod q \\ &\equiv \left(\textstyle\sum_{i=1}^{w} x_i v^i\right)\left(\textstyle\sum_{i=1}^{w} f_i v^{-i}\right) - a_0 \cdot v^0 \bmod q \\ &\equiv \mathsf{vk}_f \cdot c - \textstyle\sum_{i=1}^{w} f_i \cdot x_i \equiv \mathsf{vk}_f \cdot c - y \bmod q\ .\end{aligned}$$

□

Theorem 4.4.2. *Let $\mathcal{X}^* := \{x \in \mathcal{R} \mid \|x\| \le \alpha^*\}$. Orbweaver is $\mathcal{X}^*$-extractable if*

$$\begin{aligned}&{}_0 \ge \mathsf{lhl}(\mathcal{R}_q, b_0),\ \ {}_1 \ge \mathsf{lhl}(\mathcal{R}_q, b_1)\\ &\alpha^* \ge \beta_1^* \ge \delta_1\\ &\beta_0^* \ge 2 \cdot w^2 \cdot \alpha \cdot \alpha^* \cdot \beta_0 \cdot \gamma_{\mathcal{R}}^2\end{aligned}$$

where $b_0 = O(\beta_0), b_1 = O(\beta_1)$, and the k-R-$\mathsf{ISIS}_{\mathcal{R}_q,0,1,\mathcal{G}_0,g_0^*,\mathcal{D}_0,\beta_0,\beta_0^*}$ (i.e., k-P-R-ISIS) and knowledge k-R-$\mathsf{ISIS}_{\mathcal{R}_q,1,1,\mathcal{G}_1,\mathcal{D}_1,\mathcal{T},\alpha^*,\beta_1,\beta_1^*}$ (i.e., knowledge k-P-R-ISIS) assumptions hold, where $\mathcal{D}_0$ and $\mathcal{D}_1$ are such that

$$\approx \left\{\begin{array}{l|l} (\mathbf{a}_0, [\mathbf{u}_{0,g}]_{g\in\mathcal{G}_0}, v) & \mathbf{a}_0 \leftarrow\$ \mathcal{R}_q^{\ell_0};\ \ v \leftarrow\$ \mathcal{R}_q^{\times} \\ & \mathbf{u}_{0,g} \leftarrow\$ \mathcal{D}_{g,\mathbf{a}_0,v},\ \ \forall i \in P \end{array}\right\}$$
$$\left\{\begin{array}{l|l} (\mathbf{a}_0, [\mathbf{u}_{0,g}]_{g\in\mathcal{G}_0}, v) & \mathbf{a}_0 \leftarrow\$ \mathcal{R}_q^{\ell_0};\ \ v \leftarrow\$ \mathcal{R}_q^{\times} \\ & \mathbf{u}_{0,g} \leftarrow\$ \mathsf{SampD}(1^{\ell_0}, \mathcal{R}_q, \beta_0) : \langle \mathbf{a}_0,\ \mathbf{u}_{0,g}\rangle \equiv v^i \bmod q,\ \ \forall i \in P \end{array}\right\}$$

and

$$\approx \left\{\begin{array}{l|l} (\mathbf{a}_1, t_i, [\mathbf{u}_{1,g}]_{g\in\mathcal{G}_1}, v) & \mathbf{a}_1 \leftarrow\$ \mathcal{R}_q^{\ell_1};\ \ t \leftarrow\$ \mathcal{T};\ \ v \leftarrow\$ \mathcal{R}_q^{\times} \\ & \mathbf{u}_{1,g} \leftarrow\$ \mathcal{D}_{g,\mathbf{a}_1,t,v},\ \ \forall i \in [w] \end{array}\right\}$$
$$\left\{\begin{array}{l|l} (\mathbf{a}_1, t_i, [\mathbf{u}_{1,g}]_{g\in\mathcal{G}_1}, v) & \mathbf{a}_1 \leftarrow\$ \mathcal{R}_q^{\ell_1};\ \ t \leftarrow\$ \mathcal{T};\ \ v \leftarrow\$ \mathcal{R}_q^{\times} \\ & \mathbf{u}_{1,g} \leftarrow\$ \mathsf{SampD}(1^{\ell_1}, \mathcal{R}_q, \beta_1) : \langle \mathbf{a}_1,\ \mathbf{u}_{1,g}\rangle \equiv v^i \cdot t \bmod q,\ \ \forall i \in [w] \end{array}\right\}\ .$$

Proof. Suppose there exists a PPT adversary $\mathcal{A}$ that, on input $\mathsf{ck} \leftarrow \mathsf{Setup}(1^\lambda, 1^w)$, outputs (f, c, y, π_1, π_0) such that $\mathsf{PreVerify}(\mathsf{ck}, f)$ does not abort and $\mathsf{Verify}(\mathsf{vk}_f, c, y, \pi_1, \pi_0) = 1$ with non-negligible probability in λ over any randomness it uses and the choice of ck. We construct an extractor $\mathcal{E}_{\mathcal{A}}$ that, input ck and given black-box access to $\mathcal{A}$ and any randomness it uses, returns $\mathbf{x}^* \in (\mathcal{X}^*)^w$ such that $c = \mathsf{Com}(\mathsf{ck}, \mathbf{x}^*)$ and $f(\mathbf{x}^*) = y$ with all but neglibible probability in λ when $\mathcal{A}$ succeeds.

For commitment key $\mathsf{ck} = (\mathbf{a}_0, [\mathbf{u}_{0,i}]_{i\in P}, \mathbf{a}_1, t, [\mathbf{u}_{1,i}]_{i\in[w]}, v)$, define $\mathsf{ck}_0 := (\mathbf{a}_0, [\mathbf{u}_{1,i}]_{i\in[w]}, v)$ and $\mathsf{ck}_1 := (\mathbf{a}_1, t, [\mathbf{u}_{1,i}]_{i\in[w]}, v)$. Let $\mathcal{B}_{\mathcal{A}}$ be an adversary with black-box access to $\mathcal{A}$ that runs on input

ck_1' sampled according to the knowledge k-R-$\mathsf{ISIS}_{\mathcal{R}_q,1,\mathcal{G}_1,\mathcal{D}_1,\mathcal{T},\alpha^*,\beta_1,\beta_1^*}$ definition (definition 4.2.15). First, $\mathcal{B}_{\mathcal{A}}$ samples ck_0' according to the k-R-$\mathsf{ISIS}_{\mathcal{R}_q,0,1,\mathcal{G}_0,g_0^*,\mathcal{D}_0,\beta_0,\beta_0^*}$ definition (definition 4.2.13), except fixing $v' = v$. Then $\mathcal{B}_{\mathcal{A}}$ runs $\mathcal{A}$ on $(\mathsf{ck}_0', \mathsf{ck}_1)$ (passing its own random input tape) to obtain (f, c, y, π_1, π_0), which it parses to output (c, π_1).

Let $\mathcal{E}_{\mathcal{B}_{\mathcal{A}}}$ be the extractor promised by our knowledge assumption for $\mathcal{B}_{\mathcal{A}}$. Extractor $\mathcal{E}_{\mathcal{A}}$, on input $\mathsf{ck} = (\mathsf{ck}_0, \mathsf{ck}_1)$, runs $\mathcal{E}_{\mathcal{B}_{\mathcal{A}}}(\mathsf{ck}_1)$ to obtain $\mathbf{x}^*$. Next consider the following hybrid experiments:

1. Hyb_0: the real experiment, where $\mathsf{ck} \leftarrow \mathsf{Setup}(1^\lambda, 1^w)$, $(f, c, y, \pi_1, \pi_0) \leftarrow \mathcal{A}(\mathsf{ck})$, and $\mathbf{x}^* \leftarrow \mathcal{E}_{\mathcal{A}}(\mathsf{ck})$.
2. Hyb_1: the same as Hyb_0, but $\mathsf{ck} = (\mathsf{ck}_0, \mathsf{ck}_1')$, where ck_1' is sampled according to the knowledge k-R-$\mathsf{ISIS}_{\mathcal{R}_q,1,\mathcal{G}_1,\mathcal{D}_1,\mathcal{T},\alpha^*,\beta_1,\beta_1^*}$ definition, conditioned on $v' = v$.
3. Hyb_2: the same as Hyb_1, but $\mathsf{ck} = (\mathsf{ck}_0'', \mathsf{ck}_1')$, where ck_1' is sampled as in Hyb_1, but ck_0'' is sampled according to the k-R-$\mathsf{ISIS}_{\mathcal{R}_q,0,1,\mathcal{G}_0,g_0^*,\mathcal{D}_0,\beta_0,\beta_0^*}$ definition, again conditioned on $v'' = v'$.

By our assumption on $\mathcal{D}_1$ (resp., $\mathcal{D}_0$) hybrids 0 and 1 (resp., 1 and 2) are statistically close. Therefore, they are all statistically close. This implies if a property holds with respect to the output of one hybrid with a certain probability, it will hold with respect to the output of the others with statistically close probability.

From our knowledge assumption, we know that in Hyb_1 the extractor outputs $\mathbf{x}^*$ satisfying $c \equiv \sum_{i=1}^{w} x_i \cdot v^i \bmod q$ and $\|x_i\| \leq \alpha^*$ for all $i \in [w]$ with probability all but negligible in λ. Now consider the output in Hyb_2 satisfies $c \equiv \sum_{i=1}^{w} x_i \cdot v^i \bmod q$ for $\mathbf{x}^* \in \mathcal{X}^*$, but $f(\mathbf{x}^*) \neq y$. When this happens we can break k-R-$\mathsf{ISIS}_{\mathcal{R}_q,0,1,\mathcal{G}_0,g_0^*,\mathcal{D}_0,\beta_0,\beta_0^*}$ as follows. Let $y^* := f(\mathbf{x}^*)$ and $\pi_0^* \leftarrow \mathsf{Open}(\mathsf{ck}, f, \mathbf{x}^*)$. Then we obtain the solution $(y^* - y, \pi_0 - \pi_0^*)$ for k-R-ISIS, where

$$\langle \mathbf{a}_0,\ \pi_0 - \pi_0^* \rangle \equiv \mathsf{vk}_f \cdot c - y - (\mathsf{vk}_f \cdot c - y^*) \equiv (y^* - y) \cdot 1 \equiv (y^* - y) \cdot g^*(v) \bmod q\ .$$

To see that the norm bounds hold, we can bound $\delta_M^* := \|y^*\| \leq w\alpha^*\alpha\gamma_{\mathcal{R}}$ and $\delta_0^* := \|\pi_0^*\| \leq w^2\alpha^*\alpha\beta_0\gamma_{\mathcal{R}}^2 \leq \beta_0^*/2$ by modifying the calculations of δ_M and δ_0 in the correctness proof. Therefore, this can only happen with negligible probability in λ. Together, these two properties about the results of Hyb_1 and Hyb_2 show that $\mathsf{Orbweaver}$ is $\mathcal{X}^*$-extractable. □

Theorem 4.4.3. *Orbweaver is succinct (definition 4.2.9).*

Proof. We first show Orbweaver has $O(\log w + \log \alpha)$ size commitments and proofs. From our construction we have that $|c| = n\lceil \log(q) \rceil$, $|\pi_0| = n_0 \lceil \log \delta_0 \rceil$, and $|\pi_1| = n_1 \lceil \log \delta_1 \rceil$. We set a base $b_0 = O(\beta_0), b_1 = O(\beta_1)$ for trapdoor generation and let ${}_0 := \mathsf{lhl}(\mathcal{R}_q, b_0) = O(\log(q)/\log(\beta_0))), {}_1 := \mathsf{lhl}(\mathcal{R}_q, b_1) = O(\log(q)/\log(\beta_1))$. We consider the ring degree n to be a function of the security parameter and treat it as a constant for the following analysis.

As implicated by our extractability theorem (theorem 4.4.2) together with our assumptions relating the hardness of plain and knowledge k-P-R-ISIS to R-SIS (Assumptions 2 and 3), we must have that $R\text{-}\mathsf{SIS}_{\mathcal{R}_q,\ell,\beta_0^*}$ and $R\text{-}\mathsf{SIS}_{\mathcal{R}_q,\ell,\delta_1}$ are hard. Recall $\beta_0^* = 2w^2\alpha\alpha^*\beta_0\gamma_{\mathcal{R}}^2$. Using $\alpha^* = \delta_1 = w\alpha\beta_1\gamma_{\mathcal{R}}$, we can write $\beta_0^* = 2w^3\alpha^2\beta_0\beta_1\gamma_{\mathcal{R}}^3$. Since $\delta_1 < \beta_0^*$, we can just pick q s.t. $R\text{-}\mathsf{SIS}_{\mathcal{R}_q,\ell,\beta_0^*}$ is hard.

In particular, we let $q \approx \beta_0^* n \log(n)$ such that $\log(q) = O(\log \beta_0^*) = O(\log(w) + \log(\alpha) + \log(\beta_0) + \log(\beta_1))$ since for power-of-two cyclotomic rings $\gamma_{\mathcal{R}} \leq n$ (theorem 4.2.2). Setting $\beta_0 := \max\{w, \alpha\}$ and $\beta_1 := \max\{w, \alpha\}$, we obtain logarithmic commitment and proof sizes:

$$
\begin{aligned}
|c| &= n\lceil \log(q) \rceil \\
&\approx n \log(n \log(n) \beta_0^*) \\
&= O(\log \beta_0^*) \\
&= O(\log w + \log \alpha)
\end{aligned}
$$

$$
\begin{aligned}
|\pi_0| &= n_0 \lceil \log \delta_0 \rceil \\
&= 2\lceil \log(q)/\log(\beta_0) \rceil n \lceil \log \delta_0 \rceil \\
&\approx 2n \log(\beta_0^* n \log n) \log \delta_0 / \log \beta_0 \\
&= O(\log(\beta_0^*) \cdot \log(\delta_0)/\log \beta_0) \\
&= O(\log w + \log \alpha)
\end{aligned}
$$

$$
\begin{aligned}
|\pi_1| &= n_0 \lceil \log \delta_1 \rceil \\
&= 2\lceil \log(q)/\log(\beta_1) \rceil n \lceil \log \delta_1 \rceil \\
&\approx 2n \log(\beta_0^* n \log n) \log \delta_1 / \log \beta_1 \\
&= O(\log \beta_0^* \cdot \log \delta_1 / \log \beta_1) \\
&= O(\log w + \log \alpha)
\end{aligned}
$$

Next, we show that the Orbweaver verifier runs in time $O(\log(w\alpha) \log\log(w\alpha))$ time. The verifier work is dominated by computing the inner products $\langle \mathbf{a}_1,\ \pi_1 \rangle$ and $\langle \mathbf{a}_0,\ \pi_0 \rangle$, where $|\mathbf{a}_i| = \ell_i$. This work is dominated by $\ell_0 + \ell_1$ multiplications in the ring $\mathcal{R}_q$ of degree n. Using the NTT, this can be computed with $(\ell_0 + \ell_1)(n \log n)$ multiplications in $\mathbb{Z}_q$. Though since we've defined n in terms of λ, which itself is treated as a constant in our asymptotic analysis, we can simplify this to $O(\ell_0 + \ell_1) = O(\log q/\log \beta_0 + \log q/\log \beta_1) = O(\log \beta_0^*/\log \beta_0 + \log \beta_0^*/\log \beta_1) = O((\log w + \log \alpha)/\log \beta_0 + (\log w + \log \alpha)/\log \beta_1) = O(1)$ operations over the ring $\mathbb{Z}_q$, where elements have bit length $\log q = O(\log w + \log \alpha)$. Asymptotically, $\mathbb{Z}_q$ multiplications can be performed in $\log(q) \log\log(q)$ time

[**Harvey2021IntegerMI**], giving verifier complexity is $O(\log(w\alpha)\log\log(w\alpha))$. [4] $\square$

4.4.1 Extensions

This section discuss extensions to Orbweaver including support for proof aggregation, an inner product argument, and recursive proof composition. Even if the reader is only interested in using Orbweaver just for ring relations, we suggest looking at the section on linear functional commitments for $\mathbb{Z}_M$ (section 4.4.2) anyway as many of the optimizations introduced there including ternary decomposition can be applied in the ring setting as well.

Universal SRS. In order to support all instances of length $w \leq W$, observe it suffices to run the protocol without modification using only powers in $\mathbb{Z}(w)$. Since parameters are set such that R-SIS is hard for β_0^* based on W, it is not an issue that the extractor outputs an input of length W.

Public proof aggregation. Orbweaver can be extended to support public proof aggregation (i.e., the aggregator can be an untrusted third party) by taking advantage of the linearly homomorphic property of the opening proofs. Using short challenges $[[h_{i,j}]_{i=1}^{t}]_{j=1}^{T[i]}$ we can check that commitments $[c_i]_{i=1}^{t}$, evaluated at function sets $[f_j]_{j=1}^{T[i]}$, output values $[[y_{i,j}]_{i=1}^{t}]_{j=1}^{T[i]}$, respectively. The verifier checks the opening proof equation

$$\langle \mathbf{a},\ \pi_0 \rangle \equiv \sum_{i=1}^{t} \sum_{j=1}^{T[i]} h_{i,j} \cdot \left(c_i \cdot \mathsf{ck}_{f_{i,j}} - y_{i,j} \right) \bmod q \ \ , \text{ where } \ \ \pi_0 = \sum_{i=1}^{t} \sum_{j=1}^{T[i]} h_{i,j} \pi_{0,i,j} \quad \bmod q$$

We give an instantiation of $\mathcal{H}$ from exceptional sets over $\mathcal{R}_q$ in section 4.2.1. With this challenge set, the protocol becomes interactive: the aggregator sends $[(c_i, \pi_{1,i}, [y_{i,j}]_{j=1}^{T[i]})]_{i=1}^{t}$, the verifier samples $[h_{i,j}]_{i=1,j=1}^{t,T[i]} \leftarrow\$ \mathcal{H}$, and the aggregator responds with π_0. Using an R-SIS instance as a fixed challenge string as introduced in [**C:ACLMT22**] provides a way to maintain non-interactivity, while using an exceptional set provides better performance.

To see how security holds, after using our knowledge assumption extractor to obtain the $[\mathbf{x}_i^*]_{i=1}^{t}$,

[4] We note [**Harvey2021IntegerMI**] is a galactic algorithm, i.e., it is only efficient at instance sizes so large it is never used in practice. For the integer sizes we care about (several thousand bits) recent works have shown NTT-based approaches, with complexity $O(\log w \log\log w \log\log\log w)$, concretely most efficient [**bhkpy22**].

we have by k-P-R-ISIS that

$$\sum_{i=1}^{t}\sum_{j=1}^{T[i]} h_{i,j} \cdot \left(\langle f_{i,j},\ \mathbf{x}_i^*\rangle - y_{i,j}\right) \equiv 0 \bmod q$$

When interactively sampling challenges from an exceptional set $\mathcal{H}$ over $\mathcal{R}_q$, we obtain that $\langle f_{i,j},\ \mathbf{x}_i^*\rangle = y_{i,j}$ for all i, j except with probability $1/|\mathcal{H}|$ by theorem 4.2.4. If instead we use a fixed R-SIS instance $\mathbf{h} \leftarrow\$ \mathcal{R}_p$ for $\alpha \ll p \ll q$, then if $\langle f_{i,j},\ \mathbf{x}_i^*\rangle \neq y_{i,j}$ for some i, j, we obtain a non-trivial R-SIS solution.

Observe that because we extract the $\mathbf{x}_i^*$ directly from the knowledge proofs, we do not need to extract the hypothetical $\pi_{0,i,j}$ inside π_0 for our security reduction. This lets use an exceptional set over $\mathcal{R}_q$, which can be exponential in size (see section 4.2.1 for an explicit construction). We can then invoke the Alon–Füredi Theorem theorem 4.2.4 to say the adversary succeeds with. On the other hand, when extraction is required current techniques rely on using exceptional sets over $\mathcal{R}$, and there's evidence such sets are all polynomial-sized for rings of interest [**C:AlbLai21**].

When using exceptional sets our interactive protocol is public-coin and we may therefore apply the Fiat–Shamir transform to once again obtain a non-interactive protocol. The interactive protocol is round-by-sound since Σ-protocols are, and since its relation (finding a non-zero polynomial such that the hash of its commitment is one of its zeros) is sparse, the non-interactive version is secure in the random oracle model [**canettichlrr18**]. Security in the standard model is also possible if there is a correlation intractable hash function family for the relation.

As introduced in [**C:ACLMT22**], it's also possible to use a public R-SIS vector $\mathbf{h}$ over a $\mathcal{R}_{c'}$ for $c' \ll q$ as a fixed challenge vector rather than sampling a fresh $\mathbf{h}$ each time from an exceptional set as above. This requires no interaction, but comes at the cost of a much larger operator norm $c = c' \cdot \gamma_{\mathcal{R}}$ versus the exceptional sets approach.

Private aggregation. If the aggregator is the prover itself, they may compute the linear combination of the $a_{i,j,k}$ values first to obtain a vector $\mathbf{a}'$ and then compute the aggregate proof $\pi_0 = \sum_i a_i' \cdot \mathbf{u}_{0,i}$ thereby reducing the amount of costly ring operations.

Inner product argument. We can easily construct an inner product argument (IPA) using $\mathsf{Orbweaver}$, allowing a prover to prove the inner product $y := \langle \mathbf{x},\ \mathbf{x}'\rangle \bmod q$ between two committed vectors. The prover runs commit twice to produce c, c', π_1, and π_1', and runs $\mathsf{Open}(\mathsf{ck}, \mathbf{x}', \mathbf{x})$ to

obtain π_0. The verifier checks the norm bounds of all three proofs, verifies both knowledge proofs as in Verify, and then checks the opening proof satisfies $\langle \mathbf{a},\ \pi_0 \rangle \equiv c' \cdot c - y \bmod q$.

It should be noted that this requires a modification of the norm bound β_0^* for which R-SIS must be hard. In the modified proof of theorem 4.4.2 the extractor would generate π_0^* using extracted values $\mathbf{x}^*$ and f^*, both of norm α^* (whereas in the linear function case they use f of norm α). Then we must set $\beta_0^* = w^2(\alpha^*)^2\beta_0\gamma_{\mathcal{R}}^2 = w^4\alpha^2\beta_1^2\beta_0\gamma_{\mathcal{R}}^4$.

Recursive proof composition. Observe that $\mathsf{vk}_f \cdot \mathsf{ck} - y \bmod q$ and $c \cdot t \bmod q$ are Ajtai R-SIS commitments to π_0 and π_1 for respective public vectors $\mathbf{a}_0$ and $\mathbf{a}_1$. Using any proof-succinct argument of knowledge for R-SIS commitment preimages (e.g., [**BS22**]), we can achieve better proof sizes by instead sending an AoK of π_0 and π_1. The composed scheme will still have log-time verification even if the outer argument has a linear-time verifier.

One may also recursively compose $\mathsf{Orbweaver}$ with itself by sending commitments to the proofs and proving they satisfy the (linear) verification equations. To deal with the norm bound checks, the only non-structure-preserving part of verification, one could send a random Johnson-Lindenstrauss projection, a linear operation which requires only constant communication, then prove it was computed correctly with respect to the committed proofs. At the 128-bit security level, an optimized version of $\mathsf{Orbweaver}$ achieves witness compression starting at sizes a little under 668KiB, or 165KiB if we forego extractability and only require evaluation binding; achieving smaller proof sizes will require switching to recursion using another (possibly even linear verifier) scheme at this point, as described above.

Linear functional commitments for integers. We can also extended $\mathsf{Orbweaver}$ to support the input alphabets $\mathcal{X} = \{z \in \mathbb{Z} \mid |z| < \alpha\}$ and, using the centered representation, $\mathcal{X} = \{z \in \mathbb{Z} \mid 0 \le z \le 2\alpha\}$. One way to do this is to use the coefficient embedding to encode $\hat{f}, \hat{\mathbf{x}} \in \mathbb{Z}^N$ as $\mathbf{x}, f' \in \mathcal{R}^w$ where $N = wn$. We further set $f = [\sigma_{-1}(f'_i)]_{i \in [w]}$, where $\sigma_{-1} \in \mathsf{aut}(\mathcal{R}_q)$ is the Galois automorphism corresponding to $\sigma_{-1}(X) = X^{-1}$ (see section 4.2). We then have that $\langle \hat{f},\ \hat{\mathbf{x}} \rangle \equiv \mathsf{ct}(\langle f,\ \mathbf{x} \rangle) \bmod q$, where ct returns the constant coefficient. We run $\mathsf{Orbweaver}$ with commitments to $f, \mathbf{x}$ and with $y = \langle f,\ \mathbf{x} \rangle$. While y is part of the instance over $\mathcal{R}$, only $\mathsf{ct}(y)$ is in the instance over $\mathbb{Z}$ in this shim protocol, while the non-constant coefficients of y are instead part of the proof for the relation over $\mathbb{Z}$.

4.4.2 Sublogarithmic proofs in the input norm

We present a functional commitment based on Orbweaver that achieves $O(\log w + \log\log\alpha)$ commitment and proof size by combining ternary decomposition and proof aggregation. The construction below is restricted to the input alphabets consisting of bounded-size balanced integers $\mathcal{X} = \{z \in \mathbb{Z} \mid |z| < \alpha\}$ and (using the centered representation) unbalanced integers $\mathcal{X} = \{z \in \mathbb{Z} \mid 0 \le z \le 2\alpha\}$. While our techniques should extend to subsets of cyclotomic rings more generally, we leave this to future work.

Let $t = \log_3(\alpha)$. Ternary decomposition allows us express a linear map over length-N (where $N = wn$, a notation switch that will soon be helpful) vectors of norm α as $2t-1$ linear maps of length-(Nt) vectors of norm 1. Using the aggregation techniques in section 4.4.1, we can create a single compact aggregate proof for k separate linear maps of size $O(\log(N\alpha k))$. For $k = 2t-1$ and $\alpha = 1$ this gives $O(\log N + \log t)$ proofs. In terms of proof norms, we trade a power of α for a higher power of $\log_3(\alpha)$. Similarly, committing to the ternary decomposition of the witness results in a $O(\log(Nt) + \log\alpha) = O(\log N + \log t)$ sized commitment. Experimentally, at the 128-bit security level decomposition and aggregation performed better starting around $\alpha = 2^{16}$ for all input lengths of interest for this scheme, i.e., for input vectors of length at least 2^{16} where we begin to achieve witness compression.

We preface our result below with a few remarks. The requirement $\alpha = 3^t$ is not strict and the proceeding construction works if we set $t = \lceil \log_3(\alpha) \rceil$. This is more than just with respect to correctness: since Orbweaver proves $\langle f,\ \mathbf{x}\rangle = y$ over the integers, it will also hold $\bmod\ \alpha$. It is always possible to include in a commitment some elements of norm larger than correctness is guaranteed for. When taking the result $\bmod\ \alpha$ by committing to these values the prover is effectively committing to their coset representative.

Theorem 4.4.4. *Let $\hat{\mathcal{X}} = \{z \in \mathbb{Z} \mid |z| \le \alpha = 3^t\}$, and for $N = wn$ let $\hat{f}, \hat{\mathbf{x}} \in \hat{\mathcal{X}}^N$. We obtain a succinct functional commitment for $\langle \hat{f},\ \hat{\mathbf{x}}\rangle = y$ where the size of c, π_1, and π_0 are all $O(\log N + \log\log\alpha)$.*

Proof (informal). We can write

$$\langle \hat{f},\ \hat{\mathbf{x}}\rangle = \sum_{i=0}^{2t-2} 3^i \langle f'_i,\ \mathbf{x}'\rangle$$

where $[f'_i]_{i=1}^t, \mathbf{x}' \in \{-1,0,1\}^{Nt}$. We then combine the techniques from "private aggregation" and "linear functional commitments for bounded integers" above to compute this using Orbweaver.

If we view each f'_i as a column vector and, abusing notation, let $f'_{i,j}$ refer to j-th (for $j \in \{0,\dots,N-1\}$) sequential t-length subvector of f'_i we see the following form

$$[f'_{0,j} \cdots f'_{2t-1,j}] = \begin{bmatrix} b_0 & b_1 & b_2 & \cdots & b_{t-1} & 0 & 0 & \cdots & 0 \\ 0 & b_0 & b_1 & \cdots & b_{t-2} & b_{t-1} & 0 & \cdots & 0 \\ \vdots & & & & & & & & \\ 0 & 0 & \cdots & 0 & b_0 & b_1 & b_2 & \cdots & b_{t-1} \end{bmatrix} \in \{-1,0,1\}^{t\times(2t-1)}$$

where $[b_k]_{k=0}^{t-1}$ is the ternary decomposition of $\hat{f}_j$. For simplicity, let $n = rt$ (otherwise, padding may be applied) for $r \in Z^+$, thus supporting decomposition for norm bounds up to 3^{512} (where, again, modifications enable more generality). Let $f_i \in \mathcal{R}^{wt}$ denote the coefficient embedding of f'_i. Then we have for

$$\tilde{f} = \sum_{i=0}^{2t-2} h_i f_i$$

that $\left\|\tilde{f}_j\right\| \leq \mathsf{min}\{c,r\} \cdot t^2 \leq ct^2$ for all $j \in [wt]$, where c is the operator norm of the challenge set $\mathcal{H}$ used. This follows from the fact we are taking the sum over $2t-1$ vectors $h_i f_i$, where when averaging over the i the norm bound of each element is $\mathsf{min}\{c,r\} \cdot t^2/(2t-1)$. The norm of $\tilde{f}$ is now larger than $\mathbf{x} \in \mathcal{R}^{wt}$, and for a more tight result we recompute the norm bounds with this in mind.

First, recall $y_i = \langle f_i,\ \mathbf{x}\rangle$. For $i \in \{0,\dots,t-1\}$ we then obtain

$$\|y_i\| \leq (i+1)rwt = (i+1)N = \delta_{M,i}$$

and for $i \in \{t,\dots,2t-1\}$ we have

$$\|y_i\| \leq (2n-(i+1)r)wt = (2t-i-1)N = \delta_{M,i}$$

Next, $\pi_1 = \sum_{i=1}^w x_i \mathbf{u}_{1,i} \bmod q$, where $\left\|x_i \mathbf{u}_{1,i}\right\| \leq \beta_1 \gamma_{\mathcal{R}}$ for all i. It follows

$$\|\pi_1\| \leq wt\beta_1\gamma_{\mathcal{R}} = Nt\beta_1 = \delta_1$$

Then, $\pi_0 = \sum_{k\in P} a_k \cdot \mathbf{u}_{0,k} \bmod q$, where $a_{-wt+1}, \dots, a_{wt-1}$ denote the $2wt-2$ coefficients of $\mathbf{x}(v)\tilde{f}(v) =$

$\left(\sum_{j=1}^{wt} x_j v^j\right)\left(\sum_{j=1}^{wt} \tilde{f}_j v^{-j}\right)$ and $P := \{-wt+1, \dots, wt-1\} \setminus \{0\}$. Since $\|\mathbf{u}_{0,k}\| \leq \beta_0$ for all k, $\|\pi_0\| \leq \beta_0 \gamma_{\mathcal{R}} \sum_{k \in P} a_k$. Then $\|a_{|k|}\| \leq (wt-k)ct^2\gamma_{\mathcal{R}}$ for $k \in [w-1]$. We then have

$$\sum_{k \in P} a_k = 2ct^2\gamma_{\mathcal{R}} \sum_{\kappa=1}^{wt-1} (wt-\kappa) = ct^2\gamma_{\mathcal{R}}(w^2t^2 - wt)$$

Thus,

$$\begin{aligned} \|\pi_0\| &\leq \beta_0 ct^2\gamma_{\mathcal{R}}^2((N/n)^2t^2 - (N/n)t) = N^2ct^4\beta_0 - Nct^3\beta_0 n \\ &= \delta_0 \\ &\leq (N^2 - N)ct^4\beta_0 \leq N^2ct^4\beta_0 \end{aligned}$$

Last, we consider δ_0^*. Replacing, $\mathbf{x}$ with the extracted $\mathbf{x}^*$ in the analysis above, where $\|x_j^*\| \leq \delta_1 = Nt\beta_1$ for all $j \in [wt]$, gives

$$\|\pi_0^*\| \leq Nct^3\beta_0\beta_1\gamma_{\mathcal{R}}^2((N/n)^2t^2 - (N/n)t) = N^3ct^5\beta_0\beta_1 - N^2ct^4\beta_0\beta_1 n$$

Finally,

$$\begin{aligned} \|\pi_0 - \pi_0^*\| &\leq N^2ct^4\beta_0 - Nct^3\beta_0 n + N^3ct^5\beta_0\beta_1 - N^2ct^4\beta_0\beta_1 n \\ &= \delta_0^* \\ &\leq N^3ct^5\beta_0\beta_1 \end{aligned}$$

□

4.4.3 Polynomial commitments for integers

We show how to construct $O(\log N)$ proof size and $O(\log N \log\log N)$ verifier time polynomial commitment (PC) for both degree-N univariate polynomials and N-linear polynomials over $\mathbb{Z}$ using Orbweaver. How to do this while preserving a quasilog-verifier is not immediate. In a PC protocol based on Orbweaver, the prover commits to the coefficients of the polynomial and the function f becomes the powers of a challenge evaluation point z. Previously we considered the preprocessing setting, where we assume the same function will be evaluated repeatedly on different commitments. In this amortized setting the cost to run PreVerify once is considered a constant. On the other hand, in the PC setting f is defined by z, which is usually a fresh challenge sampled uniformly at random from $\mathbb{Z}_M$—a setting in which preprocessing no longer makes sense.

We present the univariate degree-N PC. Omitting a couple details, by replacing z^{2^i} with z_i for all i below we obtain the N-linear version. Again, we choose $\alpha \geq (M-1)/2$ and assume wlog that

$\alpha = 3^t$, $wn = N$, and w is a power of 2. We show how to compute, in log-time, a commitment key $\mathsf{ck}'_z \equiv \mathsf{ck}_z \bmod M$, where ck_z is the commitment key corresponding to running $\mathsf{PreVerify}$ on the function $f = (f_1, ..., f_w) \in \mathbb{Z}_M^w$, where $\sigma_{-1}(f_i) \in \mathcal{R}_q$ is equal to the coefficient embedding of the vector $z^{n(i-1)} \cdot (z^0, z^{1+n(i-1)}, \dots, z^{n-1}) := z^{n(i-1)} \cdot f_\heartsuit \in \mathbb{Z}_M^n$. This choice of f follows the techniques we used in section 4.4.2 such that the evaluation of the committed polynomial at z will be equal to the constant coefficient of y. The verifier computes

$$\mathsf{ck}'_z := f_\heartsuit \cdot v^{-w} \cdot \prod_{i=0}^{\log(w)-1} ((z^{n \cdot 2^i} \bmod M) + v^{2^i}) = \sum_{i=1}^{w} f_\heartsuit \cdot \hat{z}_{i-1} \cdot v^{-i} \equiv \mathsf{ck}_z \bmod M$$

In particular this holds because each $\hat{z}_i \equiv z^{in} \bmod M$. The downside of computing the key this way is that it corresponds to a commitment key for the function $\hat{f} := f_\heartsuit \cdot \hat{z} \in \mathcal{R}_q^w$, where $\left\|\hat{f}\right\| \leq \alpha^{\log w+1}$ since the norm of $\hat{z}_i$ is bounded by α to the power of the Hamming weight of i. Roughly speaking, computing the commitment key this way comes at a cost of a $\log \alpha$ times larger opening proof.

4.5 Evaluation

We used SageMath to compute proof and CRS sizes for an optimized version $\mathsf{Orbweaver}$ for integers in section 4.5.2, and in section 4.5.2 we compare an unoptimized version for cyclotomic rings to previous work [**C:ACLMT22**]. Before getting there, we describe the additional optimizations in section 4.5.1.

4.5.1 Implementation optimizations

Sending only π_1 and not c. We can slightly reduce communication and both prover and verifier complexity by only sending π_1 and not c. The verifier first computes $\langle \mathbf{a}_1,\ \pi_1 \rangle = c_t$, then checks $c_t \in \langle t \rangle$,[5] and verifies

$$\langle \mathbf{a}_0,\ \pi_0 \rangle \cdot t = c_t \cdot \mathsf{ck}_f - z \cdot t$$

[5]Recall that we pick t by setting half its NTT coefficients t_i to 0 and picking the rest at random from $\mathbb{Z}_q$. The condition $b \in \langle t \rangle$ is equivalent to $b_i \in \langle t_i \rangle$ (in the NTT basis), which can be verified by checking $b_i = 0$ for all i such that $t_i = 0$ since $b_i, t_i \in \mathbb{F}_q$.

as well as the standard norm checks. This works because

$$t(\langle \mathbf{a},\ \pi_0\rangle - c \cdot \mathsf{ck}_f + z) \equiv 0 \bmod q$$

implies

$$\langle \mathbf{a}_0,\ \pi_0\rangle = c \cdot \mathsf{ck}_f - z$$

by assumption (2) Since c is a deterministic function of π_1, an extractor can still obtain $\mathbf{x}^*$ given π_1. To see the evaluation proof check still holds, note that $t \leftarrow\$ \mathcal{T}$ is sampled such that finding $s' = 0$ and $s't = 0$ is hard. This implies that if $lt = rt$, then $l - r = 0$.

Distinct vs. equal a_0 and a_1. We choose to fix a single public R-SIS vector $\mathbf{a}$ with respect to which preimages for v^i and $v^i t$ are generated, as this enables better verifier complexity. We note that with distinct trapdoor SIS vectors $\mathbf{a}_0, \mathbf{a}_1$, we can further optimize proof size, but only very slightly in our experiments.

The most expensive part for the verifier is computing $\langle \mathbf{a}_0,\ \pi_0\rangle$ and $\langle \mathbf{a}_1,\ \pi_1\rangle$, which requires $O(n \log n \cdot (\ell_0 + \ell_1))$ multiplications between the arbitrary $\mathcal{R}_q$ elements in the $\mathbf{a}_i$ and the bounded norm elements in the π_i. For concrete parameters, we believe these multiplications might be computed most efficiently using an NTT-based algorithm [**bhkpy22**]. Either way, by fixing a single $\mathbf{a} = \mathbf{a}_0 = \mathbf{a}_1$ the verifier runtime can be nearly halved using the verification equation

$$\langle \mathbf{a},\ (\mathsf{ck}_f \cdot \pi_1 - t \cdot \pi_0)\rangle = z \cdot t$$

Computationally indistinguishable trapdoor matrix. Recall that trapdoor public RSIS vectors in [**EC:MicPei12**] are of the form $\mathbf{a} = [\hat{\mathbf{a}} | \mathbf{g} - \mathbf{r}\hat{\mathbf{a}}]$, where $\hat{\mathbf{a}} \leftarrow\$ \mathcal{R}_q^\ell$, $\mathbf{g}$ is the public gadget vector and $\mathbf{r}$ is the trapdoor/ In prior work [**C:ACLMT22**], $\ell = \lceil \log_\beta q \rceil$ was chosen, where it can be shown using the leftover hash lemma the resulting $\mathbf{a}$ is indistinguishable from random. We can instead use $\ell = 2$, where one can see the resulting $\mathbf{a}$ is an RLWE sample, $[\hat{\mathbf{a}}, \mathbf{g} - \hat{A}\mathbf{r}]$ where $\hat{\mathbf{a}} = [1 | \hat{\mathbf{a}}^\top \leftarrow \mathcal{R}_q^{\log_b q}] \in \mathcal{R}_q^{\log_b q \times 2}$, for some $b = O(\beta)$ and $\mathbf{r} \leftarrow\$ \mathcal{D}$ where $\mathcal{D}$ is some distribution for the RLWE secret and error. Experimentally, we find the best distribution, b, β such that the proof size is minimized and the RLWE problem has 128-bit hardness.

Other trapdoor optimizations. We only consider one optimization to the computationally

indistinguishable variant of [**EC:MicPei12**]: the heuristic from [**gmpw20**] for computing s_1. However, works focused on improving lattice signature sizes have introduced optimizations we estimate together could reduce our proof sizes by 5–10× [**cgm19**; **dsh21**; **jht22**]. We leave parameterizing such variants as future work.

4.5.2 Proof and CRS sizes

In the tables below we present results for our optimized bounded integer maps for input alphabets $\mathcal{X}_1 = \{0, 1, 2\}$ and $\mathcal{X}_{32} = \{0, \ldots, 2^{32} + 1\}$, both with and without extractability, at the 128 bit classical security level. Our proof sizes are competitive with other non-recursive post-quantum arguments, especially for witnesses of length $> 2^{25}$. The verifier keys remain very small even up to 2^{30}, while for larger instances the prover key grows quite large.

Sparse inputs. We note that the proof sizes below are actually upper bounds. When working with sparse f and/or $\mathbf{x}$, as is common in practice, the proof size will be dependent on the number of non-zero entries

$\mathcal{X}_{32}$, extractable	2^{18}	2^{19}	2^{20}	2^{21}	2^{22}	2^{23}	2^{24}	2^{25}	2^{26}	2^{27}	2^{28}	2^{29}	2^{30}
commitment (B)	147	167	154	163	173	188	185	200	211	227	243	236	252
total proof size (KiB)	845	904	977	1,028	1,081	1,137	1,197	1,257	1,315	1,376	1,438	1,505	1,571
witness compression	1.2×	2.3×	4.2×	8.0×	15×	28×	54×	104×	199×	381×	729×	1,394×	2,670×
verifier key size (MiB)	28	29	31	32	33	34	36	37	38	40	41	42	43
prover key size (GiB)	0.3	0.7	1.3	2.8	5.9	12.8	25.3	53.2	111	234	494	982	2,070

$\mathcal{X}_{32}$, binding	2^{17}	2^{18}	2^{19}	2^{20}	2^{21}	2^{22}	2^{23}	2^{24}	2^{25}	2^{26}	2^{27}	2^{28}	2^{29}	2^{30}
commitment (B)	51	56	57	60	65	66	71	75	76	82	83	92	93	95
total proof size (KiB)	264	278	294	311	327	344	360	379	398	415	434	453	474	494
witness compression	1.9×	3.7×	7.0×	13×	25×	48×	91×	173×	329×	632×	1207×	2314×	4427×	8483×
verifier key size (MiB)	1.7	1.9	2.1	2.2	2.4	2.6	2.8	3.0	3.2	3.4	3.7	3.9	4.2	4.4
prover key size (GiB)	< 0.1	0.1	0.2	0.3	0.7	1.5	3.1	6.6	13.5	28.2	57.7	124	253	517

$\mathcal{X}_1$, extractable	2^{22}	2^{23}	2^{24}	2^{25}	2^{26}	2^{27}	2^{28}	2^{29}	2^{30}
commitment (B)	106	118	129	145	162	188	166	180	178
total proof size (KiB)	668	712	757	805	862	931	986	1,038	1,096
witness compression	1.2×	2.3×	4.3×	8.1×	15×	28×	53×	100×	190×
verifier key size (MiB)	2.6	2.8	3.0	3.2	3.4	3.7	3.9	4.2	4.4
prover key size (GiB)	3.8	8.3	17.7	38.4	83.7	187	360	764	1,545

$\mathcal{X}_1$, binding	2^{20}	2^{21}	2^{22}	2^{23}	2^{24}	2^{25}	2^{26}	2^{27}	2^{28}	2^{29}	2^{30}
commitment (B)	35	38	42	42	49	50	51	55	59	63	65
total proof size (KiB)	165	178	189	202	215	229	243	256	272	286	302
witness compression	1.2×	2.3×	4.3×	8.0×	15×	28×	54×	101×	191×	364×	689×
verifier key size (MiB)	1.0	1.1	1.3	1.4	1.5	1.7	1.8	1.9	2.1	2.3	2.5
prover key size (GiB)	0.2	0.5	1.0	2.1	4.5	9.3	19.1	40.5	85.5	180	371

Comparison to [C:ACLMT22]

For sake of better comparison with [**C:ACLMT22**], in Figure 4.2 we consider an unoptimized version of Orbweaver (as trit decomposition in section 4.4.2, better trapdoor sampling, and many other types of optimizations in section 4.5.1 can also be applied to [**C:ACLMT22**]). We also note that due to errors in their code, they (1) set a root Hermite factor much smaller than currently achievable (resulting in a security level higher than 128 bits); and (2) overestimate the desired preimage quality, i.e., the shortness/norm bound of the preimages. In the comparison below, we apply the same errors that "oversecure" our scheme for $\lambda = 128$.

In order to make the comparison as fair as possible, we also modified the code of [**C:ACLMT22**] to remove the overhead induced by their scheme's support for batching and higher degree polynomial maps. While our unoptimized proof sizes are about $2 - 3\times$ smaller, perhaps even more important practically was our reduction of the CRS size and prover time from quadratic to quasilinear in w and α. The Orbweaver CRS size is calculated as $3wn \log \beta$ bits, as it consists of $3w$ vectors of ring elements, each of which with bit length bounded by by $\log(\beta)$. In contrast, [**C:ACLMT22**] requires about $w^2 \ell n \log \beta$ bits for the SRS.

Acknowledgments

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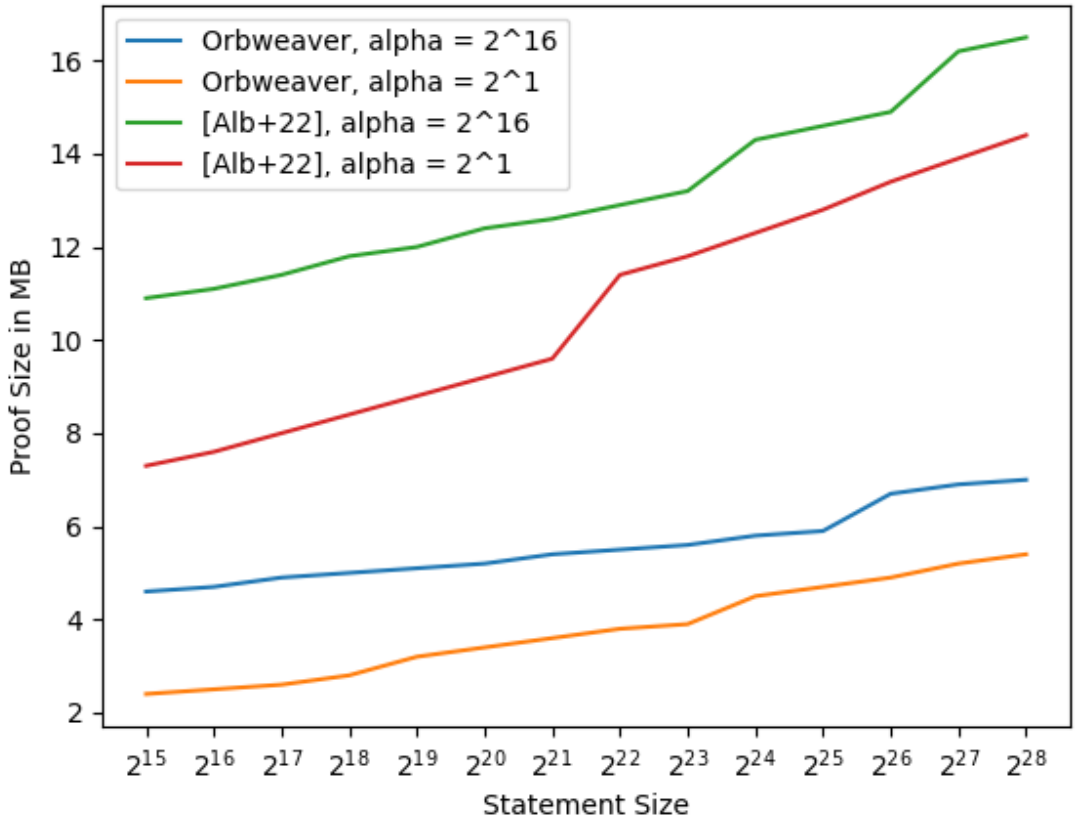

Figure 4.2: Comparison of combined commitment and proof size for an unoptimized version of Orbweaver and [**C:ACLMT22**] with $n \geq 2^9$. We have purposely left errors in the code when producing the data in this figure for better comparison with [**C:ACLMT22**] (see details at the start of section 4.5.2).

.1 Sumcheck algorithms

This section includes an optimized sumcheck protocol in Figure 3 and a linear time and space prover algorithm for it in Figure 4, both from from [**BagadDaoDombThaler25**]. We include these both for self-containedness and because these particular protocols provide a launching off point for some further sumcheck optimizations introduced in our paper.

In the following version of sumcheck the polynomial s_i in each round i is specified by its evaluation on the $d+1$ points $\widehat{U}_d = \{\infty\} \cup \{0, \dots, d\}$ (where the point ∞ is explained just below). The evaluation $s_i(1)$ can actually be omitted, using the equation $s_i(1) = \sigma_{i-1} - s_i(0)$.

For a degree d univariate polynomial $s(X) = a_d X^d + \cdots$ we use a variant of Lagrange interpolation

where the highest degree coefficient a_d is sent in addition to evaluations of the polynomial $t(X) = s(X) - a_d \cdot \prod_{k=0}^{d-1}(X - x_k)$ over the domain $\{0, \dots, d\}$. We note that $t(X)$ is of degree $d-1$, and equal to $s(x)$ over $\{0, \dots, d\}$. Defining $s(\infty) = a_d$, we recover s via evaluations of

$$s(X) = a_d \cdot \prod_{k=1}^{d}(X - x_k) + \sum_{k=1}^{d} t(x_k) \cdot \mathcal{L}_{\{0,\dots,d\},k}(X)$$

over $\widehat{U}_d$, where $\mathcal{L}_{\{0,\dots,d\},k}$ is the kth Lagrange basis polynomial over the interpolation set $\{0, \dots, d\}$.

If $s(X) = \prod_{i=1}^{d} p_i(X)$ is a product of linear polynomials, then

$$s(\infty) = \prod_{i=1}^{d} p_i(\infty) = \prod_{i=1}^{d}(p_i(1) - p_i(0)) \quad .$$

More generally, when $s(X) = G(p_1(X), \dots, p_d(X))$ for some polynomial G and linear polynomials $p_1, \dots, p_d$, we can calculate $s(\infty)$ by evaluating only the terms of G of highest total degree. Then, like $s(1)$, the prover can generally omit sending the evaluation $s(\infty)$, instead computing it using the equation above.

Figure 3: The sumcheck protocol

Setup: Assume that the verifier has oracle access to the polynomial g, and knows the value σ_0 claimed to equal

$$\sum_{x \in \{0,1\}^v} g(x) = \sigma_0.$$

For each round $i = 1, \dots, v$:

1. The prover P sends the univariate polynomial $s_i(X)$ claimed to equal

$$\sum_{(x_{i+1},\dots,x_v) \in \{0,1\}^{v-i}} g(\alpha_1, \dots, \alpha_{i-1}, X, x_{i+1}, \dots, x_v).$$

   P does so by sending the evaluations $\{s_i(u) : u \in \widehat{U}_d\}$ to the verifier V.

2. The verifier V sends a random challenge $\alpha_i \leftarrow \mathbb{F}$ to P.

3. V derives $s_i(1) := \sigma_{i-1} - s_i(0)$, then sets $\sigma_i := s_i(\alpha_i)$.

After v rounds, we reduce to the claim that

$$g(\alpha_1, \dots, \alpha_v) = \sigma_v.$$

V checks this claim by making a single oracle query to g.

A linear time and space prover for this sumcheck version may be implemented as follows. The prover maintains d arrays $P_1, \dots, P_d$, which initially store all evaluations of $p_1, \dots, p_d$ over $\{0,1\}^v$; we have $P_k[x] = p_k(x)$ for all $k \in [d]$ and $x \in \{0,1\}^v$. The prover halves the size of the arrays after each round i, via "binding" the arrays $[P_k]_{k \in [d]}$ to the challenge α_i received from the verifier.

Figure 4: A linear time and space prover implementation of sumcheck

Setup: Length-2^v arrays $P_1^{(0)}, \dots, P_d^{(0)}$ such that

$$P_k^{(0)}[x] = p_k(x) \text{ for all } k = 1, \dots, d \text{ and } x \in \{0,1\}^v.$$

For each round $i = 1, \dots, v$:

1. For $u \in U_d^b$, compute $s_i(u)$ using the formula

$$s_i(u) := \sum_{x' \in \{0,1\}^{v-i}} \prod_{k=1}^{d} \Big((P_k^{(i)}[1, x'] - P_k^{(i)}[0, x']) \cdot u + P_k^{(i)}[0, x'] \Big).$$

2. Send $\{ s_i(u) : u \in U_d^b \}$ to the verifier.
3. Receive challenge $\alpha_i \in F$ from the verifier.
4. For $k = 1, \dots, d$ and $x' \in \{0,1\}^{v-i}$, update arrays:

$$P_k^{(i+1)}[x'] := (P_k^{(i)}[1, x'] - P_k^{(i)}[0, x']) \cdot \alpha_i + P_k^{(i)}[0, x'].$$

`// Invariant:` $P_k^{(i)}[x'] = p_k(r[1:i], x')$ for all $k \in [d]$, $i \in [0, v]$, $x' \in \{0,1\}^{v-i}$.